Book II of Euclid's Elements

Formal Reconstruction — Technical Documentation

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Document Status

This volume is the technical documentation on formal reconstruction of Book II of Euclid's *Elements*. It records the mathematical architecture of the reconstruction, the Lean formalization, the dependency structure of individual propositions, the reusable rectangle and scissors-congruence infrastructure, and the relation between the formal development and the classical sources.

The documentation accompanies the formal development. The Lean source is authoritative for theorem statements, hypotheses, dependencies, and proof status.

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Proposition II.1

1.1 Introduction

Proposition II.1 opens Book II with a change of mathematical language. Book I has developed a synthetic theory of congruence, parallels, parallelograms, squares, and equal content. Book II begins to combine these ingredients in a systematic calculus of rectangles. A modern reader immediately recognizes in II.1 the pattern of a distributive law. That recognition is useful, but it must be stated carefully: Euclid does not introduce variables for numerical lengths, does not define multiplication of lengths, and does not prove an identity in a number field.

The primitive object in the proposition is a geometric figure: a rectangle "contained by" two straight lines. When one of the two lines is divided into parts, the large rectangle is divided into smaller rectangles having the same other side. The equality asserted by the proposition is therefore first of all a statement about decomposition of figures.

This is a natural point at which to speak of the beginning of the *geometrical algebra* characteristic of Book II. The phrase is an interpretive description of a stable pattern of geometric operations, not a claim that Euclid was secretly manipulating modern symbolic formulas. In the present formal reconstruction this distinction is especially sharp: the code contains rectangles, segment congruence, betweenness, parallelograms, right angles, and scissors relations, but no multiplication operation on segment lengths and no numerical area function.

The formal development separates three levels:

1. a concrete rectangle and its geometric cuts;
2. the Euclidean phrase "rectangle contained by" two given segments;
3. independence of the particular rectangle chosen to represent that phrase.

The final theorem is stronger than a bare equal-content statement. Its conclusion is the exact relation

```
HilbertScissorsEq
```

for the whole rectangle and the formal sum of its parts.

1.2 Euclid's "Rectangle Contained By"

1.2.1 The original geometric meaning

Book II begins with the terminology that a rectangular parallelogram is "contained by" the two straight lines which contain one of its right angles. Thus the expression does not initially denote a product of two numbers. It names a rectangle by two adjacent sides.

Suppose a rectangle has adjacent sides congruent to two given segments PQ and RS. Euclid's phrase

$$\text{the rectangle contained by } PQ, RS$$

should be read geometrically: choose a right-angled parallelogram whose two adjacent sides have those magnitudes. The rectangle itself remains a plane figure.

For exposition only, we will occasionally write

$$\mathrm{Rect}(PQ, RS).$$

This is documentary shorthand. It is not a Lean definition, it is not a number-valued function, and the comma is not multiplication.

1.2.2 Why this already looks algebraic

If BC is divided into consecutive parts BM, MN, and NC, II.1 says in this shorthand

$$\mathrm{Rect}(BC, PQ) \; = \; \mathrm{Rect}(BM, PQ) + \mathrm{Rect}(MN, PQ) + \mathrm{Rect}(NC, PQ).$$

A modern algebraist naturally sees the shape

$$(b_1 + b_2 + b_3)a = b_1 a + b_2 a + b_3 a.$$

The formal proof, however, establishes the geometric statement on the first line, not the numerical identity on the second. The second formula is only a modern mnemonic for the combinatorial pattern of the dissection.

This is consistent with the modern synthetic reading of Euclid's area theory. Hartshorne treats Book II as a collection of equal-content statements about figures and describes the resulting manipulation as "geometrical algebra." The point is precisely that the manipulation precedes any assignment of numerical areas to the figures.

1.3 The Synthetic Rectangle Layer

1.3.1 Rectangles

The reconstruction introduces a small reusable rectangle API in `HilbertRectangle.lean`. A rectangle is a parallelogram with one specified right angle:

```
def IsRectangle
    (A B C D : Geo.Point) : Prop :=
  IsParallelogram Geo A B C D /\
  HilbertRightAngle Geo A B C
```

The parallelogram structure supplies the opposite-side relations, while the right-angle component distinguishes rectangles from arbitrary parallelograms. The remaining right angles can then be propagated synthetically.

1.3.2 The formal meaning of "contained by"

The Euclidean phrase is represented by the predicate

```
def IsRectangleContainedBy
    (A B C D P Q R S : Geo.Point) : Prop :=
  IsRectangle Geo A B C D /\
  Geo.Congruent A B P Q /\
  Geo.Congruent B C R S
```

Thus

```
IsRectangleContainedBy Geo A B C D P Q R S
```

means exactly that $ABCD$ is a concrete rectangle, $AB \cong PQ$, and $BC \cong RS$. Nothing in this definition assigns a number to PQ or RS. The two given segments specify the two adjacent geometric magnitudes of a representative rectangle.

The layer also proves existence of such a representative and, more importantly for the semantics of Book II, uniqueness of its scissors content. The theorem

```
rectangle_contained_by_unique
```

says that any two rectangles contained by the same ordered pair of segments satisfy the exact scissors relation. Consequently Euclid's definite description "the rectangle contained by PQ, RS" becomes well-defined at the scissors level even though many different concrete quadrilaterals can represent it.

1.3.3 No segment multiplication and no numerical area

There are two tempting modernizations that the present reconstruction explicitly avoids.

First, we do not define a product of segment lengths $|PQ|\,|RS|$. The pair (PQ, RS) remains a pair of geometric segments, and the rectangle is related to that pair by congruence of adjacent sides.

Second, we do not map a rectangle to a real number called its area. The content of the proposition is expressed through relations between finite formal sums of triangles. This permits addition, dissection, transport, and comparison of figures at the synthetic level.

This is also conceptually distinct from Hilbert's later arithmetic of segments, where arithmetic operations are introduced as an additional construction. No such field or segment-arithmetic layer is imported into II.1.

1.4 Exact Scissors Equality and Equal Content

The project uses two related but different content relations. They should not be conflated.

1.4.1 Exact scissors equality

`HilbertScissorsEq` records an exact finite cut-and-paste relation between formal scissors terms. In II.1 it is produced by explicit splits of the rectangle into smaller rectangles and by normalization of the triangulated parallelogram terms. Schematically,

$$P \simeq_{\mathrm{sc}} Q + R$$

means that the two sides can be matched by the exact scissors machinery already developed in the project.

The rectangle transport theorem is already available in this stronger form:

```
rectangle_transport_scissorsEq
```

for rectangles with congruent corresponding adjacent sides.

1.4.2 The more general equal-content relation

The broader project notion corresponding to equal content is

```
HilbertScissorsEquicomplementable
```

It allows suitable complementary terms to be adjoined before exact scissors equality is established. Thus it is designed to capture a more general synthetic equal-content relation.

The theorem

```
rectangle_transport
```

is obtained by weakening the exact transport theorem from `HilbertScissorsEq` to equicomplementability.

For II.1 no such weakening is necessary. The final theorem proves `HilbertScissorsEq` itself. Therefore II.1 certainly yields an equal-content consequence, but its formal statement retains the stronger fact that the displayed equality comes from an exact dissection.

1.5 The Classical and Formal Configurations

1.5.1 The classical three-part picture

Euclid's source proof starts from one undivided straight line and a second straight line divided into consecutive parts. The undivided magnitude fixes the common transverse side of the rectangle. Parallels through the division points carry the partition across the rectangle, so the three resulting subrectangles literally constitute the whole.

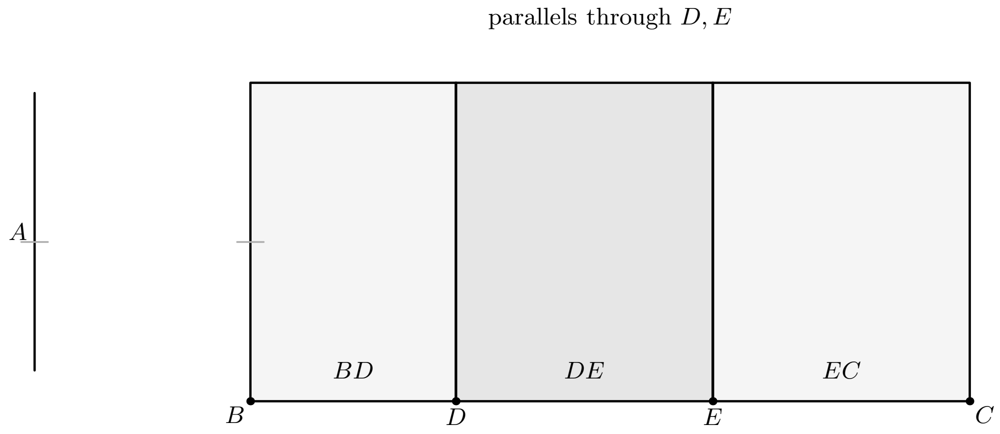


Figure 1.1: The source-level configuration of Euclid II.1. The separate straight line A supplies the common transverse magnitude. The other line is divided in the order $B - D - E - C$, and the parallels through D and E partition the outer rectangle into the three rectangles corresponding to BD, DE, and EC. No diagonal intersection witnesses are part of Euclid's local argument.

1.5.2 The formal binary split

The reusable core is a one-cut theorem. Let $B - M - C$ and let the corresponding point L divide the opposite side DE. The rectangle $BCED$ is cut into the rectangles $BMLD$ and $MCEL$:

$$BCED \simeq_{\text{sc}} BMLD + MCEL.$$

In Lean the theorem is

```
rectangle_split
```

and it returns both the rectangle structure of the two pieces and the exact scissors equality.

The formal configuration contains one additional point X. It satisfies $D - X - C$ and $M - X - L$. This is not decorative data: the inherited exact split lemma from the I.47 scissors infrastructure uses the intersection of the cut line with a diagonal. The classical rectangle picture suppresses this witness; the formal theorem makes it explicit.

1.5.3 The formal three-part split

The three-part form applies the binary split twice. First cut at $M - L$; then cut the remaining rectangle at $N - K$. The result is

$$\begin{aligned} BCED \simeq_{\text{sc}} & \ BMLD \\ & + MNKL \\ & + NCEK. \end{aligned}$$

The second split has its own diagonal-intersection witness Y, with $L - Y - C$ and $N - Y - K$.

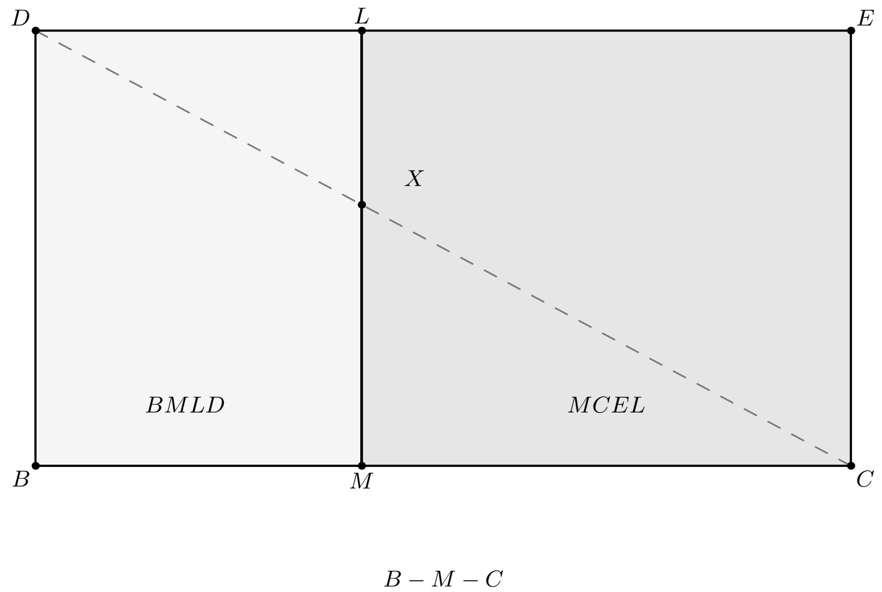
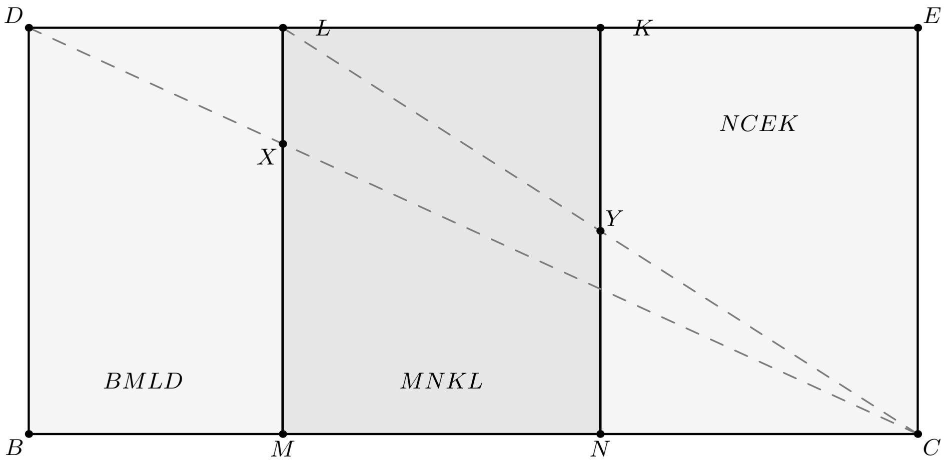


Figure 1.2: The geometric state consumed by `rectangle_split`. The strict order $B-M-C$ and the cut ML divide $BCED$ into the two rectangles $BMLD$ and $MCEL$. The dashed diagonal DC meets the cut at X, recording the hidden exact-dissection witness required by the inherited scissors interface.



first cut $M - L$; second cut $N - K$

Figure 1.3: The second genuinely new formal state. After the first cut $M - L$, the right-hand remainder is cut again at $N - K$. The three concrete rectangles are $BMLD$, $MNKL$, and $NCEK$. The dashed diagonals display the two hidden witnesses X and Y used by the two successive exact split calls.

The actual three-part theorem is

```
rectangle_split_three
```

Its proof is structurally transparent: obtain the first binary equality, split the remaining right-hand rectangle, add the unchanged left piece to that second equality, and compose by transitivity of `HilbertScissorsEq`.

Diagrammatic audit. Figures 1.1–1.3 exhaust the geometric states that need separate pictures. The source construction and the formal binary and iterated cuts are genuinely different configurations. By contrast, attaching the "contained by" predicates only transports the fixed side magnitude through already drawn rectangles, while representative independence replaces one representative by another without creating a new cut. Those later steps are therefore documented in prose rather than by new figures.

1.5.4 Scope of the present formal statement

Euclid states II.1 for one line cut into an arbitrary finite number of parts. The current Lean file does not yet package an n-ary list-valued theorem. It proves the binary split and the explicit three-part form used in the standard II.1 configuration. The binary theorem is the reusable kernel from which a general finite version could later be obtained by iteration.

This is a deliberate statement-fidelity point. The documentation records Euclid's general formulation, but the final formal statement displayed below is the representative-independent three-part theorem actually present in `Proposition2_1.lean`.

1.6 From a Concrete Diagram to Euclid's Language

The geometric split alone produces concrete rectangles. Euclid II.1 is more abstract: it speaks of the rectangles *contained by* the whole segment and the several parts together with one fixed segment.

The theorem

```
rectangle_split_three_contained_by
```

bridges these levels. If $CE \cong PQ$, it packages the four concrete rectangles as

$$\begin{array}{rcl} BCED & : & \mathrm{Rect}(BC, PQ), \\ BMLD & : & \mathrm{Rect}(BM, PQ), \\ MNKL & : & \mathrm{Rect}(MN, PQ), \\ NCEK & : & \mathrm{Rect}(NC, PQ). \end{array}$$

The opposite-side congruences of the parallelograms are what transport the fixed side PQ through the two cuts. The conclusion still contains the same exact scissors equality between the concrete quadrilateral terms.

At this stage the geometry corresponding to Euclid's wording is visible, but the theorem still uses one particular drawn rectangle and its particular cut points. The final step removes this representational dependence.

1.7 Representative Independence

1.7.1 Why representatives matter formally

A phrase such as

$$\mathrm{Rect}(BC, PQ)$$

does not identify four vertices. Lean, however, must manipulate an actual term such as

```
hilbertParallelogramTerm Geo U0 U1 U2 U3
```

with named vertices. The final theorem therefore accepts arbitrary representatives for the whole rectangle and for each of the three part rectangles.

The concrete dissection first proves

$$BCED \simeq_{\mathrm{sc}} BMLD + MNKL + NCEK.$$

Then four calls to

```
rectangle_contained_by_unique
```

transport the whole and the three pieces to arbitrary representatives U, V, W, and Z. Compatibility of `HilbertScissorsEq` with formal addition transports the nested sum. Transitivity completes the chain

$$\begin{gathered}\text{arbitrary whole representative}\\ \Downarrow\\ \text{concrete whole rectangle}\\ \Downarrow\\ \text{concrete three-part dissection}\\ \Downarrow\\ \text{arbitrary representatives of the three parts.}\end{gathered}$$

This is the formal content of representative independence. It is not merely a convenience for Lean. It explains why the Euclidean expression “the rectangle contained by two segments” can function as a stable geometric magnitude without first quotienting segments into numerical lengths or assigning numerical areas to rectangles.

1.8 Commutativity by Rectangle Rotation

The second algebra-like feature is symmetry in the two containing segments. For a numerical product one would write $ab = ba$. The synthetic proof does not appeal to a commutative multiplication law. Instead it rotates the presentation of a rectangle.

If $ABCD$ is contained by PQ, RS, then the cyclically reoriented rectangle $BCDA$ has adjacent sides BC and CD. The first is already congruent to RS; the second is congruent to AB, hence to PQ. Thus the same geometric rectangle, with its vertices read from the next corner, is contained by RS, PQ.

The exact theorem available in the rectangle layer is

```
rectangle_contained_by_swap
```

It returns both the swapped `IsRectangleContainedBy` fact and a `HilbertScissorsEq` identifying the two quadrilateral presentations. The current rectangle API also packages the representative-independent consequence as

```
rectangle_contained_by_comm
```

so the commutativity needed by later Book II propositions is a theorem of rectangle geometry rather than an assumed law of segment multiplication.

Diagrammatic audit. No separate picture is retained for this step. The swap operation does not change the geometric rectangle or introduce a new cut: it changes only which vertex is read first, and hence which adjacent side is assigned the first role in `IsRectangleContainedBy`. Drawing the same quadrilateral twice would therefore turn a representation normalization into an apparent new geometric state. The cyclic order change is recorded by the theorem statement and the exact scissors transport instead.

This lemma is already operational in the next proposition. II.2 naturally writes rectangles in the order “whole side, part”, whereas the II.1 cut lemma produces them in the order “part, fixed side”. The implementation of II.2 uses the swap theorem to pass between these two presentations before applying the binary form of II.1.

1.9 Proof Architecture of the Reconstruction

The Lean proof is best read as five mathematical stages.

1.9.1 Step 1: exact binary dissection

```
rectangle_split_scissorsEq
rectangle_split_left
rectangle_split_right
rectangle_split
```

The inherited I.47 split theorem supplies the exact scissors decomposition. The two local rectangle helpers prove that each piece remains a rectangle by propagating right angles through the relevant parallelograms and along the same rays determined by betweenness. The resulting one-cut state is exactly Figure 1.2.

1.9.2 Step 2: iterate the split

```
rectangle_split_three
```

The right-hand remainder of the first split is itself a rectangle, so the same binary theorem can be applied again. Addition and transitivity of exact scissors equality compose the two decompositions. The only new geometry is the second cut shown in Figure 1.3.

1.9.3 Step 3: attach the "contained by" data

```
rectangle_split_three_contained_by
```

Opposite-side congruence transports the fixed segment PQ through all three pieces. Each concrete piece is thereby promoted from `IsRectangle` to `IsRectangleContainedBy`.

1.9.4 Step 4: remove the choice of representatives

```
rectangle_contained_by_unique
euclid_proposition_2_1_three
```

The uniqueness theorem replaces the concrete whole and each concrete part by an arbitrary rectangle contained by the same pair of segments. This is the semantic step that makes the final statement faithful to Euclid's definite phrase "the rectangle contained by."

1.9.5 Step 5: expose the symmetric side order

```
rectangle_contained_by_swap
rectangle_contained_by_comm
```

Cyclic rotation of the rectangle exchanges the two containing segments. This is not needed to establish the displayed three-part split itself, but it is an important part of the Book II rectangle interface and is immediately useful in II.2 and later propositions.

Diagrammatic audit. Steps 3–5 introduce no further geometric configuration. Step 3 certifies side-congruence data on the three already drawn pieces; Step 4 transports their exact scissors content to arbitrary representatives; Step 5 cyclically rereads a rectangle. None of these operations warrants an additional picture under the Book II visual audit criterion.

1.10 Classical Proof and Formal Proof

1.10.1 Euclid's proof pattern

In the classical three-part configuration, take one undivided segment A and a segment BC divided at D and E. Erect a perpendicular at one endpoint of BC, lay off on it a segment equal to A, complete the outer rectangle, and through the division points draw parallels to the perpendicular side. The outer rectangle is visibly partitioned into three rectangles, as in Figure 1.1. Each smaller rectangle is contained by A and one of the three consecutive parts of BC. Common Notion 2 then combines the parts into the whole equality.

The classical DAG is therefore extremely small:

$$\begin{gathered}\text{divide one side}\\ \Downarrow\\ \text{draw the corresponding parallels}\\ \Downarrow\\ \text{obtain three subrectangles}\\ \Downarrow\\ \text{whole rectangle} = \text{sum of the three subrectangles.}\end{gathered}$$

1.10.2 The formal DAG

The current reconstruction expands two things that the classical diagram treats as immediate: exact scissors bookkeeping and independence of the chosen rectangle representatives.

$$\begin{gathered}\text{rectangle geometry} + \text{betweenness} + \text{cut parallelograms}\\ \Downarrow\\ \texttt{rectangle\_split}\\ \Downarrow\\ \texttt{rectangle\_split\_three}\\ \Downarrow\\ \texttt{rectangle\_split\_three\_contained\_by}\\ \Downarrow\\ \texttt{rectangle\_contained\_by\_unique}\\ \Downarrow\\ \texttt{euclid\_proposition\_2\_1\_three}.\end{gathered}$$

The proof is therefore not a mechanical transcription of Euclid. It preserves the geometric core, but makes the semantic and representational layers explicit.

1.11 Geometry, Content, and Representation

1.11.1 Geometry

The geometric data are substantial even though the proposition is visually simple. The current theorem receives the outer rectangle, strict betweenness relations on the divided side and its opposite, two diagonal-intersection witnesses, and four parallelogram facts for the two successive cuts. Right angles of the two resulting rectangles are not read from the picture; they are proved by right-angle transport and parallelogram geometry.

No coordinates are introduced. The cut points are controlled by `Between`, the side lengths by segment congruence, and the rectangular structure by parallels and synthetic right angles.

1.11.2 Content

Content enters in its strongest local form. The core statements are exact `HilbertScissorsEq` decompositions. The three-part theorem is composed from two exact binary splits by addition and transitivity. Representative transport is also exact scissors equality.

The more general equicomplementability relation is part of the surrounding Book I content theory, but II.1 does not need to weaken its conclusion to that level.

1.11.3 Representation

Representation is central to the formal meaning of the proposition. A rectangle contained by PQ, RS is not one canonical four-tuple of vertices. The predicate `IsRectangleContainedBy` describes a representative; `rectangle_contained_by_unique` proves that different representatives have the same exact scissors content.

There is a second representation issue inside the split itself. The inherited scissors theorem and the Book II statement use different cyclic orders for some parallelogram terms. The proof normalizes these by

```
parallelogram_term_rotateOne
```

before composing the scissors equalities. These cyclic changes are bookkeeping of a geometric representation, not new area facts.

1.12 Hidden Configuration Data

The classical drawing makes several facts appear automatic. Lean records them explicitly:

- $B - M - C$ for the first cut and $M - N - C$ for the second;
- $D - L - E$ and $L - K - E$ on the opposite side;

- $D - X - C$ together with $M - X - L$ for the first diagonal/cut intersection;
- $L - Y - C$ together with $N - Y - K$ for the second diagonal/cut intersection;
- explicit parallelogram witnesses for the left and right pieces of each split;
- congruence $CE \cong PQ$, which identifies the fixed side used in the "contained by" descriptions;
- four arbitrary rectangle representatives in the final theorem.

The points X and Y do not appear in Euclid's statement and need not be emphasized in an elementary picture of II.1. They are nevertheless real data of the current exact-dissection API, inherited from the triangulated proof of the rectangle split.

1.13 Axiomatic Status

The public II.1 theorems assume

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

so II.1 is Euclidean in the current project API. Group IV is available through this typeclass and is consumed transitively by the parallelogram and parallel infrastructure on which the rectangle layer and inherited split theorem depend. The proposition itself does not introduce a new parallel axiom.

No continuity typeclass, Archimedean axiom, numerical area function, or segment field is introduced in `Proposition2_1.lean`. No proposition-local `axiom` or `sorry` occurs in the audited source. The modular builds for `HilbertRectangle` and `Proposition2_1` are recorded as clean at the present project checkpoint.

The content status is also worth separating from the later converse arguments of Book I. II.1 proceeds from geometry to an exact dissection. It does not infer metric geometry from an equal-content hypothesis, so it does not need a proper-part or De Zolt-type strictness argument.

1.14 Relationship with Book I and the Following Propositions

The exact split theorem reuses the scissors infrastructure established during I.47. In particular, the underlying theorem historically named

```
i47_square_split
```

actually requires only parallelogram structure, and II.1 repackages it as a general rectangle split.

The rectangle layer also uses Book I results about parallelograms, opposite sides, congruent right angles, SAS/SSS transport, and the scissors representation of parallelograms. Thus Book II does not start a new foundational geometry; it starts a new way of organizing geometric magnitudes on top of the completed Book I infrastructure.

II.2 immediately imports II.1. Its formal proof illustrates why the symmetric rectangle API matters: Euclid's order of the two containing sides is opposite to the order produced by the binary cut theorem, so the representatives are swapped by rectangle rotation before II.1 is applied. II.3 continues the same rectangle calculus. In this sense II.1 is not only the first theorem of Book II but the API-level distributive kernel for the propositions that follow.

1.15 Source Audit

1.15.1 Euclid

Book II Definition 1 supplies the semantic key: the two lines "containing" a rectangle are the adjacent sides that contain a right angle. Proposition II.1 then divides one of two given lines into several parts and identifies the whole rectangle with the sum of the rectangles formed from the undivided line and those parts.

The proof is geometric throughout. The division points on one side are carried across the outer rectangle by parallels. The resulting subrectangles literally partition the original rectangle. Nothing in the proposition requires a numerical assignment to a segment or a plane figure.

1.15.2 Hartshorne

Hartshorne's modern synthetic discussion is a useful semantic control. He interprets Euclid's equalities of figures through equal content rather than numerical area, and notes that all the Book II propositions are valid at that level. He also uses "geometrical algebra" for the systematic manipulation of such geometric contents.

Our reconstruction is deliberately slightly sharper for II.1: the final result is not only equicomplementability but exact `HilbertScissorsEq`. This is possible because II.1 is itself a direct dissection theorem.

1.15.3 Hilbert

Hilbert is relevant here mainly as a warning against collapsing distinct layers. His later arithmetic of segments introduces arithmetic operations only after substantial synthetic development. The current Book II reconstruction does not import that arithmetic and does not use Hilbert's later numerical surface-area theory. It remains in the incidence/order/congruence plus scissors language already available after Book I.

1.15.4 The phrase "geometric algebra"

The phrase is therefore appropriate only with a qualification. II.1 exhibits an algebraic *pattern*: partition of one side corresponds to additive decomposition of a rectangle; exchanging the two containing sides corresponds to cyclic rotation; a square is the special case in which the two containing segments are the same. These operations anticipate the formal laws of a product-like calculus.

But the objects being manipulated by Euclid are still geometric figures and relations of equality between them. The documentation should therefore resist rewriting the theorem as if Euclid had stated

$$a(b + c + d) = ab + ac + ad$$

in symbolic algebra. That equation is a modern shadow of the configuration, not the proposition itself.

1.16 Dependency Summary

The actual formal dependency chain is

$$\begin{gathered}
\text{Hilbert incidence + order + congruence + Group IV} \\
\Downarrow \\
\text{right angles + parallels + parallelograms + opposite-side congruence} \\
\Downarrow \\
\text{Book I scissors terms and exact triangle/parallelogram splitting} \\
\Downarrow \\
\texttt{IsRectangle} \\
\Downarrow \\
\texttt{IsRectangleContainedBy} \\
\Downarrow \\
\texttt{rectangle_contained_by_unique} \\
\Downarrow \\
\texttt{rectangle_split} \\
\Downarrow \\
\texttt{rectangle_split_three} \\
\Downarrow \\
\texttt{rectangle_split_three_contained_by} \\
\Downarrow \\
\texttt{euclid_proposition_2_1_three}.
\end{gathered}$$

The symmetric side-order branch is

$$\begin{gathered}
\texttt{IsRectangleContainedBy}(PQ, RS) \\
\Downarrow \\
\text{cyclic rectangle rotation} \\
\Downarrow \\
\texttt{rectangle_contained_by_swap} \\
\Downarrow \\
\texttt{rectangle_contained_by_comm}.
\end{gathered}$$

New geometric axiom in II.1:	no
New proposition-local axiom:	no
New rectangle abstraction:	yes, in `HilbertRectangle`
New reusable split helper:	yes
New representative-independence theorem:	yes
Exact scissors dissection used:	yes
General equicomplementability required by II.1:	no
Numerical area semantics:	no
Segment multiplication / field theory:	no
Group IV available/used transitively:	yes
Continuity assumption in the file:	no
Proposition-local `sorry`/`admit`:	no
Modular build:	clean

1.17 Conclusion

Proposition II.1 is the first clean instance in Book II of a product-like law being realized entirely by geometry. A segment is divided; a rectangle is cut along corresponding parallels; exact scissors equality identifies the whole with the sum of the pieces. The rectangle layer then removes the accidental choice of a concrete drawing, so Euclid's phrase "the rectangle contained by" becomes a representative-independent synthetic magnitude.

The formalization also isolates the two structural laws that make the language look algebraic. Additivity comes from actual dissection. Commutativity comes from cyclic rotation of the rectangle. A square is the special case in which the two containing segments agree. None of these steps requires a product of segment lengths or a numerical area map.

Thus the "geometric algebra" of II.1 is best understood not as hidden modern symbolism but as a calculus of geometric magnitudes whose algebraic laws emerge from constructions, congruence, and finite decomposition.

The representative-independent three-part theorem at the end of the current formalization is:

```
theorem euclid_proposition_2_1_three
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    -- Concrete dissection diagram.
    (B C D E L M K N X Y P Q : Geo.Point)
    (hRect : IsRectangle Geo B C E D)
    (hCE_PQ : Geo.Congruent C E P Q)

    -- First cut.
    (hBMC : Geo.Between B M C)
    (hDLE : Geo.Between D L E)
    (hDXC : Geo.Between D X C)
    (hMXL : Geo.Between M X L)
    (hLeftPar :
      IsParallelogram Geo L D B M)
    (hRightPar :
      IsParallelogram Geo C E L M)

    -- Second cut.
    (hMNC : Geo.Between M N C)
    (hLKE : Geo.Between L K E)
    (hLYC : Geo.Between L Y C)
    (hNYK : Geo.Between N Y K)
    (hMiddlePar :
      IsParallelogram Geo K L M N)
    (hFinalPar :
      IsParallelogram Geo C E K N)

    -- Arbitrary representatives of the four rectangles.
    (U0 U1 U2 U3 : Geo.Point)
    (V0 V1 V2 V3 : Geo.Point)
    (W0 W1 W2 W3 : Geo.Point)
    (Z0 Z1 Z2 Z3 : Geo.Point)

    (hWhole :
      IsRectangleContainedBy Geo
        U0 U1 U2 U3 B C P Q)
```

```
  (hPart1 :
    IsRectangleContainedBy Geo
      V0 V1 V2 V3 B M P Q)

  (hPart2 :
    IsRectangleContainedBy Geo
      W0 W1 W2 W3 M N P Q)

  (hPart3 :
    IsRectangleContainedBy Geo
      Z0 Z1 Z2 Z3 N C P Q) :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo U0 U1 U2 U3)
    (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
     (hilbertParallelogramTerm Geo W0 W1 W2 W3 +
      hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3)) := by
```

Chapter 2

Proposition II.2

2.1 Introduction

Proposition II.2 is the first place where the rectangle calculus established in II.1 is reused rather than developed further. The classical proposition is visually elementary: a square on a straight line is cut by a parallel through an arbitrary point of that line, and the two resulting rectangles fill the square. The formal reconstruction preserves exactly that geometric content, but its proof architecture is deliberately different from Euclid's.

Euclid proves II.2 directly from a square construction and a parallel through the point of division. The current Lean theorem instead imports `Proposition2_1` and applies the representative-independent binary form of II.1. One additional operation is needed: Euclid names each part rectangle with the whole side first, whereas the II.1 cut theorem returns the divided part first. The proof therefore rotates each rectangle representative by one vertex, applies II.1, and then transports the result back to Euclid's original order.

This makes II.2 a particularly clean test of the Book II API. Two algebra-like laws are already available, but neither is primitive algebra:

- additivity is realized by an exact rectangle dissection;
- exchange of the two containing segments is realized by cyclic rotation of a rectangle.

No multiplication of segment lengths and no numerical area function occurs in the theorem.

2.2 Euclid's Statement and Classical Configuration

2.2.1 The statement

In Heath's wording, Proposition II.2 states:

> If a straight line be cut at random, the rectangle contained by the whole and both of the segments is equal to the square on the whole.

Here "both of the segments" means the two rectangles obtained by pairing the whole line with each of the two parts. If AB is cut at C, the geometric statement is therefore

$$\mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, CB) \;=\; \mathrm{Sq}(AB).$$

The notation is documentary shorthand only. It does not introduce a number-valued rectangle function.

2.2.2 The classical diagram

Euclid takes a straight line AB cut at C, constructs the square $ADEB$ on AB by I.46, and through C draws CF parallel to the two opposite sides AD and BE by I.31 [1]. The square is thereby partitioned into two rectangles. One is contained by the whole AB and the segment AC; the other is contained by the whole AB and the segment CB.

$$A - C - B$$

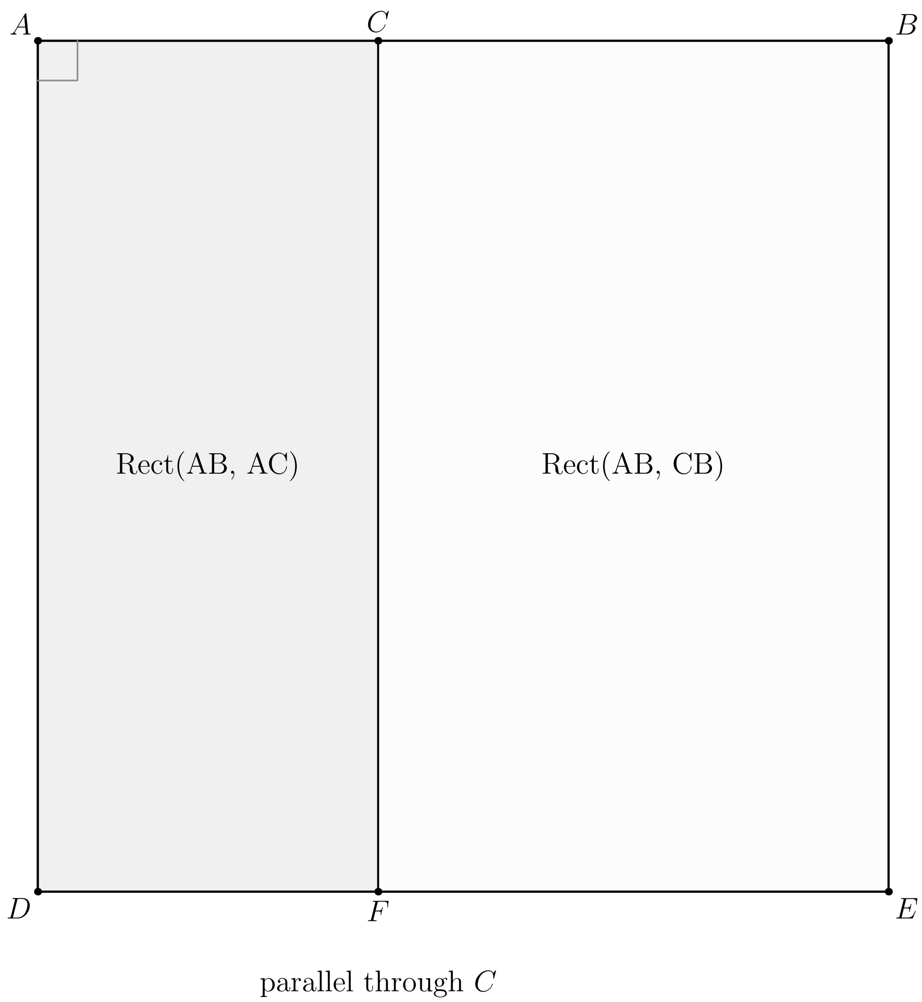


Figure 2.1: The classical source configuration for II.2. The square on AB is cut at C by the parallel CF. The two visible subfigures are the rectangles contained by AB, AC and AB, CB. No diagonal or intersection witness is part of Euclid's local argument.

The current Lean configuration uses different letters:

classical role	Lean role
AB	BC
C	M
$ADEB$	$BCED$
CF	ML

Thus $B - M - C$, the outer figure $BCED$ is a square, and the cut ML divides it into the left and right rectangular pieces. The formal exact-dissection interface additionally carries the diagonal/cut intersection X; that genuinely new auxiliary state is shown later, at the II.1 call where it is used.

Diagrammatic audit. Three pictures are retained. Figure 2.1 records Euclid's source construction. The proof architecture later shows the two cyclic rectangle-representative changes because those are the distinctive local representation step in Lean, and it separately shows the configured II.1 square cut with the hidden witness X. There is no picture for recognizing the square as a rectangle contained by BC, BC, because that changes only the predicate under which the same square is viewed. Likewise the final addition, transitivity, and symmetry introduce no new geometric configuration.

2.3 Classical Proof

The classical proof has only four mathematical steps.

First, construct the square on the whole segment. Second, draw through the point of section a line parallel to the appropriate pair of sides of the square. Third, observe that this line partitions the square into two rectangles. Finally, identify those rectangles by Book II Definition 1: their adjacent sides are the whole segment together with the two parts of the divided segment.

In the standard lettering this gives

$$\mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, CB) = \mathrm{Sq}(AB).$$

Euclid does not invoke II.1 in this proof. The local classical dependency graph is instead

$$\begin{array}{c} \text{I.46: construct the square on } AB \\ \Downarrow \\ \text{I.31: draw the parallel through } C \\ \Downarrow \\ \text{II.Def.1: identify the two rectangles} \\ \Downarrow \\ \text{the two pieces constitute the square.} \end{array}$$

This fact is important for documentation. The formal theorem proves the same geometric identity, but it is not a line-by-line formalization of Euclid's historical derivation.

2.4 The Geometric Identity

In the current Lean lettering the square side is BC and the cut point is M. The two Euclidean rectangles are therefore

$$\mathrm{Rect}(BC, BM), \qquad \mathrm{Rect}(BC, MC),$$

and the theorem asserts the exact scissors relation

$$\mathrm{Rect}(BC, BM) + \mathrm{Rect}(BC, MC) \simeq_{\mathrm{sc}} \mathrm{Sq}(BC).$$

A modern algebraic shadow is obtained by writing $x = y + z$:

$$x^2 = xy + xz.$$

This is useful as a mnemonic for the pattern, but it is not the object proved by Lean and it is not a claim that Euclid has defined multiplication on segment lengths. The formal terms remain rectangle representatives and their exact scissors relation.

2.5 Reusable Book II Infrastructure

II.2 imports

```
import CGJteamLab.Proposition2_1
```

and introduces no new rectangle abstraction. Its proof consumes four pieces of the existing Book II infrastructure.

2.5.1 A square as a rectangle contained by the same side twice

From

```
hSquare : IsSquare Geo B C E D
```

the theorem

```
square_is_rectangle_contained_by_side
```

produces

```
IsRectangleContainedBy Geo B C E D B C B C
```

so the concrete square is recognized synthetically as a rectangle contained by BC, BC.

2.5.2 The binary form of II.1

The reusable distributive kernel is

```
euclid_proposition_2_1_two
```

whose geometric content is

$$\mathrm{Rect}(BC, PQ) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, PQ) + \mathrm{Rect}(MC, PQ)$$

when $B - M - C$ and the corresponding rectangular cut data are supplied.

For II.2 the fixed segment is specialized to $PQ = BC$. Thus the right-hand pieces naturally arrive in the order

$$\mathrm{Rect}(BM, BC), \qquad \mathrm{Rect}(MC, BC).$$

2.5.3 Exchanging the containing-side order

Euclid's statement uses the opposite order:

$$\mathrm{Rect}(BC, BM), \qquad \mathrm{Rect}(BC, MC).$$

The current proof does not treat this as "commutativity of multiplication." It calls

```
rectangle_contained_by_swap
```

on each arbitrary representative.

The theorem rotates a rectangle

$$A\,B\,C\,D \longmapsto B\,C\,D\,A,$$

returns the corresponding `IsRectangleContainedBy` fact with the two containing segments exchanged, and simultaneously returns a `HilbertScissorsEq` relating the two cyclic presentations.

The higher-level theorem `rectangle_contained_by_comm` belongs to the same rectangle API, but the public II.2 proof uses `rectangle_contained_by_swap` directly because its returned data are exactly the swapped representatives and exact scissors transports needed for the subsequent II.1 call.

2.6 Proof Architecture of the Reconstruction

The Lean proof has four stages.

2.6.1 Step 1: recognize the square

The square hypothesis is converted to the "contained by" language:

```
have hSquareContained :
    IsRectangleContainedBy Geo
      B C E D B C B C :=
  square_is_rectangle_contained_by_side
    Geo B C E D hSquare
```

The rectangle structure and the side congruence $CE \cong BC$ are then extracted from this fact.

2.6.2 Step 2: rotate the two arbitrary part representatives

The two hypotheses supplied in Euclid's order are

```
hPart1 :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B C B M

hPart2 :
  IsRectangleContainedBy Geo
```

```
    W0 W1 W2 W3 B C M C
```

The current source then applies `rectangle_contained_by_swap` separately to `hPart1` and `hPart2`. Each call returns a swapped `IsRectangleContainedBy` fact and an exact scissors equality for the cyclically rotated representative. The resulting representatives are $V_1V_2V_3V_0$ and $W_1W_2W_3W_0$.

The mathematical result is that the rotated representatives now have the side orders required by the II.1 binary theorem.

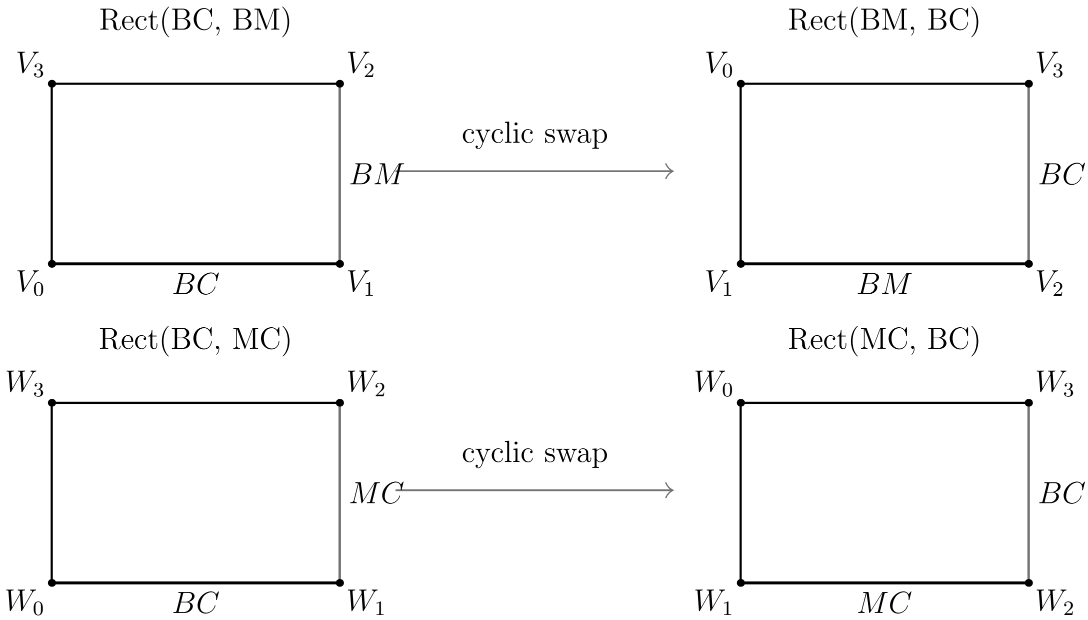


Figure 2.2: The two independent representative changes performed in Step 2. The first changes the presentation of a rectangle contained by BC, BM to one contained by BM, BC; the second does the same for BC, MC. The quadrilateral is not replaced by a new metric figure: its vertices are cyclically re-indexed, and `rectangle_contained_by_swap` supplies the exact scissors certificate for that re-presentation.

2.6.3 Step 3: apply II.1 with fixed segment BC

The central call is

```
have hSplit :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo B C E D)
      (hilbertParallelogramTerm Geo V1 V2 V3 V0 +
       hilbertParallelogramTerm Geo W1 W2 W3 W0) :=
  euclid_proposition_2_1_two
    Geo
    B C D E L M X
    B C
    hRect
    hCE_BC
    hBMC
    hDLE
    hDXC
```

```
    hMXL
    hLeftPar
    hRightPar
    B C E D
    V1 V2 V3 V0
    W1 W2 W3 W0
    hSquareContained
    hPart1Swap
    hPart2Swap
```

This is the whole mathematical core of the reconstruction:

$$\mathrm{Sq}(BC) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, BC) + \mathrm{Rect}(MC, BC).$$

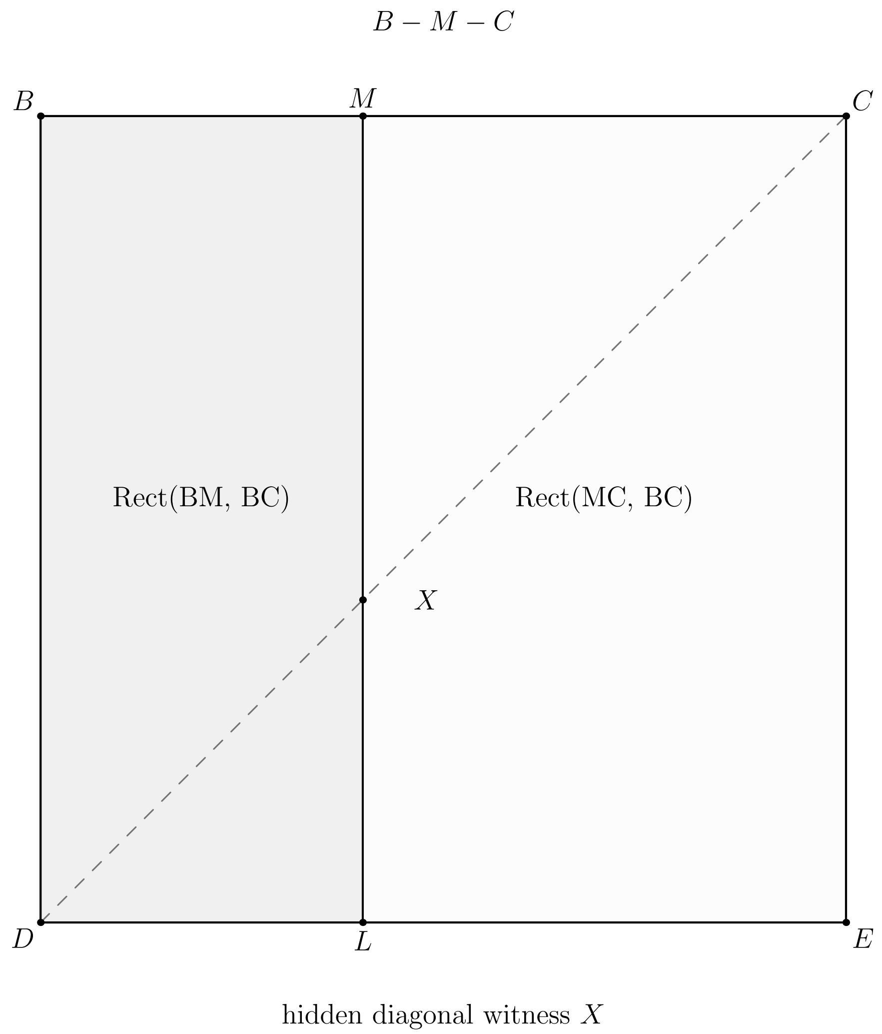


Figure 2.3: The actual configured square-cut state consumed by the II.1 binary theorem. The order is $B - M - C$, the cut reaches the opposite side at L, and the dashed diagonal DC meets that cut at X. Thus $D - X - C$ and $M - X - L$ are visible here as the hidden formal geometry needed by the inherited exact scissors decomposition. The two pieces are read in the II.1 side order BM, BC and MC, BC.

2.6.4 Step 4: return to Euclid's ordering

The two exact rotation equalities are added:

```
have hPartsBack :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo V1 V2 V3 V0 +
       hilbertParallelogramTerm Geo W1 W2 W3 W0)
      (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
       hilbertParallelogramTerm Geo W0 W1 W2 W3) :=
  HilbertScissorsEq.add
    (Geo := Geo)
    (HilbertScissorsEq.symm
      (Geo := Geo) hRotate1)
    (HilbertScissorsEq.symm
      (Geo := Geo) hRotate2)
```

Transitivity composes `hSplit` with `hPartsBack`, and the final symmetry reverses the result into the order used by the public theorem: sum of the two Euclidean rectangles on the left, square on the right.

Diagrammatic audit. No additional figure is inserted for Step 4. At this point the geometric cut and both representative rotations have already been displayed. The remainder is only exact-scissors addition of the two rotation certificates, transitivity with the II.1 split, and a final symmetry; drawing a new physical arrangement would incorrectly suggest a further dissection that does not occur in the source.

2.7 Lean Formalization

The current public theorem is the following instantiated diagram theorem:

```
theorem euclid_proposition_2_2
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (B C D E L M X : Geo.Point)

    (hSquare : IsSquare Geo B C E D)

    (hBMC : Geo.Between B M C)
    (hDLE : Geo.Between D L E)
    (hDXC : Geo.Between D X C)
    (hMXL : Geo.Between M X L)
    (hLeftPar :
      IsParallelogram Geo L D B M)
    (hRightPar :
      IsParallelogram Geo C E L M)

    (V0 V1 V2 V3 : Geo.Point)
    (W0 W1 W2 W3 : Geo.Point)

    (hPart1 :
```

```
      IsRectangleContainedBy Geo
        V0 V1 V2 V3 B C B M)

    (hPart2 :
      IsRectangleContainedBy Geo
        W0 W1 W2 W3 B C M C) :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
       hilbertParallelogramTerm Geo W0 W1 W2 W3)
      (hilbertParallelogramTerm Geo B C E D) := by
```

Three aspects of the signature are mathematically significant.

First, the theorem assumes an instantiated square and cut diagram. It is not an existence theorem constructing the square and the parallel cut from only a segment and a cut point. Euclid performs those constructions in the classical proof; the present public Lean theorem starts after they have been supplied.

Second, the two part rectangles are arbitrary representatives satisfying `IsRectangleContainedBy`. Their exact placement in the plane is not fixed by the statement.

Third, the conclusion is the exact relation `HilbertScissorsEq`. It is not merely a conclusion in the more general `HilbertScissorsEquicomplementable` relation. II.2 remains at the exact-dissection level because its proof is ultimately the binary exact split of II.1 plus exact representation transports.

2.8 Relationship with Proposition II.1

The contrast between the classical and formal dependency graphs is especially clear here.

Euclid's II.2 is proved directly from the square and the parallel cut. The current Lean file instead begins with

```
import CGJteamLab.Proposition2_1
```

and the public proof calls

```
euclid_proposition_2_1_two
```

once.

Thus II.2 is an explicit formal corollary of the binary II.1 API. No new dissection lemma is proved. The only additional mathematical work is conversion between the two possible orders of the containing segments.

This reuse also explains why the symmetric rectangle interface added at II.1 is not cosmetic. Without the side-order transport, the II.1 theorem would produce perfectly correct rectangles whose formal orientation does not match the language of II.2.

2.9 Relationship with Book I Infrastructure

The direct import is II.1, not a Book I proposition. Nevertheless the inherited chain underneath II.1 contains the Book I geometry on which the rectangle calculus rests.

The exact binary split is built over the scissors decomposition infrastructure developed around I.47. Rectangle structure uses Euclidean parallelogram theory and synthetic right angles; the square hypothesis uses the square infrastructure established in Book I. Consequently the formal proof of II.2 does not replay I.31 and I.46 as Euclid does. Those classical constructions have been absorbed into the established geometric API and into the hypotheses of the instantiated theorem.

2.10 Formalization Notes and Hidden Geometric Data

The visible square cut hides several obligations that are explicit parameters of the current theorem:

- $B - M - C$, fixing the orientation of the cut on the square side;
- $D - L - E$, locating the opposite endpoint of the cut;
- $D - X - C$ and $M - X - L$, recording the intersection X of the cut with the diagonal used by the inherited exact scissors split;
- `IsParallelogram Geo L D B M` for the left piece;
- `IsParallelogram Geo C E L M` for the right piece;
- two arbitrary representatives of the Euclidean part rectangles;
- two cyclic representative changes, each accompanied by an exact scissors equality.

The point X has no role in the elementary verbal statement. It belongs to the current exact-dissection interface inherited from the triangulated scissors proof. It is therefore documented as hidden formal geometry rather than presented as part of Euclid's original construction.

There is also an important negative observation. No `SameRay`, `SameSide`, or `OppositeSide` obligation appears in `Proposition2_2.lean` itself. The corresponding low-level geometry has already been discharged inside the reusable rectangle and II.1 theorems.

2.11 Geometry, Content, Representation, and Later Arithmetic

2.11.1 Geometry

The geometric layer consists of a square, betweenness on two opposite sides, the diagonal/cut witness, and two parallelogram pieces. Congruence of the square side with itself identifies the square as a rectangle contained by BC, BC.

2.11.2 Content / scissors

The conclusion and every transport used in the proof are exact `HilbertScissorsEq` relations. The proof uses symmetry, transitivity, and addition of exact scissors equalities. It does not weaken to the more general equicomplementability relation.

2.11.3 Representation

Representation is the distinctive issue in II.2. The theorem accepts arbitrary rectangles for

$$\mathrm{Rect}(BC, BM) \quad \text{and} \quad \mathrm{Rect}(BC, MC).$$

The binary II.1 theorem wants the factors in the reverse order. Cyclic rotation of each representative supplies the required conversion while preserving exact scissors content.

2.11.4 Later arithmetic / numerical area

Neither layer is used. In particular, the proof does not define a segment product, does not invoke Hilbert's later segment field, and does not assign real areas to the square or rectangles.

2.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore works in the same Euclidean project layer as II.1. No new proposition-specific geometric axiom or content axiom is declared in `Proposition2_2.lean`. Inspection of the current 162-line source shows one theorem declaration and no proposition-local `axiom`, `sorry`, or `admit`.

This should not be described as "axiom-free." The theorem inherits the axioms and typeclass assumptions of the established Hilbert/Euclidean/scissors library transitively. No current `#print axioms` output was supplied with the source used for this documentation pass, so the transitive axiom list is not reproduced here.

The implementation checkpoint records that the II.2 Lean build had already been checked during theorem development. This documentation workspace does not contain the complete Lean project, so the Lean build is not re-run here; the LaTeX and Asymptote checks are reported separately.

2.13 Hartshorne's Semantic Control

Hartshorne places the end of Book I and all of Book II inside Euclid's theory of area, interpreted synthetically as equal content rather than as an unexplained numerical area function [5]. He also describes Book II as part of an "algebra of areas": rectangles can play a product-like role because Euclid has no operation that multiplies two line segments to produce another segment.

II.2 fits that interpretation exactly, provided the levels are kept separate. The square on the whole and the two rectangles are geometric figures; their equality is a content statement; the modern identity $x^2 = xy + xz$ is only an interpretive shadow.

The current Lean theorem is locally stronger than the weakest equal-content reading because its configured proof stays at exact scissors equality. That strength comes from the explicit binary dissection inherited from II.1, not from a general principle identifying every equal-content pair with an exact dissection.

2.14 Hilbert's Layering

Hilbert supplies a second semantic guardrail. His theory separates synthetic plane content from the later arithmetic of segments. Segment multiplication is introduced only in a later arithmetic construction, with a fixed unit segment and parallel geometry; it is not primitive rectangle notation.

The II.2 proof remains entirely on the earlier side of this boundary:

$$\begin{gathered}\text{square and rectangle geometry}\\ \Downarrow\\ \text{exact scissors dissection}\\ \Downarrow\\ \text{representative rotation and transport.}\end{gathered}$$

The apparent commutativity required by the formula $xy = yx$ is realized geometrically by `rectangle_contained_by_swap`, not imported from Hilbert's later segment arithmetic.

2.15 Dependency Summary

The actual Lean dependency chain is

$$\begin{gathered}\text{Hilbert incidence + Euclidean plane layer}\\ \Downarrow\\ \text{Book I right-angle, square, parallelogram, and scissors infrastructure}\\ \Downarrow\\ \texttt{HilbertRectangle}: \texttt{IsRectangleContainedBy}, \texttt{rectangle\_contained\_by\_swap}\\ \Downarrow\\ \text{Proposition II.1 binary API}\\ \Downarrow\\ \texttt{euclid\_proposition\_2\_1\_two}\\ \Downarrow\\ \text{two representative rotations + exact scissors addition/transitivity}\\ \Downarrow\\ \texttt{euclid\_proposition\_2\_2}.\end{gathered}$$

The classical local chain is different:

$$\text{I.46} \longrightarrow \text{I.31} \longrightarrow \text{square cut into two rectangles} \longrightarrow \text{II.2.}$$

New geometric axiom in II.2:	no
New content axiom in II.2:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New `HilbertRectangle` theorem introduced by II.2:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses an earlier Book II proposition:	yes, II.1
Direct side-order transport used:	yes, `rectangle_contained_by_swap`
Exact `HilbertScissorsEq`:	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem build:	previously checked; not re-run here
Full relevant Lean build:	previously checked; not re-run here
Proposition-local `sorry`/`admit`:	no

2.16 Conclusion

Proposition II.2 is mathematically a square cut into two rectangles, but formally it does more than repeat the picture. It demonstrates that the II.1 rectangle calculus is already reusable: the binary dissection theorem supplies additivity, and cyclic rectangle rotation reconciles the two possible orders of the containing sides.

This is also the first sharp divergence between Euclid's local proof graph and the formal Book II graph. Euclid proves II.2 directly from I.46 and I.31. Lean uses II.1 as a library theorem and turns II.2 into a short transport-and-reuse argument. The theorem is therefore not a literal transcription of the classical proof, but it remains entirely synthetic and geometric.

The result can be remembered by the modern shadow

$$x^2 = xy + xz, \qquad x = y + z,$$

but the formal content is more precise: arbitrary representatives of the two rectangles contained by the whole side and the two parts are exactly scissors-equal, in sum, to the concrete square on the whole side. No numerical area and no multiplication of segment lengths is required.

Chapter 3

Proposition II.3

3.1 Introduction

Proposition II.3 is the second immediate reuse of the rectangle calculus established in Proposition II.1. Its classical form is again a statement about a straight line cut at an arbitrary point, but now one of the two resulting segments plays a distinguished role. The rectangle contained by the whole line and that segment is decomposed into a rectangle contained by the two parts and the square on the distinguished segment.

The current Lean reconstruction makes this structure unusually transparent. Its only direct import is the II.1 module:

```
import CGJteamLab.Proposition2_1
```

The proof first recognizes the supplied square on MC as a rectangle contained by MC, MC, and then applies the binary form of II.1 with the fixed segment specialized to MC. No new dissection theorem, side-order swap, or proposition-local helper theorem is needed.

This is an important contrast with II.2. There the Euclidean order of the two containing segments did not match the order produced by the binary II.1 theorem, so cyclic rectangle rotation was required. In II.3 the order already matches:

$$\mathrm{Rect}(BC, MC) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, MC) + \mathrm{Rect}(MC, MC).$$

The square on MC is then simply a chosen representative of the final $\mathrm{Rect}(MC, MC)$ term.

Thus II.3 is not formalized through multiplication of segment lengths or through a numerical area function. Its apparently algebraic shape is realized by an exact geometric cut together with representative independence in the rectangle API.

3.2 Euclid's Statement and Classical Configuration

3.2.1 The statement

In Heath's wording, Proposition II.3 states:

> If a straight line be cut at random, the rectangle contained by the whole and one of

the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.

If AB is cut at C and the distinguished part is BC, the geometric identity is

$$\mathrm{Rect}(AB, BC) = \mathrm{Rect}(AC, CB) + \mathrm{Sq}(BC).$$

As throughout the Book II documentation, Rect and Sq are expository shorthand for geometric figures and their content. They are not number-valued functions.

3.2.2 The classical diagram

Euclid constructs the square $CDEB$ on the distinguished segment CB by Proposition I.46. The side ED is produced, and through A a line is drawn parallel to the corresponding sides of the square by Proposition I.31 [1]. The resulting larger rectangle has adjacent side magnitudes AB and BC. The line through C separates it into a rectangle with side magnitudes AC, CB and the square on CB.

The classical state is shown first in its own lettering. The square $CDEB$ on CB is already one of the two final pieces, while the parallel through A extends it to the whole rectangle on AB and BC.

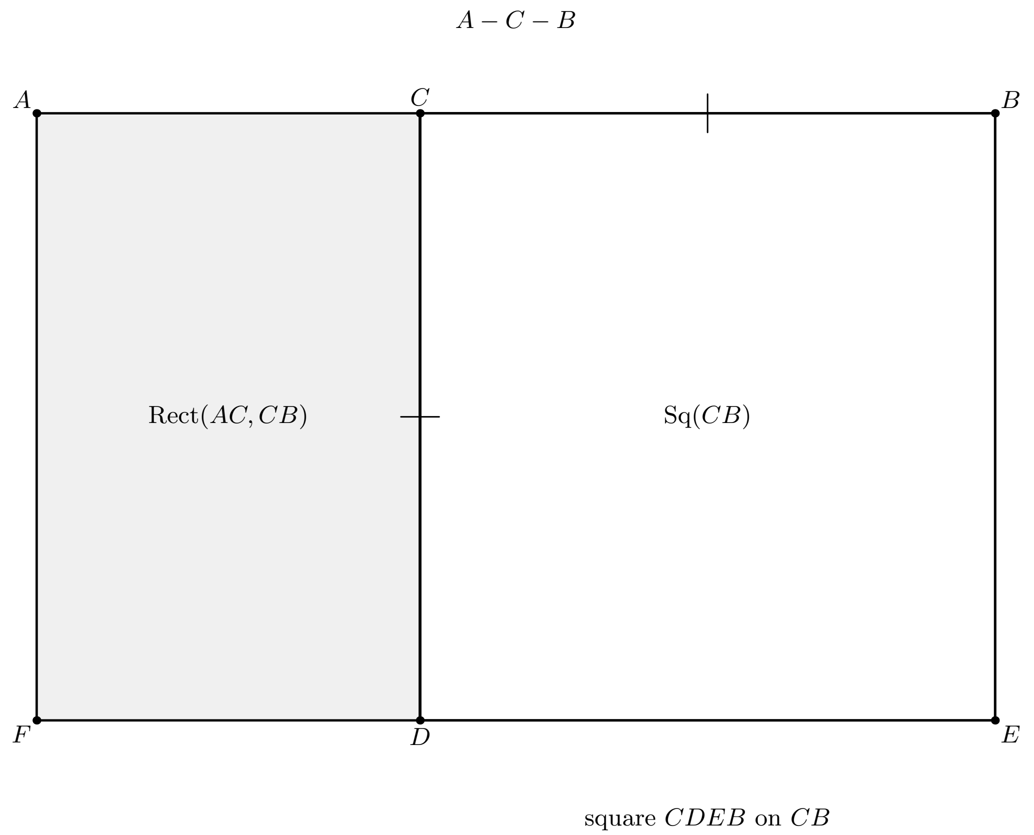


Figure 3.1: The classical configuration of Euclid II.3. The line is cut as $A - C - B$. The square $CDEB$ is erected on the distinguished part CB, and its lower side is produced to F while AF is drawn parallel to the square sides. The resulting whole rectangle $ABEF$ is visibly the union of the rectangle contained by AC, CB and the square on CB. No diagonal or scissors-refinement witness is shown, because those belong to the present formal interface rather than to Euclid's local construction.

The current Lean lettering is related to the classical roles by

classical role	Lean role
AB	BC
C	M
AC	BM
CB	MC

so the formal cut relation is $B - M - C$ and the distinguished segment is MC. The configured outer rectangle is $BCED$, with $CE \cong MC$, while the cut through M meets the opposite side at L. That supplied Lean configuration is shown later, exactly where the II.1 binary split is invoked.

Diagrammatic audit. Two pictures are sufficient for II.3. Figure 3.1 records the source construction and its immediate rectangle-plus-square partition. The second picture is deferred to the formal proof step because it contains additional interface data—in particular the diagonal intersection witness X and a square representative that need not be the literal internal cut piece. Recognizing that supplied square as a rectangle contained by MC, MC does not change the geometry, so it does not merit a separate snapshot.

3.3 Classical Proof

Let AB be cut at C, and choose BC as the segment named in the proposition. Construct the square $CDEB$ on CB by I.46. Produce one side of the square and through A draw a line parallel to the opposite sides by I.31. This forms a larger rectangle whose adjacent side magnitudes are AB and BC.

The line through C partitions that larger rectangle into two pieces. The first has adjacent side magnitudes AC and CB, and the second is precisely the square on CB. Hence

$$\mathrm{Rect}(AB, BC) = \mathrm{Rect}(AC, CB) + \mathrm{Sq}(BC).$$

This is Euclid's entire content argument: the whole rectangle is constituted by the two displayed subfigures.

The local classical dependency graph is therefore

$$\begin{gathered}\text{I.46: construct the square on } CB \\ \Downarrow \\ \text{I.31: draw the required parallel through } A \\ \Downarrow \\ \text{Book II Definition 1: identify the rectangles} \\ \Downarrow \\ \text{whole rectangle} = \text{rectangle} + \text{square} \\ \Downarrow \\ \text{II.3.}\end{gathered}$$

Euclid does not cite Proposition II.1 in this proof. The current Lean proof, by contrast, is explicitly organized as a corollary of the reusable II.1 binary theorem. The classical and formal dependency graphs therefore prove the same geometric identity through different library organizations.

3.4 The Geometric Identity

In the current Lean lettering, $B - M - C$. The theorem fixes the segment MC and applies it to the whole BC and the two parts BM, MC. Its exact geometric content is

$$\mathrm{Rect}(BC, MC) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, MC) + \mathrm{Sq}(MC).$$

A modern algebraic shadow is obtained by writing

$$x = y + z,$$

where x corresponds to BC, y to BM, and z to MC. Then the pattern is

$$xz = yz + z^2,$$

or equivalently

$$(y + z)z = yz + z^2.$$

This formula is useful only as a mnemonic. The Lean proof never forms the products xz or yz as segment-arithmetic objects. The actual objects are rectangle representatives and a square, related by exact scissors equality.

3.5 Reusable Book II Infrastructure

The current file begins with

```
import CGJteamLab.Proposition2_1
```

and introduces no new rectangle abstraction.

3.5.1 The binary form of II.1

The reusable theorem is

```
euclid_proposition_2_1_two
```

with geometric content

$$\mathrm{Rect}(BC, PQ) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, PQ) + \mathrm{Rect}(MC, PQ)$$

whenever $B - M - C$ and the required rectangular cut data are supplied.

For II.3 the fixed segment is specialized directly to

$$PQ = MC.$$

Therefore II.1 returns

$$\mathrm{Rect}(BC, MC) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, MC) + \mathrm{Rect}(MC, MC),$$

which already has exactly the containing-side order required by the public II.3 theorem.

3.5.2 A square as $\mathrm{Rect}(MC, MC)$

The square hypothesis

```
hSquare : IsSquare Geo M C F G
```

is converted by

```
square_is_rectangle_contained_by_side
```

into

```
IsRectangleContainedBy Geo M C F G M C M C
```

so the supplied square is a concrete representative of the final $\mathrm{Rect}(MC, MC)$ term.

3.5.3 What II.3 does not use

Unlike II.2, the current proof does not call

```
rectangle_contained_by_swap
rectangle_contained_by_comm
```

The absence is mathematically informative. The binary II.1 theorem already places BM and MC first and the fixed segment MC second, exactly as II.3 needs. No geometric analogue of exchanging factors is required.

3.6 Proof Architecture of the Reconstruction

The public proof has only two mathematical stages.

3.6.1 Step 1: recognize the square

The first local fact is

```
have hSquareContained :
    IsRectangleContainedBy Geo
      M C F G M C M C :=
  square_is_rectangle_contained_by_side
    Geo M C F G hSquare
```

This moves from the square-specific geometric predicate to the general "rectangle contained by" language used by II.1.

Figure 3.2 records the configured cut that the next step will pass to the binary II.1 theorem.

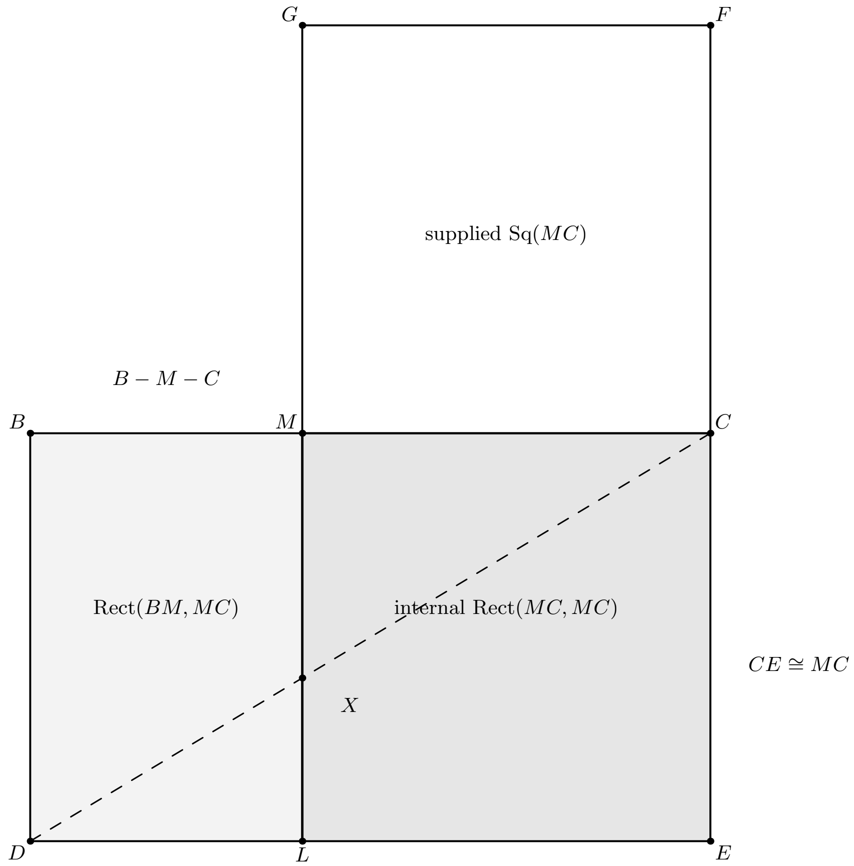


Figure 3.2: The configured geometric state consumed by the single call to `euclid_proposition_2_1_two`. The outer rectangle $BCED$ is cut at $B - M - C$ by ML, producing a left piece representing Rect(BM, MC) and an internal right piece representing Rect(MC, MC). The congruence $CE \cong MC$ provides the fixed second-side magnitude. The dashed diagonal DC meets the cut at X, the hidden exact-dissection witness inherited from II.1. The square $MCFG$ is deliberately drawn separately: Step 1 certifies it as another representative of Rect(MC, MC), so the public conclusion need not reuse the literal internal piece.

3.6.2 Step 2: instantiate II.1 with the fixed segment MC

The remainder of the proof is one direct call:

```
exact
  euclid_proposition_2_1_two
    Geo
    B C D E L M X
    M C
    hRect
    hCE_MC
    hBMC
    hDLE
    hDXC
    hMXL
    hLeftPar
    hRightPar
    U0 U1 U2 U3
    V0 V1 V2 V3
    M C F G
```

```
    hWhole
    hProduct
    hSquareContained
```

Diagrammatic audit. Figure 3.2 is the only genuinely new formal state. Step 1 changes only the predicate under which the already supplied square is read; Step 2 is one binary II.1 dissection. After that call there is no local transitivity chain, summand rearrangement, side-order rotation, cancellation, or additional construction. A third figure would therefore depict only a change of representation or notation rather than new geometry.

No scissors transitivity, addition, symmetry, or representative rotation is performed locally in II.3. Those mechanisms have already been packaged inside the representative-independent binary theorem of II.1.

This makes the formal DAG exceptionally short:

$$\begin{gathered}\text{square } MCFG \\ \Downarrow \\ \texttt{square\_is\_rectangle\_contained\_by\_side} \\ \Downarrow \\ \mathrm{Rect}(MC, MC) \text{ representative} \\ \Downarrow \\ \texttt{euclid\_proposition\_2\_1\_two} \text{ with } PQ = MC \\ \Downarrow \\ \texttt{euclid\_proposition\_2\_3}.\end{gathered}$$

3.7 Lean Formalization

The current public theorem is:

```
theorem euclid_proposition_2_3
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (B C D E L M X : Geo.Point)
    (hRect : IsRectangle Geo B C E D)
    (hCE_MC : Geo.Congruent C E M C)

    (hBMC : Geo.Between B M C)
    (hDLE : Geo.Between D L E)
    (hDXC : Geo.Between D X C)
    (hMXL : Geo.Between M X L)
    (hLeftPar :
      IsParallelogram Geo L D B M)
    (hRightPar :
      IsParallelogram Geo C E L M)

    (F G : Geo.Point)
    (hSquare : IsSquare Geo M C F G)

    (U0 U1 U2 U3 : Geo.Point)
    (V0 V1 V2 V3 : Geo.Point)
```

```
    (hWhole :
      IsRectangleContainedBy Geo
        U0 U1 U2 U3 B C M C)

    (hProduct :
      IsRectangleContainedBy Geo
        V0 V1 V2 V3 B M M C) :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo U0 U1 U2 U3)
      (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
       hilbertParallelogramTerm Geo M C F G) := by
```

Several features of this signature are mathematically significant.

First, the theorem assumes a configured outer rectangle and its cut. It does not construct that diagram from the bare data $B - M - C$. Euclid performs the square and parallel constructions in his proof; the formal theorem receives the corresponding geometry as explicit hypotheses.

Second, $U_0U_1U_2U_3$ and $V_0V_1V_2V_3$ are arbitrary representatives of

$$\mathrm{Rect}(BC, MC) \quad \text{and} \quad \mathrm{Rect}(BM, MC).$$

The conclusion is therefore not restricted to the literal pieces drawn in the configured cut.

Third, the square $MCFG$ is concrete, but the II.1 representative-independence machinery permits it to stand for the $\mathrm{Rect}(MC, MC)$ piece even when it is not the internal right-hand rectangle of the supplied cut diagram.

Fourth, the conclusion is exactly

```
HilbertScissorsEq
```

rather than the more general relation

```
HilbertScissorsEquicomplementable
```

and it is not equality of real numbers.

3.8 Relationship with Earlier Book II Propositions

3.8.1 Proposition II.1

II.3 is a direct formal corollary of II.1. The import is

```
import CGJteamLab.Proposition2_1
```

and the only substantial theorem called in the final proof is

```
euclid_proposition_2_1_two
```

The specialization $PQ = MC$ converts the general binary distributive pattern into II.3 immediately.

The relationship can be written schematically as

$$\mathrm{Rect}(BC, PQ) = \mathrm{Rect}(BM, PQ) + \mathrm{Rect}(MC, PQ)$$

followed by

$$PQ := MC.$$

The final term then becomes $\mathrm{Rect}(MC, MC)$, represented by the square on MC.

3.8.2 Proposition II.2

The current Lean file does *not* import Proposition II.2, and its public theorem is not used in the proof. II.2 and II.3 are therefore sibling corollaries of the II.1 binary API rather than a formal chain II.1 $\to$ II.2 $\to$ II.3.

The contrast is nevertheless useful. II.2 required cyclic representative rotation because its chosen side order was opposite to the order returned by II.1. II.3 chooses the fixed segment MC in precisely the position expected by II.1, so even that transport disappears. This is evidence that the II.1 API is not merely sufficient but well aligned with the geometry of the next Book II identities.

3.9 Relationship with Book I Infrastructure

Euclid's local proof explicitly calls I.46 to construct the square and I.31 to draw the parallel. The current Lean theorem does not replay those constructions. Instead it assumes the resulting rectangle, cut, parallelograms, side congruence, and square as data.

Below II.1, however, the formal dependency chain still rests on the Book I infrastructure for rectangles, right angles, squares, parallelograms, and exact scissors decomposition. The binary split ultimately uses the exact rectangle cut machinery developed over the Book I scissors layer. Thus the difference is one of modular organization, not a change from synthetic to numerical geometry.

3.10 Formalization Notes and Hidden Geometric Data

The elementary picture hides several obligations that appear explicitly in the current theorem signature:

- $B - M - C$, fixing the cut point and its orientation on the whole side;
- $D - L - E$, locating the opposite endpoint of the cut;
- $D - X - C$ and $M - X - L$, recording the intersection X of the diagonal and the cut line used by the inherited exact scissors theorem;
- `IsParallelogram Geo L D B M` for the left piece;
- `IsParallelogram Geo C E L M` for the right piece;

- $CE \cong MC$, making the outer rectangle have the required fixed second side magnitude;
- a supplied square $MCFG$;
- arbitrary representatives of the whole rectangle and the BM-by-MC product rectangle.

The point X is invisible in Euclid's verbal statement. It belongs to the triangulated exact-dissection interface inherited from II.1. It should therefore be read as hidden formal geometry rather than as an additional step in Euclid's classical construction.

There are also useful negative observations. No `SameRay`, `SameSide`, or `OppositeSide` hypothesis occurs in `Proposition2_3.lean` itself, and no side-order swap is performed. Those lower-level obligations have already been absorbed by the reusable rectangle and II.1 layers.

3.11 Geometry, Content, Representation, and Later Arithmetic

3.11.1 Geometry

The geometric layer consists of the rectangle $BCED$, the cut point M, the opposite cut point L, the diagonal witness X, the two parallelogram pieces, the congruence $CE \cong MC$, and the square $MCFG$.

3.11.2 Content / scissors

The conclusion is the exact relation `HilbertScissorsEq`. The public proof does not weaken the result to general equicomplementability. Exactness is inherited from the binary rectangle split and representative transports already proved in II.1.

3.11.3 Representation

Representation is present but simpler than in II.2. The theorem accepts an arbitrary rectangle representative for $\mathrm{Rect}(BC, MC)$ and an arbitrary representative for $\mathrm{Rect}(BM, MC)$, together with a concrete square on MC. The binary II.1 theorem is already independent of the particular concrete rectangles appearing in the cut, so no additional representation transport is required locally.

3.11.4 Later arithmetic / numerical area

Neither layer is used. No segment product is defined, no fixed unit segment is introduced, and no real-valued area function appears. The expression $xz = yz + z^2$ is only the modern algebraic shadow of a geometric scissors identity.

3.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
```

```
[HilbertEuclideanPlane Geo]
```

and therefore lives in the same Euclidean project layer as II.1 and II.2. Inspection of the current 96-line source shows one public theorem declaration and no proposition-local `axiom`, `sorry`, or `admit`.

This should not be described as "axiom-free." The result inherits the axioms and typeclass assumptions of the established Hilbert/Euclidean/scissors library transitively. No current `#print axioms` output for II.3 was supplied with the documentation sources, so a transitive axiom list is not asserted here.

The implementation checkpoint records that the II.3 Lean theorem had already been build-checked during formalization. The present documentation pass uses the current source as authoritative but does not contain the full Lean project needed to repeat that build locally.

3.13 Hartshorne's Semantic Control

Hartshorne's general analysis of Euclid's area theory states that the results of Book II are valid when Euclid's equality of figures is read as equality of content [5]. His condensed list of selected Book II propositions does not separately single out II.3, so the source should not be cited as providing a special proof of this proposition.

The general semantic point is sufficient. II.3 compares a rectangle with the union of another rectangle and a square. In the present reconstruction this comparison is sharpened, for this configured case, to exact scissors equality. The modern identity

$$xz = yz + z^2$$

therefore belongs to Hartshorne's "algebra of areas" only as an interpretive shadow of operations on geometric figures.

II.3 also clarifies the representative issue. The term called "the square on MC" need not be the literal internal square-shaped piece of the cut; the formal theorem accepts the separately supplied square $MCFG$ because the rectangle calculus already proves independence of representatives at the scissors level.

3.14 Hilbert's Layering

Hilbert provides the complementary warning against reading later segment arithmetic into the proof. His synthetic theory of plane content distinguishes finite equidecomposition from the more general content relations, while his arithmetic of segments is introduced later and requires a fixed unit segment and additional parallel constructions [2].

II.3 remains entirely below that arithmetic boundary. Its formal chain is

$$\begin{gathered}\text{rectangle and square geometry}\\ \Downarrow\\ \text{exact binary dissection from II.1}\\ \Downarrow\\ \text{representative-independent exact scissors equality.}\end{gathered}$$

In fact II.3 is even more direct than II.2: no geometric commutativity theorem is needed because the ordered containing segments already agree with the II.1 binary interface.

3.15 Dependency Summary

The actual Lean dependency chain is

Hilbert incidence + Euclidean plane layer

$\Downarrow$

Book I right-angle, square, parallelogram, and scissors infrastructure

$\Downarrow$

`HilbertRectangle`

`IsRectangleContainedBy`

`square_is_rectangle_contained_by_side`

$\Downarrow$

Proposition II.1 binary API

`euclid_proposition_2_1_two`

specialized by $PQ = MC$

$\Downarrow$

`euclid_proposition_2_3`.

The classical local chain is different:

$$\text{I.46} \longrightarrow \text{I.31} \longrightarrow \text{rectangle cut into rectangle + square} \longrightarrow \text{II.3.}$$

New geometric axiom in II.3:	no
New content axiom in II.3:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New `HilbertRectangle` theorem introduced by II.3:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses an earlier Book II proposition:	yes, II.1
Imports II.2:	no
Direct side-order transport used:	no
Exact `HilbertScissorsEq`:	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem build:	previously checked; not re-run here
Full relevant Lean build:	previously checked; not re-run here
Proposition-local `sorry`/`admit`:	no

3.16 Conclusion

Proposition II.3 is one of the cleanest tests of the Book II rectangle API. Classically, Euclid constructs a square on one segment and enlarges it by a parallel construction until the whole rectangle contained by the original line and that segment is obtained. Formally, the same identity is already contained in the binary theorem of II.1.

The decisive specialization is simply

$$PQ = MC.$$

It turns the second part rectangle of II.1 into $\mathrm{Rect}(MC, MC)$, and the supplied square on MC provides exactly that representative. Nothing else is needed.

This also explains the precise relation between II.2 and II.3. Both are formal corollaries of II.1, but II.2 needs geometric side-order transport while II.3 does not. The result is therefore a very short Lean proof without being an algebraic proof in the modern numerical sense: the content remains rectangles, squares, congruent sides, and exact scissors equality throughout.

Chapter 4

Proposition II.4

4.1 Introduction

Proposition II.4 is the first proposition of Book II in the present reconstruction whose formal proof is naturally expressed as a composition of *two different earlier Book II propositions.* The classical theorem says that when a segment is cut into two parts, the square on the whole is equal to the two squares on the parts together with twice the rectangle contained by the parts.

The modern algebraic shadow is the familiar identity

$$(y+z)^2 = y^2 + z^2 + 2yz.$$

The formal proof, however, does not define a product of segment lengths and does not evaluate a numerical area. It remains entirely inside the rectangle and exact-scissors layer established at the beginning of Book II.

The current Lean source begins with

```
import CGJteamLab.Proposition2_2
import CGJteamLab.Proposition2_3
```

and the proof uses those imports exactly as the architecture suggests:

1. Proposition II.2 splits the square on the whole segment into two rectangles pairing the whole with the two parts;
2. Proposition II.3 expands each of those rectangles into one square and one cross rectangle;
3. representation transport identifies the two cross rectangles as two representatives of the same “rectangle contained by” the two parts;
4. addition and transitivity in `HilbertScissorsEq` combine the results.

Thus the substantial new fact at II.4 is not a new geometric construction or a new arithmetic law. It is the observation that the Book II rectangle API is already compositional enough to express the square-of-a-sum pattern using previously proved exact scissors theorems.

4.2 Euclid's Statement and Classical Configuration

4.2.1 The statement

In Heath's wording, Proposition II.4 states:

> If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.

Let AB be cut at C. In the geometric shorthand used throughout this volume, the statement is

$$\mathrm{Sq}(AB) = \mathrm{Sq}(AC) + \mathrm{Sq}(CB) + 2\,\mathrm{Rect}(AC, CB).$$

Here Sq and Rect are expository names for geometric figures and their content. The symbol 2 means that two equal rectangle contents occur in the decomposition; it is not scalar multiplication inside the Lean segment language.

4.2.2 The classical diagram

Euclid describes the square $ADEB$ on AB by I.46 and joins the diagonal BD. Through the cut point C he draws CF parallel to the vertical sides of the square, and through the intersection G of CF with BD he draws HK parallel to AB and DE by I.31 [1]. The square is thereby divided into four subfigures named AG, CK, HF, and GE.

The figure is the classical Euclidean configuration. It is deliberately not presented as the literal Lean input diagram. The formal theorem receives three configured rectangle-cut diagrams inherited from II.2 and II.3, rather than reconstructing this single classical picture point by point.

Diagrammatic audit. Figure 4.1 is retained because it is the genuine new classical state of II.4: one square, its diagonal, and the two auxiliary parallels. It should not also carry the formal proof DAG. The current Lean proof first consumes an II.2 square cut and then two independent II.3 rectangle cuts, one in the reversed orientation $B - C - A$ and one in the direct orientation $A - C - B$. Those three states are therefore shown later, at the steps where the corresponding imported theorem is actually used. Pure endpoint reversal, representative rotation, addition of scissors equalities, and transitivity do not receive separate geometric figures.

4.3 Classical Proof

Euclid's proof first identifies the lower-right parallelogram $CGKB$ as a square on CB. Since $CF \parallel AD$, the diagonal BD gives the relevant angle equality by I.29. The square has $BA = AD$, so I.5 gives equal base angles in triangle ADB. Combining these angle equalities and using I.6 yields

$$BC = CG.$$

Opposite sides in the surrounding parallelograms are equal by I.34, so the four sides of $CGKB$ are equal. Right-angle information from the parallels and the square then shows that $CGKB$ is a square on CB [1].

The same reasoning identifies HF as the square on AC. The two remaining subfigures AG and GE are complements about the diagonal of the outer square, and Proposition I.43 gives

$$AG = GE.$$

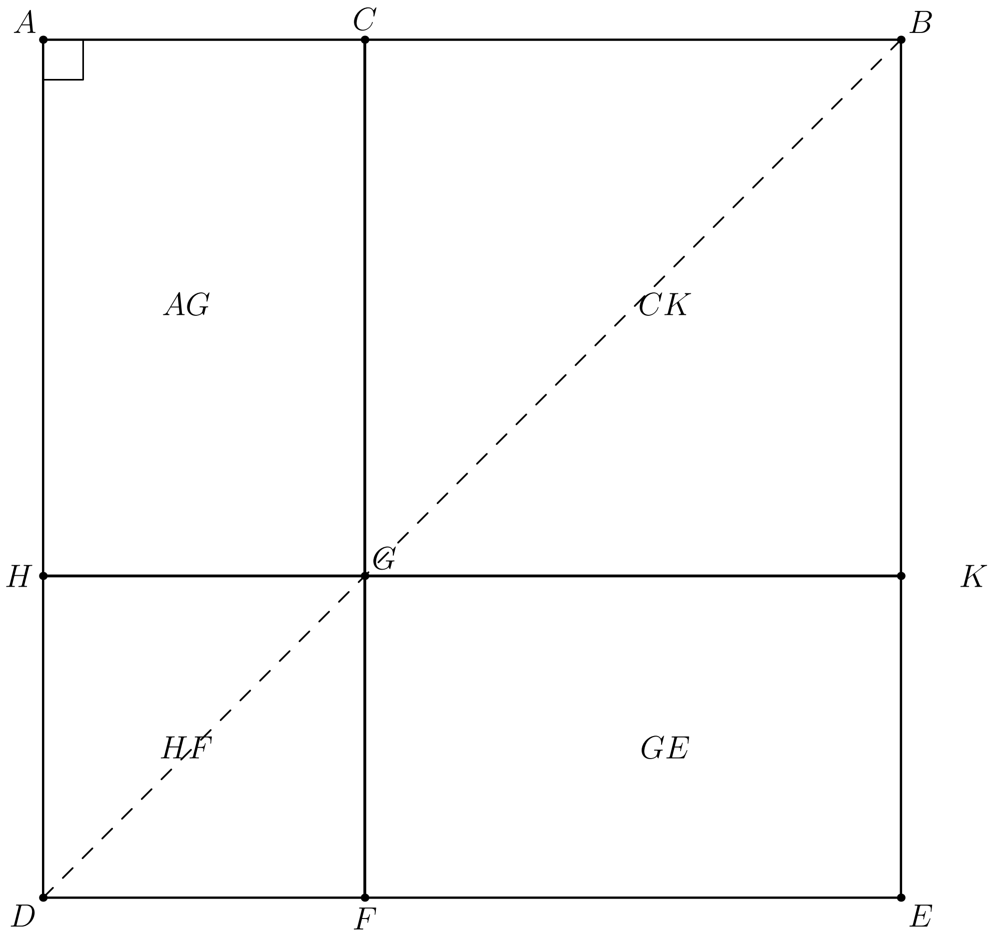


Figure 4.1: The classical configuration of Euclid II.4. The square $ADEB$ is cut at C by CF, the diagonal BD meets it at G, and the parallel HK through G produces four subfigures. Euclid identifies HF and CK as the squares on AC and CB, and I.43 compares the two remaining complements AG and GE. This picture records Euclid's source geometry; it does not identify the three separate configured cut diagrams supplied to the current Lean theorem.

Moreover AG has adjacent side magnitudes AC and CG, while $CG = CB$. Hence AG is the rectangle contained by AC, CB, and $AG + GE$ is twice that rectangle. Since the four subfigures constitute the whole square $ADEB$, Euclid obtains the required equality [1].

The local classical dependency graph is therefore

$$\begin{gathered}
\text{I.46: square on } AB \quad + \quad \text{I.31: required parallels} \\
\Downarrow \\
\text{I.29 + I.5 + I.6: } BC = CG \\
\Downarrow \\
\text{I.34: recognize the two smaller squares} \\
\Downarrow \\
\text{I.43: complements } AG = GE \\
\Downarrow \\
\mathrm{Sq}(AB) = \mathrm{Sq}(AC) + \mathrm{Sq}(CB) + 2\,\mathrm{Rect}(AC, CB) \\
\Downarrow \\
\text{II.4.}
\end{gathered}$$

This classical DAG is important because it is *not* the DAG of the Lean proof. In particular, the formal reconstruction does not locally call I.43 and does not prove the two cross rectangles equal by diagonal complements. That equality has already been absorbed into the representative-independent rectangle calculus developed after Book I.

4.4 The Geometric Identity in the Current Formalization

The public Lean theorem starts with a strict cut

$$A - C - B.$$

It receives a square on AB, two square representatives on the two parts, and two arbitrary representatives of the cross rectangle contained by AC and CB.

At the level of exact scissors terms the final conclusion has the form

$$\mathrm{Sq}(AB) \simeq_{\mathrm{sc}} (\mathrm{Rect}(AC, CB) + \mathrm{Sq}(CA)) + (\mathrm{Rect}(AC, CB) + \mathrm{Sq}(CB)).$$

The use of CA rather than AC in the first square is only an endpoint orientation choice. Segments are unoriented for congruence purposes, so this is the same geometric side magnitude.

The two occurrences of the cross rectangle are deliberately represented by independent quadrilaterals. The theorem therefore says more than "the same literal rectangle occurs twice in one drawing." It says that any two representatives satisfying

```
IsRectangleContainedBy Geo ... A C C B
```

may occupy the two cross-term positions, because the rectangle layer has already proved representative independence at the exact scissors level.

A modern mnemonic is

$$x = y + z \qquad \Longrightarrow \qquad x^2 = y^2 + z^2 + 2yz.$$

This is useful for recognizing the combinatorial pattern only. No object corresponding to yz is constructed by `HilbertSegmentMul` in the II.4 proof.

4.5 Reusable Book II Infrastructure

4.5.1 Proposition II.2: split the square on the whole

The first reusable theorem gives

$$\mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, CB) \simeq_{\mathrm{sc}} \mathrm{Sq}(AB).$$

In the current source it is called exactly as

```
have hBCA : Geo.Between B C A :=
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2

-----------------------------------------------------------------
-- II.2:
--
--   Rect(AB,AC) + Rect(AB,CB) = Square(AB).
```

```
------------------------------------------------------------

have hII2 :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo U0 U1 U2 U3 +
       hilbertParallelogramTerm Geo V0 V1 V2 V3)
      (hilbertParallelogramTerm Geo A B E0 D0) :=
  euclid_proposition_2_2
    Geo
    A B D0 E0 L0 C X0
    hSquareAB
    hACB
    hD0L0E0
    hD0X0B
    hCX0L0
    hII2LeftPar
    hII2RightPar
    U0 U1 U2 U3
    V0 V1 V2 V3
    hAB_AC
    hAB_CB
```

The final II.4 proof uses the symmetry of this relation, so the square becomes the starting point and the two whole-by-part rectangles become the two branches to be expanded.

4.5.2 Proposition II.3: expand a whole-by-part rectangle

Proposition II.3 has the generic geometric pattern

$$\mathrm{Rect}(XY, ZY) \simeq_{\mathrm{sc}} \mathrm{Rect}(XZ, ZY) + \mathrm{Sq}(ZY)$$

for a strict division point Z on XY, with the configured rectangle cut supplied explicitly.

II.4 calls this theorem twice. The right branch uses the original orientation $A - C - B$ directly:

$$\mathrm{Rect}(AB, CB) \simeq_{\mathrm{sc}} \mathrm{Rect}(AC, CB) + \mathrm{Sq}(CB).$$

The left branch reads the same line in the reverse direction $B - C - A$:

$$\mathrm{Rect}(BA, CA) \simeq_{\mathrm{sc}} \mathrm{Rect}(BC, CA) + \mathrm{Sq}(CA).$$

After endpoint reversal and rectangle rotation, its cross term is another representative of $\mathrm{Rect}(AC, CB)$.

4.5.3 Rectangle rotation and endpoint reversal

The only side-order normalization that is mathematically visible in II.4 is

```
rectangle_contained_by_swap
```

applied to the second cross-rectangle representative. It cyclically rotates the concrete rectangle and exchanges the roles of its containing sides while returning an exact scissors certificate for the change of presentation.

Endpoint reversal is handled by the already established congruence theorem

```
CongruentSwapSecond
```

so that AB may be read as BA, AC as CA, and CB as BC when the reversed II.3 invocation requires that orientation. These steps are representation transport, not new metric statements.

4.5.4 Exact scissors addition

Once the two II.3 branches are available, the constructor

```
HilbertScissorsEq.add
```

combines them componentwise. Transitivity then attaches the resulting four-term expression to the square through the symmetric II.2 equality.

No subtraction, cancellation, positivity, content order, or numerical area operation occurs.

4.6 Proof Architecture of the Reconstruction

4.6.1 Step 1: reverse the cut orientation for the left branch

From $A - C - B$ the proof derives $B - C - A$:

```
have hBCA : Geo.Between B C A :=
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2
```

This is the only explicit betweenness reversal performed locally.

4.6.2 Step 2: apply II.2 to the square on the whole

The theorem `euclid_proposition_2_2` produces the exact decomposition of the square into the two rectangles $\mathrm{Rect}(AB, AC)$ and $\mathrm{Rect}(AB, CB)$. Nothing about the smaller squares is used yet.

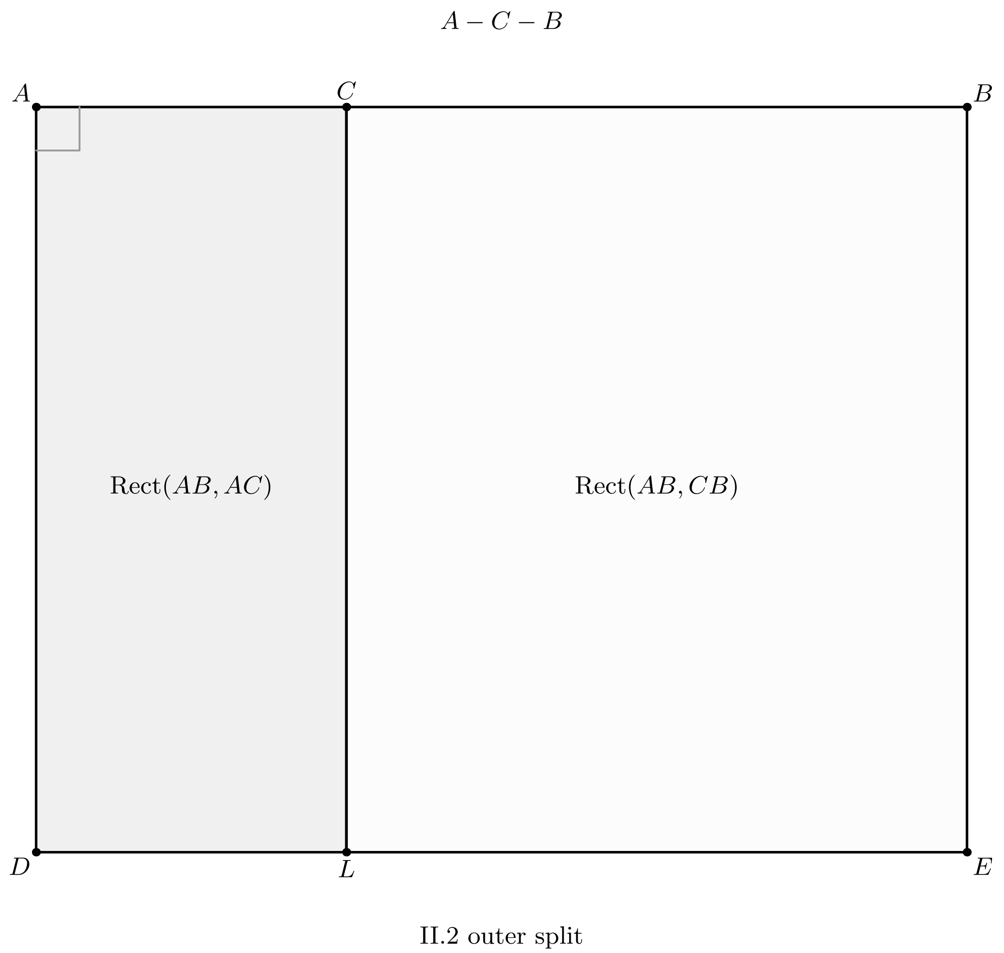


Figure 4.2: The first formal geometric state consumed by II.4. Proposition II.2 cuts the supplied square on AB at the strict division point $A - C - B$ and identifies its content with two whole-by-part rectangles, $\mathrm{Rect}(AB, AC)$ and $\mathrm{Rect}(AB, CB)$. No square on either part and no cross rectangle has entered yet. The drawing shows the dissection certified by II.2, not the arbitrary representatives U and V used by the public Lean signature.

4.6.3 Step 3: prepare the reversed left II.3 call

The first II.2 rectangle is formally contained by AB, AC, while the reversed II.3 call expects BA, CA. The same concrete rectangle is repackaged using endpoint reversal. The second final cross-rectangle representative is then rotated from the side order AC, CB to CB, AC and read, after endpoint reversal, as BC, CA.

The current source makes these transports explicit:

```
have hAB_AC_rev :
    IsRectangleContainedBy Geo
      U0 U1 U2 U3 B A C A :=
  And.intro
    hAB_AC.1
    (And.intro
      (CongruentSwapSecond
        Geo U0 U1 A B hAB_AC.2.1)
      (CongruentSwapSecond
        Geo U1 U2 A C hAB_AC.2.2))

-------------------------------------------------------------
-- Rotate the second cross rectangle:
--
--   Rect(AC,CB) -> Rect(CB,AC).
--
-- Endpoint reversal then reads the rotated representative as
-- Rect(BC,CA), exactly what the reversed II.3 needs.
-------------------------------------------------------------

have hSwapResult :=
  rectangle_contained_by_swap
    Geo
    T20 T21 T22 T23
    A C C B
    hCross2

have hCross2Swap :=
  hSwapResult.1

have hRotateCross2 :=
  hSwapResult.2

have hCross2ForLeft :
    IsRectangleContainedBy Geo
      T21 T22 T23 T20 B C C A :=
  And.intro
    hCross2Swap.1
    (And.intro
      (CongruentSwapSecond
        Geo T21 T22 C B hCross2Swap.2.1)
      (CongruentSwapSecond
        Geo T22 T23 A C hCross2Swap.2.2))
```

4.6.4 Step 4: expand the left rectangle by II.3

The reversed invocation gives

$$\mathrm{Rect}(BA, CA) \simeq_{\mathrm{sc}} \mathrm{Rect}(BC, CA) + \mathrm{Sq}(CA).$$

The rotated cross term is then transported back to the arbitrary final AC, CB representative. The exact source block is:

```
have hLeftRaw :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo U0 U1 U2 U3)
      (hilbertParallelogramTerm Geo T21 T22 T23 T20 +
       hilbertParallelogramTerm Geo C A F G) :=
  euclid_proposition_2_3
    Geo
    B A D1 E1 L1 C X1
    hRectLeft
    hAE1_CA
    hBCA
    hD1L1E1
    hD1X1A
    hCX1L1
    hLeftLeftPar
    hLeftRightPar
    F G
    hSquareCA
    U0 U1 U2 U3
    T21 T22 T23 T20
    hAB_AC_rev
    hCross2ForLeft

-------------------------------------------------------------
-- Return the left cross term to the original arbitrary
-- representative T20-T21-T22-T23.
-------------------------------------------------------------

have hRotateCross2Back :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo T21 T22 T23 T20)
      (hilbertParallelogramTerm Geo T20 T21 T22 T23) :=
  HilbertScissorsEq.symm
    (Geo := Geo)
    hRotateCross2

have hLeftTarget :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo U0 U1 U2 U3)
      (hilbertParallelogramTerm Geo T20 T21 T22 T23 +
       hilbertParallelogramTerm Geo C A F G) :=
  HilbertScissorsEq.trans
    (Geo := Geo)
    hLeftRaw
    (HilbertScissorsEq.add
      (Geo := Geo)
      hRotateCross2Back
      (HilbertScissorsEq.refl
        (Geo := Geo)
```

```
          (hilbertParallelogramTerm Geo C A F G)))
```

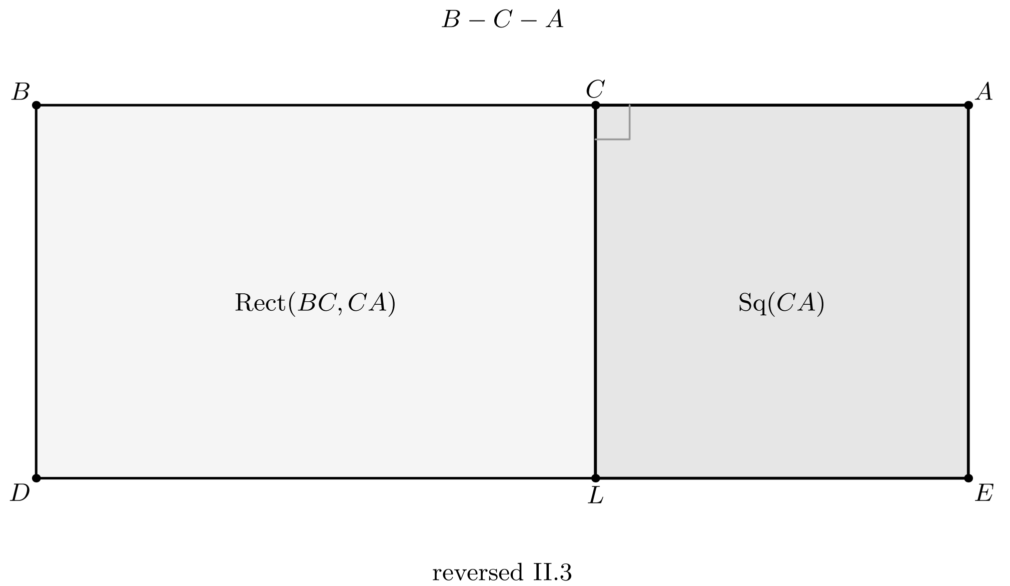


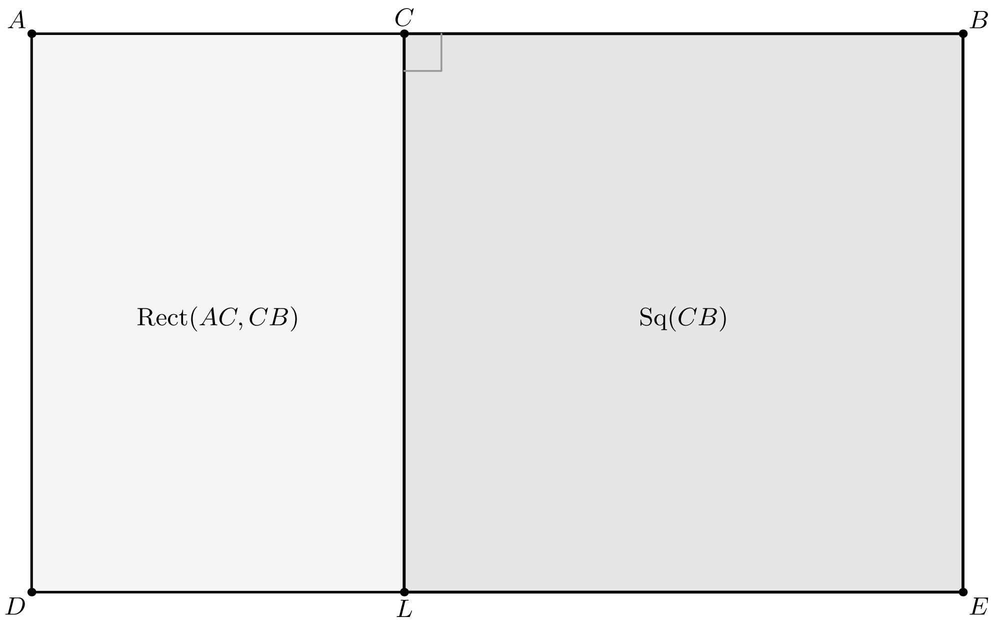
Figure 4.3: The left branch is II.3 read on the same divided segment in the reverse order $B-C-A$. The rectangle Rect(BA, CA) is cut at C into Rect(BC, CA) and the square on CA. In the Lean proof the cross term is supplied by an arbitrary representative of Rect(AC, CB); cyclic rectangle rotation and endpoint reversal make that representative fit the displayed BC, CA side order. Those representation transports do not change the geometric dissection and are therefore explained in the caption rather than drawn as a second state.

4.6.5 Step 5: expand the right rectangle by II.3

The right branch needs no reversal of the whole segment. It directly produces

$$\mathrm{Rect}(AB, CB) \simeq_{\mathrm{sc}} \mathrm{Rect}(AC, CB) + \mathrm{Sq}(CB).$$

$A - C - B$



direct II.3

Figure 4.4: The second branch uses II.3 directly in the original orientation $A - C - B$. Cutting the supplied $\mathrm{Rect}(AB, CB)$ at C gives the cross rectangle $\mathrm{Rect}(AC, CB)$ together with the square on CB. Unlike the left branch, no reversal of the divided whole segment is required. The cross rectangle shown here is a geometric content type; the theorem permits it to be represented by a quadrilateral independent of the cross representative used in the reversed branch.

4.6.6 Step 6: add the branches and attach them to the square

The two II.3 relations are added. Finally the result is composed with the symmetric II.2 relation. The closing source is:

```
have hRight :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo V0 V1 V2 V3)
      (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
       hilbertParallelogramTerm Geo C B H K) :=
  euclid_proposition_2_3
    Geo
    A B D2 E2 L2 C X2
    hRectRight
    hBE2_CB
    hACB
    hD2L2E2
    hD2X2B
    hCX2L2
    hRightLeftPar
    hRightRightPar
    H K
    hSquareCB
    V0 V1 V2 V3
    T10 T11 T12 T13
    hAB_CB
    hCross1

-----------------------------------------------------------------
-- Expand both II.2 summands simultaneously.
-----------------------------------------------------------------

have hExpanded :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo U0 U1 U2 U3 +
       hilbertParallelogramTerm Geo V0 V1 V2 V3)
      ((hilbertParallelogramTerm Geo T20 T21 T22 T23 +
        hilbertParallelogramTerm Geo C A F G) +
       (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
        hilbertParallelogramTerm Geo C B H K)) :=
  HilbertScissorsEq.add
    (Geo := Geo)
    hLeftTarget
    hRight

-----------------------------------------------------------------
-- Square(AB)
--   = Rect(AB,AC) + Rect(AB,CB)
--   = the four terms of II.4.
-----------------------------------------------------------------

exact
  HilbertScissorsEq.trans
    (Geo := Geo)
    (HilbertScissorsEq.symm
      (Geo := Geo)
      hII2)
```

```
      hExpanded
```

Diagrammatic audit. Figures 4.2–4.4 exhaust the new geometric states of the formal proof: the outer II.2 split and the two II.3 branch dissections. Step 1 merely reverses the already known betweenness $A - C - B$; Step 3 repackages and cyclically rotates rectangle representatives; and Step 6 applies `HilbertScissorsEq.add` followed by symmetry and transitivity. Drawing a fourth formal picture for the final four-term sum would incorrectly suggest a newly constructed common dissection, whereas the Lean theorem intentionally allows the two cross rectangles to be independent representatives of the same contained-by data.

The actual Lean DAG is therefore

$$
\begin{array}{c}
\texttt{euclid\_proposition\_2\_2} \\
\Downarrow \\
\mathrm{Sq}(AB) \simeq_{\mathrm{sc}} \mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, CB) \\
\swarrow \qquad\qquad \searrow \\
\text{reversed II.3} \qquad\qquad \text{direct II.3} \\
\searrow \qquad\qquad \swarrow \\
\texttt{HilbertScissorsEq.add} \\
\Downarrow \\
\text{two squares + two cross rectangles} \\
\Downarrow \\
\texttt{euclid\_proposition\_2\_4}.
\end{array}
$$

This is substantially shorter conceptually than Euclid's local geometric DAG, but it is not an algebraic shortcut. The earlier propositions already package the relevant geometric dissections as reusable exact-scissors theorems.

4.7 Lean Formalization

The current public theorem signature is copied from the compiling `Proposition2_4.lean` source:

```
theorem euclid_proposition_2_4
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (A B C : Geo.Point)

    ------------------------------------------------------------------
    -- II.2 diagram: square on AB cut at C.
    ------------------------------------------------------------------

    (D0 E0 L0 X0 : Geo.Point)
    (hSquareAB : IsSquare Geo A B E0 D0)

    (hACB : Geo.Between A C B)
    (hD0L0E0 : Geo.Between D0 L0 E0)
    (hD0X0B : Geo.Between D0 X0 B)
    (hCX0L0 : Geo.Between C X0 L0)

    (hII2LeftPar :
```

```
    IsParallelogram Geo L0 D0 A C)
  (hII2RightPar :
    IsParallelogram Geo B E0 L0 C)

  -- Arbitrary representatives of Rect(AB,AC) and Rect(AB,CB).
  (U0 U1 U2 U3 : Geo.Point)
  (V0 V1 V2 V3 : Geo.Point)

  (hAB_AC :
    IsRectangleContainedBy Geo
      U0 U1 U2 U3 A B A C)

  (hAB_CB :
    IsRectangleContainedBy Geo
      V0 V1 V2 V3 A B C B)

  ------------------------------------------------------------------
  -- II.3 diagram for the left term.
  --
  -- We read the divided whole segment in the reverse direction:
  --
  --   B-C-A.
  --
  -- Thus II.3 gives
  --
  --   Rect(BA,CA) = Rect(BC,CA) + Square(CA).
  ------------------------------------------------------------------

  (D1 E1 L1 X1 : Geo.Point)

  (hRectLeft :
    IsRectangle Geo B A E1 D1)

  (hAE1_CA :
    Geo.Congruent A E1 C A)

  (hD1L1E1 : Geo.Between D1 L1 E1)
  (hD1X1A : Geo.Between D1 X1 A)
  (hCX1L1 : Geo.Between C X1 L1)

  (hLeftLeftPar :
    IsParallelogram Geo L1 D1 B C)
  (hLeftRightPar :
    IsParallelogram Geo A E1 L1 C)

  -- Square on CA, i.e. on the same unoriented segment as AC.
  (F G : Geo.Point)
  (hSquareCA : IsSquare Geo C A F G)

  ------------------------------------------------------------------
  -- II.3 diagram for the right term:
  --
  --   Rect(AB,CB) = Rect(AC,CB) + Square(CB).
  ------------------------------------------------------------------

  (D2 E2 L2 X2 : Geo.Point)

  (hRectRight :
```

```
    IsRectangle Geo A B E2 D2)

  (hBE2_CB :
    Geo.Congruent B E2 C B)

  (hD2L2E2 : Geo.Between D2 L2 E2)
  (hD2X2B : Geo.Between D2 X2 B)
  (hCX2L2 : Geo.Between C X2 L2)

  (hRightLeftPar :
    IsParallelogram Geo L2 D2 A C)
  (hRightRightPar :
    IsParallelogram Geo B E2 L2 C)

  -- Square on CB.
  (H K : Geo.Point)
  (hSquareCB : IsSquare Geo C B H K)

  ------------------------------------------------------------
  -- Two arbitrary representatives of the repeated cross rectangle
  -- Rect(AC,CB).
  ------------------------------------------------------------

  (T10 T11 T12 T13 : Geo.Point)
  (T20 T21 T22 T23 : Geo.Point)

  (hCross1 :
    IsRectangleContainedBy Geo
      T10 T11 T12 T13 A C C B)

  (hCross2 :
    IsRectangleContainedBy Geo
      T20 T21 T22 T23 A C C B) :

  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo A B E0 D0)
    ((hilbertParallelogramTerm Geo T20 T21 T22 T23 +
      hilbertParallelogramTerm Geo C A F G) +
     (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
      hilbertParallelogramTerm Geo C B H K)) := by
```

Several aspects of the signature deserve explicit interpretation.

First, one strict division relation $A - C - B$ is shared by the theorem, but the public statement receives *three* configured cut diagrams:

- the II.2 square-cut diagram, with points D_0, E_0, L_0, X_0;
- the left II.3 rectangle-cut diagram, with points D_1, E_1, L_1, X_1;
- the right II.3 rectangle-cut diagram, with points D_2, E_2, L_2, X_2.

The theorem does not construct these configurations from A, C, B alone. Rather, it proves the content identity once the corresponding geometric witnesses have been supplied.

Second, $U_0U_1U_2U_3$ and $V_0V_1V_2V_3$ are arbitrary representatives of the two rectangles produced at the II.2 level. They need not be literal subrectangles of the square ABE_0D_0.

Third, the cross terms are represented independently by $T_{10}T_{11}T_{12}T_{13}$ and $T_{20}T_{21}T_{22}T_{23}$. Both satisfy the same `IsRectangleContainedBy` predicate for AC, CB. This makes the formal "twice the rectangle" statement representative-independent rather than tied to one duplicated quadrilateral.

Fourth, the conclusion is exactly

```
HilbertScissorsEq Geo ...
```

and not the weaker

```
HilbertScissorsEquicomplementable Geo ...
```

relation. Exactness is inherited from II.2, II.3, rectangle rotation, formal addition, and transitivity.

4.8 Relationship with Earlier Book II Propositions

4.8.1 Proposition II.1

II.4 does not import `Proposition2_1` directly. Nevertheless II.1 is the underlying distributive core because both II.2 and II.3 are themselves built from its binary rectangle split. The dependency is therefore transitive:

$$\text{II.1} \longrightarrow \text{II.2 and II.3} \longrightarrow \text{II.4}.$$

This is the first point where the investment in II.1 becomes visibly compositional at the proposition level. II.2 and II.3 were sibling specializations of the binary II.1 API; II.4 uses both siblings together.

4.8.2 Proposition II.2

II.2 supplies exactly the outer decomposition needed by II.4:

$$\mathrm{Sq}(AB) \simeq_{\mathrm{sc}} \mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, CB).$$

The current II.4 source imports II.2 explicitly and calls its public theorem once.

4.8.3 Proposition II.3

II.3 supplies the expansion of each whole-by-part rectangle. The current proof calls the same public theorem twice, once after reversing the line orientation and once directly. Thus II.4 is a genuine composition theorem over the public Book II API rather than a restatement of one earlier helper.

4.9 Relationship with Book I Infrastructure

Euclid's own local proof depends explicitly on I.46, I.31, I.29, I.5, I.6, I.34, and I.43. The formal II.4 theorem does not replay any of those arguments locally. Their geometric content has entered the current project through the established square, parallel, parallelogram, rectangle, and scissors layers and through the already proved II.2 and II.3 theorems.

The contrast with I.43 is especially instructive. Euclid needs I.43 to prove that the two complements AG and GE in one concrete square are equal. Lean instead receives two arbitrary cross-rectangle representatives and normalizes one of them with `rectangle_contained_by_swap`. Representative independence is already built into the rectangle API, so no local complement theorem is required.

The formal proof is therefore not a transcription of Euclid's diagram. It is a proof over a stronger reusable library whose theorems were themselves proved synthetically from the established Hilbert/Book I geometry.

4.10 Formalization Notes and Hidden Geometric Data

The polished identity hides a substantial amount of explicit positional data. The current signature contains:

- $A - C - B$, fixing the strict order of the original segment;
- a complete square-cut configuration for the II.2 call;
- two complete rectangle-cut configurations for the two II.3 calls;
- three diagonal/cut intersection witnesses X_0, X_1, X_2;
- six parallelogram hypotheses certifying the two pieces in each cut;
- one square on the whole and one square on each part;
- two arbitrary II.2 rectangle representatives;
- two arbitrary representatives of the repeated cross rectangle.

The proof then adds two representational obligations not visible in Euclid's statement:

- reverse $A - C - B$ to $B - C - A$ for the left II.3 call;
- rotate one cross-rectangle representative and reverse segment endpoints so that the ordered side data match the reversed II.3 interface.

There is also useful negative information. No `SameRay`, `SameSide`, or `OppositeSide` datum occurs in the public II.4 proof. No new right-angle propagation theorem, no new parallel theorem, and no new scissors constructor is introduced. Those obligations are encapsulated inside the imported rectangle and Book II layers.

4.11 Geometry, Content, Representation, and Later Arithmetic

4.11.1 Geometry

The geometry consists of one segment cut $A - C - B$, one square on the whole, two squares on the parts, and three configured rectangle dissections. The classical source packages these ideas in one square diagram; the Lean theorem packages them as three reusable instances of already formalized cut theorems.

4.11.2 Content / scissors

The conclusion is exact `HilbertScissorsEq`. The proof only uses reflexivity, symmetry, transitivity, compatibility with addition, and exact scissors certificates supplied by II.2, II.3, and rectangle rotation. General equicomplementability is not required.

4.11.3 Representation

Representation is the genuinely nontrivial local layer. Endpoint order on the containing segments must be normalized for the reversed II.3 branch, and one cross rectangle must be cyclically rotated before it can be used in that branch. The final theorem nevertheless permits two completely independent representatives of $\mathrm{Rect}(AC, CB)$.

This is the precise formal content of "twice the rectangle" in the current architecture: two scissors terms representing the same geometric rectangle magnitude occur as separate summands.

4.11.4 Later arithmetic / numerical area

Neither later layer is used. The proof imports no segment multiplication module, chooses no unit segment, proves no distributive law for a product of segment classes, and evaluates no real-valued area function.

The formula

$$(y + z)^2 = y^2 + z^2 + 2yz$$

should therefore remain explicitly labelled as the modern algebraic shadow of the synthetic construction.

4.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore belongs to the same Euclidean project layer as II.1–II.3.

The current `Proposition2_4.lean` file introduces no proposition-local `axiom`, `sorry`, or `admit`. The theorem was compiled successfully in the project. The supplied transitive axiom audit is:

```
#print axioms Geometry.euclid_proposition_2_4

'Geometry.euclid_proposition_2_4' depends on axioms:
[propext, Classical.choice, bookZero_nullSegment1, Quot.sound]
```

Accordingly II.4 should not be described as "axiom-free." The precise statement is that it introduces no new proposition-specific axiom and no `sorry`; its transitive dependencies are exactly the four declarations listed above. In particular the list contains no `sorryAx`.

The compile result reported for the current theorem is clean. A separate full `lake build` result was not part of the documentation input used here, so the status table below does not claim that stronger check.

4.13 Hartshorne's Semantic Control

Hartshorne's Section 22 supplies the appropriate general semantic control for Book II: Euclid's equalities of plane figures can be interpreted through equal content without first assigning numerical areas [5]. He also explains why the algebraic-looking manipulations of rectangles and squares may be regarded as a kind of geometrical algebra.

II.4 is a particularly vivid example of that interpretation. Its modern shadow is a quadratic identity, but Euclid's objects are still a large square, two smaller squares, and two rectangles. The current formalization sharpens the configured result from general equal content to exact scissors equality, because every step is built from exact dissection and exact representation transport.

Hartshorne's "Brief Euclid" explicitly lists II.4 in the form that the square on the whole line equals the squares on its two segments plus twice the rectangle on the two segments [5]. Thus II.4 has a more direct Hartshorne-side textual counterpart than II.2 or II.3. The surrounding Section 22 discussion still supplies the essential semantic interpretation: Euclid's equality is read as equal content, not as prior equality of numerical areas.

The stronger exact-scissors certificate remains a fact about the present Lean architecture. It should not be attributed to Hartshorne's general content relation.

4.14 Hilbert's Layering

Hilbert provides the complementary architectural warning. His plane-content theory and his later arithmetic of segments are distinct layers. Segment multiplication is introduced only later, using a fixed unit segment and parallel constructions, after which commutativity and distributivity become actual arithmetic theorems [2].

II.4 may look like the first place where one should invoke those later laws, because the shadow

$$(y+z)^2 = y^2 + z^2 + 2yz$$

contains both distributivity and commutativity. The compiling source shows that this move is unnecessary. The proof stays below segment arithmetic:

$$\begin{array}{c}\text{rectangle/square geometry}\\ \Downarrow \\ \text{exact II.2 and II.3 scissors identities}\\ \Downarrow \\ \text{geometric representation transport}\\ \Downarrow \\ \text{exact II.4 scissors identity.}\end{array}$$

This is a useful architectural checkpoint. Even a full quadratic-looking Book II identity can be derived from the geometric rectangle calculus without crossing into Hilbert's later segment field.

4.15 Dependency Summary

The actual Lean dependency chain is

Hilbert incidence + Euclidean plane layer
$\Downarrow$
Book I square, parallelogram, right-angle, and scissors infrastructure
$\Downarrow$
`HilbertRectangle` + Proposition II.1 binary API
$\Downarrow$
`euclid_proposition_2_2` + `euclid_proposition_2_3`
$\Downarrow$
endpoint reversal + `rectangle_contained_by_swap`
$\Downarrow$
`HilbertScissorsEq.add` + `trans`
$\Downarrow$
`euclid_proposition_2_4`.

The classical local chain is different:

$$\text{I.46} + \text{I.31} + \text{I.29} + \text{I.5} + \text{I.6} + \text{I.34} + \text{I.43} \longrightarrow \text{II.4}.$$

New geometric axiom in II.4:	no
New content axiom in II.4:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New `HilbertRectangle` theorem introduced by II.4:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses earlier Book II propositions:	yes, II.2 and II.3
Direct import of II.1:	no; inherited through II.2 and II.3
Direct side-order transport used:	yes, `rectangle_contained_by_swap`
Endpoint-reversal transport used:	yes
Exact `HilbertScissorsEq`:	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem compile:	yes, clean
Full relevant `lake build`:	not separately reported here
Proposition-local `sorry`/`admit`:	no
`sorryAx` in reported axiom list:	no

4.16 Conclusion

Proposition II.4 is the first strong compositional test of the Book II API. Euclid proves the theorem inside one carefully constructed square: two subfigures are shown to be squares, and I.43 identifies the two remaining complements. The current Lean proof instead works one level higher. It takes II.2 as the outer square decomposition and applies II.3 to each of the two resulting rectangles.

The only local difficulty is representational. One II.3 call uses the reversed order $B - C - A$, and one cross rectangle must be rotated so that its ordered containing sides match that call. Once these transports are made, `HilbertScissorsEq.add` and transitivity close the theorem.

Consequently II.4 requires no new rectangle theorem, no new scissors theorem, no new proposition-local helper, and no segment multiplication. Its apparent quadratic algebra is already present in the synthetic rectangle calculus as a composition of exact geometric decompositions. The reported axiom audit adds no new proposition-specific debt and contains no `sorryAx`.

Chapter 5

Proposition II.5

5.1 Introduction

Proposition II.5 is the first Book II theorem in the present reconstruction to combine a *midpoint condition* with the rectangle/scissors calculus. Let C bisect AB, and let D be a second cut point. In the ordering chosen by the Lean theorem,

$$A - D - C - B,$$

so that C is the midpoint of AB and D lies in the left half. The theorem asserts, in geometric content language,

$$\mathrm{Rect}(AD, DB) + \mathrm{Sq}(CD) = \mathrm{Sq}(CB).$$

The familiar modern algebraic shadow is obtained by writing the half-length as a and the displacement of the unequal cut from the midpoint as x:

$$(a - x)(a + x) + x^2 = a^2.$$

This formula is only an interpretive mnemonic. The Lean proof does not define multiplication of segment lengths and does not evaluate numerical area. Its objects are concrete rectangle and square representatives, while equality is expressed by exact scissors equivalence.

The current source begins with

```
import CGJteamLab.Proposition2_4
```

but this import fact should not be confused with the proof DAG. The proof does not invoke `euclid_proposition_2_4`. It directly reuses the binary rectangle theorem from II.1 and Proposition II.3, together with the reusable rectangle representation API. Thus II.5 provides an especially useful example of the difference between module dependency and mathematical proof dependency.

5.2 Euclid's Statement and Classical Configuration

5.2.1 The statement

In the notation used here, Euclid's proposition may be stated as follows. A straight line AB is divided equally at C and unequally at D. Then the rectangle contained by AD and DB, together with the square on CD, is equal in content to the square on the half CB [1].

The phrase "rectangle contained by AD, DB" names a plane figure whose adjacent sides have those magnitudes. It is not a product already formed in a segment field. Likewise, the addition in the displayed identity describes the content of two figures taken together.

5.2.2 The classical diagram

The principal figure below follows the standard Euclidean configuration with the unequal cut D chosen in the right half, so its baseline order is $A - C - D - B$. This is the mirror version of the Lean convention $A - D - C - B$; the content identity is unchanged. Euclid erects the square $CEFB$ on the half CB, joins BE, draws the parallel through D, then the horizontal through its intersection H with the diagonal, and finally the parallel through A [1].

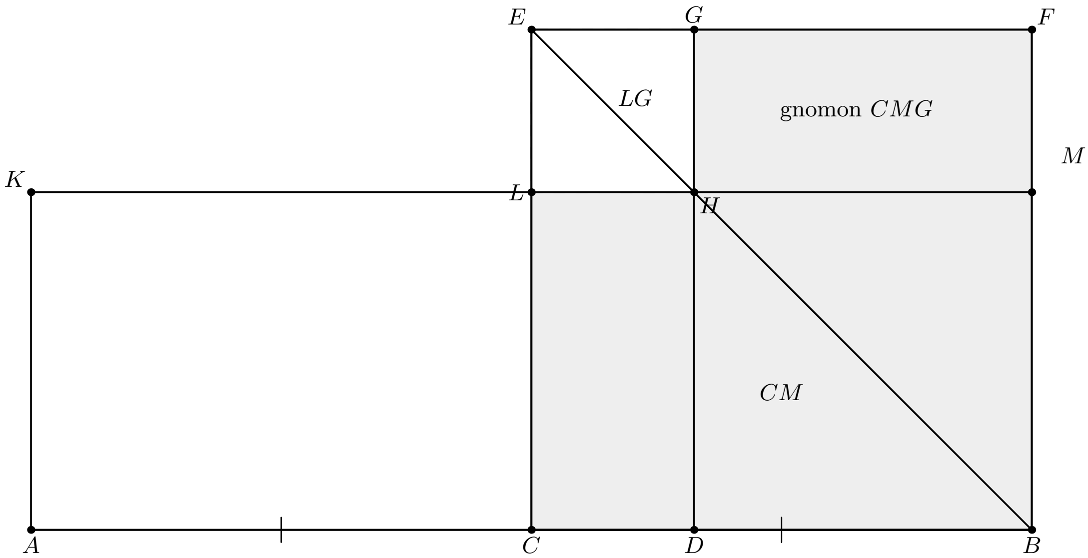


Figure 5.1: The classical configuration of Euclid II.5. The square $CEFB$ is built on the half CB in the customary mirror orientation $A - C - D - B$. The diagonal BE and the parallels through the unequal cut D form the complementary-parallelogram configuration; LG is the small square on CD, and the shaded remainder is Euclid's gnomon. This figure is retained because it is the actual source geometry. The later formal figures use the Lean orientation $A - D - C - B$ and do not introduce a primitive gnomon object.

Diagrammatic audit. The classical gnomon is retained as a source diagram, but it is no longer asked to stand in for the Lean proof. The reconstruction fixes the mirror order $A - D - C - B$ and consumes three configured rectangle cuts: II.3 on AC at D, II.1 on DB at C, and II.1 again on AC at D after side-order rotation. Those genuinely different dissection states are shown below at the points where they enter. No separate picture is used merely for the derived order $D - C - B$, for rectangle representative transport, or for formal reassociation.

5.3 Classical Proof

The classical proof uses the gnomon structure of Figure 5.1. The diagonal BE and the parallels create complementary parallelograms. By I.43 the complements CH and HF are equal. Adding the same rectangle DM to the relevant figures yields an equality in which the rectangle CM can be replaced by AL: these are equal parallelograms on equal bases AC and CB and in the same parallels, using I.36 and the midpoint condition $AC = CB$ [1].

The construction also gives $DH = DB$ by the square-of-a-sum geometry already available at II.4 (in Euclid's classical dependency chain). Consequently the figure AH is the rectangle contained by the unequal parts AD and DB. Euclid identifies that rectangle with the gnomon obtained by removing the small corner square LG from the square $CEFB$. Finally LG is the square on the segment CD, and adjoining it to the gnomon recovers the entire square on CB.

Thus the classical local DAG is schematically

$$\begin{array}{c}
\text{I.43: equal complements} \quad + \quad \text{I.36: equal parallelograms on equal bases} \\
\text{together with the II.4 corollary} \\
\Downarrow \\
\text{rectangle on } AD, DB \;=\; \text{gnomon in the square on } CB \\
\Downarrow \\
\text{add the square on } CD \\
\Downarrow \\
\mathrm{Rect}(AD, DB) + \mathrm{Sq}(CD) = \mathrm{Sq}(CB).
\end{array}$$

This is *not* the Lean DAG. In particular, the current formal proof does not reconstruct the classical gnomon, does not call I.43 locally, and does not call II.4 despite importing the II.4 module.

5.4 The Geometric Identity in the Current Formalization

The public Lean theorem chooses the strict order

$$A - D - C - B$$

through two hypotheses:

```
(hMidC : HilbertIsMidpoint Geo C A B)
(hADC : Geo.Between A D C)
```

The midpoint hypothesis supplies $A - C - B$ and $AC \cong CB$. From $A - D - C$ and $A - C - B$, a private order helper derives $D - C - B$ and $A - D - B$. Thus the full four-point order is a theorem of the hypotheses, not an additional assumption.

At the exact-scissors level the target is

$$\mathrm{Rect}(DB, AD) + \mathrm{Sq}(DC) \simeq_{\mathrm{sc}} \mathrm{Sq}(CB).$$

The reversed order DB, AD rather than AD, DB is immaterial at the geometric rectangle level because side-order exchange is itself proved by rectangle rotation/transport. It is not treated as primitive commutativity of a segment product.

A notable semantic point is that no gnomon type occurs in the source. The classical gnomon is represented extensionally by formal sums of rectangle and square scissors terms. The proof then rearranges and composes those sums by `HilbertScissorsEq.add`, symmetry, transitivity, and the existing rectangle transport lemmas.

5.5 Reusable Book II Infrastructure

5.5.1 The binary rectangle split from II.1

The theorem `euclid_proposition_2_1_two` is used twice. First it splits DB at C, with AD held as the other containing side:

$$\mathrm{Rect}(DB, AD) \simeq_{\mathrm{sc}} \mathrm{Rect}(DC, AD) + \mathrm{Rect}(CB, AD).$$

Later the same binary theorem, now applied to AC split at D, assembles

$$\mathrm{Rect}(CB, AD) + \mathrm{Rect}(CB, DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(CB, AC),$$

after geometric side-order transports.

5.5.2 Proposition II.3

Proposition II.3 supplies the first quadratic-looking decomposition:

$$\mathrm{Rect}(AC, DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(AD, DC) + \mathrm{Sq}(DC).$$

The current source calls `euclid_proposition_2_3` directly. The midpoint congruence $AC \cong CB$ is then used to transport the left-hand rectangle to a representative contained by CB, DC.

5.5.3 Rectangle representation and transport

The proof relies on the established Book II rectangle API rather than numerical products. In particular it uses

- `rectangle_transport_scissorsEq` to change containing segments along segment congruences;
- `rectangle_contained_by_comm` and `rectangle_contained_by_swap` to exchange the two adjacent side roles geometrically;
- `rectangle_contained_by_unique` to compare arbitrary representatives of the same contained-by data;
- `square_is_rectangle_contained_by_side` to view a square as a rectangle contained by one side twice.

5.6 Proof Architecture of the Reconstruction

The production file preserves the private helper names created during the incremental proof search. The names contain `testXX`, but all these lemmas are `private`; they are implementation history, not public API. Their mathematical roles are stable and form the following chain.

5.6.1 Step 1: recover the full order and midpoint congruence

`proposition2_5_test01_order` derives

$$D - C - B, \qquad A - D - B, \qquad AC \cong CB, \qquad CA \cong CB.$$

This is the only place where the four-point order is consolidated.

5.6.2 Step 2: apply II.3 on the left half

`proposition2_5_test02_left_decomposition` proves

$$\mathrm{Rect}(AC, DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(AD, DC) + \mathrm{Sq}(DC).$$

5.6.3 Step 3: transport the midpoint side

`proposition2_5_test03_midpoint_rectangle_transport` uses $AC \cong CB$ to prove

$$\mathrm{Rect}(AC, DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(CB, DC).$$

By symmetry and transitivity, `proposition2_5_test04_first_assembled_identity` obtains

$$\mathrm{Rect}(CB, DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(AD, DC) + \mathrm{Sq}(DC).$$

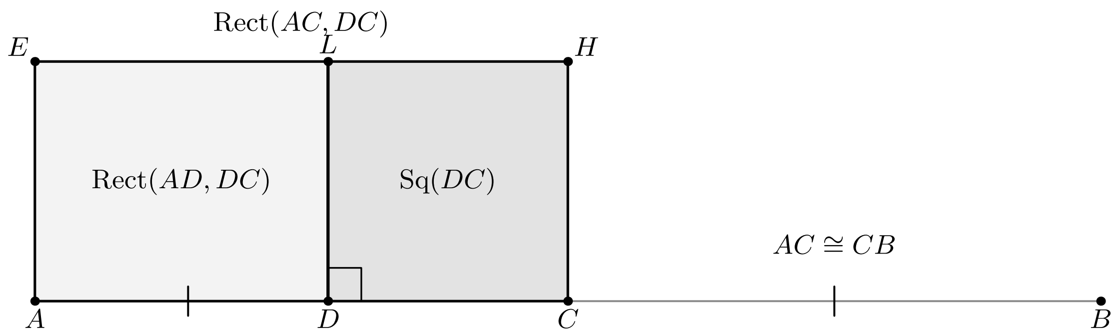


Figure 5.2: The first formal dissection in II.5, shown in the Lean orientation $A - D - C - B$. Proposition II.3 cuts the rectangle contained by AC, DC at D into a rectangle contained by AD, DC and a square-shaped DC piece. The continuation to B and the matching marks record the independent midpoint fact $AC \cong CB$, which transports the whole rectangle to one contained by CB, DC. The separately supplied square and rectangle representatives used by Lean are deliberately omitted: they do not change this geometric cut.

5.6.4 Step 4: split the unequal part DB

`proposition2_5_test05_db_decomposition` applies the binary II.1 theorem to $D - C - B$:

$$\mathrm{Rect}(DB, AD) \simeq_{\mathrm{sc}} \mathrm{Rect}(DC, AD) + \mathrm{Rect}(CB, AD).$$

The next two helpers use geometric commutativity to normalize the first term:

$$\mathrm{Rect}(DB, AD) \simeq_{\mathrm{sc}} \mathrm{Rect}(AD, DC) + \mathrm{Rect}(CB, AD).$$

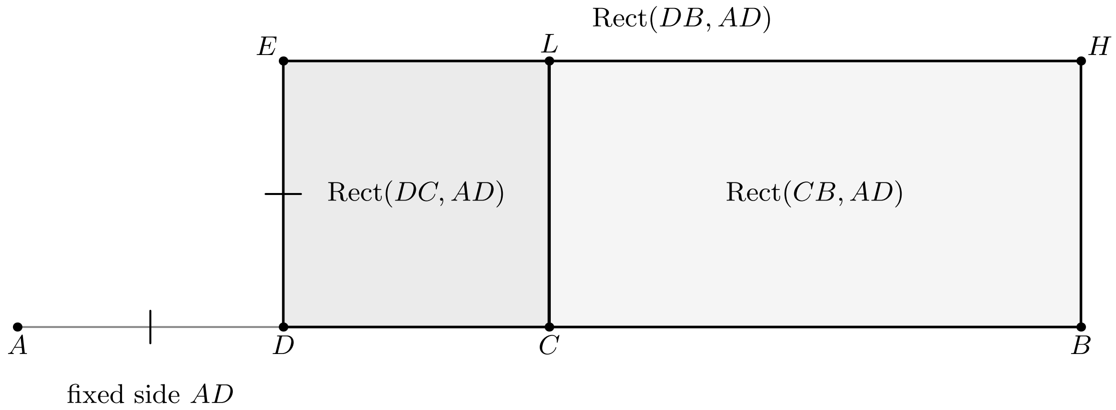


Figure 5.3: The second formal dissection. The derived order $D - C - B$ allows the binary II.1 theorem to cut the rectangle contained by DB, AD into the two visible pieces contained by DC, AD and CB, AD. The small segment AD at the left records the fixed second-side magnitude. The subsequent replacement of $\mathrm{Rect}(DC, AD)$ by $\mathrm{Rect}(AD, DC)$ is a rotation/contained-by normalization of a representative, so it receives no additional geometric picture.

5.6.5 Step 5: the scissors bridge

The private lemma `proposition2_5_test08_scissors_bridge` is an abstract content calculation. Given

$$P \simeq X + Y, \qquad Z \simeq X + S,$$

it proves

$$P + S \simeq Y + Z.$$

Crucially, this is *not* a cancellation principle. It is obtained by adding exact scissors equalities, reassociating and commuting the formal multiset sum, and using transitivity. Applying it to the two preceding identities gives

$$\boxed{\mathrm{Rect}(DB, AD) + \mathrm{Sq}(DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(CB, AD) + \mathrm{Rect}(CB, DC).}$$

5.6.6 Step 6: assemble the right side

`proposition2_5_test09_assemble_right_side` rotates the two right-hand rectangle representatives, applies the binary II.1 split to $A - D - C$, and rotates the result back. It proves

$$\mathrm{Rect}(CB, AD) + \mathrm{Rect}(CB, DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(CB, AC).$$

5.6.7 Step 7: recognize the half-square

Finally `proposition2_5_test10_half_rectangle_is_square` uses $AC \cong CB$ and representative uniqueness to identify

$$\mathrm{Rect}(CB, AC) \simeq_{\mathrm{sc}} \mathrm{Sq}(CB).$$

This is where the midpoint hypothesis enters a second time: first it controlled order, and now it turns the final rectangle into the square on the half.

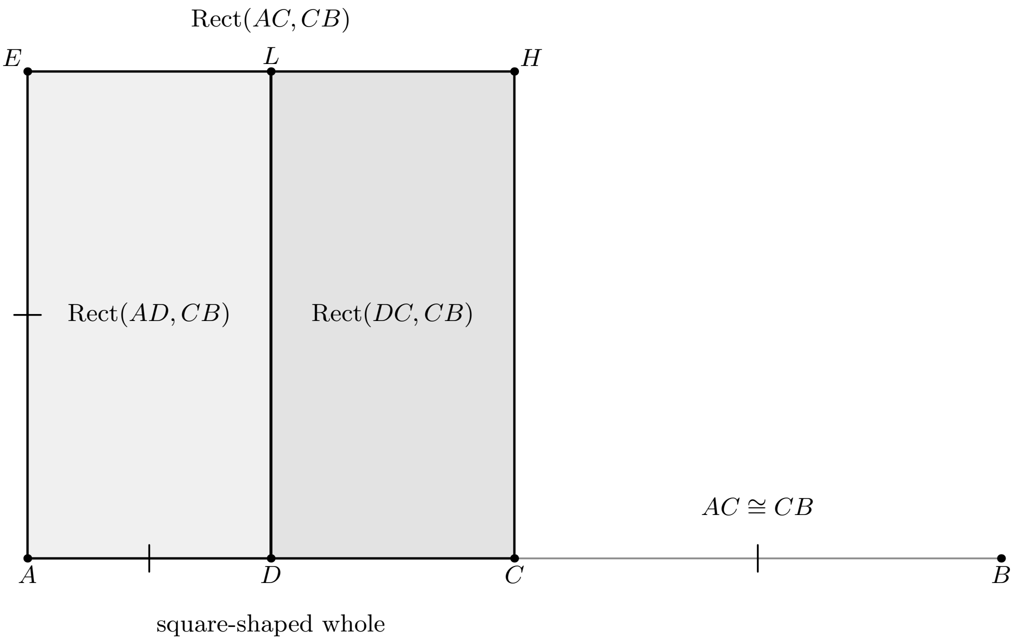


Figure 5.4: The final geometric assembly used by the reconstruction. After cyclically rotating the two incoming representatives, II.1 reads the order $A - D - C$ as an actual cut of a rectangle with common side magnitude CB: the two pieces contained by AD, CB and DC, CB fill the rectangle contained by AC, CB. The congruence $AC \cong CB$ marks its adjacent side magnitudes as equal, explaining the final square recognition. Lean may compare this whole with a different supplied square representative by uniqueness; that representative bookkeeping does not create another spatial state.

Diagrammatic audit. No figure is added for the private scissors bridge in Step 5: it only adds an existing square term and reassociates/commutes a formal multiset sum. Likewise, no extra snapshot is added for the final representative-uniqueness call. The three formal figures above exhaust the proposition-local geometry consumed by the current proof; everything between them is transport or exact-scissors bookkeeping.

5.7 Lean Formalization

5.7.1 The public theorem signature

The following is the current compiling public signature. It is intentionally large because the theorem receives the concrete rectangle-cut configurations and representatives required by the earlier Book II APIs rather than hiding them behind choice constructions.

```
theorem euclid_proposition_2_5
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (A B C D : Geo.Point)
    (hMidC : HilbertIsMidpoint Geo C A B)
    (hADC : Geo.Between A D C)

    ------------------------------------------------------------
    -- Data used by test 04 inside test 08.
    ------------------------------------------------------------

    (E4 H4 L4 X4 : Geo.Point)
    (hRect4 : IsRectangle Geo A C H4 E4)
    (hCH_DC : Geo.Congruent C H4 D C)
    (hELE4 : Geo.Between E4 L4 H4)
    (hEXC4 : Geo.Between E4 X4 C)
    (hDXL4 : Geo.Between D X4 L4)
    (hLeftPar4 : IsParallelogram Geo L4 E4 A D)
    (hRightPar4 : IsParallelogram Geo C H4 L4 D)

    -- Square on DC.
    (F G : Geo.Point)
    (hSquareDC : IsSquare Geo D C F G)

    -- Rect(AC,DC).
    (R0 R1 R2 R3 : Geo.Point)
    (hAC_DC :
      IsRectangleContainedBy Geo
        R0 R1 R2 R3 A C D C)

    -- Common Rect(AD,DC).
    (T0 T1 T2 T3 : Geo.Point)
    (hAD_DC :
      IsRectangleContainedBy Geo
        T0 T1 T2 T3 A D D C)

    -- Rect(CB,DC).
    (Z0 Z1 Z2 Z3 : Geo.Point)
    (hCB_DC :
      IsRectangleContainedBy Geo
```

```
    Z0 Z1 Z2 Z3 C B D C)

------------------------------------------------------------
-- Data used by test 07 inside test 08.
------------------------------------------------------------

(E7 H7 L7 X7 : Geo.Point)
(hRect7 : IsRectangle Geo D B H7 E7)
(hBH_AD : Geo.Congruent B H7 A D)
(hELE7 : Geo.Between E7 L7 H7)
(hEXB7 : Geo.Between E7 X7 B)
(hCXL7 : Geo.Between C X7 L7)
(hLeftPar7 : IsParallelogram Geo L7 E7 D C)
(hRightPar7 : IsParallelogram Geo B H7 L7 C)

-- Rect(DB,AD), the unequal-parts rectangle.
(P0 P1 P2 P3 : Geo.Point)
(hDB_AD :
  IsRectangleContainedBy Geo
    P0 P1 P2 P3 D B A D)

-- Rect(DC,AD), before commutativity normalization.
(Q0 Q1 Q2 Q3 : Geo.Point)
(hDC_AD :
  IsRectangleContainedBy Geo
    Q0 Q1 Q2 Q3 D C A D)

-- Rect(CB,AD).
(Y0 Y1 Y2 Y3 : Geo.Point)
(hCB_AD :
  IsRectangleContainedBy Geo
    Y0 Y1 Y2 Y3 C B A D)

------------------------------------------------------------
-- Data for test 09:
-- assemble Rect(CB,AD) + Rect(CB,DC) into Rect(CB,AC).
------------------------------------------------------------

(E9 H9 L9 X9 : Geo.Point)
(hRect9 : IsRectangle Geo A C H9 E9)
(hCH_CB9 : Geo.Congruent C H9 C B)
(hELE9 : Geo.Between E9 L9 H9)
(hEXC9 : Geo.Between E9 X9 C)
(hDXL9 : Geo.Between D X9 L9)
(hLeftPar9 : IsParallelogram Geo L9 E9 A D)
(hRightPar9 : IsParallelogram Geo C H9 L9 D)

-- Rect(CB,AC).
(W0 W1 W2 W3 : Geo.Point)
(hCB_AC :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 C B A C)

------------------------------------------------------------
-- Data for test 10: square on CB.
------------------------------------------------------------

(S0 S1 S2 S3 : Geo.Point)
```

```
    (hSquareCB : IsSquare Geo S0 S1 S2 S3)
    (hS01_CB : Geo.Congruent S0 S1 C B) :

    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo P0 P1 P2 P3 +
       hilbertParallelogramTerm Geo D C F G)
      (hilbertParallelogramTerm Geo S0 S1 S2 S3) := by
```

Several layers of data are visible in the signature.

1. A, B, C, D and the midpoint/betweenness hypotheses encode the one-dimensional configuration.
2. The groups labelled in the source as data for tests 04, 07, and 09 are concrete rectangle-cut witnesses needed by II.3 or II.1.
3. F, G give a square on DC.
4. $P_0P_1P_2P_3$ represents the rectangle contained by DB, AD.
5. $W_0W_1W_2W_3$ represents the intermediate rectangle contained by CB, AC.
6. $S_0S_1S_2S_3$ is an arbitrary square representative whose side is congruent to CB.

Thus the theorem proves representative-independent content equality for supplied geometric witnesses; it does not claim that all of those witnesses are constructed internally from four points alone.

5.7.2 The final three-stage assembly

After all private helpers have been established, the public theorem is short in conceptual structure. The current source builds exactly three equations:

1. `h08`: unequal-parts rectangle plus the DC square equals two rectangles with common side CB;
2. `h09`: those two rectangles assemble into the rectangle $\mathrm{Rect}(CB, AC)$;
3. `h10`: midpoint congruence turns that rectangle into the square on CB.

The final transitivity block is:

```
  have h08 :
      HilbertScissorsEq Geo
        (hilbertParallelogramTerm Geo P0 P1 P2 P3 +
         hilbertParallelogramTerm Geo D C F G)
        (hilbertParallelogramTerm Geo Y0 Y1 Y2 Y3 +
         hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3) :=
    proposition2_5_test08_core_identity
      Geo
      A B C D
      hMidC hADC
      E4 H4 L4 X4
      hRect4 hCH_DC hELE4 hEXC4 hDXL4
```

```
    hLeftPar4 hRightPar4
    F G hSquareDC
    R0 R1 R2 R3
    hAC_DC
    T0 T1 T2 T3
    hAD_DC
    Z0 Z1 Z2 Z3
    hCB_DC
    E7 H7 L7 X7
    hRect7 hBH_AD hELE7 hEXB7 hCXL7
    hLeftPar7 hRightPar7
    P0 P1 P2 P3
    hDB_AD
    Q0 Q1 Q2 Q3
    hDC_AD
    Y0 Y1 Y2 Y3
    hCB_AD

-------------------------------------------------------------
-- test09:
-- Rect(CB,AD) + Rect(CB,DC) = Rect(CB,AC).
-------------------------------------------------------------

have h09 :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo Y0 Y1 Y2 Y3 +
       hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3)
      (hilbertParallelogramTerm Geo W0 W1 W2 W3) :=
  proposition2_5_test09_assemble_right_side
    Geo
    A B C D
    hADC
    E9 H9 L9 X9
    hRect9 hCH_CB9 hELE9 hEXC9 hDXL9
    hLeftPar9 hRightPar9
    Y0 Y1 Y2 Y3
    hCB_AD
    Z0 Z1 Z2 Z3
    hCB_DC
    W0 W1 W2 W3
    hCB_AC

-------------------------------------------------------------
-- test10:
-- Rect(CB,AC) = Square(CB).
-------------------------------------------------------------

have h10 :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo W0 W1 W2 W3)
      (hilbertParallelogramTerm Geo S0 S1 S2 S3) :=
  proposition2_5_test10_half_rectangle_is_square
    Geo
    A B C D
    hMidC hADC
    W0 W1 W2 W3
    hCB_AC
    S0 S1 S2 S3
```

```
    hSquareCB
    hS01_CB

------------------------------------------------------------------
-- Full II.5.
------------------------------------------------------------------

exact
  HilbertScissorsEq.trans
    (Geo := Geo)
    h08
    (HilbertScissorsEq.trans
      (Geo := Geo)
      h09
```

The formal result is therefore exact `HilbertScissorsEq`, not the weaker relation `HilbertScissorsEquicomplementable`, and not equality of real numbers.

5.8 Relationship with Earlier Book II Propositions

5.8.1 Proposition II.1

II.1 is the basic additive engine of the proof. Its binary theorem is called directly in the decomposition of DB and again in the assembly of AC. Thus II.5 genuinely reuses the rectangle split calculus rather than reproving the corresponding dissections from Book I geometry.

5.8.2 Proposition II.3

II.3 is also called directly. It supplies the identity

$$\mathrm{Rect}(AC, DC) = \mathrm{Rect}(AD, DC) + \mathrm{Sq}(DC),$$

at the exact scissors level. This is the step that inserts the square on the difference between the two section points.

5.8.3 Proposition II.4

The source file imports `CGJteamLab.Proposition2_4`, so II.4 is a direct *module* dependency. Nevertheless the theorem `euclid_proposition_2_4` is not invoked in the proof term. This sharply contrasts with Euclid's classical proof, where a corollary of II.4 participates in the identification of the relevant rectangle and small square.

This distinction should be recorded explicitly:

$$\text{module chain: II.4 module} \to \text{II.5 module},$$
$$\text{Lean theorem calls: II.1 binary API} + \text{II.3} + \text{rectangle API.}$$

5.9 Relationship with Book I Infrastructure

The Book I geometry is present below the reusable rectangle layer. Rectangles, squares, right angles, parallel sides, parallelograms, and finite scissors terms all ultimately depend on the established Hilbert/Book I development. II.5 does not locally replay Euclid's diagonal-complement argument from I.43. Instead, the effect of those earlier geometric facts has already been packaged into `HilbertRectangle`, the scissors API, and the earlier Book II theorems.

This is an architectural gain: a proposition whose classical proof is diagram specific becomes a composition of stable content-level interfaces.

5.10 Formalization Notes and Hidden Geometric Data

5.10.1 Order orientation

The classical diagram commonly places the unequal cut in the right half $A - C - D - B$, whereas the Lean theorem fixes $A - D - C - B$. These are mirror cases. The source makes the chosen orientation explicit through `hADC` and then derives $D - C - B$ from the midpoint betweenness. No appeal to a picture is allowed to supply this ordering silently.

5.10.2 Midpoint as geometry, not arithmetic

`HilbertIsMidpoint` contributes both betweenness and segment congruence. The proof uses those components separately. Betweenness creates the required subsegment orders; congruence transports rectangle sides and eventually recognizes the square on the half. There is no variable standing for a numerical half-length.

5.10.3 Representative independence

The objects named "the rectangle contained by" particular segments are concrete quadrilateral representatives. Different helpers may receive different representatives of the same pair of side magnitudes. `rectangle_transport_scissorsEq`, `rectangle_contained_by_unique`, and the side-order transport theorems are therefore mathematically substantive parts of the formal statement, not mere syntactic rewrites.

5.10.4 No gnomon primitive

Euclid's proof naturally names a gnomon. The present library does not define a `GnomonTerm` or another primitive gnomon object for II.5. No such new abstraction is needed here: the required content is expressed as a formal sum of scissors terms. The private scissors bridge then performs the exact content rearrangement that the classical gnomon packages visually.

5.10.5 No cancellation axiom

At an intermediate stage the two identities contain the common rectangle $\mathrm{Rect}(AD, DC)$. It might be tempting to introduce a cancellation principle for content. The actual source deliberately avoids this. The scissors bridge combines equalities additively and rearranges formal sums, so no new positivity or cancellation theorem is required.

5.11 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and the production file introduces no new proposition-specific `axiom`, `sorry`, or `admit`. The user-verified module build is clean. The reported transitive axiom audit is

```
#print axioms Geometry.euclid_proposition_2_5

'Geometry.euclid_proposition_2_5' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

Accordingly II.5 should not be described as "axiom-free." The precise claim is that II.5 introduces no new proposition-specific axiom and no `sorry`; its transitive dependencies are the four declarations shown above, and the list contains no `sorryAx`.

5.12 Hartshorne's Semantic Control

Hartshorne's treatment of equal content provides the appropriate semantic background for the algebraic-looking identities of Book II. The relevant point for II.5 is unchanged from II.1–II.4: equality of the plane figures may be treated synthetically without first assigning numerical areas, while the familiar algebraic formula remains an interpretive shadow [5].

The present Lean theorem is stronger than a bare equal-content statement in one specific sense: it produces exact `HilbertScissorsEq` for the supplied configurations. That strengthening belongs to the project-level scissors architecture and should not be attributed automatically to Hartshorne's general content relation.

5.13 Hilbert's Layering

Hilbert supplies the complementary warning against importing later arithmetic too early. His synthetic theory of plane content and his later arithmetic of segments are separate layers; genuine segment multiplication requires its own construction and algebraic theorems [2]. II.5 remains entirely below that boundary. In particular, the identity

$$(a - x)(a + x) + x^2 = a^2$$

is not the proof object. The actual chain is

$$\begin{gathered}\text{order + congruence}\\ \Downarrow \\ \text{rectangle representatives}\\ \Downarrow \\ \text{exact scissors decompositions and transports.}\end{gathered}$$

5.14 Dependency Summary

The actual Lean dependency chain is best displayed as

$$\begin{gathered}\text{Hilbert incidence + Euclidean plane layer}\\ \Downarrow \\ \text{Book I square, rectangle, parallelogram, and scissors infrastructure}\\ \Downarrow \\ \text{HilbertRectangle representative/transport API}\\ \Downarrow \\ \text{II.1 binary split} \quad + \quad \text{II.3 rectangle-plus-square theorem}\\ \Downarrow \\ \text{private II.5 order, transport, bridge, and assembly helpers}\\ \Downarrow \\ \text{public theorem } \texttt{euclid\_proposition\_2\_5}.\end{gathered}$$

The classical local chain is different:

$$\text{I.43 + I.36 + II.4 corollary} \longrightarrow \text{gnomon identification} \longrightarrow \text{II.5.}$$

The formal proof is therefore not a line-by-line translation of the classical one. It reconstructs the same geometric content through a different, reusable DAG.

New geometric axiom in II.5:	no
New content axiom in II.5:	no
New proposition-local helper theorem:	yes, private helpers
New reusable public helper theorem:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.1 and II.3
Imports II.4 module:	yes
Calls `euclid_proposition_2_4`:	no
Uses exact `HilbertScissorsEq`:	yes
Uses general equicomplementability:	no
Uses numerical area:	no
Uses segment multiplication:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Proposition-local `sorry`/`admit`:	no
Transitive `sorryAx`:	absent in reported audit

5.15 Conclusion

Proposition II.5 is an important architectural checkpoint for Book II. The classical proof organizes the equality around a gnomon inside the square on the half. The formal proof shows that a separate gnomon primitive is not needed to recover the theorem: the same content can be assembled from the binary II.1 rectangle split, II.3, midpoint congruence, exact rectangle transport, and the additive structure of `HilbertScissorsEq`.

The result also clarifies the role of the modern quadratic identity. It is a useful shadow for recognizing the proposition, but the verified argument stays synthetic throughout. Geometry supplies the order and congruences; representation supplies rectangles and squares; scissors equality supplies the content calculus. Numerical area and segment multiplication remain outside the proof.

Chapter 6

Proposition II.6

6.1 Introduction

Proposition II.6 is the external-section companion of Proposition II.5. A straight line AB is bisected at C and then produced beyond B to D. Thus the one-dimensional order used by the formal theorem is

$$A - C - B - D.$$

The proposition asserts, in geometric content language,

$$\mathrm{Sq}(BC) + \mathrm{Rect}(AD, BD) = \mathrm{Sq}(CD).$$

The familiar modern algebraic shadow is obtained by writing the half-length $AC = CB$ as a and the added segment BD as x:

$$a^2 + (2a + x)x = (a + x)^2.$$

Equivalently, if $y = CD$ and $z = CB$, one sees the difference-of-squares pattern

$$(y + z)(y - z) + z^2 = y^2.$$

Both formulas are documentary mnemonics only. The Lean proof does not define a multiplication of segment lengths and does not evaluate a numerical area. Its objects are concrete rectangle and square representatives, and the conclusion is exact scissors equality.

The production source begins with

```
import CGJteamLab.Proposition2_5
```

so Proposition II.5 is a module dependency. It is not, however, a theorem dependency of the proof term. The current II.6 proof directly calls the binary theorem from II.1, Proposition II.3, Proposition II.4, and the rectangle-transport/scissors API. This distinction between the module chain and the actual theorem-call DAG is important in the present reconstruction.

6.2 Euclid's Statement and Classical Configuration

6.2.1 The statement

In Heath's wording, the proposition begins:

> If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.

With AB bisected at C and BD added in a straight line, the concrete statement is

$$\mathrm{Rect}(AD, DB) + \mathrm{Sq}(CB) = \mathrm{Sq}(CD) \qquad [1].$$

As throughout the present Book II documentation, the notation Rect$(-,-)$ and Sq$(-)$ is documentary shorthand for geometric figures. It is not a definition of a numerical product or numerical square.

6.2.2 The classical diagram

Euclid constructs the square $CEFD$ on CD, joins the diagonal DE, draws through B a parallel to CE (or DF), then through the resulting intersection H a line parallel to the baseline, and finally through A a parallel to the vertical sides [1]. The construction is shown in Figure 6.1.

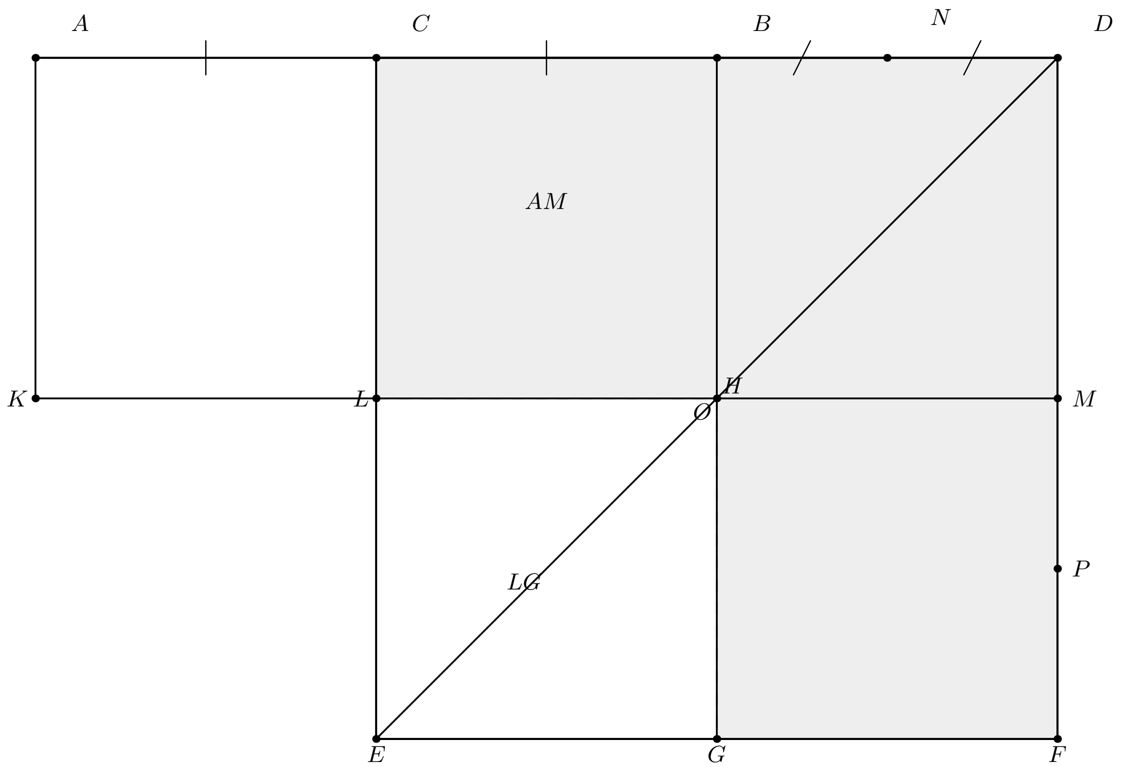


Figure 6.1: The classical mechanism of Euclid II.6. The baseline has order $A - C - B - D$ and C bisects AB. The square $CEFD$ is built on CD. The dashed square LG is equal to the square on BC; the shaded remainder is Euclid's gnomon NOP. The rectangle AM has adjacent side magnitudes AD and DB.

The same picture exposes the central geometry. Since $AC = CB$, the parallelograms AL and CH have equal content by I.36. The diagonal construction makes CH and HF complementary parallelograms, so I.43 gives their equality. Thus $AL = HF$. Adding the same rectangle CM identifies the whole rectangle AM with the gnomon NOP.

Diagrammatic audit. The classical gnomon figure is retained because it is the genuine Euclidean construction used by the source proof. No second picture is added merely for the derived order $C - B - D$ and $A - C - D$: those facts certify the already drawn baseline $A - C - B - D$ rather than create a new geometric state. The formal proof then changes viewpoint. Its genuinely visible content states are the rectangle dissection assembled from II.3 and II.1 and the independent II.4 square dissection on CD; those two states are displayed below at the exact steps where Lean uses them.

6.3 Classical Proof

The classical proof can be organized into four content steps.

First, the midpoint condition gives

$$AC = CB.$$

The parallelograms AL and CH stand on these equal bases and lie between the same parallels, so I.36 yields

$$AL = CH.$$

Second, CH and HF are complements about the diagonal DE of the relevant parallelogram configuration, hence I.43 gives

$$CH = HF.$$

Therefore

$$AL = HF.$$

Third, Euclid adds the same rectangle CM to both sides. The whole figure AM is therefore equal to the gnomon NOP. The construction gives $DM = DB$, so AM is the rectangle contained by AD and DB. Hence

$$\mathrm{Rect}(AD, DB) = \text{gnomon } NOP.$$

Finally, the figure LG is the square on BC. Adding it to both sides gives

$$\mathrm{Rect}(AD, DB) + \mathrm{Sq}(BC) = \text{gnomon } NOP + \mathrm{Sq}(BC).$$

The right-hand side is the whole square $CEFD$ on CD, proving II.6 [1].

The classical local DAG is therefore

$$\begin{array}{c} \text{I.36: equal parallelograms on equal bases} \quad + \quad \text{I.43: equal complements} \\ \Downarrow \\ \mathrm{Rect}(AD, DB) = \text{gnomon } NOP \\ \Downarrow \\ \text{add the square on } BC \\ \Downarrow \\ \mathrm{Rect}(AD, DB) + \mathrm{Sq}(BC) = \mathrm{Sq}(CD). \end{array}$$

This is not the Lean DAG. The production proof does not construct a gnomon object and does not call I.36 or I.43 locally.

6.4 The Geometric Identity in the Current Formalization

The public theorem receives

```
(A B C D : Geo.Point)
(hMidC : HilbertIsMidpoint Geo C A B)
(hABD : Geo.Between A B D)
```

so C is the midpoint of AB and B lies between A and D. The midpoint hypothesis gives

$$A - C - B \qquad \text{and} \qquad AC \cong CB.$$

Combining $A - C - B$ with $A - B - D$ by Hilbert betweenness transitivity yields

$$C - B - D \qquad \text{and} \qquad A - C - D.$$

Thus the full order

$$A - C - B - D$$

is derived, not read from a diagram.

At the exact-scissors level the public target is

$$\mathrm{Sq}(BC) + \mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} \mathrm{Sq}(CD).$$

The order of the two summands on the left is the one present in the actual Lean conclusion. The theorem proves `HilbertScissorsEq`; it does not merely prove the weaker general content relation and it does not assert an equality of real numbers.

A second formal distinction is important. The square on BC in the public conclusion is the concrete square BCS_0S_1. The rectangle contained by AD, BD is represented by the supplied quadrilateral $P_0P_1P_2P_3$. The statement is therefore about concrete representatives whose side magnitudes satisfy the required contained-by predicates.

6.5 Reusable Book II Infrastructure Used

6.5.1 Proposition II.3: extend the unequal-parts rectangle

The first major exact dissection is obtained by applying `euclid_proposition_2_3` to AD cut at B, with BD as the fixed second side:

$$\mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} \mathrm{Rect}(AB, BD) + \mathrm{Sq}(BD).$$

This is the external-section analogue of introducing a square term in the earlier Book II identities.

6.5.2 Proposition II.1: split the old segment at its midpoint

The binary theorem `euclid_proposition_2_1_two` is then applied to AB cut at C, again with BD as the fixed side:

$$\mathrm{Rect}(AB, BD) \simeq_{\mathrm{sc}} \mathrm{Rect}(AC, BD) + \mathrm{Rect}(CB, BD).$$

Composing with the II.3 decomposition gives

$$\mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} (\mathrm{Rect}(AC, BD) + \mathrm{Rect}(CB, BD)) + \mathrm{Sq}(BD).$$

6.5.3 Midpoint transport

The midpoint congruence $AC \cong CB$ is converted into equality of rectangle content by `rectangle_transport_scissorsEq`:

$$\mathrm{Rect}(AC, BD) \simeq_{\mathrm{sc}} \mathrm{Rect}(CB, BD).$$

Consequently

$$\mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} (\mathrm{Rect}(CB, BD) + \mathrm{Rect}(CB, BD)) + \mathrm{Sq}(BD).$$

No numerical factor 2 is introduced. The phrase "twice the rectangle" means two scissors terms with the same geometric content.

6.5.4 Proposition II.4 on the produced segment

The decisive second decomposition calls `euclid_proposition_2_4` directly on CD cut at B. The formal substitution is

$$A_{\mathrm{II.4}} := C, \qquad B_{\mathrm{II.4}} := D, \qquad C_{\mathrm{II.4}} := B.$$

The required order hypothesis is exactly the derived betweenness $C - B - D$. The result is

$$\mathrm{Sq}(CD) \simeq_{\mathrm{sc}} (\mathrm{Rect}(CB, BD) + \mathrm{Sq}(BC)) + (\mathrm{Rect}(CB, BD) + \mathrm{Sq}(BD)).$$

This use of II.4 is one of the sharpest differences between II.5 and II.6 in the current formal library. II.5 imported the II.4 module but did not call `euclid_proposition_2_4`; II.6 calls it explicitly.

6.6 Proof Architecture of the Reconstruction

The production file preserves the names of the private incremental lemmas used during proof development. Their `testXX` names are implementation history only; because the declarations are `private`, they do not form part of the public API.

6.6.1 Step 1: recover the full order

`proposition2_6_test01_order` derives

$$C - B - D, \qquad A - C - D, \qquad AC \cong CB, \qquad CA \cong CB.$$

The only input is the midpoint structure together with $A - B - D$.

6.6.2 Step 2: decompose the rectangle on AD, BD

`proposition2_6_test02_ad_decomposition` calls II.3 and proves

$$\mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} \mathrm{Rect}(AB, BD) + \mathrm{Sq}(BD).$$

6.6.3 Step 3: split AB at C

`proposition2_6_test03_ab_decomposition` calls the binary II.1 theorem:

$$\mathrm{Rect}(AB, BD) \simeq_{\mathrm{sc}} \mathrm{Rect}(AC, BD) + \mathrm{Rect}(CB, BD).$$

6.6.4 Step 4: compose the two decompositions

`proposition2_6_test04_compose_decompositions` uses the same concrete representative of $\mathrm{Rect}(AB, BD)$ in both preceding lemmas. Therefore no representative-uniqueness transport is needed at this junction. The result is

$$\mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} (\mathrm{Rect}(AC, BD) + \mathrm{Rect}(CB, BD)) + \mathrm{Sq}(BD).$$

6.6.5 Step 5: use the midpoint congruence

`proposition2_6_test05_midpoint_rectangle_transport` proves

$$\mathrm{Rect}(AC, BD) \simeq_{\mathrm{sc}} \mathrm{Rect}(CB, BD)$$

by transporting the first adjacent side along $AC \cong CB$ while the second side remains congruent to BD.

6.6.6 Step 6: normalize the left decomposition

`proposition2_6_test06_double_cross` inserts the midpoint transport into Step 4:

$$\boxed{\mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} (\mathrm{Rect}(CB, BD) + \mathrm{Rect}(CB, BD)) + \mathrm{Sq}(BD).}$$

The proof uses only addition, reflexivity, and transitivity of exact scissors equality.

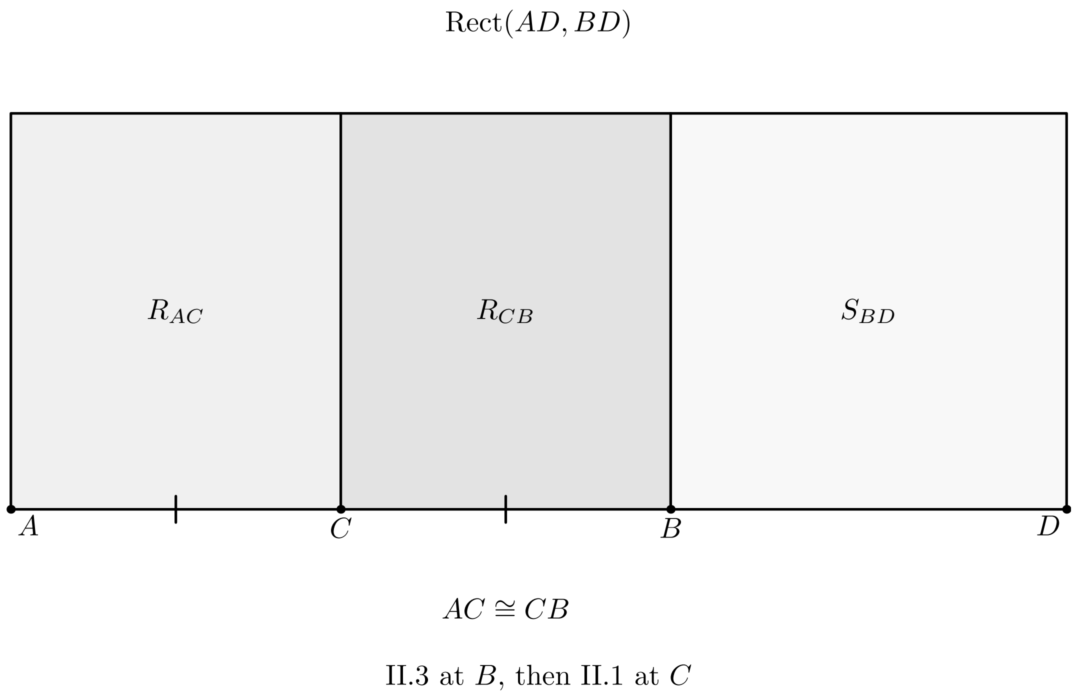


Figure 6.2: The visible content state behind Steps 2–6. Proposition II.3 first cuts the rectangle contained by AD, BD at B, producing the square S_{BD} and the rectangle contained by AB, BD. Proposition II.1 then cuts that remaining rectangle at the midpoint C. The two left strips have containing sides AC, BD and CB, BD; the midpoint congruence $AC \cong CB$ transports the first to the content of the second. The drawing therefore records the genuine two-stage dissection whose normalized scissors form is $R + R + S_{BD}$. The actual Lean theorem uses supplied rectangle representatives, so the labels denote content roles rather than a claim that one particular coordinate realization occurs in the proof term.

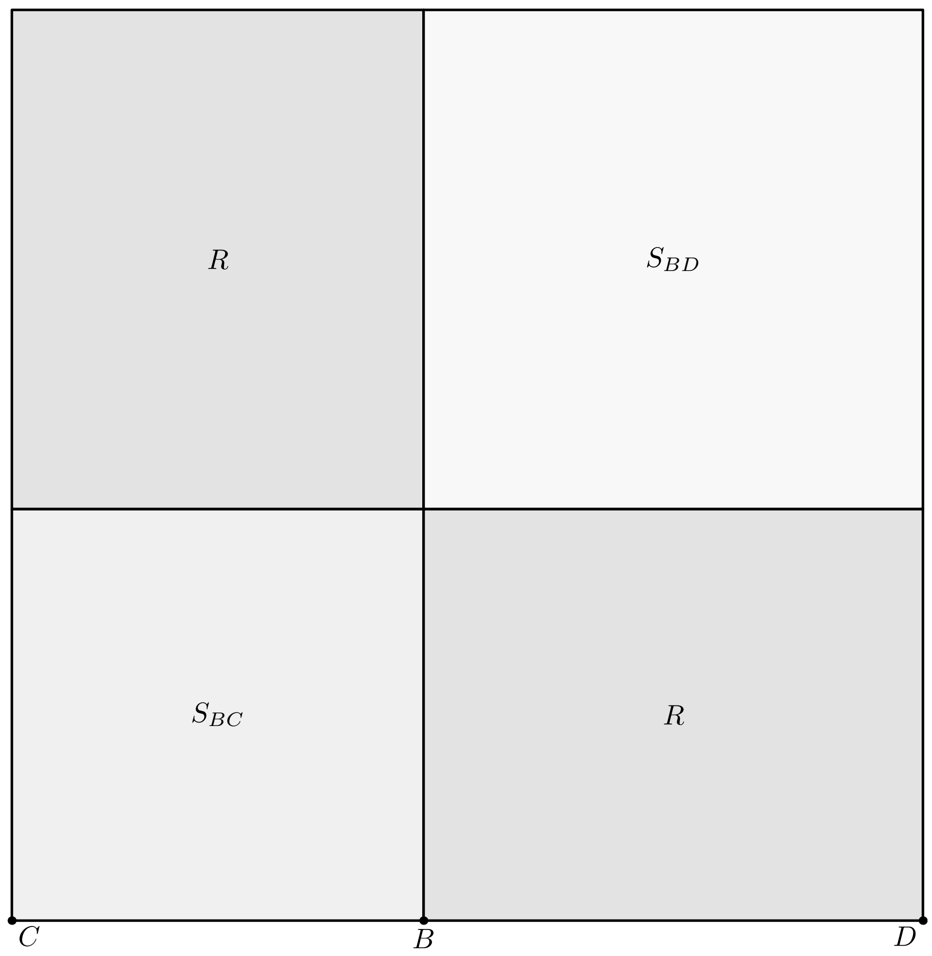
6.6.7 Step 7: expand the square on CD by II.4

`proposition2_6_test07_square_cd_decomposition` applies II.4 to the segment CD cut at B:

$$\boxed{\mathrm{Sq}(CD) \simeq_{\mathrm{sc}} (\mathrm{Rect}(CB, BD) + \mathrm{Sq}(BC)) + (\mathrm{Rect}(CB, BD) + \mathrm{Sq}(BD)).}$$

A useful implementation decision is visible here. The final assembly supplies the *same* rectangle representative $Z_0Z_1Z_2Z_3$ to both occurrences of $\mathrm{Rect}(CB, BD)$, and it reuses the same supplied square on BD as in Step 6. Hence the following algebraic bridge can match the two decompositions literally.

Sq(CD)



II.4 on $C - B - D$

Figure 6.3: The second geometric normal form in the production proof. Proposition II.4 is applied to the segment CD with $C - B - D$. The square on CD is split into the square S_{BC}, the square S_{BD}, and two cross rectangles with containing side magnitudes CB, BD. The same rectangle representative R is supplied to both cross positions in the final assembly, and the square S_{BD} is the same square already used in Figure 6.2. This literal representative reuse is why the following scissors bridge can be purely associative-commutative.

6.6.8 Step 8: the abstract scissors bridge

`proposition2_6_test08_scissors_bridge` is purely algebraic at the level of formal scissors sums. For terms P, Q, R, S, T, it assumes

$$P \simeq (R + R) + T, \qquad Q \simeq (R + S) + (R + T),$$

and proves

$$S + P \simeq Q.$$

The only rearrangement is associativity and commutativity of the formal sum, implemented by `ac_rfl`. The scissors equalities themselves are combined by `HilbertScissorsEq.add`, reflexivity, symmetry, and transitivity.

There is no cancellation theorem here. In particular, the proof does not infer equality by subtracting the common copies of R or T. It simply adds and rearranges exact scissors equalities.

Diagrammatic audit. No further figure is introduced for Steps 8–9. Once Figures 6.2 and 6.3 are available, the remaining operation creates no new point, line, cut, or common refinement. Lean adds the already supplied square S_{BC} to the first normal form and rearranges the formal scissors sum until it has exactly the second normal form. Drawing that regrouping as another plane dissection would falsely suggest new geometry where the proof performs only exact-scissors addition, symmetry, transitivity, and associativity/commutativity.

6.6.9 Step 9: assemble II.6

`proposition2_6_test09_assembly` makes the literal substitution

$$\begin{array}{rcl} P & := & \mathrm{Rect}(AD, BD), \\ Q & := & \mathrm{Sq}(CD), \\ R & := & \mathrm{Rect}(CB, BD), \\ S & := & \mathrm{Sq}(BC), \\ T & := & \mathrm{Sq}(BD), \end{array}$$

into the abstract bridge. Steps 6 and 7 are exactly its two hypotheses, so the conclusion is the proposition:

$$\mathrm{Sq}(BC) + \mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} \mathrm{Sq}(CD).$$

6.7 Lean Formalization

6.7.1 The public theorem signature

The following is the current compiling public signature, copied from the production source. It is intentionally explicit: the theorem receives the concrete cut diagrams and rectangle/square representatives required by the reused Book II APIs.

```
theorem euclid_proposition_2_6
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (A B C D : Geo.Point)
```

```
(hMidC : HilbertIsMidpoint Geo C A B)
(hABD : Geo.Between A B D)

------------------------------------------------------------
-- Data for test 06.
------------------------------------------------------------

(E2 H2 L2 X2 : Geo.Point)
(hRect2 : IsRectangle Geo A D H2 E2)
(hDH_BD2 : Geo.Congruent D H2 B D)
(hELE2 : Geo.Between E2 L2 H2)
(hEXD2 : Geo.Between E2 X2 D)
(hBXL2 : Geo.Between B X2 L2)
(hLeftPar2 : IsParallelogram Geo L2 E2 A B)
(hRightPar2 : IsParallelogram Geo D H2 L2 B)

-- Square on BD, shared by tests 06 and 07.
(F G : Geo.Point)
(hSquareBD : IsSquare Geo B D F G)

-- Rect(AD,BD).
(P0 P1 P2 P3 : Geo.Point)
(hAD_BD :
  IsRectangleContainedBy Geo
    P0 P1 P2 P3 A D B D)

-- Rect(AB,BD).
(V0 V1 V2 V3 : Geo.Point)
(hAB_BD :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 A B B D)

(E3 H3 L3 X3 : Geo.Point)
(hRect3 : IsRectangle Geo A B H3 E3)
(hBH_BD3 : Geo.Congruent B H3 B D)
(hELE3 : Geo.Between E3 L3 H3)
(hEXB3 : Geo.Between E3 X3 B)
(hCXL3 : Geo.Between C X3 L3)
(hLeftPar3 : IsParallelogram Geo L3 E3 A C)
(hRightPar3 : IsParallelogram Geo B H3 L3 C)

-- Rect(AC,BD).
(Y0 Y1 Y2 Y3 : Geo.Point)
(hAC_BD :
  IsRectangleContainedBy Geo
    Y0 Y1 Y2 Y3 A C B D)

-- The single representative R = Rect(CB,BD).
(Z0 Z1 Z2 Z3 : Geo.Point)
(hCB_BD :
  IsRectangleContainedBy Geo
    Z0 Z1 Z2 Z3 C B B D)

------------------------------------------------------------
-- Data for test 07: II.4 on CD cut at B.
------------------------------------------------------------

(D0 E0 L0 X0 : Geo.Point)
```

```
    (hSquareCD : IsSquare Geo C D E0 D0)

    (hD0L0E0 : Geo.Between D0 L0 E0)
    (hD0X0D : Geo.Between D0 X0 D)
    (hBX0L0 : Geo.Between B X0 L0)

    (hII4LeftPar :
      IsParallelogram Geo L0 D0 C B)

    (hII4RightPar :
      IsParallelogram Geo D E0 L0 B)

    -- Representatives required internally by II.4.
    (J0 J1 J2 J3 : Geo.Point)
    (K0 K1 K2 K3 : Geo.Point)

    (hCD_CB :
      IsRectangleContainedBy Geo
        J0 J1 J2 J3 C D C B)

    (hCD_BD :
      IsRectangleContainedBy Geo
        K0 K1 K2 K3 C D B D)

    ------------------------------------------------------------
    -- Left II.3 decomposition inside II.4.
    ------------------------------------------------------------

    (D1 E1 L1 X1 : Geo.Point)

    (hRectLeft :
      IsRectangle Geo D C E1 D1)

    (hCE1_BC :
      Geo.Congruent C E1 B C)

    (hD1L1E1 : Geo.Between D1 L1 E1)
    (hD1X1C : Geo.Between D1 X1 C)
    (hBX1L1 : Geo.Between B X1 L1)

    (hLeftLeftPar :
      IsParallelogram Geo L1 D1 D B)

    (hLeftRightPar :
      IsParallelogram Geo C E1 L1 B)

    -- Square on BC.
    (S0 S1 : Geo.Point)
    (hSquareBC : IsSquare Geo B C S0 S1)

    ------------------------------------------------------------
    -- Right II.3 decomposition inside II.4.
    ------------------------------------------------------------

    (D2 E4 L4 X4 : Geo.Point)

    (hRectRight :
      IsRectangle Geo C D E4 D2)
```

```
    (hDE4_BD :
      Geo.Congruent D E4 B D)

    (hD2L4E4 : Geo.Between D2 L4 E4)
    (hD2X4D : Geo.Between D2 X4 D)
    (hBX4L4 : Geo.Between B X4 L4)

    (hRightLeftPar :
      IsParallelogram Geo L4 D2 C B)

    (hRightRightPar :
      IsParallelogram Geo D E4 L4 B) :

    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo B C S0 S1 +
       hilbertParallelogramTerm Geo P0 P1 P2 P3)
      (hilbertParallelogramTerm Geo C D E0 D0)
```

The large signature falls into three geometric groups.

1. The points A, B, C, D, midpoint data, and $A - B - D$ encode the baseline configuration.
2. The first block supplies the two exact decompositions used to reach Step 6: II.3 on AD cut at B, II.1 on AB cut at C, a square on BD, and representatives of the three required rectangles.
3. The second block supplies the II.4 diagram on CD cut at B, including its two internal II.3 configurations, the square on BC, and the intermediate rectangle representatives required by II.4.

The conclusion itself is much smaller than the witness data:

```
HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo B C S0 S1 +
   hilbertParallelogramTerm Geo P0 P1 P2 P3)
  (hilbertParallelogramTerm Geo C D E0 D0)
```

and is exactly the formal version of

$$\mathrm{Sq}(BC) + \mathrm{Rect}(AD, BD) = \mathrm{Sq}(CD).$$

6.7.2 The final bridge call

After the two large decompositions have been established, the conceptual end of the proof is only the following substitution into the abstract bridge:

```
exact
  proposition2_6_test08_scissors_bridge
    Geo
    (hilbertParallelogramTerm Geo P0 P1 P2 P3)
    (hilbertParallelogramTerm Geo C D E0 D0)
```

```
    (hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3)
    (hilbertParallelogramTerm Geo B C S0 S1)
    (hilbertParallelogramTerm Geo B D F G)
    h06
    h07
```

This makes the proof architecture unusually transparent: geometry produces two normal forms, and the last step is an exact-scissors rearrangement of those normal forms.

6.8 Relationship with Earlier Book II Propositions

6.8.1 Proposition II.1

II.1 is called directly through `euclid_proposition_2_1_two`. Its role is to split $\mathrm{Rect}(AB, BD)$ at the midpoint C. Thus the basic rectangle-distribution theorem remains the additive engine of Book II.

6.8.2 Proposition II.3

II.3 is also called directly. Applied to AD cut at B, it introduces the square on the added segment BD:

$$\mathrm{Rect}(AD, BD) = \mathrm{Rect}(AB, BD) + \mathrm{Sq}(BD)$$

at the exact-scissors level.

6.8.3 Proposition II.4

II.4 is a direct theorem dependency and supplies the expansion of the square on CD after B has been proved to lie between C and D. Since the current II.4 theorem itself composes earlier Book II infrastructure, II.6 inherits that established internal structure rather than rebuilding it.

6.8.4 Proposition II.5

The production file imports `CGJteamLab.Proposition2_5`, so the II.5 module lies immediately below II.6 in the module graph. Nevertheless `euclid_proposition_2_5` is not called by any of the II.6 private helpers or by the public theorem.

Mathematically the two propositions are companions: II.5 concerns an unequal cut inside a bisected segment, whereas II.6 concerns an extension beyond the endpoint. Formally their proof DAGs differ. The current II.5 proof does not call II.4, while II.6 uses II.4 directly as one of its two main normal-form decompositions.

6.9 Relationship with Book I Infrastructure

The classical proof uses I.36 and I.43 explicitly. The Lean proof does not replay those steps proposition-locally. Their geometric effect has already been absorbed into the rectangle, parallelogram, square, and scissors infrastructure on which II.1–II.4 are built.

Book I geometry still remains foundational below the current theorem: right-angle propagation, squares, parallelograms, parallel lines, and the finite-scissors representation all depend on the established Hilbert/Book I layers. The gain is architectural: II.6 can be expressed as a composition of stable Book II content interfaces rather than as a second formalization of Euclid's gnomon diagram.

6.10 Formalization Notes and Hidden Geometric Data

6.10.1 The extension order is proved

The diagram suggests $A - C - B - D$, but the public theorem assumes only midpoint data for C together with $A - B - D$. The private order lemma explicitly proves $C - B - D$ and $A - C - D$. This matters because II.4 can be applied to CD cut at B only after $C - B - D$ is available as a theorem.

6.10.2 Midpoint data have two independent roles

`HilbertIsMidpoint` is not a numerical assertion that C has coordinate one half. Its betweenness component supplies $A - C - B$, while its congruence component supplies $AC \cong CB$. The former feeds the order DAG; the latter feeds rectangle transport.

6.10.3 Representative reuse is deliberate

In the final II.4 call the same Z representative is used twice for $\mathrm{Rect}(CB, BD)$, and the same square $BDFG$ is shared with the Step 6 decomposition. This is not a mathematical restriction on the content identity. It is a deliberate proof-engineering choice that lets the final abstract scissors bridge identify the repeated terms literally and avoids unnecessary representative transports.

6.10.4 The square terms remain geometric representatives

The terms called $\mathrm{Sq}(BC)$, $\mathrm{Sq}(BD)$, and $\mathrm{Sq}(CD)$ are concrete squares supplied through `IsSquare`. They are inserted into the scissors calculus as parallelogram terms. No operation sends a segment to a real number or squares a numerical length.

6.10.5 No gnomon primitive

As in II.5, the formal library does not introduce a special gnomon type. Euclid's shaded remainder is represented extensionally by a formal sum of rectangle and square terms. In II.6 the combination of II.4 with the abstract scissors bridge replaces the gnomon bookkeeping completely.

6.10.6 No cancellation or subtraction of content

The identity strongly resembles a difference-of-squares calculation, but the proof does not subtract a square or cancel equal rectangles. The abstract bridge is additive: it uses only exact scissors equality, addition, associativity/commutativity of the formal sum, symmetry, and transitivity.

6.11 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and the production file introduces no new proposition-specific `axiom`, `sorry`, or `admit`. Both the module build and the full project build were reported clean.

The user-verified transitive axiom audit is

```
#print axioms Geometry.euclid_proposition_2_6

'Geometry.euclid_proposition_2_6' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The precise status is therefore not "axiom-free." Rather, II.6 introduces no new proposition-specific axiom and no `sorry`; its reported transitive dependency list is exactly the four declarations above and contains no `sorryAx`.

Semantically, the theorem lies in the exact finite-scissors layer. It proves `HilbertScissorsEq` for the supplied configurations. It does not invoke a numerical area function, a segment multiplication operation, or a general cancellation principle for equal content.

6.12 Dependency Summary

The actual theorem-call DAG can be displayed as

$$\begin{gathered}\text{Hilbert incidence} + \text{Euclidean plane layer}\\ \Downarrow \\ \text{Book I square, rectangle, parallelogram, and scissors infrastructure}\\ \Downarrow \\ \text{HilbertRectangle representative/transport API}\\ \Downarrow \\ \text{II.1 binary split} \quad + \quad \text{II.3 rectangle-plus-square} \quad + \quad \text{II.4 square expansion}\\ \Downarrow \\ \text{private II.6 order, transport, composition, and bridge helpers}\\ \Downarrow \\ \text{public theorem } \texttt{euclid\_proposition\_2\_6}.\end{gathered}$$

The module chain is slightly different:

$$\text{Proposition2\_5 module} \longrightarrow \text{Proposition2\_6 module},$$

but `euclid_proposition_2_5` itself is not called.

The classical local chain is different again:

$$\text{I.36} + \text{I.43} \longrightarrow \text{gnomon identification} \longrightarrow \text{add square on } BC \longrightarrow \text{II.6}.$$

The formal reconstruction therefore proves the same geometric-content identity through a different DAG rather than translating Euclid line by line.

New geometric axiom in II.6:	no
New content axiom in II.6:	no
New proposition-local helper theorem:	yes, nine private helpers
New reusable public helper theorem:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.1, II.3, II.4
Imports II.5 module:	yes
Calls `euclid_proposition_2_5`:	no
Uses exact `HilbertScissorsEq`:	yes
Uses general equicomplementability:	no
Uses numerical area:	no
Uses segment multiplication:	no
Uses content cancellation:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Full project build:	clean (user verified)
Proposition-local `sorry`/`admit`:	no
Transitive `sorryAx`:	absent in reported audit

6.13 Conclusion

Proposition II.6 is a strong checkpoint for the reuse of the Book II rectangle calculus. Classically, Euclid proves the identity by constructing a gnomon inside the square on CD and comparing its pieces through I.36 and I.43. The current Lean proof reaches the same content equality without a gnomon primitive and without numerical algebra.

Its architecture is instead a composition of three already established Book II mechanisms: II.3 decomposes the rectangle on AD, BD, II.1 splits the old segment AB at its midpoint, and II.4 expands the square on CD at the point B. Midpoint congruence normalizes the repeated cross rectangles, and a small abstract scissors lemma performs the final associative-commutative rearrangement. The result is an exact `HilbertScissorsEq` proof with no new axiom, no `sorry`, no segment multiplication, and no numerical area.

Chapter 7

Proposition II.7

7.1 Introduction

Proposition II.7 returns from the midpoint identities II.5–II.6 to an arbitrary division point. Let C lie between A and B. Euclid's statement permits either of the two parts to be chosen as the "said segment." The current Lean theorem chooses AC. In geometric-content notation its conclusion is

$$\mathrm{Sq}(AB) + \mathrm{Sq}(AC) = 2\ \mathrm{Rect}(AB, AC) + \mathrm{Sq}(CB).$$

The modern algebraic shadow is immediate if $AC = a$ and $CB = b$:

$$(a + b)^2 + a^2 = 2(a + b)a + b^2.$$

This formula is only a mnemonic for the geometric pattern. Neither Euclid's proposition nor the current Lean theorem defines a numerical multiplication of segment lengths. The formal objects are concrete square and rectangle representatives, and the conclusion is an exact `HilbertScissorsEq`.

The production source begins with

```
import CGJteamLab.Proposition2_4
```

so Proposition II.4 is the immediate module dependency. At theorem-call level the proof uses `euclid_proposition_2_4` and `euclid_proposition_2_3`. It does not call II.1 or II.2 directly; their rectangle-dissection machinery is inherited through II.4.

7.2 Euclid's Statement and Classical Configuration

7.2.1 The statement

Euclid considers a straight line cut at an arbitrary point. The square on the whole together with the square on one chosen part equals twice the rectangle contained by the whole and that chosen part, together with the square on the remaining part [1].

A common classical presentation chooses BC as the selected part and writes

$$\mathrm{Sq}(AB) + \mathrm{Sq}(BC) = 2\ \mathrm{Rect}(AB, BC) + \mathrm{Sq}(CA).$$

The current Lean theorem instead chooses AC:

$$\mathrm{Sq}(AB) + \mathrm{Sq}(AC) = 2\ \mathrm{Rect}(AB, AC) + \mathrm{Sq}(CB).$$

These are the two symmetric readings already allowed by the wording of the proposition. The formal theorem therefore does not alter the mathematical content; it fixes one of Euclid's two admissible choices.

7.2.2 The classical diagram

In the classical proof one constructs the square on AB, draws the diagonal, and then draws parallels through the division point C. The resulting configuration contains two complementary parallelograms about the diagonal, a gnomon, and the squares on the two parts. Euclid uses I.43 to compare the complements and then adds the same figures to equal figures until the desired whole-square identity is obtained [1].

The current formal proof does not reproduce that gnomon bookkeeping line-by-line. The first figure therefore records the classical geometric state itself: the square on AB, its diagonal, and the two parallels through the division point and the diagonal intersection. The two normal forms actually consumed by Lean are shown later, at the proof steps where they enter.

Diagrammatic audit. The classical construction deserves its own figure because the diagonal and parallel lines are genuine geometric data in Euclid's proof. The formal proof then changes viewpoint: its geometrically meaningful states are the II.4 square dissection and the cyclic reorientation needed before the II.3 call. Betweenness reversal, endpoint-orientation changes of an unchanged rectangle, and the final associative-commutative regrouping do not create new geometry and are therefore not given separate figures.

7.3 Classical Proof

A convenient reading of Euclid's proof takes the selected part to be BC. Construct the square on AB and the usual parallels through C. The diagonal of the square produces complementary parallelograms; I.43 identifies the relevant pair. After adding the same adjacent rectangle to both, Euclid obtains two equal larger figures, so their sum is twice one of them.

That doubled figure is identified with twice the rectangle contained by AB and BC, because one adjacent side in the constructed rectangle is equal to BC. The remainder of the large square is organized as a gnomon together with the square on BC. Finally, adding the square on the remaining part AC converts that gnomon expression into the whole square on AB plus the square on BC. Thus

$$\mathrm{Sq}(AB) + \mathrm{Sq}(BC) = 2\ \mathrm{Rect}(AB, BC) + \mathrm{Sq}(AC).$$

The classical local dependency picture is therefore schematically

$$\begin{gathered}
\text{I.46 and I.31: construct square and parallels}\\
\Downarrow\\
\text{I.43: complementary parallelograms are equal}\\
\Downarrow\\
\text{gnomon and common-figure additions}\\
\Downarrow\\
\text{II.7.}
\end{gathered}$$

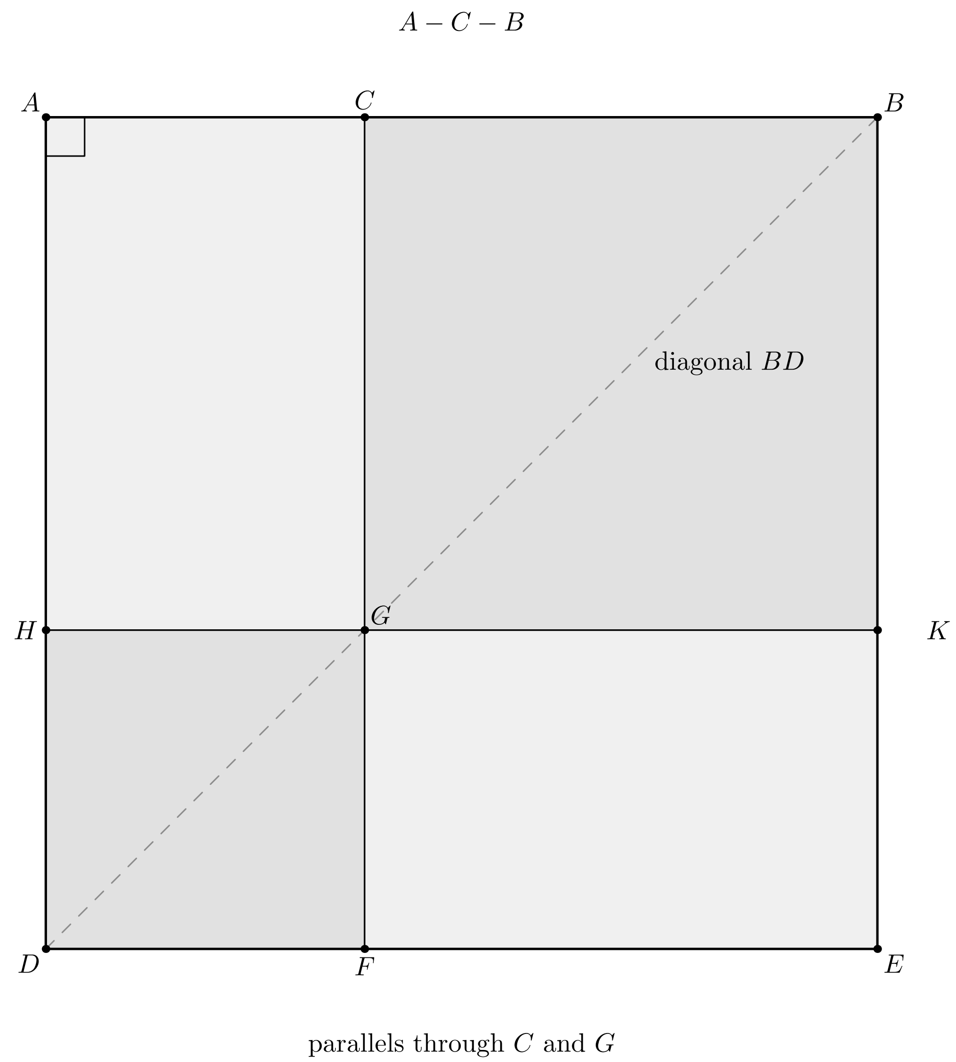


Figure 7.1: The classical square construction underlying Euclid II.7. The arbitrary cut is $A - C - B$; the diagonal BD and the parallels through C and G create the complementary parallelogram configuration used in the gnomon argument. The light shading separates the four regions without assigning them numerical areas. This is the source geometry; the current Lean proof later consumes already proved II.4 and II.3 normal forms instead of replaying this gnomon bookkeeping.

The present Lean proof has a different DAG. It consumes already formalized Book II identities instead of reconstructing the gnomon from Book I each time.

7.4 The Geometric Identity in the Current Formalization

The baseline assumption is

```
(A B C : Geo.Point)
(hACB : Geo.Between A C B)
```

so the formal order is exactly

$$A - C - B.$$

No midpoint condition is present. This is the main geometric difference from II.5 and II.6.

The public conclusion is

$$\begin{aligned}&\mathrm{Sq}(AB) + \mathrm{Sq}(AC)\\ &\qquad \simeq_{\mathrm{sc}} (\mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, AC)) + \mathrm{Sq}(CB).\end{aligned}$$

The notation 2 $\mathrm{Rect}(AB, AC)$ in the prose means two copies in a formal scissors sum. There is no scalar multiplication operation on content terms in the proof.

The actual Lean conclusion is even more concrete. The square on AB is the supplied square ABE_0D_0; the square on AC is supplied in reversed orientation as $CAFG$; the square on CB is $CBHK$; and the two final rectangles on AB, AC are literally the same supplied representative $U_0U_1U_2U_3$ written twice.

7.5 Reusable Book II Infrastructure Used

7.5.1 Proposition II.4: expand the square on the whole

The first normal form comes directly from `euclid_proposition_2_4`. With

$$W := \mathrm{Sq}(AB), \qquad X := \mathrm{Rect}(AC, CB), \qquad S := \mathrm{Sq}(AC), \qquad T := \mathrm{Sq}(CB),$$

the proof obtains

$$\boxed{W \simeq_{\mathrm{sc}} (X + S) + (X + T).}$$

A deliberate implementation choice makes this especially useful: the same concrete representative $T_0T_1T_2T_3$ of $\mathrm{Rect}(AC, CB)$ is supplied to both cross-rectangle positions inside II.4. Thus the two occurrences of X are literally the same scissors term, not merely representatives later proved equivalent.

7.5.2 Proposition II.3: normalize the rectangle on the whole and one part

The second normal form is

$$\boxed{R \simeq_{\mathrm{sc}} X + S,}$$

where

$$R := \mathrm{Rect}(AB, AC).$$

It comes from `euclid_proposition_2_3`, but not in the original orientation $A - C - B$. The proof first reverses the baseline to

$$B - C - A$$

and applies II.3 to that divided segment.

This reversal is geometric bookkeeping, not algebra. The contained-by data for R are reoriented from (AB, AC) to (BA, CA) by congruence symmetry. The cross rectangle is then rotated by `rectangle_contained_by_swap`, producing both the contained-by data needed by II.3 and an exact scissors equality between the rotated and original representatives.

7.5.3 The rectangle swap is geometric, not multiplicative commutativity

The call

```
have hSwapResult :=
  rectangle_contained_by_swap
    Geo
    T0 T1 T2 T3
    A C C B
    hCross
```

is the key representation step. It does not say that a numerical product $AC \cdot CB$ is commutative. It says that cyclically rotating a concrete rectangle changes the ordered side presentation while preserving its exact scissors term.

This is precisely the distinction established in the Book II rectangle API: geometric side-order transport precedes, and is independent of, any later arithmetic of segment lengths.

7.6 Proof Architecture of the Reconstruction

Unlike II.6, the production file introduces no private proposition-local helper declarations. The whole reconstruction is one public theorem containing local `let` abbreviations and `have` steps. The proof architecture is therefore visible directly in the theorem body.

7.6.1 Step 1: name the five scissors terms

The proof introduces

$$\begin{aligned} W &:= \mathrm{Sq}(AB), \\ R &:= \mathrm{Rect}(AB, AC), \\ X &:= \mathrm{Rect}(AC, CB), \\ S &:= \mathrm{Sq}(AC), \\ T &:= \mathrm{Sq}(CB). \end{aligned}$$

These are abbreviations for supplied `HilbertScissorsTerm` objects, not new geometric constructions.

7.6.2 Step 2: obtain the II.4 normal form

The local fact `hW` calls II.4 and proves

$$W \simeq_{\mathrm{sc}} (X + S) + (X + T).$$

The same X representative is passed twice to the II.4 call.

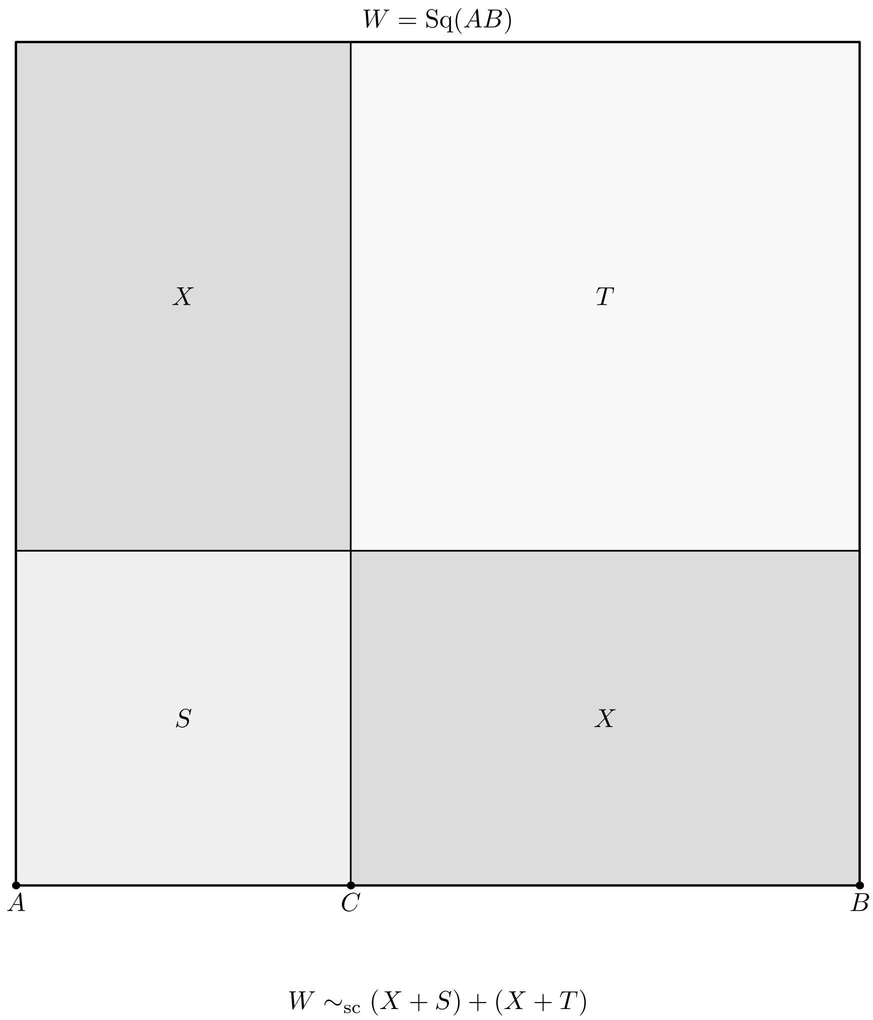


Figure 7.2: The first formal geometric state used by II.7. Proposition II.4 decomposes the supplied square W on AB at the cut C into the square term S on AC, the square term T on CB, and two literal copies of the same cross-rectangle term X. Thus the picture encodes $W \simeq_{\mathrm{sc}} (X + S) + (X + T)$ as an actual square dissection. No final II.7 regrouping is shown here.

7.6.3 Step 3: reverse the betweenness orientation

From
$$A - C - B$$
the proof derives
$$B - C - A$$
using `HilbertOrder.between_incidence`. This is the exact order required by the chosen II.3 instantiation.

7.6.4 Step 4: reorient the contained-by data for R

The representative $U_0U_1U_2U_3$ initially satisfies
$$\mathrm{Rect}(AB, AC).$$
For the II.3 call its side data are re-read as
$$\mathrm{Rect}(BA, CA).$$
The underlying rectangle is unchanged; only the orientations of the two unoriented segment congruences are reversed with `CongruentSwapSecond`.

7.6.5 Step 5: rotate the cross rectangle

The call to `rectangle_contained_by_swap` produces a cyclic rotation
$$T_0T_1T_2T_3 \longmapsto T_1T_2T_3T_0.$$
The rotated rectangle can then be presented with the ordered containing sides required by II.3. The same call also supplies an exact scissors equality between the two cyclic presentations.

7.6.6 Step 6: apply II.3 on $B - C - A$

The local fact `hRawR` proves
$$R \simeq_{\mathrm{sc}} X_{\mathrm{rot}} + S.$$
The subsequent facts `hRotateCrossBack` and `hR_concrete` transport this to the original cross representative:
$$\boxed{R \simeq_{\mathrm{sc}} X + S.}$$
No representative-uniqueness theorem is needed. The proof knows exactly how the rotated rectangle is related to the original one.

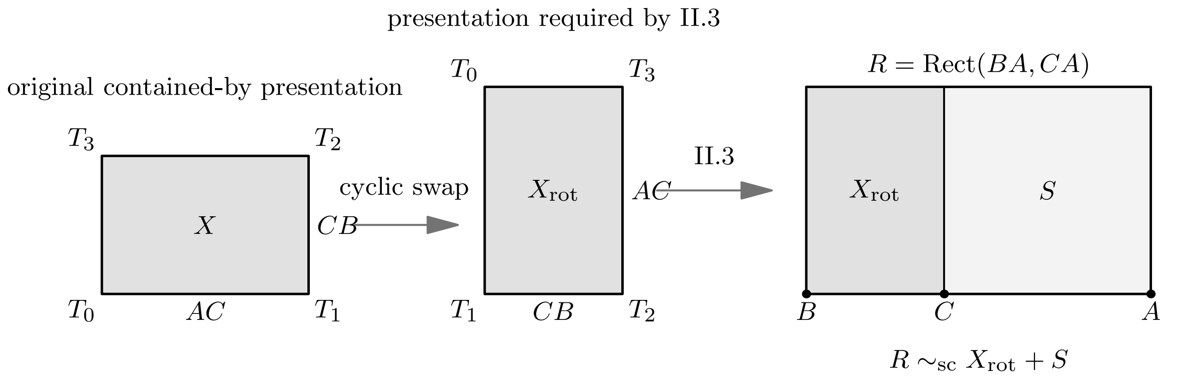


Figure 7.3: The second formal geometric state. On the left, the concrete cross rectangle X is shown with its original ordered side presentation. The cyclic swap changes only the vertex presentation and exchanges the ordered containing sides, producing the form required by II.3. On the right, II.3 is applied to the reversed baseline $B - C - A$: the rectangle R contained by BA, CA is split at C into the rotated cross rectangle and the square term S on CA. This is the geometric content behind $R \simeq_{\mathrm{sc}} X + S$ after transport back to the original cross representative.

7.6.7 Step 7: add the extra square S

From the II.4 normal form, `hAddS` gives

$$W + S \simeq_{\mathrm{sc}} ((X + S) + (X + T)) + S.$$

7.6.8 Step 8: regroup the formal sum

The only literal equality of terms used in the final bridge is

$$((X + S) + (X + T)) + S = ((X + S) + (X + S)) + T.$$

Lean proves it by

```
have hRegroup :
    ((X + S) + (X + T)) + S
      =
    ((X + S) + (X + S)) + T := by
  ac_rfl
```

so this step uses only associativity and commutativity of the formal sum.

7.6.9 Step 9: replace the two copies of $X + S$

Since `hR` states

$$R \simeq_{\mathrm{sc}} X + S,$$

its symmetry can replace each copy of $X + S$ by R. The proof uses `HilbertScissorsEq.add` twice and obtains

$$((X + S) + (X + S)) + T \simeq_{\mathrm{sc}} (R + R) + T.$$

7.6.10 Step 10: compose

Transitivity finally gives

$$\boxed{W + S \simeq_{\mathrm{sc}} (R + R) + T.}$$

Expanding the abbreviations is exactly the public II.7 conclusion.

This proof is additive from beginning to end. It never cancels a common content term, subtracts one figure from another, or invokes a positivity principle.

Diagrammatic audit. No additional picture is introduced for Steps 7–10. Once the II.4 state in Figure 7.2 and the II.3 state in Figure 7.3 are available, the remaining proof consists of adding the already supplied square term, reassociating and commuting a formal sum, replacing two exact-scissors-equal subterms, and composing the resulting relations. Drawing a new region arrangement for those operations would suggest a new physical dissection that the Lean proof does not construct.

7.7 Lean Formalization

7.7.1 The public theorem signature

The following is the current compiling public signature copied from `CGJteamLab/Proposition2_7.lean`. The amount of witness data reflects the fact that II.7 reuses the configured II.4 and II.3 interfaces rather than hiding their geometry behind an existential wrapper.

```
theorem euclid_proposition_2_7
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (A B C : Geo.Point)

    ------------------------------------------------------------------
    -- II.2 / II.4 diagram: square on AB cut at C.
    ------------------------------------------------------------------

    (D0 E0 L0 X0 : Geo.Point)
    (hSquareAB : IsSquare Geo A B E0 D0)

    (hACB : Geo.Between A C B)
    (hD0L0E0 : Geo.Between D0 L0 E0)
    (hD0X0B : Geo.Between D0 X0 B)
    (hCX0L0 : Geo.Between C X0 L0)

    (hII2LeftPar :
      IsParallelogram Geo L0 D0 A C)
    (hII2RightPar :
      IsParallelogram Geo B E0 L0 C)

    -- Representatives of Rect(AB,AC) and Rect(AB,CB) used by II.4.
    (U0 U1 U2 U3 : Geo.Point)
    (V0 V1 V2 V3 : Geo.Point)

    (hAB_AC :
      IsRectangleContainedBy Geo
        U0 U1 U2 U3 A B A C)

    (hAB_CB :
      IsRectangleContainedBy Geo
        V0 V1 V2 V3 A B C B)

    ------------------------------------------------------------------
    -- II.3 diagram for the left II.4 term, read on B-C-A.
    ------------------------------------------------------------------

    (D1 E1 L1 X1 : Geo.Point)

    (hRectLeft :
      IsRectangle Geo B A E1 D1)

    (hAE1_CA :
      Geo.Congruent A E1 C A)

    (hD1L1E1 : Geo.Between D1 L1 E1)
```

```
  (hD1X1A : Geo.Between D1 X1 A)
  (hCX1L1 : Geo.Between C X1 L1)

  (hLeftLeftPar :
    IsParallelogram Geo L1 D1 B C)
  (hLeftRightPar :
    IsParallelogram Geo A E1 L1 C)

  -- Square on CA, i.e. the same unoriented segment as AC.
  (F G : Geo.Point)
  (hSquareCA : IsSquare Geo C A F G)

  ------------------------------------------------------------------
  -- II.3 diagram for the right II.4 term.
  ------------------------------------------------------------------

  (D2 E2 L2 X2 : Geo.Point)

  (hRectRight :
    IsRectangle Geo A B E2 D2)

  (hBE2_CB :
    Geo.Congruent B E2 C B)

  (hD2L2E2 : Geo.Between D2 L2 E2)
  (hD2X2B : Geo.Between D2 X2 B)
  (hCX2L2 : Geo.Between C X2 L2)

  (hRightLeftPar :
    IsParallelogram Geo L2 D2 A C)
  (hRightRightPar :
    IsParallelogram Geo B E2 L2 C)

  -- Square on CB.
  (H K : Geo.Point)
  (hSquareCB : IsSquare Geo C B H K)

  ------------------------------------------------------------------
  -- One representative of Rect(AC,CB), used twice in II.4.
  ------------------------------------------------------------------

  (T0 T1 T2 T3 : Geo.Point)

  (hCross :
    IsRectangleContainedBy Geo
      T0 T1 T2 T3 A C C B) :

  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo A B E0 D0 +
     hilbertParallelogramTerm Geo C A F G)
    ((hilbertParallelogramTerm Geo U0 U1 U2 U3 +
      hilbertParallelogramTerm Geo U0 U1 U2 U3) +
     hilbertParallelogramTerm Geo C B H K)
```

The signature has four geometric blocks.

1. The points A, B, C and `hACB` specify the arbitrary internal cut.

2. The D_0, E_0, L_0, X_0 block supplies the square-on-AB diagram required by II.4, together with representatives of $\mathrm{Rect}(AB, AC)$ and $\mathrm{Rect}(AB, CB)$.

3. The D_1, E_1, L_1, X_1 and D_2, E_2, L_2, X_2 blocks are the two II.3 subconfigurations consumed inside II.4, together with the squares on CA and CB.

4. The T_0, T_1, T_2, T_3 block supplies one representative of $\mathrm{Rect}(AC, CB)$, deliberately reused twice.

The conclusion contains only the square on AB, the square on AC, two copies of the same rectangle on AB, AC, and the square on CB. Much of the signature is therefore construction witness data inherited from the reusable Book II APIs rather than conceptual data in the final identity.

7.7.2 The two decisive local facts

The mathematical core of the production proof is compact:

```
have hW :
    HilbertScissorsEq Geo
      W
      ((X + S) + (X + T)) := by
  ...
  exact euclid_proposition_2_4 ...

...

have hR :
    HilbertScissorsEq Geo
      R
      (X + S) := by
  ...
```

Everything after these two normal forms is an exact-scissors rearrangement. This is the central architectural fact of II.7.

7.8 Relationship with Earlier Book II Propositions

7.8.1 Proposition II.3

II.3 is called directly on the reversed divided segment $B - C - A$. Its role is to convert the rectangle contained by the whole AB and the selected part AC into

$$\mathrm{Rect}(AC, CB) + \mathrm{Sq}(AC).$$

Because the current II.3 interface has ordered contained-by side data, the proof must explicitly reverse segment orientations and rotate the cross rectangle before the call.

7.8.2 Proposition II.4

II.4 is the main normal-form theorem:

$$\mathrm{Sq}(AB) \simeq_{\mathrm{sc}} (\mathrm{Rect}(AC, CB) + \mathrm{Sq}(AC)) + (\mathrm{Rect}(AC, CB) + \mathrm{Sq}(CB)).$$

This is exactly the decomposition needed for II.7 once one recognizes the first parenthesis as $\mathrm{Rect}(AB, AC)$ by II.3.

7.8.3 Propositions II.1 and II.2

Neither `euclid_proposition_2_1_two` nor `euclid_proposition_2_2` is called directly by II.7. They are, however, below II.4 in the established Book II dependency chain. Thus II.7 inherits their rectangle split and square decomposition indirectly.

This distinction matters: the module/theorem DAG should not be replaced by a vague statement that II.7 "uses all previous propositions."

7.8.4 Propositions II.5 and II.6

II.5 and II.6 are not dependencies of II.7. They form the preceding midpoint block, whereas II.7 again assumes only an arbitrary cut. The production module therefore imports II.4 directly rather than continuing the linear module chain through II.6.

7.9 Relationship with Book I Infrastructure

Classically, II.7 is a gnomon argument centered on I.43 and on the square and parallel constructions supplied by I.46 and I.31. The current Lean theorem does not call those Book I propositions locally. Their geometric consequences have already been absorbed into the square, rectangle, parallelogram, and scissors infrastructure used by II.3 and II.4.

This is another clear example of the difference between the classical DAG and the formal library DAG. The reconstruction proves the same content identity without requiring every proposition to replay Euclid's local construction.

7.10 Formalization Notes and Hidden Geometric Data

7.10.1 The chosen segment differs from a common printed diagram

Many printed versions instantiate the proposition with BC as the selected part. The current Lean theorem selects AC. Because Euclid's statement says "one of the segments," this is not a mathematical strengthening or weakening. It is a concrete orientation choice that must nevertheless be recorded in technical documentation.

7.10.2 Betweenness reversal is explicit

The source assumption is `Geo.Between A C B`. II.3 is used on $B - C - A$, so the reversed betweenness statement is derived explicitly. The formal proof cannot infer the desired orientation merely from the picture.

7.10.3 Squares use unoriented side congruence

The final square on the selected part is supplied as

```
(hSquareCA : IsSquare Geo C A F G)
```

although the prose calls it the square on AC. This is legitimate because the underlying segment is unoriented at the congruence level, but the orientation is visible in the concrete Lean arguments.

7.10.4 The unused final representative V

The II.4 interface requires representatives of both $\mathrm{Rect}(AB, AC)$ and $\mathrm{Rect}(AB, CB)$. The latter, represented by $V_0V_1V_2V_3$, is internal to the II.4 call and does not appear in the final II.7 conclusion. This is an example of witness data required by a reused theorem but not by the mathematical endpoint.

7.10.5 Representative reuse avoids uniqueness transport

The same T rectangle is supplied twice to II.4. The same U rectangle is written twice in the final target. These choices make repeated terms literal. The proof therefore does not call `rectangle_contained_by_unique`.

This is not a restriction on the proposition. It is a proof-engineering normalization that removes unnecessary representative bookkeeping.

7.10.6 No cancellation principle

The algebraic shadow strongly invites subtraction. The Lean proof does not subtract X, S, or any square. It adds S, rearranges a commutative formal sum, and replaces equal-scissors subterms. No proper-part principle, positivity theorem, or content cancellation axiom enters II.7.

7.11 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and the production file introduces no new proposition-specific `axiom`, `sorry`, or `admit`.

The user-verified transitive axiom audit is

```
#print axioms Geometry.euclid_proposition_2_7
```

```
'Geometry.euclid_proposition_2_7' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

Hence the precise status is not "axiom-free." Proposition II.7 adds no new axiom and no `sorry`; its reported transitive dependencies are the four declarations displayed above, with no `sorryAx`.

The public module and the full relevant project build were reported clean after the theorem was added to the project. Semantically, the theorem lies in the exact finite-scissors layer. It proves `HilbertScissorsEq` and does not invoke a numerical area function, segment multiplication, or general equicomplementability as its conclusion.

7.12 Dependency Summary

The theorem-call DAG is

$$
\begin{array}{c}
\text{Hilbert incidence} + \text{Euclidean plane layer} \\
\Downarrow \\
\text{Book I square, rectangle, parallelogram, and scissors infrastructure} \\
\Downarrow \\
\text{HilbertRectangle contained-by and swap API} \\
\Downarrow \\
\text{II.3 rectangle-plus-square} \quad + \quad \text{II.4 square expansion} \\
\Downarrow \\
\text{betweenness/orientation transport} + \text{rectangle rotation} \\
\Downarrow \\
\text{exact-scissors addition} + \text{AC regrouping} \\
\Downarrow \\
\text{public theorem } \texttt{euclid\_proposition\_2\_7}.
\end{array}
$$

The immediate module chain is simply

$$\text{Proposition2\_4 module} \longrightarrow \text{Proposition2\_7 module}.$$

The classical local chain is different:

$$\text{I.46} + \text{I.31} + \text{I.43} \longrightarrow \text{gnomon comparison} \longrightarrow \text{common-figure additions} \longrightarrow \text{II.7}.$$

New geometric axiom in II.7:	no
New content axiom in II.7:	no
New proposition-local helper declaration:	no; local `haves` only
New reusable public helper theorem:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.3 and II.4
Directly calls II.1 or II.2:	no
Imports II.5 or II.6:	no
Uses exact `HilbertScissorsEq`:	yes
Uses general equicomplementability as conclusion:	no
Uses numerical area:	no
Uses segment multiplication:	no
Uses content cancellation/subtraction:	no
Uses representative uniqueness:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Full relevant project build:	clean (user verified)
Proposition-local `sorry`/`admit`:	no
Transitive `sorryAx`:	absent in reported audit

7.13 Conclusion

Proposition II.7 is an especially transparent demonstration of the reusable Book II rectangle calculus. The classical proof organizes the square on the whole through a gnomon and complementary parallelograms. The current Lean proof instead starts from two already established exact-scissors normal forms.

II.4 gives

$$W \simeq_{\mathrm{sc}} (X + S) + (X + T),$$

and II.3, after reversing the divided segment and rotating a rectangle representative, gives

$$R \simeq_{\mathrm{sc}} X + S.$$

Adding S, regrouping the formal sum by associativity and commutativity, and replacing two copies of $X + S$ by R yields

$$W + S \simeq_{\mathrm{sc}} (R + R) + T.$$

Thus the formal proof preserves the synthetic geometric-content character of Book II while avoiding numerical area, segment multiplication, subtraction, and cancellation. II.7 introduces no new axiom and no new public infrastructure; its value is architectural: it shows that II.3 and II.4 are already strong enough to recover another classical gnomon identity by exact scissors composition.

Chapter 8

Proposition II.8

8.1 Introduction

Proposition II.8 is the first proposition in the current Book II reconstruction whose classical statement explicitly asks for four copies of one rectangle. Let C lie between A and B. Euclid extends the line beyond B to a point D such that $BD = BC$. The proposition then asserts

$$4\,\mathrm{Rect}(AB, BC) + \mathrm{Sq}(AC) = \mathrm{Sq}(AD).$$

Here the expression "four times" is geometric language for four equal plane figures. In the Lean theorem it becomes four literal copies of one concrete scissors term; there is no scalar multiplication operation on content terms.

A modern algebraic shadow is obtained by writing $AC = a$ and $CB = b$. Then $AB = a + b$ and the produced condition $BD = BC$ gives $AD = a + 2b$, so the mnemonic is

$$4(a + b)b + a^2 = (a + 2b)^2.$$

This identity is documentary only. Neither the classical proof nor the current formal proof introduces numerical area or multiplication of segment lengths.

The production source begins with

```
import CGJteamLab.Proposition2_7
```

so Proposition II.7 is the immediate module dependency. At theorem-call level the proof uses `euclid_proposition_2_4` and `euclid_proposition_2_7`, together with the established rectangle representative API and exact scissors addition, symmetry, and transitivity.

8.2 Euclid's Statement and Classical Configuration

8.2.1 The statement

Euclid's statement says that if a straight line is cut at random, four times the rectangle contained by the whole and one segment, together with the square on the remaining segment, is equal to the square described on the whole and the chosen segment taken together as one straight line [1].

With the current lettering, $A - C - B$ and the chosen part is BC. Euclid produces D beyond B with $BD = BC$. Thus the final square is the square on AD, and the geometric identity is

$$4\ \mathrm{Rect}(AB, BC) + \mathrm{Sq}(AC) = \mathrm{Sq}(AD).$$

The phrase "on the whole and the chosen part as one straight line" is therefore not an abstract addition of lengths: it is represented by the actually produced collinear segment AD.

8.2.2 The classical diagram

Euclid produces BD in a straight line with AB, makes $BD = CB$, describes the square on AD, and draws the network of parallels usually called the "double" figure. The proof identifies two families of four equal regions by I.34, I.36, and I.43. Together the eight regions form a gnomon equal to four copies of the rectangle contained by AB and BD. Adding the remaining square on AC fills the square on AD [1].

Figure 8.1 records only the first complete geometric state: the produced baseline and the square on AD. The two imported Book II normal forms are shown later, exactly where they enter the formal proof, rather than being compressed into one three-panel summary.

$A - C - B - D$

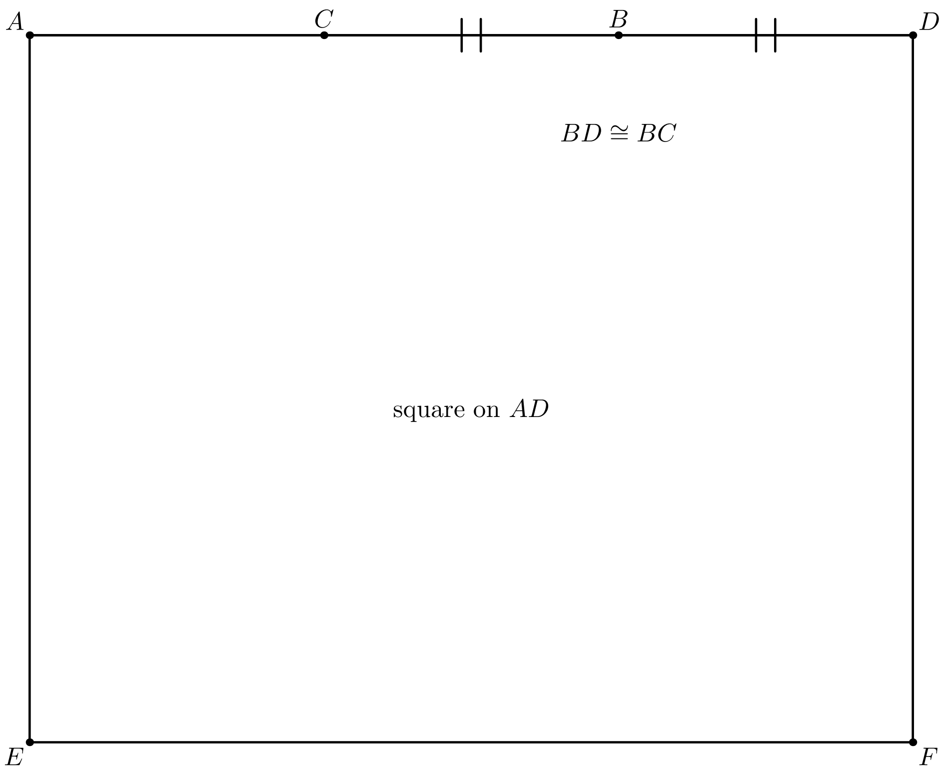


Figure 8.1: The initial geometric state for Proposition II.8. The order is $A - C - B - D$, the produced segment satisfies $BD \cong BC$, and the square on AD has been constructed. No II.4 or II.7 subdivision is shown yet. In particular, the figure records the geometry that makes the later representative transports meaningful without pretending that the Lean proof reconstructs Euclid's full eight-region gnomon.

Diagrammatic audit. The proof below changes visible state only when it imports the two geometric normal forms already proved as II.4 and II.7. These are therefore drawn as separate figures at Steps 3 and 4. The reversal $A - C - B \mapsto B - C - A$, the re-reading of one rectangle representative, the transport $S_{BD} \simeq_{\text{sc}} S_{CB}$, and the final associative-commutative regrouping do not create a new geometric configuration, so no additional figure is introduced for those bookkeeping steps.

8.3 Classical Proof

Start with $A-C-B$ and produce D beyond B so that $BD = BC$. Construct the square $AEFD$ on AD and draw Euclid's double network of parallels. The proof then has two four-region stages [1].

First, the equalities $CB = BD$ are propagated through opposite sides of the constructed parallelograms. I.36 converts the equal bases into equal parallelograms, while I.43 identifies the relevant complements. Euclid obtains four equal regions DK, CK, GR, RN.

A second use of the same mechanism gives four equal regions AG, MQ, QL, RF. The eight regions together form the gnomon STU, and Euclid proves that this gnomon is four times the region AK. The latter is a rectangle contained by AB and BD, because one of its adjacent sides is equal to BD. Consequently

$$4\ \mathrm{Rect}(AB, BD) = \text{gnomon } STU.$$

The remaining region OH is the square on AC. Adding it to both sides gives

$$4\ \mathrm{Rect}(AB, BD) + \mathrm{Sq}(AC) = \mathrm{Sq}(AD).$$

Finally $BD = BC$, so the rectangle may be read as the rectangle contained by AB and BC.

The classical local dependency picture is therefore

$$\begin{array}{c}
\text{I.3: lay off } BD = BC \quad + \quad \text{I.46: square on } AD \\
\Downarrow \\
\text{I.31: the parallel network} \\
\Downarrow \\
\text{I.34 and I.36: side and parallelogram equalities} \\
\Downarrow \\
\text{I.43: equal complements} \\
\Downarrow \\
\text{eight-region gnomon and common-figure addition} \\
\Downarrow \\
\text{II.8.}
\end{array}$$

The Lean DAG below is substantially shorter because the relevant Book II normal forms have already been proved.

8.4 The Geometric Identity in the Current Formalization

The public theorem receives the three baseline facts

```
(A B C D : Geo.Point)
(hACB : Geo.Between A C B)
(hABD : Geo.Between A B D)
(hBD_BC : Geo.Congruent B D B C)
```

so the intended order is

$$A - C - B - D, \qquad BD \cong BC.$$

Only the two displayed betweenness relations are primitive inputs; the proof reverses $A - C - B$ explicitly when II.7 needs $B - C - A$.

Let

$$R := \mathrm{Rect}(AB, BD).$$

The same concrete quadrilateral is then re-read as $\mathrm{Rect}(BA, BC)$ using endpoint reversal on the first containing side and the hypothesis $BD \cong BC$ on the second. The public conclusion is exactly

$$(R + R) + (R + R) + \mathrm{Sq}(CA) \simeq_{\mathrm{sc}} \mathrm{Sq}(AD).$$

There is no hidden coefficient object: four occurrences of R appear literally in the Lean term.

This conclusion is `HilbertScissorsEq`, not merely `HilbertScissorsEquicomplementable`. The theorem therefore certifies an exact finite-scissors relation for the configured representatives. It does not assert equality of real numbers.

8.5 Reusable Book II Infrastructure Used

8.5.1 Proposition II.4 on the produced segment

The first normal form is obtained by applying II.4 to the whole segment AD cut at B:

$$\mathrm{Sq}(AD) \simeq_{\mathrm{sc}} (R + \mathrm{Sq}(BA)) + (R + \mathrm{Sq}(BD)).$$

In this invocation the same concrete representative $T_0T_1T_2T_3$ is supplied to both cross-rectangle positions. Consequently the two occurrences of R are literally identical scissors terms.

8.5.2 Proposition II.7 on the reversed cut

From $A - C - B$ the proof derives $B - C - A$ and applies II.7 with the whole segment read as BA. The result is

$$\mathrm{Sq}(BA) + \mathrm{Sq}(CB) \simeq_{\mathrm{sc}} (R + R) + \mathrm{Sq}(CA).$$

The square on BA is deliberately the same concrete square $BAFG$ already used inside the II.4 call. This avoids a second representative-transport step.

8.5.3 Square transport through the rectangle API

II.4 supplies a square on BD, whereas II.7 uses a square on CB. The proof converts both squares to `IsRectangleContainedBy` data with `square_is_rectangle_contained_by_side`. The congruence $BD \cong BC$ is then used to read the CB square as another representative of the same rectangle contained by BD, BD. Finally `rectangle_contained_by_unique` gives

$$\mathrm{Sq}(BD) \simeq_{\mathrm{sc}} \mathrm{Sq}(CB).$$

This is representative transport in the established geometric rectangle API; it is not an appeal to numerical equality of squared lengths.

8.5.4 Exact scissors addition and regrouping

After the three normal forms are available, the remainder is formal addition, transitivity, symmetry, and associativity/commutativity of the multiset sum. No cancellation, subtraction, positivity theorem, or numerical content map is used.

8.6 Proof Architecture of the Reconstruction

The production proof can be read in seven steps.

8.6.1 Step 1: reverse the internal cut

The source assumption $A - C - B$ is converted to $B - C - A$ by

```
have hBCA : Geo.Between B C A :=
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2
```

This is the orientation required by the II.7 invocation.

8.6.2 Step 2: re-read the repeated rectangle

The concrete rectangle $T_0T_1T_2T_3$ initially satisfies

$$\mathrm{Rect}(AB, BD).$$

The proof reverses AB to BA with `CongruentSwapSecond` and transports BD to BC by congruence transitivity. The same quadrilateral therefore satisfies

$$\mathrm{Rect}(BA, BC).$$

No scissors theorem is needed because the geometric representative itself is unchanged.

8.6.3 Step 3: apply II.4 to $A - B - D$

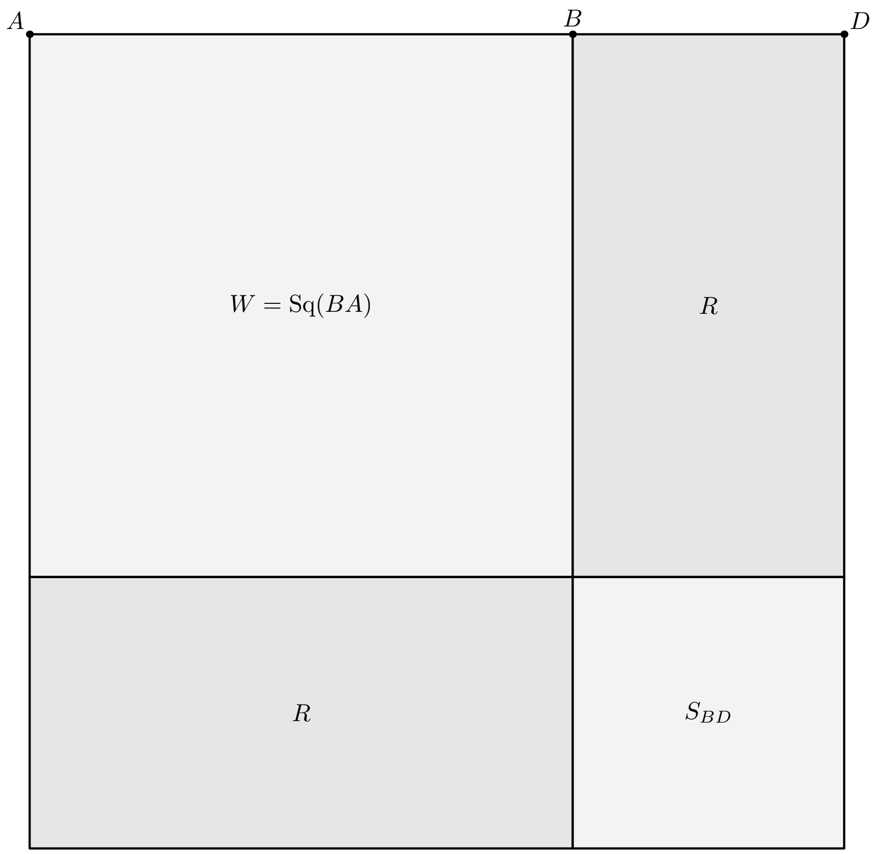
The theorem `euclid_proposition_2_4` produces

$$\mathrm{Sq}(AD) \simeq_{\mathrm{sc}} (R + W) + (R + S_{BD}),$$

where

$$W := \mathrm{Sq}(BA), \qquad S_{BD} := \mathrm{Sq}(BD).$$

II.4 on $A - B - D$



$$S_{AD} \simeq_{\mathrm{sc}} (R + W) + (R + S_{BD})$$

Figure 8.2: The II.4 normal form used at Step 3. The square on AD is cut at B. The large square W is the square on BA, the small square S_{BD} is the square on the produced part BD, and the two off-diagonal pieces are two occurrences of the same rectangle term R. Thus the visible partition realizes $S_{AD} \simeq_{\mathrm{sc}} (R + W) + (R + S_{BD})$ before any use of II.7.

8.6.4 Step 4: apply II.7 to $B - C - A$

The theorem `euclid_proposition_2_7` produces

$$W + S_{CB} \simeq_{\mathrm{sc}} (R + R) + S_{CA},$$

where

$$S_{CB} := \mathrm{Sq}(CB), \qquad S_{CA} := \mathrm{Sq}(CA).$$

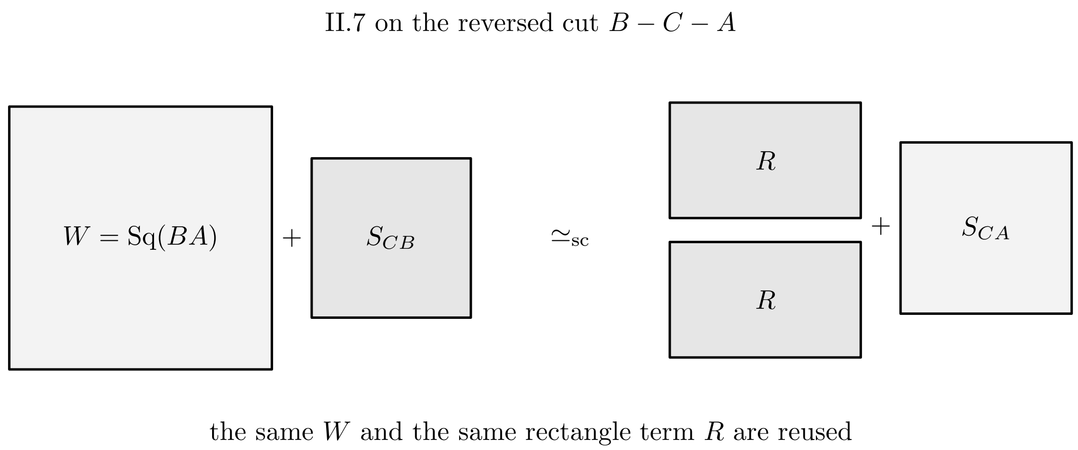


Figure 8.3: The II.7 normal form used at Step 4 after the original order $A - C - B$ has been reversed to $B - C - A$. On the left are the already reused square W on BA and the square S_{CB} on the selected part. II.7 replaces this pair, at the exact-scissors level, by two copies of the same rectangle term R together with the square S_{CA}. This is a schematic display of the proved scissors identity, not a claim that the five boxes form one literal planar dissection.

8.6.5 Step 5: transport the square on the congruent produced segment

Using the square-as-rectangle conversion and `rectangle_contained_by_unique`, the proof obtains

$$S_{BD} \simeq_{\mathrm{sc}} S_{CB}.$$

This is the formal counterpart of Euclid's final use of $BD = BC$.

8.6.6 Step 6: substitute and regroup

Scissors addition first replaces S_{BD} by S_{CB} inside the II.4 normal form. Associativity and commutativity then expose the summand $W + S_{CB}$:

$$(R + W) + (R + S_{CB}) = (R + R) + (W + S_{CB}).$$

The proof uses `ac_rfl`; this is equality of the formal multiset sum, not a new geometric theorem.

8.6.7 Step 7: replace the II.7 block and close

Replacing $W + S_{CB}$ by the II.7 normal form gives

$$\begin{aligned}\mathrm{Sq}(AD) &\simeq_{\mathrm{sc}} (R + R) + ((R + R) + S_{CA}) \\ &= ((R + R) + (R + R)) + S_{CA}.\end{aligned}$$

A final symmetry puts the four rectangles plus the square on the left, exactly in the orientation of the public theorem.

Diagrammatic audit. No further figure is needed after Figure 8.3. Step 5 changes only the representative of a square on a congruent segment; Steps 6–7 then use scissors addition, transitivity, symmetry, and `ac_rfl`. Drawing a fourth "final arrangement" would therefore suggest a new dissection that the Lean proof does not construct.

8.7 Lean Formalization

8.7.1 The public theorem signature

The current compiling source declares:

```
theorem euclid_proposition_2_8
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (A B C D : Geo.Point)

    (hACB : Geo.Between A C B)
    (hABD : Geo.Between A B D)
    (hBD_BC : Geo.Congruent B D B C)

    -- ==============================================================
    -- II.4 on A-B-D.
    -- ==============================================================

    (D40 E40 L40 X40 : Geo.Point)
    (hSquareAD : IsSquare Geo A D E40 D40)

    (hD40L40E40 : Geo.Between D40 L40 E40)
    (hD40X40D : Geo.Between D40 X40 D)
    (hBX40L40 : Geo.Between B X40 L40)

    (hII4LeftPar :
      IsParallelogram Geo L40 D40 A B)

    (hII4RightPar :
      IsParallelogram Geo D E40 L40 B)

    -- Rect(AD,AB), Rect(AD,BD).
    (U40 U41 U42 U43 : Geo.Point)
    (V40 V41 V42 V43 : Geo.Point)

    (hAD_AB :
      IsRectangleContainedBy Geo
        U40 U41 U42 U43 A D A B)

    (hAD_BD :
      IsRectangleContainedBy Geo
        V40 V41 V42 V43 A D B D)

    --------------------------------------------------------------
    -- Left II.3 branch inside II.4.
    --------------------------------------------------------------

    (D41 E41 L41 X41 : Geo.Point)

    (hRect41 :
      IsRectangle Geo D A E41 D41)

    (hAE41_BA :
      Geo.Congruent A E41 B A)
```

```
(hD41L41E41 : Geo.Between D41 L41 E41)
(hD41X41A : Geo.Between D41 X41 A)
(hBX41L41 : Geo.Between B X41 L41)

(hLeft41 :
  IsParallelogram Geo L41 D41 D B)

(hRight41 :
  IsParallelogram Geo A E41 L41 B)

--------------------------------------------------------------
-- Common square on BA: used here and again as the outer square
-- in the II.7 invocation below.
--------------------------------------------------------------

(F G : Geo.Point)
(hSquareBA : IsSquare Geo B A F G)

--------------------------------------------------------------
-- Right II.3 branch inside II.4.
--------------------------------------------------------------

(D42 E42 L42 X42 : Geo.Point)

(hRect42 :
  IsRectangle Geo A D E42 D42)

(hDE42_BD :
  Geo.Congruent D E42 B D)

(hD42L42E42 : Geo.Between D42 L42 E42)
(hD42X42D : Geo.Between D42 X42 D)
(hBX42L42 : Geo.Between B X42 L42)

(hLeft42 :
  IsParallelogram Geo L42 D42 A B)

(hRight42 :
  IsParallelogram Geo D E42 L42 B)

(H K : Geo.Point)
(hSquareBD : IsSquare Geo B D H K)

--------------------------------------------------------------
-- Common repeated rectangle:
--
--   in II.4: Rect(AB,BD),
--   in II.7: Rect(BA,BC).
--------------------------------------------------------------

(T0 T1 T2 T3 : Geo.Point)

(hCross4 :
  IsRectangleContainedBy Geo
    T0 T1 T2 T3 A B B D)

-- ==============================================================
-- II.7 on the reversed division B-C-A.
```

```
--
-- The outer square is exactly the common square B-A-F-G above.
-- ================================================================

(L70 X70 : Geo.Point)

(hGL70F : Geo.Between G L70 F)
(hGX70A : Geo.Between G X70 A)
(hCX70L70 : Geo.Between C X70 L70)

(hII7LeftPar :
  IsParallelogram Geo L70 G B C)

(hII7RightPar :
  IsParallelogram Geo A F L70 C)

-- Second outer representative Rect(BA,CA).
(V70 V71 V72 V73 : Geo.Point)

(hBA_CA :
  IsRectangleContainedBy Geo
    V70 V71 V72 V73 B A C A)

----------------------------------------------------------------
-- Left II.3 branch inside II.7.
----------------------------------------------------------------

(D71 E71 L71 X71 : Geo.Point)

(hRect71 :
  IsRectangle Geo A B E71 D71)

(hBE71_CB :
  Geo.Congruent B E71 C B)

(hD71L71E71 : Geo.Between D71 L71 E71)
(hD71X71B : Geo.Between D71 X71 B)
(hCX71L71 : Geo.Between C X71 L71)

(hLeft71 :
  IsParallelogram Geo L71 D71 A C)

(hRight71 :
  IsParallelogram Geo B E71 L71 C)

(M N : Geo.Point)
(hSquareCB : IsSquare Geo C B M N)

----------------------------------------------------------------
-- Right II.3 branch inside II.7.
----------------------------------------------------------------

(D72 E72 L72 X72 : Geo.Point)

(hRect72 :
  IsRectangle Geo B A E72 D72)

(hAE72_CA :
```

```
      Geo.Congruent A E72 C A)

    (hD72L72E72 : Geo.Between D72 L72 E72)
    (hD72X72A : Geo.Between D72 X72 A)
    (hCX72L72 : Geo.Between C X72 L72)

    (hLeft72 :
      IsParallelogram Geo L72 D72 B C)

    (hRight72 :
      IsParallelogram Geo A E72 L72 C)

    (P Q : Geo.Point)
    (hSquareCA : IsSquare Geo C A P Q)

    -------------------------------------------------------------
    -- Cross rectangle internal to II.7: Rect(BC,CA).
    -------------------------------------------------------------

    (Z0 Z1 Z2 Z3 : Geo.Point)

    (hCross7 :
      IsRectangleContainedBy Geo
        Z0 Z1 Z2 Z3 B C C A) :

    HilbertScissorsEq Geo
      (((hilbertParallelogramTerm Geo T0 T1 T2 T3 +
         hilbertParallelogramTerm Geo T0 T1 T2 T3) +
        (hilbertParallelogramTerm Geo T0 T1 T2 T3 +
         hilbertParallelogramTerm Geo T0 T1 T2 T3)) +
       hilbertParallelogramTerm Geo C A P Q)
      (hilbertParallelogramTerm Geo A D E40 D40)
```

The signature is intentionally large because the theorem is configured around concrete II.4 and II.7 diagrams. The meaningful top-level geometric data are small: $A - C - B$, $A - B - D$, and $BD \cong BC$. The remaining points certify the squares, rectangles, cuts, and parallelograms consumed by the already proved Book II propositions.

8.7.2 The decisive local composition

The two principal theorem calls are

```
have hII4 :=
  euclid_proposition_2_4 Geo A D B ...

have hII7 :=
  euclid_proposition_2_7 Geo B A C ...
```

The ellipses here are documentary abbreviations only; the full public signature above records the actual configured witnesses.

The exact square transport is obtained with

```
have hSquareTransport :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo B D H K)
      (hilbertParallelogramTerm Geo C B M N) :=
  rectangle_contained_by_unique
    Geo
    B D H K
    C B M N
    B D B D
    hSquareBDContained
    hSquareCBAsBD
```

so no new II.8-specific transport theorem is added to the library.

8.8 Relationship with Earlier Book II Propositions

8.8.1 Proposition II.4

II.4 is called directly on the produced whole AD cut at B. It supplies the square decomposition containing two copies of $\mathrm{Rect}(AB, BD)$, the square on BA, and the square on BD. This is the left normal form of the production proof.

8.8.2 Proposition II.7

II.7 is called directly on the same original arbitrary cut but in reversed orientation $B - C - A$. Its selected part is therefore BC, giving precisely

$$\mathrm{Sq}(BA) + \mathrm{Sq}(CB) \simeq_{\mathrm{sc}} 2\ \mathrm{Rect}(BA, BC) + \mathrm{Sq}(CA).$$

This is the block substituted into the II.4 expansion.

8.8.3 Propositions II.1–II.3

II.1–II.3 are not called directly by II.8. They lie below II.4 and II.7 in the established rectangle-calculus DAG. In particular, the direct proof does not need to reopen the binary rectangle split or the II.3 square specialization.

8.8.4 Propositions II.5 and II.6

II.5 and II.6 are not dependencies of II.8. Their midpoint hypotheses belong to a different Book II block. II.8 uses an arbitrary internal cut and a newly produced congruent external segment, not a midpoint.

8.9 Relationship with Book I Infrastructure

Classically, II.8 explicitly depends on I.3, I.31, I.34, I.36, I.43, and I.46: segment layoff, parallel construction, parallelogram side equalities, equal parallelograms on equal bases, equal complements, and square construction [1]. The present production theorem does not call those propositions locally. Their consequences have already been packaged in the square, rectangle, parallelogram, congruence, and exact-scissors infrastructure consumed by II.4 and II.7.

This is a substantial classical-DAG/Lean-DAG difference. The formal proof is not a transcription of the eight-region gnomon argument. It is a composition of previously verified geometric identities.

8.10 Formalization Notes and Hidden Geometric Data

8.10.1 The line order is split across two hypotheses

The public theorem does not take a single four-point order predicate. It takes $A - C - B$ and $A - B - D$. Together these encode the intended baseline $A - C - B - D$. The proof only derives the reversed $B - C - A$ fact it actually needs.

8.10.2 The same rectangle has two contained-by descriptions

The term $T_0T_1T_2T_3$ first represents $\mathrm{Rect}(AB, BD)$ and later represents $\mathrm{Rect}(BA, BC)$. The first change is endpoint orientation; the second is metric transport through $BD \cong BC$. This is representation data, not commutativity of a primitive segment product.

8.10.3 Representative reuse is deliberate

The same square on BA is supplied to both II.4 and II.7, and the same rectangle term R is supplied repeatedly. This design removes unnecessary applications of representative uniqueness from the final scissors bridge. Only the genuinely different squares on BD and CB require transport.

8.10.4 Four times means four summands

The public conclusion contains four literal occurrences of `hilbertParallelogramTerm` for the same rectangle. No natural-number scalar action is introduced. The proof therefore stays in the finite formal sum language already used throughout the scissors layer.

8.10.5 No cancellation or positivity

Although II.8 looks algebraically like an identity that could be proved by expanding and cancelling terms, the Lean proof performs no cancellation. It uses only forward exact-scissors equalities, addition, transitivity, symmetry, and regrouping. No positivity theorem enters.

8.11 Axiomatic and Semantic Status

The configured public theorem proves exact `HilbertScissorsEq`. It introduces no numerical area function and no multiplication of segment lengths. The formal "algebra" is therefore a calculus of geometric representatives and finite scissors sums.

The explicit axiom audit was

```
'Geometry.euclid_proposition_2_8' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

This is the same transitive profile observed for Proposition II.7. In particular, II.8 introduces no new proposition-specific geometric axiom, no new content axiom, and no `sorryAx` dependency.

The semantic layers should therefore be kept distinct:

$$\begin{gathered}\text{incidence/order/congruence and Euclidean rectangle geometry}\\ \Downarrow \\ \text{concrete square and rectangle representatives}\\ \Downarrow \\ \text{exact finite-scissors relation}\\ \not\Downarrow \\ \text{numerical area or segment multiplication in this proof.}\end{gathered}$$

8.12 Dependency Summary

The actual production chain is

$$\begin{gathered}\text{Hilbert incidence, order, congruence, Euclidean parallel layer}\\ \Downarrow \\ \text{Book I parallelogram/square/scissors infrastructure}\\ \Downarrow \\ \text{HilbertRectangle representative API}\\ \Downarrow \\ \text{II.1–II.3 rectangle calculus}\\ \Downarrow \\ \text{II.4 and II.7}\\ \Downarrow \\ \text{reorientation } \operatorname{Rect}(AB, BD) \to \operatorname{Rect}(BA, BC)\\ \Downarrow \\ \text{square transport } \operatorname{Sq}(BD) \simeq_{\mathrm{sc}} \operatorname{Sq}(CB)\\ \Downarrow \\ \text{exact scissors addition and regrouping}\\ \Downarrow \\ \texttt{euclid\_proposition\_2\_8}.\end{gathered}$$

The implementation checkpoint is:

- new geometric axiom: no;
- new content axiom: no;
- new proposition-local theorem declaration: no;
- new reusable helper theorem: no;

- new `HilbertRectangle` theorem: no;
- new Interface / Book Zero / Scissors theorem: no;
- uses numerical area: no;
- uses segment multiplication: no;
- public theorem compiles: yes;
- `lake build CGJteamLab.Proposition2_8`: clean;
- full `lake build`: clean;
- `sorry`/`admit`: no.

8.13 Conclusion

Proposition II.8 is a useful demonstration of the point at which the established Book II API begins to replace long classical gnomon bookkeeping. Euclid's proof constructs an eight-region configuration and proves equalities among its pieces through I.34, I.36, and I.43. The Lean proof instead takes two already verified normal forms: II.4 on the produced segment AD and II.7 on the reversed original cut $B - C - A$.

The only genuinely new local bridge is representational. The segment equality $BD \cong BC$ lets the repeated rectangle and the square on the produced segment be read in the forms required by II.7. Once those transports are explicit, the fourfold rectangle identity follows from exact scissors addition and regrouping.

Thus the algebraic-looking statement

$$4(a + b)b + a^2 = (a + 2b)^2$$

is not the proof object. It is a modern mnemonic for a theorem whose formal content remains entirely synthetic: betweenness, congruence, squares, rectangles contained by segments, and exact finite scissors equality.

Chapter 9

Proposition II.9

9.1 Introduction

Proposition II.9 is a genuine architectural change in the current reconstruction of Book II. Propositions II.1–II.8 were organized around the reusable rectangle calculus and, in their configured public forms, concluded with exact `HilbertScissorsEq`. Proposition II.9 does not call any earlier Book II proposition. Instead, the current source deliberately returns to Euclid's Book I construction and angle argument and then passes through four applications of Proposition I.47 to the scissors/content layer.

Let AB be bisected at C, and let the unequal cut point D lie in the oriented branch

$$A - C - D - B.$$

The proposition asserts that the two squares on the unequal segments AD and DB have the same content as two copies of the square on the half AC plus two copies of the square on the segment CD between the two points of section. In modern notation the familiar shadow is

$$AD^2 + DB^2 = 2(AC^2 + CD^2).$$

This formula is only a mnemonic. The public Lean conclusion neither defines a numerical area nor multiplies segment lengths. It constructs four actual square representatives and relates their formal scissors sums by `HilbertScissorsEquicomplementable`.

The production source begins with the imports

```
import CGJteamLab.Proposition2_9Construction
import CGJteamLab.HilbertAngleDecomposition
import CGJteamLab.HilbertSquareTransport
import CGJteamLab.Proposition06
import CGJteamLab.Proposition32
import CGJteamLab.Proposition34
import CGJteamLab.Proposition46
import CGJteamLab.Proposition47
```

The absence of a Book II import is intentional. The source comment describes the proof as the source-faithful synthetic proof of the oriented branch and records the classical route through I.5, I.29, I.32, I.34, I.6 and four uses of I.47, with explicit scissors bookkeeping.

9.2 Euclid's Statement and Classical Configuration

9.2.1 The statement

Euclid's Proposition II.9 says that if a straight line is cut into equal and unequal segments, then the sum of the squares on the two unequal segments of the whole is double the sum of the square on the half and the square on the straight line between the two points of section [1].

With the current lettering, C is the midpoint of AB and D is the second point of section. The formal theorem fixes the orientation $C - D - B$, so the unequal segments of the whole are AD and DB, the half is AC, and the segment between the points of section is CD. Thus the geometric statement is

$$\mathrm{Sq}(AD) + \mathrm{Sq}(DB) = 2\,\mathrm{Sq}(AC) + 2\,\mathrm{Sq}(CD),$$

where the equality is an equality of geometric content, not an equality of real-valued areas.

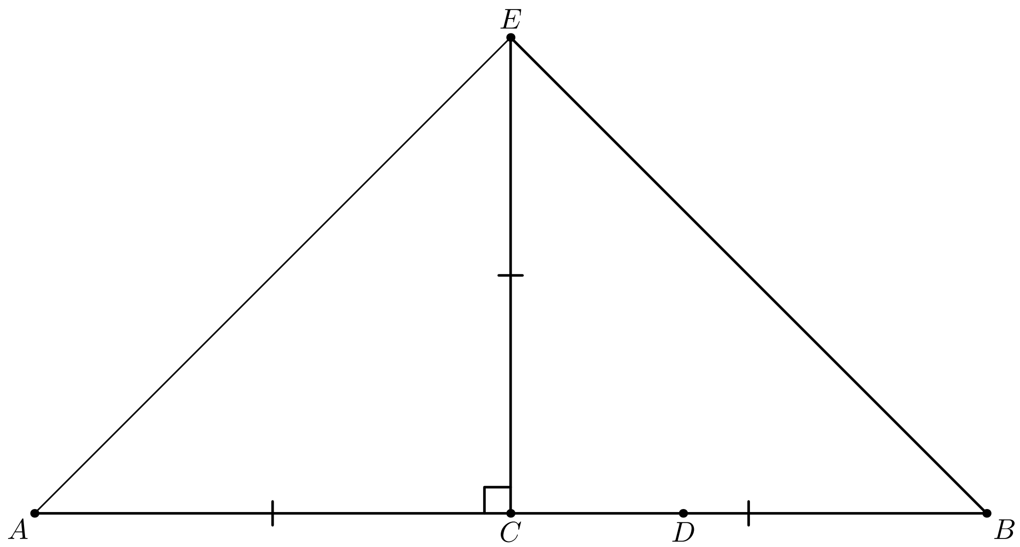
9.2.2 The classical construction

Euclid first erects CE perpendicular to AB at C and lays off

$$CE = AC = CB.$$

He then joins EA and EB [1]. This is the first complete auxiliary state: the midpoint and unequal cut are already fixed on the base, but no parallel through D has yet been introduced.

$$AC \cong CB \cong CE$$



$$A - C - D - B$$

Figure 9.1: The first geometric state of II.9. The oriented base has order $A - C - D - B$, while CE is perpendicular to the base and satisfies $CE \cong AC \cong CB$. The joins EA and EB expose the two isosceles triangles ACE and CEB used in the first angle block. The points F and G and all later metric conclusions are deliberately absent.

Through D Euclid draws the line parallel to CE until it meets EB at F. The formal construction must prove that the intersection is on the correct open segment of the carrier, namely

$$E - F - B.$$

This directional fact is later used to transport the right angle AEB to the same ray EF.

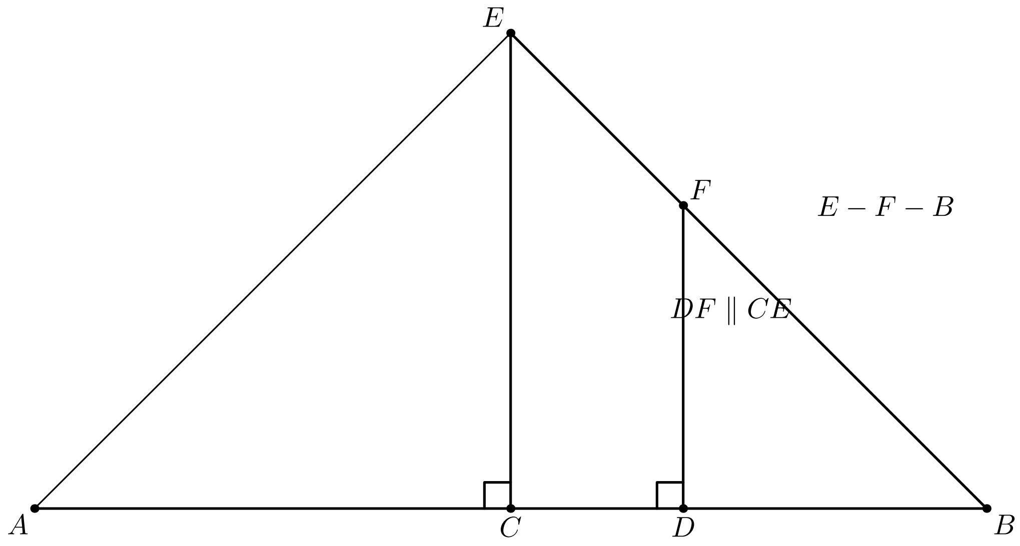


Figure 9.2: The second state of the II.9 construction. The line through D parallel to CE meets EB at F, and the strict order is $E - F - B$. The right angle at D is already forced by the parallel/perpendicular configuration. The point G has not yet been constructed, so no parallelogram or isosceles conclusion involving G is shown.

Through F draw FG parallel to the base. Its intersection with the carrier CE is the point G, and $DFGC$ is the auxiliary parallelogram. The current construction module does more than assert the intersection: a forced-Pasch argument proves

$$C - G - E.$$

Together with $E - F - B$, this is the complete directed source configuration on which the later angle transports depend.

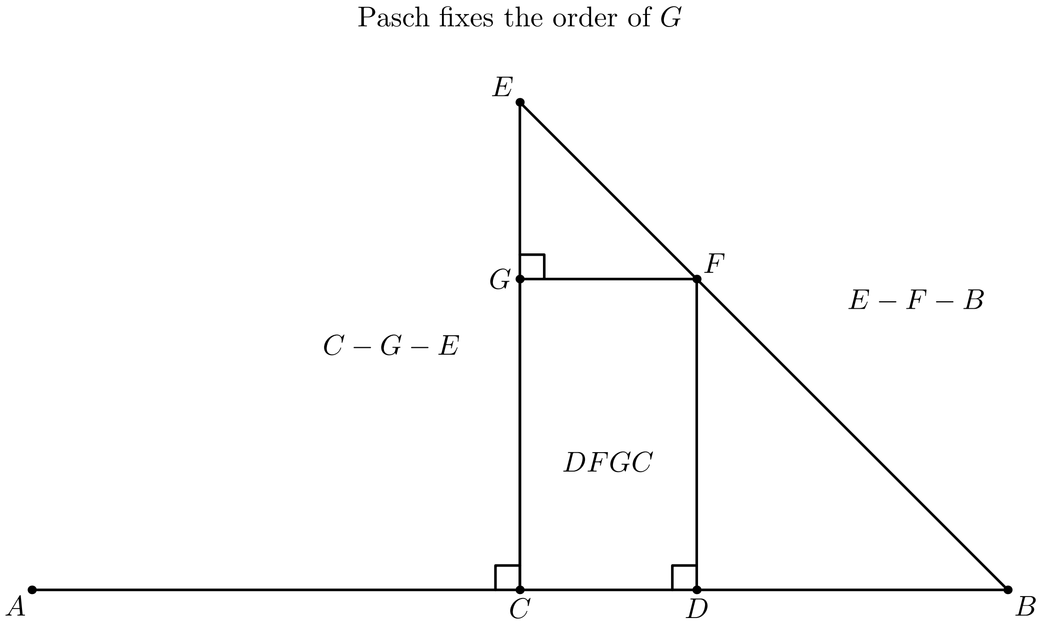


Figure 9.3: The third construction state. The horizontal through F closes the parallelogram $DFGC$ and meets CE at G. The figure records both strict orders $E - F - B$ and $C - G - E$. In the formal construction the second order is recovered by forced Pasch in the triangle EBC and line uniqueness. The later equalities $GE \cong GF$ and $DF \cong DB$ are not yet marked because they require separate angle arguments.

Finally Euclid joins AF [1].

Diagrammatic audit. The original single diagram displayed the finished construction together with metric conclusions that are proved only later. The audited sequence separates three genuine construction states: perpendicular-and-copy, the directed intersection $E - F - B$, and the Pasch-controlled completion $C - G - E$. The next figures are reserved for the angle consequences and the four Pythagorean blocks; representative transport in the scissors layer receives no geometric picture.

9.3 Classical Proof

The proof has two parts: first Euclid extracts the metric equalities required by the construction; then four Pythagorean applications identify one and the same square on AF in two ways [1].

Because $AC = CE$, Proposition I.5 gives equal base angles in triangle ACE. Since the angle at C is right, Proposition I.32 shows that each of those base angles is half a right angle. The same argument in triangle CEB shows that its two base angles are also half right angles. Hence the whole angle AEB is right, and because $E - F - B$, the angle AEF is right as well.

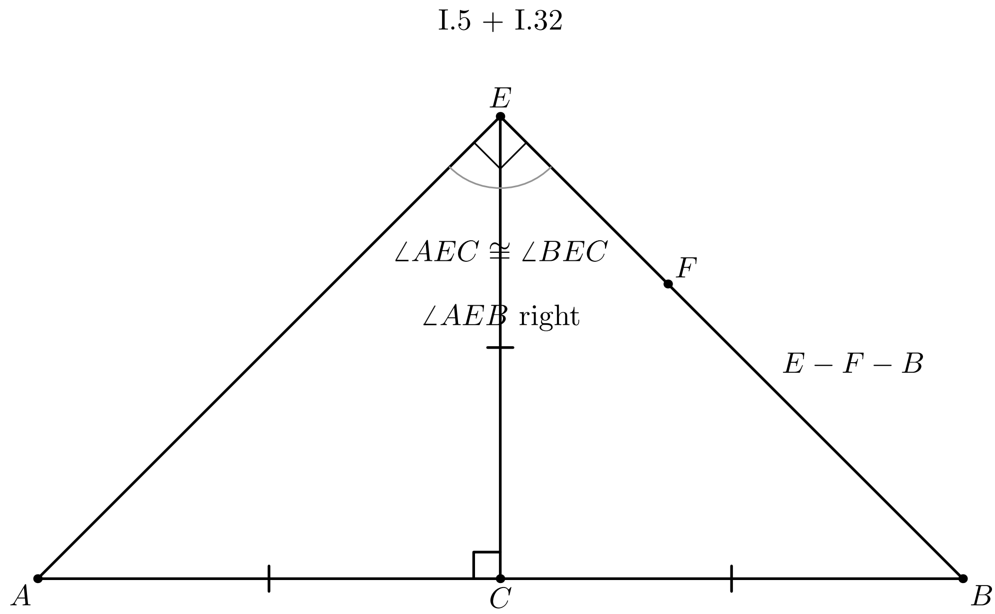


Figure 9.4: The first angle state of the proof. The isosceles triangles ACE and CEB give congruent angle components at E by I.5 and I.32. Their union is the right angle AEB, and the proven order $E - F - B$ transports that right angle to AEF. The formal witnesses R and S certify the angle decompositions but are omitted because they do not create new classical vertices.

The parallel construction gives the corresponding right-angle configurations at G and D. Using the parallel-angle theorem I.29, the triangle angle sum I.32, and the converse isosceles theorem I.6, Euclid obtains

$$GE = GF, \qquad DF = DB.$$

From the auxiliary parallelogram $DFGC$, I.34 gives

$$GF = CD.$$

Thus GE, GF, and CD represent the same segment magnitude, while DF can be replaced by DB.

I.32 + angle subtraction + I.6

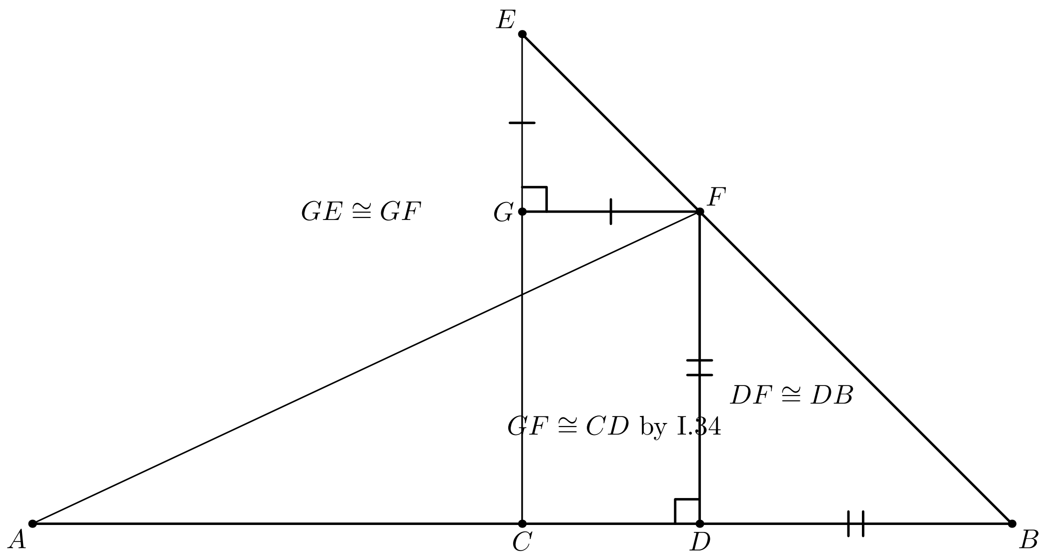


Figure 9.5: The completed geometric configuration after the two synthetic angle-subtraction arguments. I.32 and I.6 give $GE \cong GF$ and $DF \cong DB$; I.34 supplies $GF \cong CD$ from the parallelogram. The local interior-ray witnesses T and U are omitted: their role is to justify Euclid's angle subtraction, not to alter the visible configuration.

Now apply I.47 four times. In triangle ACE, right at C,

$$\mathrm{Sq}(AE) = \mathrm{Sq}(AC) + \mathrm{Sq}(CE) = 2\,\mathrm{Sq}(AC).$$

In triangle EGF, right at G,

$$\mathrm{Sq}(EF) = \mathrm{Sq}(EG) + \mathrm{Sq}(GF) = 2\,\mathrm{Sq}(CD).$$

In triangle AEF, right at E,

$$\mathrm{Sq}(AF) = \mathrm{Sq}(AE) + \mathrm{Sq}(EF) = 2\,\mathrm{Sq}(AC) + 2\,\mathrm{Sq}(CD).$$

Finally, in triangle ADF, right at D,

$$\mathrm{Sq}(AF) = \mathrm{Sq}(AD) + \mathrm{Sq}(DF) = \mathrm{Sq}(AD) + \mathrm{Sq}(DB).$$

Comparing the two descriptions of the square on AF proves II.9.

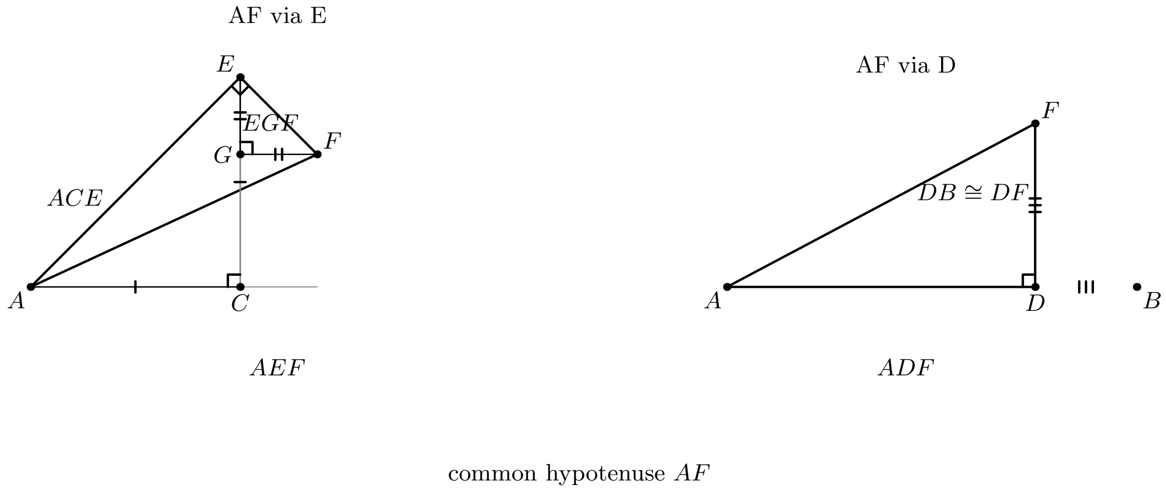


Figure 9.6: The four right triangles used by I.47, organized by the common square on AF. The left panel follows AF through the right triangle AEF, then expands its two leg squares with the right triangles ACE and EGF. The right panel gives the independent decomposition of the same hypotenuse through ADF. Segment congruences later transport DF to DB, the two $A/C/E$ legs to AC, and GE, GF to CD. No literal square representatives are drawn because the formal proof may construct them independently and then transport their content.

Diagrammatic audit. After Figure 9.6 there is no further geometric state to draw. Steps 5–7 of the formal architecture choose target square representatives, identify independent representatives of the square on AF, and compose equicomplementability relations. Those operations change the content bookkeeping, not the Euclidean configuration.

The classical local dependency picture is therefore

$$\begin{array}{c} \text{I.11} + \text{I.3: perpendicular and copied half} \\ \Downarrow \\ \text{I.31: parallel through } D \quad + \quad \text{join } EA, EB, AF \\ \Downarrow \\ \text{I.5} + \text{I.29} + \text{I.32} + \text{I.6: } GE = GF,\ DF = DB \\ \Downarrow \\ \text{I.34: } GF = CD \\ \Downarrow \\ \text{four applications of I.47} \\ \Downarrow \\ \text{II.9.} \end{array}$$

9.4 The Geometric Identity in the Current Formalization

The public theorem assumes

```
(A B C D : Geo.Point)
(hMidC : HilbertIsMidpoint Geo C A B)
(hCDB : Geo.Between C D B)
```

so the midpoint package supplies $A - C - B$ and $AC \cong CB$, while the second hypothesis selects the oriented branch $C - D - B$. The proof derives the required four-point order $A - C - D - B$ rather than taking it as a primitive datum.

The theorem then constructs concrete squares on AD, DB, AC, and CD. If their formal terms are denoted by

$$S_{AD}, \quad S_{DB}, \quad S_{AC}, \quad S_{CD},$$

the conclusion has the shape

$$S_{AD} + S_{DB} \simeq_{\text{ec}} (S_{AC} + S_{AC}) + (S_{CD} + S_{CD}).$$

Here $\simeq_{\text{ec}}$ denotes the relation formalized by `HilbertScissorsEquicomplementable`.

This distinction is essential. Unlike the configured theorems II.1–II.8, II.9 does *not* conclude `HilbertScissorsEq`. Nor does it conclude an equality of numerical areas. Its four Pythagorean blocks inherit the content strength of the current formalization of I.47 and are composed by symmetry, transitivity, additive compatibility, and transport between square representatives.

9.5 Reusable Infrastructure Used

9.5.1 The permanent II.9 construction layer

The file `Proposition2_9Construction.lean` is a separate geometric layer. Its public helpers progressively establish the source diagram rather than mixing construction with the final content proof. In particular, `proposition2_9_G_between_CE` returns points E, F, G with

$$E - F - B, \qquad C - G - E, \qquad A - C - D,$$

together with the right angle at C, $CE \cong CB$, the parallelogram $DFGC$, and its opposite-side congruences.

The construction module itself imports I.31, I.34, and I.47 infrastructure. Its initial perpendicular/copy helper packages the I.11 + I.3 construction, while its later order proofs use the established Hilbert incidence/order and Pasch machinery to locate F and G on the correct open segments.

9.5.2 Angle decomposition rather than numerical angle measure

The source proof repeatedly says that an angle is half a right angle and later subtracts equal parts from equal right angles. The formal proof does not introduce degrees or real-valued angle measures. It uses `HilbertRayMeetsSegment` as the interior-ray witness and the reusable lemmas

```
hilbert_angleDecomposition_halves_congruent_of_whole_congruent
hilbert_angleDecomposition_angle_addition_interior
hilbert_angleDecomposition_angle_subtraction
hilbert_angleDecomposition_angle_subtraction_right
```

from `HilbertAngleDecomposition.lean`.

9.5.3 Square transport

The four I.47 calls do not automatically choose the same square representative on a common side. The proof therefore uses `hilbert_square_transport` to move between squares erected on congruent segments and between different representatives of the same segment. This is most visible for the two occurrences of the square on AF and the two occurrences of the square on EF.

This is representation transport, not a hidden equation $x^2 = y^2$ in a numerical ring. The transported objects remain concrete `hilbertParallelogramTerm` values in the scissors calculus.

9.5.4 Book I square construction and Pythagoras

The public theorem constructs fresh target squares on AD, DB, AC, and CD with `euclid_proposition_46`. It then imports four already proved I.47 decompositions and transports their square terms into those target representatives. Thus I.46 controls representation while I.47 controls the content identities.

9.6 Proof Architecture of the Reconstruction

9.6.1 Step 1: recover the right-angle configuration

`proposition2_9_right_angle_configuration` starts from the permanent construction and proves the right angles needed later:

$$\angle ADF, \quad \angle EGF, \quad \angle ACE, \quad \angle BCE.$$

The proof propagates right angles through same-ray transport, opposite extensions, and adjacent angles of the parallelogram $DFGC$.

9.6.2 Step 2: reconstruct Euclid's half-right-angle block

`proposition2_9_classical_angle_block` applies I.5 in the two isosceles triangles ACE and CEB, then applies I.32 in oriented exterior-angle form. The resulting points R and S lie on EB and EA and act as explicit interior-ray witnesses for the two angle bisections at C.

Using equality of right angles and the angle-decomposition API, the proof shows that the two components AEC and BEC are congruent. Their sum is therefore a right angle, giving AEB right; the same-ray fact $E - F - B$ then gives AEF right.

9.6.3 Step 3: recover the two isosceles triangles

`proposition2_9_classical_isosceles_blocks` proves

$$GE \cong GF, \qquad DF \cong DB.$$

This is deliberately not done by segment subtraction. For each equality the proof follows the Euclidean angle route: create an I.32 decomposition witness, compare two right angles, subtract congruent components synthetically, obtain congruent base angles, and finish with `euclid_proposition_6`.

The first block introduces a witness T inside the right angle FGC; the second introduces a witness U inside the right angle CDF. These points are formal witnesses for Euclid's angle arithmetic and are not needed in the final statement.

9.6.4 Step 4: build the four Pythagorean blocks

`proposition2_9_four_pythagoras_blocks` adds the I.34 equality $GF \cong CD$ and invokes I.47 on exactly the four right triangles used in the classical proof:

$$ACE, \qquad EGF, \qquad AEF, \qquad ADF.$$

The helper returns all four existential square packages together with the segment congruences needed to identify their legs with the target sides.

9.6.5 Step 5: construct target squares

The public theorem uses I.46 four times to choose concrete squares $S_{AD}, S_{DB}, S_{AC}, S_{CD}$. This separates the statement-level representatives from whatever representatives happened to be produced inside the four I.47 applications.

9.6.6 Step 6: follow the common square on AF

The first content path begins with I.47 on ADF:

$$S_{AF} \simeq_{\text{ec}} S_{DA} + S_{DF}.$$

Square transport replaces DA by AD and DF by DB, yielding

$$S_{AF} \simeq_{\text{ec}} S_{AD} + S_{DB}.$$

The proof reverses this relation, moves to the AF representative used by the AEF block, and applies I.47 again to obtain

$$S_{AD} + S_{DB} \simeq_{\text{ec}} S_{EA} + S_{EF}.$$

9.6.7 Step 7: replace AE and EF by four leg squares

Representative transport first identifies the EA square with the AE square used by triangle ACE and the EF square with the EF square used by triangle EGF. The two remaining I.47 blocks then give

$$S_{EA} + S_{EF} \simeq_{\text{ec}} (S_{CA} + S_{CE}) + (S_{GE} + S_{GF}).$$

Finally the segment congruences

$$CA \cong AC, \quad CE \cong AC, \quad GE \cong CD, \quad GF \cong CD$$

transport these four squares to two copies of S_{AC} and two copies of S_{CD}. Additivity and transitivity close the public theorem.

9.7 Lean Formalization

9.7.1 The public theorem signature

The following is an ASCII transcription of the current public signature:

```
theorem euclid_proposition_2_9_oriented
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D : Geo.Point)
    (hMidC : HilbertIsMidpoint Geo C A B)
    (hCDB : Geo.Between C D B) :
    exists SAD0 SAD1 SDB0 SDB1 SAC0 SAC1 SCD0 SCD1 : Geo.Point,
      IsSquare Geo A D SAD0 SAD1 /\
      IsSquare Geo D B SDB0 SDB1 /\
      IsSquare Geo A C SAC0 SAC1 /\
      IsSquare Geo C D SCD0 SCD1 /\
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo A D SAD0 SAD1 +
         hilbertParallelogramTerm Geo D B SDB0 SDB1)
        ((hilbertParallelogramTerm Geo A C SAC0 SAC1 +
          hilbertParallelogramTerm Geo A C SAC0 SAC1) +
         (hilbertParallelogramTerm Geo C D SCD0 SCD1 +
          hilbertParallelogramTerm Geo C D SCD0 SCD1)) := by
```

The suffix `_oriented` is substantive documentation. The public theorem proved in this file is the branch in which the unequal point D lies strictly between the midpoint C and endpoint B. The document must therefore not silently replace the formal statement by an un-oriented theorem over all possible positions of D.

9.7.2 The proposition-local theorem chain

The main source file is organized around four named proposition-local helpers:

```
proposition2_9_right_angle_configuration
proposition2_9_classical_angle_block
proposition2_9_classical_isosceles_blocks
proposition2_9_four_pythagoras_blocks
```

They form a genuine mathematical factorization of the proof. The public statement does not expose the auxiliary points E, F, G, R, S, T, U because they are construction and angle-decomposition witnesses rather than part of the final content identity.

9.7.3 The final content bridge

The last stage contains five conceptually different transports:

(a) I.47 on ADF converts the left-hand target sum into a square on AF;

(b) two AF representatives are identified;

(c) I.47 on AEF converts that square into squares on EA and EF;

(d) I.47 on ACE and EGF expands those two squares into four leg squares;

(e) square transport identifies the four legs with two copies of AC and two copies of CD.

No cancellation of content terms occurs. The proof is a forward chain of synthetic equicomplementability relations.

9.8 Relationship with Earlier Book II Propositions

There is no earlier Book II theorem in the production dependency chain for II.9. In particular, the theorem does not call II.1, II.4, II.7, II.8, or the `HilbertRectangle` rectangle-contained-by calculus that dominated the previous block.

This is not merely an omission at theorem-call level. The production file imports `Proposition2_9Construction`, the angle-decomposition and square transport layers, and Book I propositions I.6, I.32, I.34, I.46, and I.47. The construction file itself imports Book I I.31, I.34, and I.47 infrastructure. No `Proposition2_N` module with $N < 9$ appears in these imports.

Mathematically, II.9 therefore begins a new Book II proof mode in this project: its algebraic-looking square identity is reconstructed from the classical Euclidean configuration instead of being normalized through the previously built rectangle identities.

9.9 Relationship with Book I Infrastructure

The classical and formal DAGs are close in mathematical shape but not identical line-by-line.

Classically, the construction uses I.11 and I.3 for the perpendicular and copied half, I.31 for the parallel, I.5 for isosceles base angles, I.29 for parallel angle transport, I.32 for the triangle angle-sum/exterior-angle mechanism, I.6 for the converse isosceles conclusions, I.34 for opposite sides of the parallelogram, and I.47 four times [1].

The Lean development packages some of these into reusable lower-level infrastructure. For example, "half a right angle" becomes an explicit interior-ray decomposition, and Euclid's subtraction of equal angle parts is realized by angle-decomposition lemmas. Conversely, Lean adds an explicit use of I.46 to construct the four square representatives that occur in the public statement.

Thus the source-faithful mathematical DAG is retained, but the proof is not a literal transcription of Euclid's prose.

9.10 Formalization Notes and Hidden Geometric Data

9.10.1 The oriented branch is part of the theorem

The hypothesis `hCDB : Geo.Between C D B` records the choice of the right-hand unequal cut. Euclid's printed statement is symmetric in the two possible unequal cuts, but the current public theorem proves this one branch. Any later symmetric wrapper should be documented as an additional theorem, not read into the present signature.

9.10.2 Betweenness controls same rays

The order statements $E - F - B$, $C - G - E$, $A - C - D$, and their reversed forms are used to build `HilbertSameRay` witnesses. Those witnesses justify the replacement of an angle arm by a collinear point on the same ray. Lean cannot infer these transports from the appearance of the diagram.

9.10.3 The points R, S, T, U encode angle arithmetic

The exterior-angle theorem I.32 produces explicit points used as interior rays. The first pair R, S produces the two half-right-angle decompositions needed to prove AEB right. Later T and U support the two synthetic angle-subtraction arguments that end in $GE = GF$ and $DF = DB$.

This is the main hidden geometric data of II.9. It replaces numerical statements such as "$45° + 45° = 90°$" and "subtract equal angles from equal right angles" by incidence, betweenness, ray, and angle-congruence witnesses.

9.10.4 Noncollinearity is pervasive

Every right-angle transport, angle congruence, and application of I.6 or I.47 requires nondegeneracy. The proof repeatedly derives noncollinearity from the parallelogram, the perpendicular construction, strict betweenness, and collinearity transitivity. These obligations are mathematically meaningful but are largely suppressed by the classical diagram.

9.10.5 Square identity is weaker than exact scissors equality

The final relation is equicomplementability. It is therefore incorrect to describe II.9 as an exact cut of the left pair of target squares into the four right-hand target squares. The proof composes the equicomplementability data returned by I.47 with square-representative transport. The appropriate prose is "equal content/equicomplementable", not "the same dissection", unless a later theorem strengthens the result.

9.11 Axiomatic and Semantic Status

The current source introduces no numerical area function and no multiplication of segment lengths. Its semantic layers are

$$\begin{gathered}\text{incidence/order/congruence and Euclidean parallel geometry}\\ \Downarrow\\ \text{right angles, angle decomposition, isosceles and parallelogram facts}\\ \Downarrow\\ \text{concrete square representatives}\\ \Downarrow\\ \text{I.47 equicomplementability and square transport}\\ \Downarrow\\ \text{II.9 equicomplementability conclusion.}\end{gathered}$$

A source inspection of `Proposition2_9.lean` shows no local `axiom`, `sorry`, or `admit` declaration. The present documentation pass does not contain a recorded output of

```
#print axioms Geometry.euclid_proposition_2_9_oriented
```

so no transitive axiom list is asserted here. This distinction is deliberate: "no new local axiom" is weaker than "axiom-free", and a final audit should record the actual `#print axioms` output when it is run in the project environment.

9.12 Dependency Summary

The production chain is

$$
\begin{gathered}
\text{Hilbert incidence, order, congruence, Euclidean parallel layer} \\
\Downarrow \\
\text{Book I construction/angle infrastructure: I.3, I.5, I.6, I.11, I.29, I.31, I.32, I.34} \\
\Downarrow \\
\text{Proposition2\_9Construction + HilbertAngleDecomposition} \\
\Downarrow \\
\text{right-angle configuration} \\
\Downarrow \\
\text{classical angle block} \\
\Downarrow \\
\text{classical isosceles blocks } GE = GF,\ DF = DB \\
\Downarrow \\
\text{I.34 + four I.47 blocks} \\
\Downarrow \\
\text{I.46 target squares + HilbertSquareTransport} \\
\Downarrow \\
\text{equicomplementability addition/transitivity} \\
\Downarrow \\
\texttt{euclid\_proposition\_2\_9\_oriented}.
\end{gathered}
$$

The implementation checkpoint from the inspected source is:

- new proposition-specific geometric axiom declaration: no;
- new proposition-specific content axiom declaration: no;
- proposition-local helper theorem declarations: yes;
- separate reusable infrastructure used: `HilbertAngleDecomposition` and `HilbertSquareTransport`;
- new `HilbertRectangle` theorem used or required: no;
- earlier Book II proposition used: no;
- conclusion is exact `HilbertScissorsEq`: no;
- conclusion is `HilbertScissorsEquicomplementable`: yes;
- uses numerical area: no;
- uses segment multiplication: no;
- local `sorry`/`admit` in `Proposition2_9.lean`: no;
- transitive `#print axioms` audit: not recorded in the materials available to this documentation pass.

9.13 Conclusion

Proposition II.9 is the first clear break in the current Book II documentation from the rectangle-normal-form strategy of II.1–II.8. Its formal proof is built around Euclid's own auxiliary perpendicular, parallel, and right-triangle configuration. The angle argument is reconstructed synthetically, with interior-ray witnesses replacing numerical angle arithmetic, and the two key isosceles equalities are recovered through I.32, angle subtraction, and I.6.

The four applications of I.47 then expose the real content architecture. One square on AF is simultaneously connected with the two target squares on AD, DB and with the four leg squares coming from ACE and EGF. Congruence and square transport identify those legs with two copies of AC and two copies of CD.

Accordingly, the modern identity

$$AD^2 + DB^2 = 2(AC^2 + CD^2)$$

should remain exactly what it is: a useful algebraic shadow. The theorem proved by Lean is a synthetic statement about order, perpendiculars, parallels, congruent segments, square representatives, and `HilbertScissorsEquicomplementable`; it neither invokes the earlier Book II rectangle calculus nor crosses into numerical area or segment multiplication.

Chapter 10

Proposition II.10

10.1 Introduction

Proposition II.10 is the external-section companion to Proposition II.9, but the current Lean development does not prove it by invoking II.9. Instead it reconstructs Euclid's second source configuration directly. The midpoint C lies between A and B, while the new point D lies beyond B:

$$A - C - B - D.$$

Euclid then erects the same perpendicular at C, constructs a parallel parallelogram, produces two lines to a new intersection G, proves two isosceles equalities, and uses Proposition I.47 four times.

The public theorem states that the two squares on AD and DB have the same content as two copies of the square on AC together with two copies of the square on CD. A useful modern mnemonic is

$$AD^2 + DB^2 = 2(AC^2 + CD^2).$$

If $AC = CB = a$ and $BD = x$, then $AD = 2a + x$ and $CD = a + x$, so the familiar algebraic shadow is

$$(2a + x)^2 + x^2 = 2a^2 + 2(a + x)^2.$$

Neither formula is the formal object proved by Lean. The theorem constructs four actual square representatives and relates their formal sums by `HilbertScissorsEquicomplementable`. It introduces neither numerical area nor multiplication of segment lengths.

The production source begins with

```
import CGJteamLab.Proposition2_9Construction
import CGJteamLab.Proposition34
import CGJteamLab.HilbertAngleDecomposition
import CGJteamLab.Proposition06
import CGJteamLab.Proposition32
import CGJteamLab.Proposition46
import CGJteamLab.Proposition47
import CGJteamLab.HilbertSquareTransport
```

The import of `Proposition2_9Construction` is an infrastructure reuse, not a proof by Proposition II.9. II.10 does not import `Proposition2_9.lean` and does not call `euclid_proposition`

`_2_9_oriented`. It reuses the genuinely generic perpendicular-and-copy helper extracted while II.9 was being developed.

10.2 Euclid's Statement and Classical Configuration

10.2.1 The statement

Euclid's Proposition II.10 begins with a line AB bisected at C and a segment BD added in the same straight line beyond B. In the lettering used here the order is therefore

$$A - C - B - D.$$

Euclid's claim is

$$\mathrm{Sq}(AD) + \mathrm{Sq}(DB) = 2\,\mathrm{Sq}(AC) + 2\,\mathrm{Sq}(CD),$$

where the equality is an equality of geometric content between plane figures [1].

10.2.2 The classical construction

At C erect CE perpendicular to AB and choose it so that

$$CE = AC = CB.$$

Join EA and EB. Through E draw EF parallel to AD, and through D draw DF parallel to CE. The quadrilateral $CDFE$ is therefore a parallelogram [1]. This is the first complete auxiliary state of the construction.

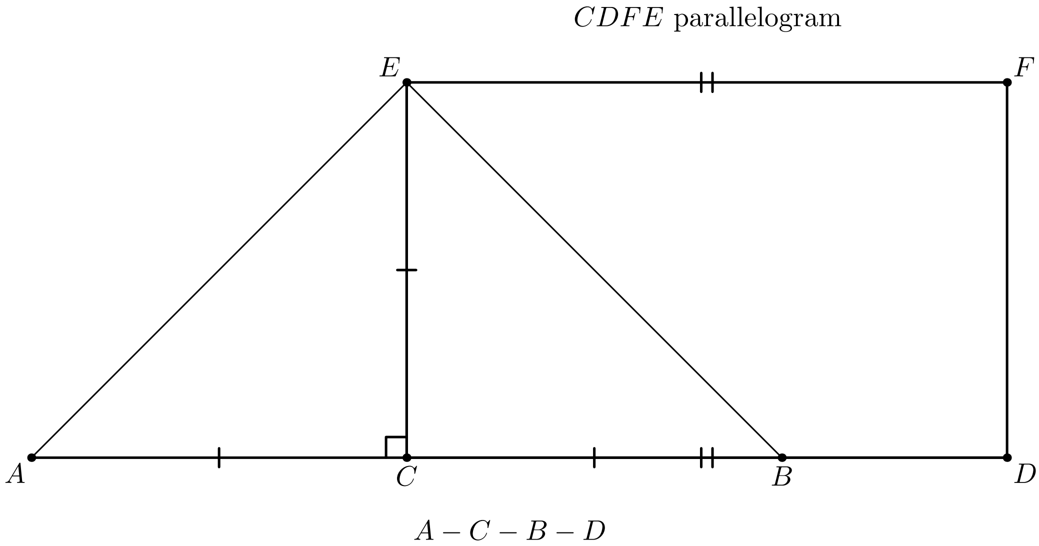


Figure 10.1: The first geometric state of II.10. The baseline has order $A - C - B - D$; the copied perpendicular satisfies $CE \cong AC \cong CB$; and F closes the auxiliary parallelogram $CDFE$. The joins EA and EB are already present. The point G and all metric conclusions depending on it are deliberately absent because they have not yet been constructed.

The next state appears when EB is produced beyond B and FD is produced beyond D until the two carriers meet at G; then AG is joined [1]. The strict positions are

$$E - B - G, \qquad F - D - G.$$

They matter formally. The first controls the extension of the right angle AEB to AEG and the vertical-angle argument at B; the second controls the right-angle and same-ray transports at D and F.

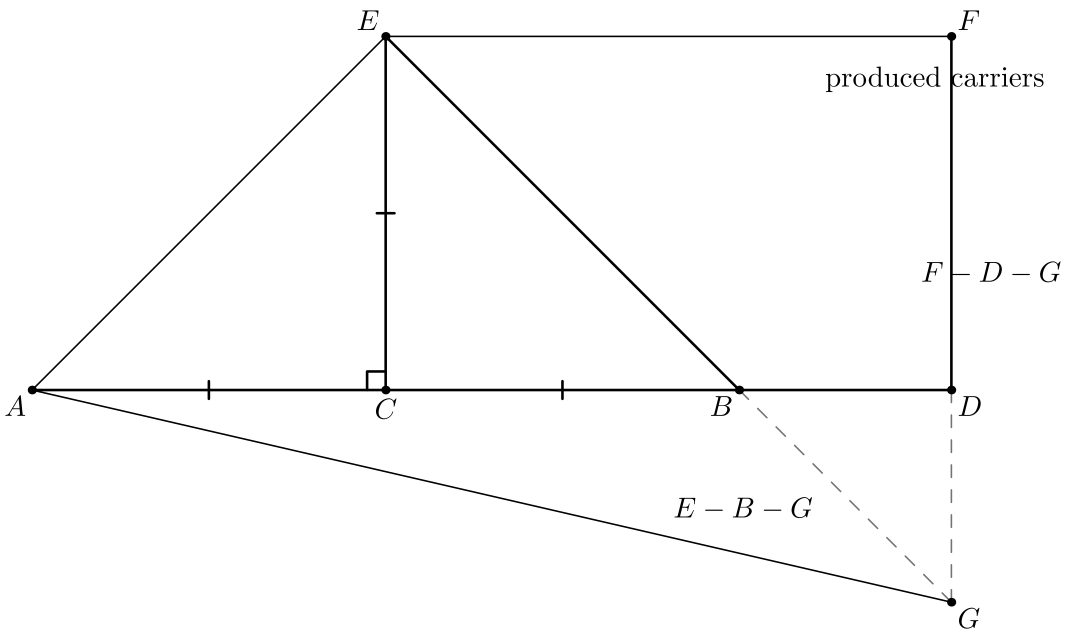


Figure 10.2: The second geometric state adds the intersection G. The dashed terminal portions of the two produced carriers emphasize that G lies beyond B on the carrier EB and beyond D on the carrier FD, giving exactly $E - B - G$ and $F - D - G$. The later conclusions $DB \cong DG$ and $GF \cong EF$ are not yet marked: their proof requires the subsequent angle blocks.

Diagrammatic audit. The original single diagram displayed the entire finished construction at once, including equalities that are proved only later. The audited sequence separates the actual proof states: first the perpendicular copy and parallelogram, then the oriented intersection, then the angle consequences, and finally the four Pythagorean triangles. Pure representative transport in the scissors layer receives no additional picture because it creates no new geometric configuration.

10.3 Classical Proof

The proof first turns the construction into two metric bridges and then compares two decompositions of the same square on AG.

Because $AC = CE$, Proposition I.5 gives

$$\angle EAC = \angle AEC.$$

Since $\angle ACE$ is right, Proposition I.32 makes each of these angles half a right angle. The same argument in the isosceles triangle CEB, using $CE = CB$, shows that $\angle CEB$ and $\angle EBC$ are also half right angles. Therefore

$$\angle AEB$$

is right.

I.5 + I.32

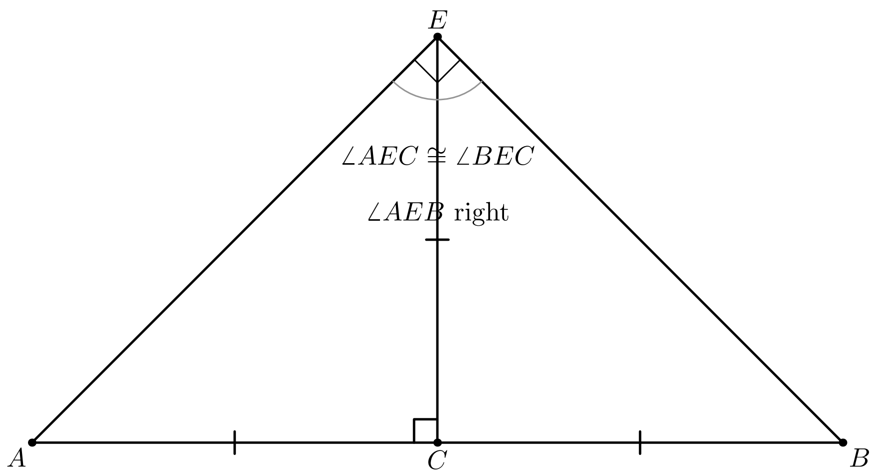


Figure 10.3: The two isosceles triangles ACE and CEB share the copied perpendicular CE. The equal segments $AC \cong CE \cong CB$ give the paired base angles by I.5; I.32 then organizes them as equal half-right components at E. Their union is the right angle AEB. The formal interior-ray witnesses R and S are omitted because their role is to certify this decomposition, not to introduce a new visible vertex of the classical configuration.

Since EB and FD have been produced to G, the vertical-angle theorem I.15 identifies the half-right angle at B with $\angle DBG$. The angle $\angle BDG$ is right by the parallel construction and I.29. Proposition I.32 then shows that the remaining angle $\angle DGB$ is half right. Hence

$$\angle DBG = \angle DGB,$$

and Proposition I.6 gives

$$DB = DG.$$

For the second isosceles bridge, the angle $\angle EGF$ is half right. The angle at F is right because it corresponds to the right angle at C in the parallelogram/parallel configuration. Proposition I.32 therefore makes $\angle FEG$ half right as well, and I.6 gives

$$GF = EF.$$

The opposite sides of parallelogram $CDFE$ are equal by I.34, so

$$EF = CD.$$

Consequently

$$GF = EF = CD.$$

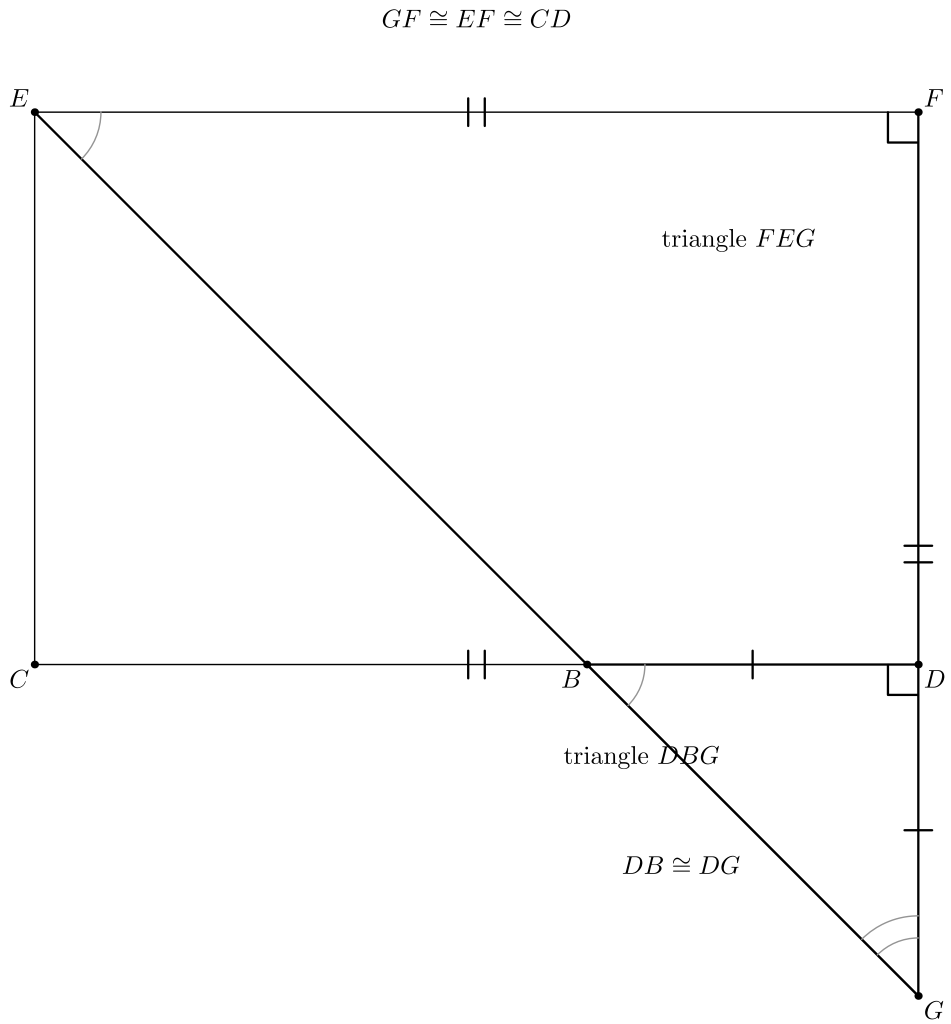


Figure 10.4: Once the orders $E - B - G$ and $F - D - G$ are known, the source angle argument acts on two visible triangles. In DBG, the vertical-angle and right-angle chain makes the angles at B and G congruent, so I.6 yields $DB \cong DG$. In FEG, the corresponding half-right-angle argument and the right angle at F give $GF \cong EF$; I.34 then transports the latter length to CD. Local angle-decomposition witnesses are omitted.

Now apply I.47 four times. In the right triangle ACE,

$$\mathrm{Sq}(AE) = \mathrm{Sq}(AC) + \mathrm{Sq}(CE) = 2\,\mathrm{Sq}(AC).$$

In the right triangle FEG,

$$\mathrm{Sq}(EG) = \mathrm{Sq}(FE) + \mathrm{Sq}(FG) = 2\,\mathrm{Sq}(CD).$$

Because $E - B - G$, the right angle AEB extends along the same ray to give $\angle AEG$ right. Thus I.47 in triangle EAG yields

$$\mathrm{Sq}(AG) = \mathrm{Sq}(AE) + \mathrm{Sq}(EG) = 2\,\mathrm{Sq}(AC) + 2\,\mathrm{Sq}(CD).$$

At D, the parallel/right-angle construction and the collinear order on the base give $\angle ADG$ right. Hence I.47 in triangle DAG gives

$$\mathrm{Sq}(AG) = \mathrm{Sq}(AD) + \mathrm{Sq}(DG) = \mathrm{Sq}(AD) + \mathrm{Sq}(DB).$$

Comparing these two descriptions of the square on AG proves II.10.

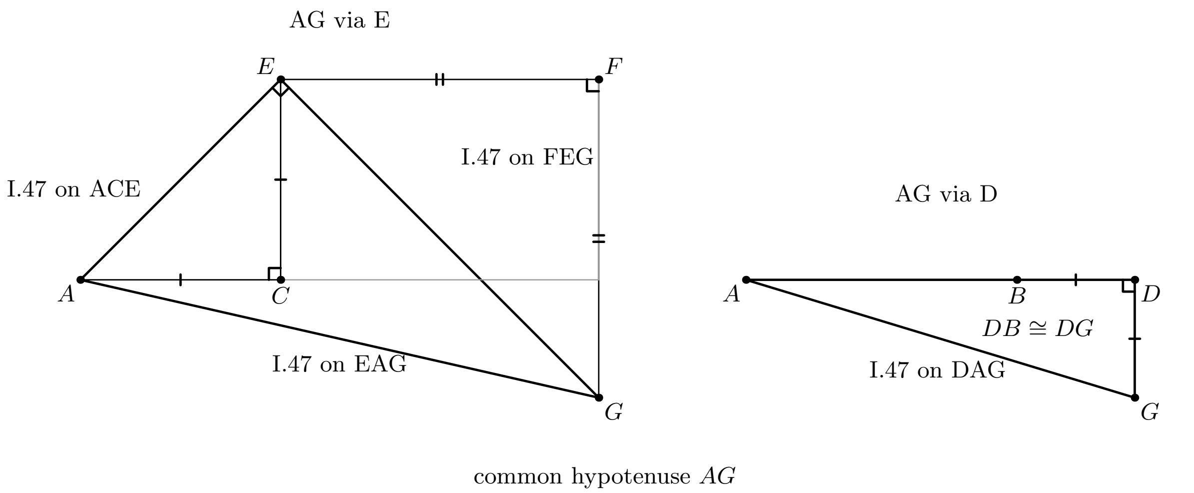


Figure 10.5: The four I.47 calls form two geometric paths to the same side AG. The left panel shows the E-path: I.47 on EAG reduces the square on AG to the squares on AE and EG, while the right triangles ACE and FEG expand those two leg squares. The right panel shows the D-path: I.47 on DAG reduces the same square on AG to the squares on AD and DG. The already proved segment congruences later transport DG to DB, the two $A/C/E$ legs to AC, and the two $F/E/G$ legs to CD. No literal square representatives are drawn: the formal proof may choose them independently and then identifies their content by square transport.

Diagrammatic audit. There is no further figure for Step 8 of the formal proof. After Figure 10.5, all spatial information needed by the argument is already visible. The remaining operations identify independent square representatives and compose equicomplementability relations; they are content bookkeeping rather than a new Euclidean state.

The classical local dependency picture is therefore

$$\begin{array}{c}
\text{I.11} + \text{I.3: perpendicular at } C \text{ and copied half} \\
\Downarrow \\
\text{I.31: parallels } EF \parallel AD,\ DF \parallel CE \\
\Downarrow \\
\text{Post.5: produced } EB \text{ and } FD \text{ meet at } G \\
\Downarrow \\
\text{I.5} + \text{I.15} + \text{I.29} + \text{I.32} + \text{I.6: } DB = DG,\ GF = EF \\
\Downarrow \\
\text{I.34: } EF = CD \\
\Downarrow \\
\text{four applications of I.47} \\
\Downarrow \\
\text{II.10.}
\end{array}$$

10.4 The Geometric Identity in the Current Formalization

The public theorem assumes

```
(A B C D : Geo.Point)
(hMidC : HilbertIsMidpoint Geo C A B)
(hABD : Geo.Between A B D)
```

The midpoint package supplies $A - C - B$ and $AC \cong CB$. The second hypothesis places D beyond B. The proof then derives, among other order facts,

$$A - C - B - D, \qquad C - B - D, \qquad A - C - D, \qquad D - B - A.$$

The theorem constructs concrete squares on AD, DB, AC, and CD. If their formal terms are denoted by

$$S_{AD}, \quad S_{DB}, \quad S_{AC}, \quad S_{CD},$$

the conclusion is

$$S_{AD} + S_{DB} \simeq_{\text{ec}} (S_{AC} + S_{AC}) + (S_{CD} + S_{CD}),$$

where $\simeq_{\text{ec}}$ denotes `HilbertScissorsEquicomplementable`.

This conclusion is not `HilbertScissorsEq`. The proof does not claim that one explicit finite dissection of the two left-hand target squares is the four right-hand target squares. Rather, the current I.47 theorem supplies equicomplementability, and II.10 composes four such Pythagorean blocks with square-representative transport.

10.5 Reusable Infrastructure Used

10.5.1 A generic perpendicular-and-copy construction

The helper

```
proposition2_9_erect_perpendicular
```

is defined in `Proposition2_9Construction.lean`, but its statement is not specific to II.9. It packages the source operations I.11 and I.3: given a nondegenerate segment, it produces a point off its carrier so that the new segment is perpendicular to the old one and congruent to a prescribed segment.

II.10 reuses this helper to construct E with $CE \perp AC$ and $CE \cong AC$. The midpoint congruence then gives $CE \cong CB$.

10.5.2 Parallelogram geometry

The formal proof constructs the fourth vertex F of the parallelogram $CDFE$ through the established Euclidean parallelogram interface. This packages the parallel-existence geometry corresponding to Euclid's I.31. The resulting `IsParallelogram` object supplies the two parallel pairs, while Proposition I.34 supplies opposite-side congruences such as

$$CD \cong FE.$$

Right-angle propagation through the parallelogram uses the reusable theorem `parallelogram_adjacent_right_angle`. This is the formal interface form of the I.29 parallel-angle mechanism needed in the source proof.

10.5.3 Angle decomposition without numerical measure

Euclid repeatedly reasons with "half a right angle" and subtracts equal angle components from equal right angles. The Lean proof introduces no degrees or real-valued angle measure. It uses explicit `HilbertRayMeetsSegment` witnesses together with

```
hilbert_angleDecomposition_halves_congruent_of_whole_congruent
hilbert_angleDecomposition_angle_addition_interior
hilbert_angleDecomposition_angle_subtraction
```

from `HilbertAngleDecomposition.lean`.

The exterior-angle form of I.32 supplies the points used to witness these decompositions. This is the formal replacement for Euclid's compressed angle arithmetic.

10.5.4 Directed intersection and plane separation

The source says that EB and FD, when produced in the directions B and D, meet by Postulate 5. The formal proof must recover more than the bare existence of an intersection: it needs

$$E - B - G, \qquad F - D - G.$$

The helper `proposition2_10_oriented_intersection` proves this directed form. It uses Euclidean parallel uniqueness together with same-side/opposite-side information, segment-line intersection, line uniqueness, and order trichotomy. This is a characteristic example of hidden formal data that the classical drawing suppresses.

10.5.5 Book I square construction, Pythagoras, and square transport

The proof invokes `euclid_proposition_47` four times and later uses `euclid_proposition_46` to choose the four square representatives appearing in the public theorem. Since independent I.47 calls may choose different squares on the same or congruent side, the reusable theorem `hilbert_square_transport` identifies their content.

This is representation transport. It is not an equation in a numerical ring and does not introduce a primitive square operation on segment lengths.

10.6 Proof Architecture of the Reconstruction

10.6.1 Step 1: normalize the external order

`proposition2_10_order` starts from the midpoint and extension hypotheses and establishes the order package needed later:

$$A - C - B, \qquad C - B - D, \qquad A - C - D, \qquad D - B - A.$$

The source statement only draws these positions; the formal proof records them individually because later ray transports depend on orientation.

10.6.2 Step 2: erect the perpendicular and copy the half

`proposition2_10_configuration` constructs E and proves

$$CE \cong AC, \qquad CE \cong CB,$$

together with noncollinearity and the right angles at C required for the two isosceles triangles ACE and ECB.

10.6.3 Step 3: build the auxiliary parallelogram

`proposition2_10_parallelogram` constructs F so that

$$CDFE$$

is a parallelogram. The helper returns the parallel pairs, the opposite-side congruences

$$CD \cong FE, \qquad DF \cong EC,$$

and the right angle at F that will later enter triangle FEG.

10.6.4 Step 4: reconstruct Euclid's first angle block

`proposition2_10_classical_angle_block` applies I.5 in triangles ACE and CEB and uses I.32 to create explicit angle-decomposition witnesses. It proves

$$\angle AEC \cong \angle BEC$$

and hence establishes the right angle AEB.

The formal witnesses R and S lie on the appropriate produced sides and encode the two half-right-angle decompositions. They are construction data, not part of the final theorem statement.

10.6.5 Step 5: produce the intersection in the correct direction

`proposition2_10_oriented_intersection` produces G and proves the strict source orders

$$E - B - G, \qquad F - D - G.$$

The proof does not accept an arbitrary intersection and then reason from the picture. It proves that the intersection lies on the required extensions.

10.6.6 Step 6: recover the two isosceles equalities

`proposition2_10_classical_isosceles_blocks` follows the source angle route to obtain

$$DB \cong DG, \qquad GF \cong EF.$$

At B, I.15 supplies the vertical-angle bridge; at D the parallel configuration gives a right angle; I.32 identifies the remaining half-right angle, and I.6 yields $DB \cong DG$.

The second block transports the corresponding half-right angle to triangle FEG, uses the right angle at F, and again finishes with I.6. Local interior-ray witnesses introduced by I.32 make Euclid's subtraction of angle parts explicit.

10.6.7 Step 7: package the four Pythagorean blocks

`proposition2_10_four_pythagoras_blocks` establishes the segment identifications

$$CA \cong CE, \qquad FE \cong CD, \qquad GF \cong CD, \qquad DB \cong DG$$

and applies I.47 to exactly four right triangles:

$$ACE, \qquad FEG, \qquad EAG, \qquad DAG.$$

The helper returns the four existential square packages and their `HilbertScissorsEquicomplementable` decompositions.

10.6.8 Step 8: follow the common square on AG

The final theorem uses I.46 to choose statement-level squares $S_{AD}, S_{DB}, S_{AC}, S_{CD}$. It begins from I.47 on DAG:

$$S_{AG} \simeq_{\mathrm{ec}} S_{DA} + S_{DG}.$$

Square transport replaces DA by AD and DG by DB, yielding the left-hand target.

A second representative of the square on AG, coming from triangle EAG, is then identified with the first. I.47 on EAG gives

$$S_{AD} + S_{DB} \simeq_{\mathrm{ec}} S_{EA} + S_{EG}.$$

The remaining I.47 blocks expand these two squares:

$$S_{EA} + S_{EG} \simeq_{\mathrm{ec}} (S_{CA} + S_{CE}) + (S_{FE} + S_{FG}).$$

Finally

$$CA \cong AC, \quad CE \cong AC, \quad FE \cong CD, \quad FG \cong CD$$

transport the four terms to two copies of S_{AC} and two copies of S_{CD}. Additive compatibility and transitivity close the proof. No content cancellation is used.

10.7 Lean Formalization

10.7.1 The public theorem signature

The following is an ASCII transcription of the current compiling source:

```
theorem euclid_proposition_2_10_oriented
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D : Geo.Point)
    (hMidC : HilbertIsMidpoint Geo C A B)
    (hABD : Geo.Between A B D) :
    exists SAD0 SAD1 SDB0 SDB1 SAC0 SAC1 SCD0 SCD1 : Geo.Point,
      IsSquare Geo A D SAD0 SAD1 /\
      IsSquare Geo D B SDB0 SDB1 /\
      IsSquare Geo A C SAC0 SAC1 /\
      IsSquare Geo C D SCD0 SCD1 /\
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo A D SAD0 SAD1 +
         hilbertParallelogramTerm Geo D B SDB0 SDB1)
        ((hilbertParallelogramTerm Geo A C SAC0 SAC1 +
          hilbertParallelogramTerm Geo A C SAC0 SAC1) +
         (hilbertParallelogramTerm Geo C D SCD0 SCD1 +
          hilbertParallelogramTerm Geo C D SCD0 SCD1)) := by
```

The suffix `_oriented` is substantive. The theorem fixes the external branch $A - C - B - D$ through the hypothesis `hABD : Geo.Between A B D`. A future theorem that packages both orientations would be a separate wrapper.

10.7.2 The proposition-local theorem chain

The production source is factored into

```
proposition2_10_order
proposition2_10_configuration
proposition2_10_parallelogram
proposition2_10_classical_angle_block
```

```
proposition2_10_oriented_intersection
proposition2_10_classical_isosceles_blocks
proposition2_10_four_pythagoras_blocks
euclid_proposition_2_10_oriented
```

This chain mirrors genuine mathematical stages rather than merely splitting a large tactic script. The hidden witnesses E, F, G, R, S and the local I.32 decomposition points are progressively introduced and then eliminated from the final public interface.

10.7.3 The final scissors bridge

The last stage uses only forward equicomplementability operations:

(a) I.47 on DAG connects a square on AG with the two left target squares after congruence transport;

(b) two independently constructed AG squares are identified;

(c) I.47 on EAG replaces AG by the EA and EG squares;

(d) I.47 on ACE and FEG expands those into four leg squares;

(e) square transport identifies the four legs with two copies of AC and two copies of CD.

No subtraction or cancellation in the scissors calculus is used.

10.8 Relationship with Earlier Book II Propositions

The production theorem does not call an earlier Book II proposition. In particular it does not use II.9 as an identity from which II.10 is deduced.

There is one important module-level nuance. The import `Proposition2_9Construction` supplies `proposition2_9_erect_perpendicular`, a generic construction helper encoding I.11 plus I.3. Thus II.10 reuses infrastructure that was extracted during the II.9 development, but it does not reuse the theorem `euclid_proposition_2_9_oriented` or any II.9 content identity.

This distinction is mathematically appropriate. Euclid's guide-level relationship between II.9 and II.10 is that the diagrams and arguments are companions: II.10 moves the unequal point beyond the end of the original segment. The formal library preserves that parallelism by sharing low-level construction machinery while proving the two proposition-level statements separately.

10.9 Relationship with Book I Infrastructure

The classical source uses I.11 and I.3 to erect and copy the perpendicular, I.31 for the parallels, I.29 and Postulate 5 for the produced intersection, I.5 for the isosceles base angles, I.15 for the vertical angle, I.32 for the half-right-angle arguments, I.6 for the two isosceles conclusions, I.34 for opposite sides of the parallelogram, and I.47 four times [1].

The Lean DAG packages several of these steps. The I.11+I.3 construction is the reusable perpendicular helper. I.31 is represented by the fourth-vertex/parallelogram interface. The I.29

right-angle consequences are available through parallelogram right-angle transport. Postulate 5 is replaced at the proof-script level by a Euclidean parallel-uniqueness contradiction together with plane-separation and order machinery that also determines the direction of the intersection.

Thus the formal proof is source-faithful in mathematical architecture but is not a line-by-line transcription of Euclid's prose.

10.10 Formalization Notes and Hidden Geometric Data

10.10.1 The external order is a theorem input

II.10 is not the internal-cut branch of II.9 with names changed. The hypothesis

```
hABD : Geo.Between A B D
```

places D strictly beyond B. From this the proof derives the other betweenness relations required to compare rays from C and D.

10.10.2 The intersection G needs orientation, not just existence

The classical phrase "produce the lines until they meet" hides a major formal obligation. Later angle identities require B to lie between E and G and D to lie between F and G. An arbitrary point lying on both carriers is insufficient.

The formal proof therefore combines side-of-line data, order trichotomy, and line uniqueness to establish the exact orders $E - B - G$ and $F - D - G$.

10.10.3 Same-ray witnesses control angle-arm replacement

Once the strict orders are known, `HilbertSameRay` witnesses justify replacing EB by EG, DB by DA or DC where appropriate, and FD by FG. These transports are invisible in the printed diagram but essential to every right-angle propagation.

10.10.4 The points R, S and later local witnesses encode angle arithmetic

The I.32 exterior-angle theorem produces concrete points that witness interior rays. The first pair supports the proof that AEB is right. Further local witnesses support the two synthetic angle-subtraction arguments leading to $DB = DG$ and $GF = EF$.

No statement of the form "$45° + 45° = 90°$" occurs in the Lean development. The information is carried by betweenness, ray intersection, and angle congruence.

10.10.5 Noncollinearity is structural data

Every use of right-angle transport, I.6, I.15, angle decomposition, and I.47 requires nondegeneracy. The proof repeatedly derives noncollinearity from strict betweenness, the perpendicular construction, and the parallelogram. These are genuine geometric facts, not merely type-checking noise.

10.10.6 Square representatives are not canonical

Four I.47 calls and four final I.46 constructions create several squares on the same or congruent segments. The proof must therefore identify their content explicitly with `hilbert_square_transport`. The final identity is independent of these representation choices, but the proof of that independence is part of the formal content layer.

10.11 Axiomatic and Semantic Status

The production theorem introduces no new proposition-specific geometric axiom and no new content axiom. It contains no `sorry` or `admit`. Its semantic layers are

$$\begin{array}{c}
\text{incidence/order/congruence and Euclidean parallel geometry} \\
\Downarrow \\
\text{perpendiculars, right angles, angle decomposition, parallelogram facts} \\
\Downarrow \\
\text{concrete square representatives} \\
\Downarrow \\
\text{I.47 equicomplementability + square transport} \\
\Downarrow \\
\text{II.10 equicomplementability conclusion.}
\end{array}$$

The transitive axiom audit was run on the compiled production theorem:

```
#print axioms Geometry.euclid_proposition_2_10_oriented
```

and reported

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are inherited project dependencies. The precise conclusion is therefore: II.10 adds no new axiom and no `sorry`; it is not literally axiom-free.

The theorem uses neither numerical area nor the later arithmetic of segment multiplication. The algebraic-looking identity remains entirely in the synthetic geometry/content architecture.

10.12 Dependency Summary

The production chain is

$$\begin{gathered}
\text{Hilbert incidence, order, congruence, Euclidean parallel layer} \\
\Downarrow \\
\text{I.11 + I.3 generic perpendicular/copy infrastructure} \\
\Downarrow \\
\text{order + perpendicular configuration + parallelogram } CDFE \\
\Downarrow \\
\text{I.5 + I.15 + I.29/I.32 + angle-decomposition machinery} \\
\Downarrow \\
\text{oriented intersection } E - B - G,\ F - D - G \\
\Downarrow \\
\text{I.6: } DB = DG,\ GF = EF \quad + \quad \text{I.34: } EF = CD \\
\Downarrow \\
\text{four I.47 blocks on } ACE, FEG, EAG, DAG \\
\Downarrow \\
\text{I.46 target squares + HilbertSquareTransport} \\
\Downarrow \\
\text{equicomplementability addition/transitivity} \\
\Downarrow \\
\texttt{euclid_proposition_2_10_oriented}.
\end{gathered}$$

The implementation checkpoint is:

New geometric axiom in II.10:	no
New content axiom in II.10:	no
New proposition-local helper theorem:	yes
New reusable public helper theorem/module:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly calls an earlier Book II proposition:	no
Imports II.9 production theorem:	no
Reuses generic helper from `Proposition2_9Construction`:	yes
Conclusion is exact `HilbertScissorsEq`:	no
Conclusion is `HilbertScissorsEquicomplementable`:	yes
Uses numerical area:	no
Uses segment multiplication:	no
Uses content cancellation:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Full project build:	clean (user verified)
Proposition-local `sorry`/`admit`:	no
Transitive `#print axioms`:	`propext`, `Classical.choice`, `bookZero_nullSegment1`, `Quot.sound`

10.13 Conclusion

Proposition II.10 confirms that the architectural change first visible in II.9 is a genuine new Book II proof mode rather than a one-off exception. The rectangle-normal-form chain of II.1–II.8 is not used. Instead the formalization follows Euclid's external-section construction through perpendiculars, parallels, produced intersections, isosceles angle arguments, and four Pythagorean content identities.

The most delicate new formal feature is the direction of the intersection G. The source drawing makes $E - B - G$ and $F - D - G$ immediate; Lean must prove both. Once that order is available, the remaining geometry follows the classical metric bridges $DB = DG$ and $GF = EF = CD$.

The final common object is the square on AG. One I.47 path connects it to the two target squares on AD and DB; the other connects it to the four leg squares that become two copies of AC and two copies of CD. Consequently the mnemonic

$$AD^2 + DB^2 = 2(AC^2 + CD^2)$$

is best understood as the algebraic shadow of a synthetic equicomplementability theorem. The current proof does not use II.9 as a proposition, numerical area, or segment multiplication, and the completed axiom audit shows that II.10 introduces no new axiom beyond the established transitive project dependencies.

Chapter 11

Proposition II.11

11.1 Introduction

Proposition II.11 is the first construction problem of Book II whose conclusion is itself a new division point on the given segment. Starting from a nondegenerate straight line AB, Euclid constructs a point H between A and B such that the rectangle contained by the whole segment AB and the smaller part HB is equal in content to the square on the remaining part AH:

$$\mathrm{Rect}(AB, HB) = \mathrm{Sq}(AH).$$

The division is the classical extreme-and-mean-ratio construction, later reused in the construction of the regular pentagon.

The current Lean theorem has the same geometric output. It constructs an interior point H, an arbitrary rectangle representative contained by AB, HB, and a square representative on AH, and proves that the two figures are `HilbertScissorsEquicomplementable`. The theorem therefore belongs to the synthetic content layer. It does not introduce numerical area, a product of segment lengths, or a field equation.

A modern algebraic shadow is obtained by writing $AB = a$ and $AH = x$. Since $HB = a - x$, the target relation has the mnemonic form

$$a(a - x) = x^2, \qquad \text{equivalently} \qquad x^2 + ax = a^2.$$

This is explanatory notation only. Neither Euclid's proposition nor the Lean statement is an equation in a previously defined arithmetic of segment lengths.

The production source begins with

```
import CGJteamLab.HilbertRectangle
import CGJteamLab.Proposition2_2
import CGJteamLab.Proposition2_3
import CGJteamLab.Proposition2_6
import CGJteamLab.Proposition10
import CGJteamLab.Proposition16
import CGJteamLab.Proposition19
import CGJteamLab.Proposition20
import CGJteamLab.Proposition46
import CGJteamLab.Proposition47
import CGJteamLab.HilbertSquareTransport
```

Thus II.11 is the first proposition after II.10 to return directly to earlier Book II rectangle identities: the content proof explicitly calls II.2, II.3, and II.6, while the construction layer uses Book I order and metric results.

11.2 Euclid's Statement and Classical Configuration

11.2.1 The statement

In Heath's wording, Proposition II.11 states:

> To cut a given straight line so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.

With the lettering used in the current formal theorem, the required point is H with

$$A - H - B$$

and the content identity is

$$\mathrm{Rect}(AB, HB) = \mathrm{Sq}(AH).$$

The order matters. The rectangle uses the *smaller* piece HB, while the square is on the *remaining* piece AH.

Hartshorne emphasizes the same interpretation in his discussion of the regular pentagon: II.11 divides a segment so that the rectangle on the whole and the smaller part has the same content as the square on the larger part [5]. This is the content form of the extreme-and-mean-ratio condition; a ratio equation is not part of the Book II statement.

11.2.2 The classical construction

The figure is normalized to the vertex order used by the Lean source.

(1) On AB construct the square $ABCD$ by I.46.

(2) Bisect the side DA at E by I.10.

(3) Join E to B.

(4) Produce DA through A to a point F such that

$$EF \cong EB.$$

(5) Transfer the length AF to the original segment, obtaining H with

$$A - H - B, \qquad AH \cong AF.$$

The characteristic source order is therefore

$$D - E - A - F, \qquad A - H - B.$$

In Euclid's diagram the square on AF and the associated parallel lines are then used to identify the final square on AH and the rectangle contained by AB, HB [1]. The current Lean proof reaches the same point H through a different synthetic existence argument; this difference is recorded below rather than hidden inside the classical picture.

Diagrammatic audit. Figure 11.1 records the completed classical construction, but it does not by itself explain the geometry that the formal reconstruction must recover or the two distinct content cancellations in the proof. The additional figures below are therefore retained only when the geometric state changes: the I.16 comparison proving $EA < EB$, the I.20 comparison proving $AF < AB$, the hidden configured rectangle cut required by the Book II APIs, the II.6–I.47 comparison through the common square on AE, and the final II.3–II.2 pair of cuts exposing the common rectangle. No separate picture is added for the final point H, since it is already visible in Figure 11.1; nor are purely formal representative transport or Common-Notion-3 witness manipulations given decorative diagrams.

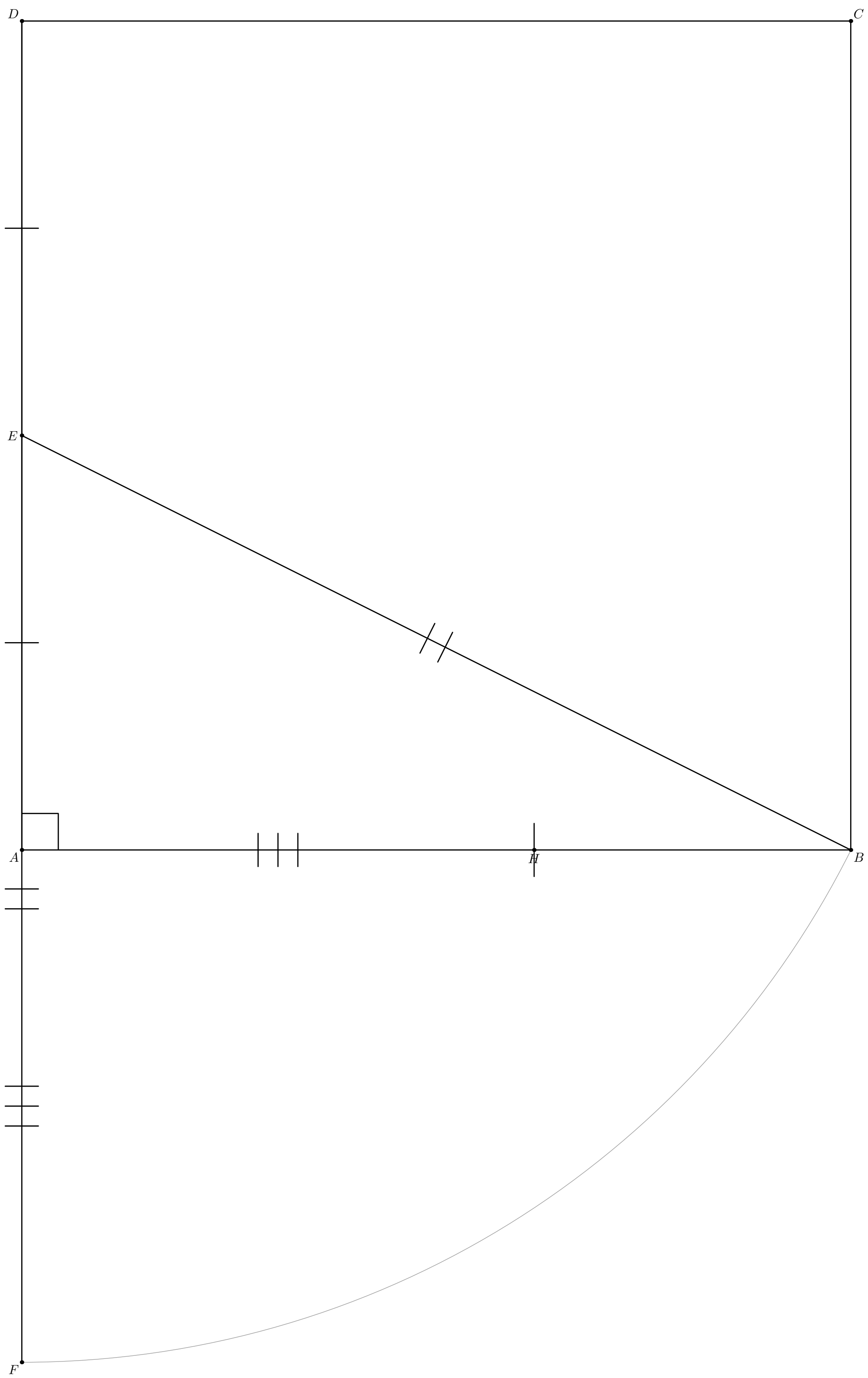


Figure 11.1: The principal configuration for Proposition II.11 in the lettering of the Lean source. $ABCD$ is the square on AB, E is the midpoint of DA, and F lies on the extension through A with $EF \cong EB$. The light construction arc centered at E records the classical layoff $EF = EB$. The final point H lies on AB and satisfies $AH \cong AF$. Thus the source orders are $D - E - A - F$ and $A - H - B$. The technical rectangle-cut witnesses used internally by Lean are deliberately omitted.

11.3 Classical Proof

The proof can be organized by comparing the exact II.6 decomposition with one Pythagorean decomposition from I.47, followed by subtraction of common figures.

Because E is the midpoint of DA and F lies beyond A, Proposition II.6 applies to the straight line DA with midpoint E and added segment AF. In the present lettering it yields

$$\mathrm{Sq}(AE) + \mathrm{Rect}(DF, AF) = \mathrm{Sq}(EF).$$

By construction $EF = EB$. The angle EAB is right because EA lies on the side AD of the square on AB. Proposition I.47 in the right triangle EAB gives

$$\mathrm{Sq}(EB) = \mathrm{Sq}(AE) + \mathrm{Sq}(AB).$$

Hence

$$\mathrm{Sq}(AE) + \mathrm{Rect}(DF, AF) = \mathrm{Sq}(AE) + \mathrm{Sq}(AB).$$

Removing the common square on AE gives

$$\mathrm{Rect}(DF, AF) = \mathrm{Sq}(AB).$$

The remaining step converts the rectangle on DF, AF into the desired rectangle on AB, HB. Euclid's square on AF and the parallel cut separate the common part geometrically. In the language of the current Book II rectangle calculus the same content bookkeeping is expressed by II.3 and II.2:

$$\mathrm{Rect}(DF, AF) = \mathrm{Rect}(DA, AF) + \mathrm{Sq}(AF),$$

while

$$\mathrm{Sq}(AB) = \mathrm{Rect}(AB, AH) + \mathrm{Rect}(AB, HB).$$

The square $ABCD$ gives $DA \cong AB$, and the construction gives $AF \cong AH$. Thus

$$\mathrm{Rect}(DA, AF) = \mathrm{Rect}(AB, AH).$$

Subtracting this common rectangle leaves

$$\mathrm{Sq}(AF) = \mathrm{Rect}(AB, HB).$$

Finally $AF \cong AH$, so the square on AF has the same content as the square on AH. Therefore

$$\mathrm{Rect}(AB, HB) = \mathrm{Sq}(AH),$$

as required.

The local classical dependency picture is therefore

$$\begin{gathered}
\text{I.46: square on } AB \quad + \quad \text{I.10: midpoint of } DA \\
\Downarrow \\
\text{I.3 / segment layoff: } EF = EB \quad + \quad \text{join } EB \\
\Downarrow \\
\text{II.6 on } D - E - A - F \quad + \quad \text{I.47 on the right triangle } EAB \\
\Downarrow \\
\text{Common Notion 3: remove the common square on } AE \\
\Downarrow \\
\text{square/rectangle decomposition on } AF \text{ and } AB \\
\Downarrow \\
\text{Common Notion 3 again} \\
\Downarrow \\
\text{II.11.}
\end{gathered}$$

11.4 The Geometric Identity in the Current Formalization

The public theorem takes only a nondegenerate segment:

```
(A B : Geo.Point)
(hAB : Ne A B)
```

and constructs all other points internally.

At the final interface Lean produces

$$A - H - B,$$

a concrete rectangle representative R contained by AB, HB, and a concrete square representative Q on AH. The conclusion is

$$R \simeq_{\mathrm{ec}} Q,$$

where $\simeq_{\mathrm{ec}}$ denotes `HilbertScissorsEquicomplementable`.

This is *not* an exact `HilbertScissorsEq` conclusion. Exact dissections occur internally, in particular in the imported II.2, II.3, and II.6 blocks. The final theorem is more general because the Pythagorean comparison enters through the current I.47 equal-content interface, and the proof performs cancellation at that level.

The representation layer is also explicit. The phrase "rectangle contained by AB, HB" is represented by `IsRectangleContainedBy`; it does not identify one canonical rectangle object. Likewise the square on AH is an existentially chosen `IsSquare` representative. The theorem proves content independence from those concrete choices through the established transport machinery.

11.5 Reusable Book II Infrastructure Used

11.5.1 Proposition II.6

The first content block is the exact identity

$$\mathrm{Sq}(AE) + \mathrm{Rect}(DF, AF) \simeq_{\mathrm{sc}} \mathrm{Sq}(EF).$$

The proposition-local II.6 wrapper

```
proposition2_11_II6_exists
```

packages the large configured API of `euclid_proposition_2_6`. It constructs the required square and rectangle representatives and all explicit cut witnesses before calling II.6.

Its current signature begins:

```
private theorem proposition2_11_II6_exists
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (D A E F : Geo.Point)
    (hMidE : HilbertIsMidpoint Geo E D A)
    (hDAF : Geo.Between D A F) :
```

and returns an exact `HilbertScissorsEq`. Thus II.6 is used at the stronger dissection level before the proof enters the broader equicomplementability chain.

11.5.2 Propositions II.3 and II.2

The second subtraction block uses II.3 to split the rectangle contained by DF, AF at A:

$$\mathrm{Rect}(DF, AF) \simeq_{\mathrm{sc}} \mathrm{Rect}(DA, AF) + \mathrm{Sq}(AF).$$

It then uses II.2 on the square $ABCD$ cut at H:

$$\mathrm{Rect}(AB, AH) + \mathrm{Rect}(AB, HB) \simeq_{\mathrm{sc}} \mathrm{Sq}(AB).$$

Both imported propositions provide exact scissors equalities. The proposition-local proof subsequently transports the first common rectangle to the second by side congruences.

11.5.3 Rectangle representative transport

The square $ABCD$ yields

$$DA \cong AB,$$

while the construction of H yields

$$AF \cong AH.$$

The theorem `rectangle_transport_scissorsEq` therefore gives an exact scissors equality between representatives of

$$\mathrm{Rect}(DA, AF) \quad \text{and} \quad \mathrm{Rect}(AB, AH).$$

This is a representation theorem, not commutativity or multiplication in a segment field.

11.5.4 Square transport

I.47 and I.46 create independent square representatives on equal or congruent segments. The proof repeatedly uses `hilbert_square_transport` to move between them. The final use transports the square on AF to the square on AH through

$$AF \cong AH.$$

Again, this is geometric/content transport between representatives rather than an arithmetic rewrite of squared numbers.

11.6 New Proposition-Local Rectangle-Cut Infrastructure

A significant formal feature of II.11 is not visible in the public theorem: the current low-level APIs of II.2, II.3, and II.6 require explicit rectangular cut diagrams. Rather than make those witnesses hypotheses of the public theorem, II.11 constructs them internally.

The proposition-local chain is

```
proposition2_11_rectangle_cut_core
proposition2_11_rectangle_cut_between
proposition2_11_diagonal_cut_intersection
proposition2_11_rectangle_cut_exists
```

`proposition2_11_rectangle_cut_core` starts with an arbitrary rectangle $BCED$ and a point M satisfying $B - M - C$. It constructs an upper point L and the two parallelograms

$$LDBM, \qquad CELM.$$

The proof uses the fourth-vertex parallelogram construction and Euclidean parallel uniqueness to show that L lies on the carrier DE.

The next helper strengthens collinearity to strict order. Opposite-side congruences in the original rectangle and the two cut parallelograms give

$$BM \cong DL, \qquad BC \cong DE, \qquad MC \cong LE.$$

Hilbert's three-point order transport then sends

$$B - M - C \quad \Longrightarrow \quad D - L - E.$$

Finally `proposition2_11_diagonal_cut_intersection` uses Pasch in the nondegenerate triangle CBD. Since the cut line ML enters through CB at M and cannot meet BD because $DB \parallel ML$, it must meet CD at an interior point N. Parallel crossing order then proves

$$D - N - C, \qquad M - N - L.$$

The final local wrapper

```
proposition2_11_rectangle_cut_exists
```

packages exactly the geometric witnesses expected by the existing rectangle split APIs.

This infrastructure is private to Proposition II.11. No new theorem is added to the public `HilbertRectangle` module. The documentation therefore records it as proposition-local reusable geometry rather than as a new public rectangle API.

11.7 Proof Architecture of the Reconstruction

11.7.1 Step 1: construct the square and midpoint

Starting from $A \neq B$, the construction helper calls I.46 to produce the square $ABCD$. It then calls I.10 on the side DA and obtains a midpoint E:

$$D - E - A, \qquad DE \cong EA.$$

Opposite sides and equal sides of the square provide

$$DA \cong AB.$$

The side EA is collinear with DA, so the right angle of the square at A is transported to

$$\angle EAB$$

as a formal `HilbertRightAngle`.

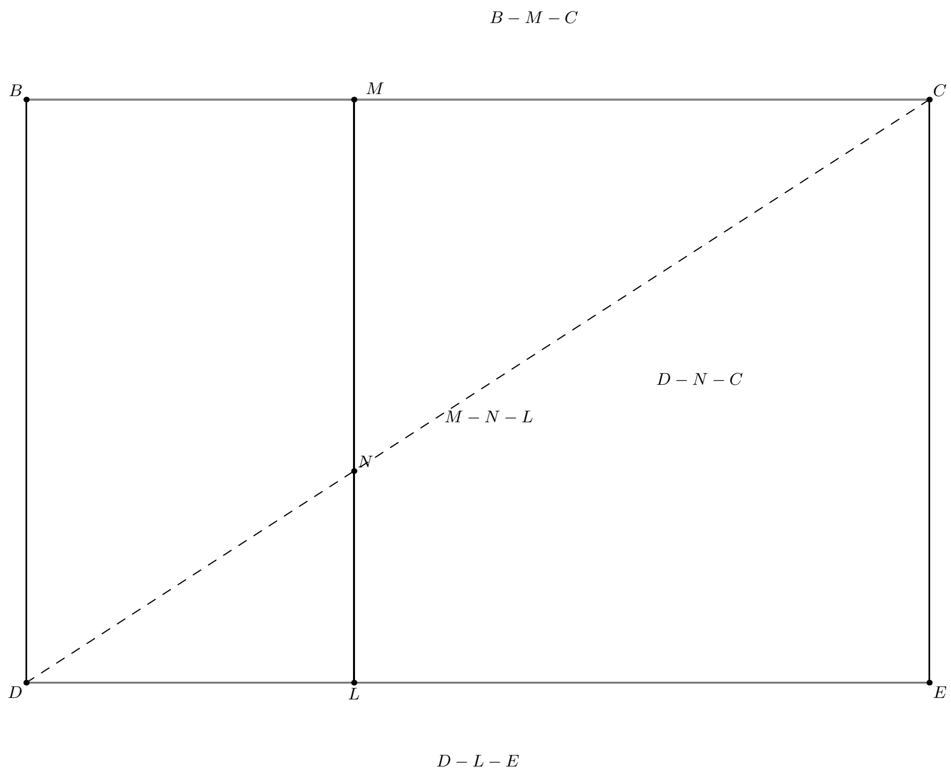


Figure 11.2: The configured rectangle split hidden behind the II.11 wrappers. The given cut $B - M - C$ is transported to the opposite side as $D - L - E$. The cut line ML cannot meet the side DB, since it is parallel to it, so Pasch forces an intersection N with the diagonal DC. The strict orders $D - N - C$ and $M - N - L$ are exactly the additional witnesses expected by the low-level Book II rectangle APIs.

11.7.2 Step 2: prove that the copied hypotenuse passes beyond A

Euclid's diagram makes it visually clear that the point F with $EF = EB$ lies beyond A. Lean proves the required inequality first.

The helper `proposition2_11_EA_lt_EB_aux` extends EA beyond A to a point X. I.16 in triangle BEA gives

$$\angle EBA < \angle BAX.$$

Because $E - A - X$ and EAB is right, the exterior angle BAX is congruent to the right angle EAB. Transporting the angle inequality and applying I.19 gives

$$EA < EB.$$

This is a genuine synthetic order argument. No square-root or numerical length comparison is used.

11.7.3 Step 3: construct F and prove the four-point order

Hilbert segment construction lays off EB on the ray EA, producing F with

$$EF \cong EB.$$

The same-ray data alone do not determine whether F lies before or beyond A. The previously proved inequality $EA < EB$, transported through $EB \cong EF$, rules out the wrong betweenness

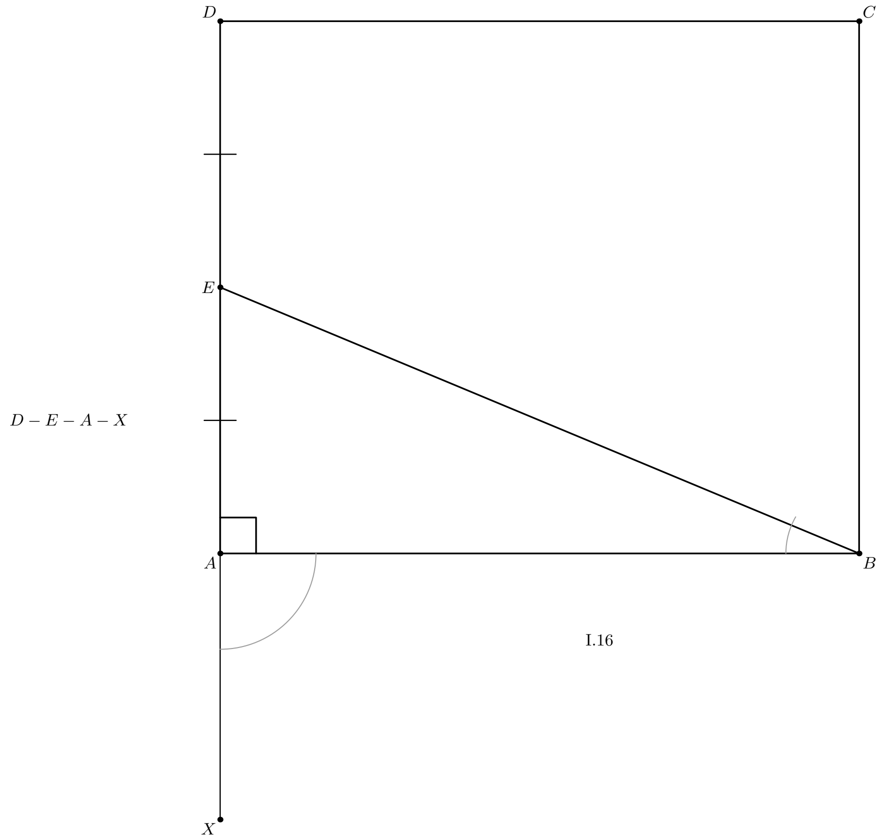


Figure 11.3: The formal comparison behind `proposition2_11_EA_lt_EB_aux`. The midpoint E lies on the square side DA, and EA is extended through A to X. In triangle BEA, Proposition I.16 compares the interior angle at B with the exterior angle BAX. Because $E - A - X$ and EAB is right, the exterior angle is right; I.19 then yields $EA < EB$. The point F is deliberately absent because it has not yet been constructed.

branches. Order trichotomy then gives

$$E - A - F.$$

Together with $D - E - A$ this yields

$$D - E - A - F.$$

11.7.4 Step 4: prove AF is shorter than AB

The formal proof does not obtain H by invoking a line-circle intersection. Instead it proves

$$AF < AB.$$

The helper `proposition2_11_AF_lt_AB_aux` applies I.20 in triangle AEB. It obtains a point T with

$$E - A - T, \qquad AT \cong AB, \qquad EB < ET.$$

Using $EF \cong EB$, the inequality becomes $EF < ET$. Unpacking the definition of strict segment comparison gives a point P on ET with $EP \cong EF$. Uniqueness of segment construction on the same ray identifies P with F. Hence

$$A - F - T,$$

so $AF < AT$. Transport through $AT \cong AB$ yields

$$AF < AB.$$

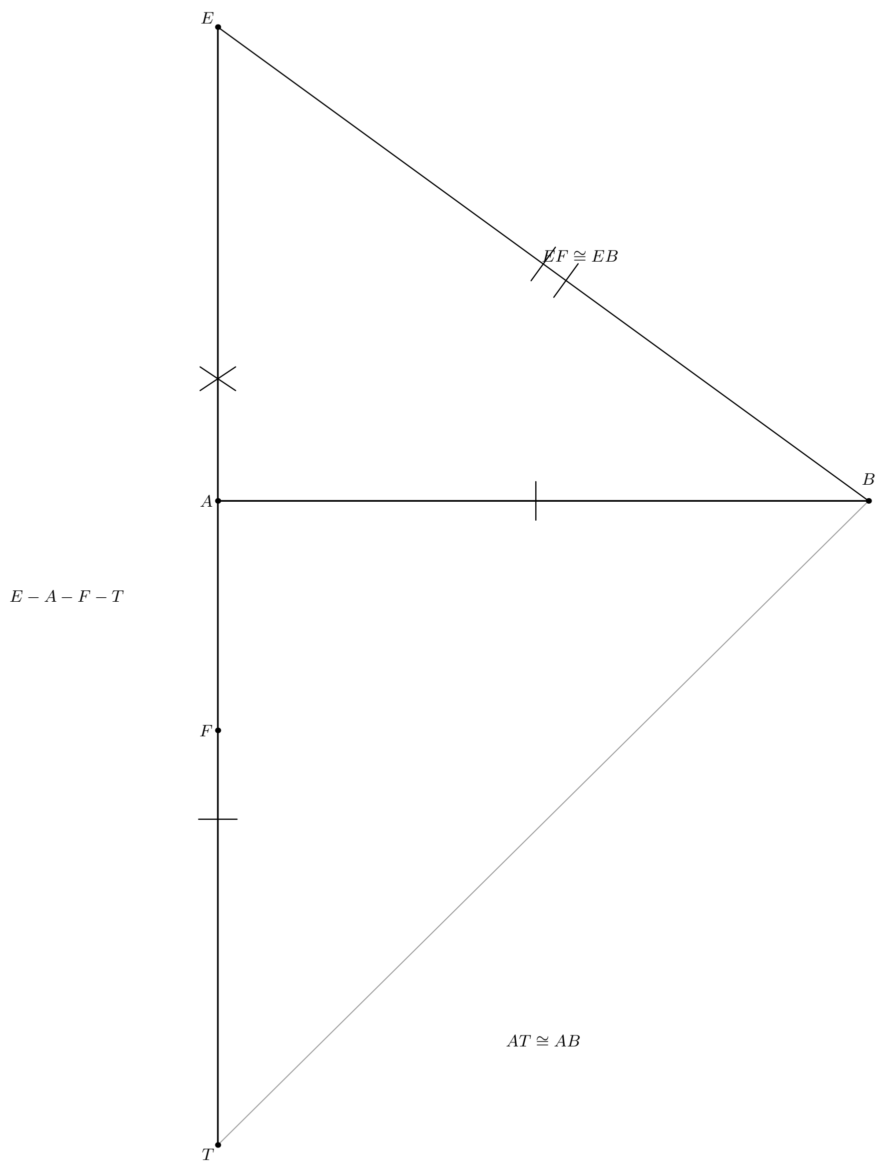


Figure 11.4: The order configuration used by `proposition2_11_AF_lt_AB_aux`. Proposition I.20 supplies $E - A - T$ with $AT \cong AB$ and $EB < ET$. Since the already constructed point F satisfies $EF \cong EB$, segment-construction uniqueness identifies the comparison witness on ET with F, giving $A - F - T$. Thus $AF < AT$, and $AT \cong AB$ transports the strict inequality to $AF < AB$. The temporary witness P is omitted because the proof immediately proves $P = F$.

11.7.5 Step 5: obtain the final cut H

By definition, `HilbertSegmentLessGeoAFAB` contains a point of the segment AB at distance AF from A. The helper `proposition2_11_construct_H` unpacks that witness and returns

$$A - H - B, \qquad AF \cong AH.$$

This is the main construction-level divergence from the classical source. Hartshorne notes that a line-circle intersection property is used in Euclid's II.11 construction [5]. The Lean proof reaches the same metric cut through segment comparison and segment-construction uniqueness. It is therefore synthetic and source-compatible in result, but not a line-by-line formalization of the historical construction.

11.7.6 Step 6: the first II.6–I.47 comparison

The helper `proposition2_11_central_equality` combines two content paths.

First, II.6 gives the exact scissors identity

$$S_{AE} + R_{DF,AF} \simeq_{\text{sc}} S_{EF}.$$

Second, I.47 in the right triangle AEB gives

$$S_{EB} \simeq_{\text{ec}} S_{AE} + S_{AB}.$$

Since $EF \cong EB$, square transport replaces S_{EB} by S_{EF}. Hence

$$S_{AE} + R_{DF,AF} \simeq_{\text{ec}} S_{AE} + S_{AB}.$$

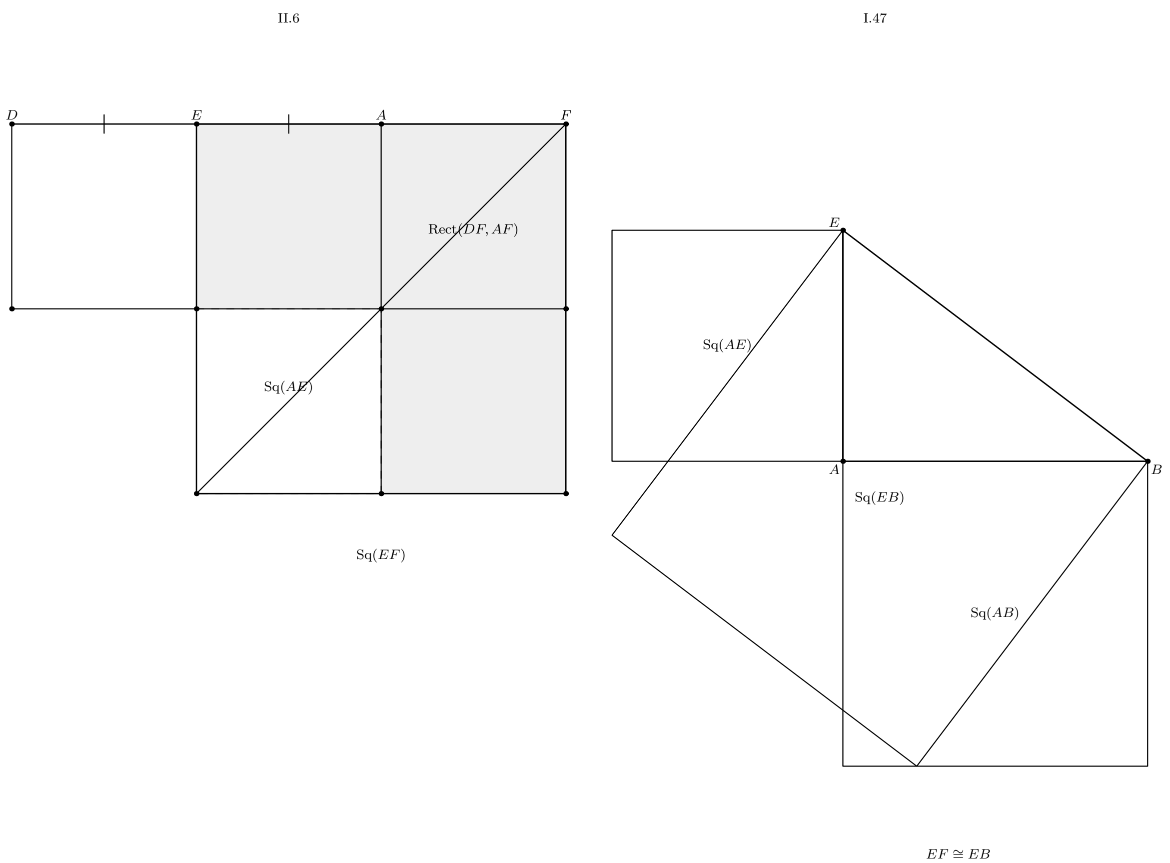


Figure 11.5: Two geometric decompositions meet at the first cancellation. On the left, Proposition II.6 is specialized to the order $D - E - A - F$: its gnomon configuration represents $S_{AE} + R_{DF,AF} \simeq_{\text{sc}} S_{EF}$. On the right, Proposition I.47 is applied to the right triangle EAB, giving $S_{EB} \simeq_{\text{ec}} S_{AE} + S_{AB}$. The construction $EF \cong EB$ identifies the two whole-square contents. The spatial point of the figure is the common S_{AE}, not an algebraic cancellation symbol.

11.7.7 Step 7: Common Notion 3 in the scissors calculus

The local helper

```
private theorem proposition2_11_common_notion_3
    (P Q R : HilbertScissorsTerm Geo)
    (hWhole :
      HilbertScissorsEquicomplementable Geo
        (R + P)
        (R + Q)) :
    HilbertScissorsEquicomplementable Geo P Q := by
```

formalizes the content-level cancellation needed by Euclid.

It does not assert subtraction in a numerical group. It unfolds the witness structure of `HilbertScissorsEquicomplementable`: if $R+P$ and $R+Q$ become scissors-equal after adding complements U, V, then P and Q become scissors-equal after adding the enlarged complements $R+U, R+V$. Associativity and commutativity of the finite formal sum provide the bookkeeping.

Applied to the previous comparison, this removes the common square on AE and gives

$$R_{DF,AF} \simeq_{\mathrm{ec}} S_{AB}.$$

11.7.8 Step 8: the second subtraction

The helper `proposition2_11_second_subtraction` now combines three exact-scissors statements:

$$\begin{aligned} R_{DF,AF} &\simeq_{\mathrm{sc}} R_{DA,AF} + S_{AF} && \text{by II.3,} \\ R_{AB,AH} + R_{AB,HB} &\simeq_{\mathrm{sc}} S_{AB} && \text{by II.2,} \\ R_{DA,AF} &\simeq_{\mathrm{sc}} R_{AB,AH} && \text{by rectangle transport.} \end{aligned}$$

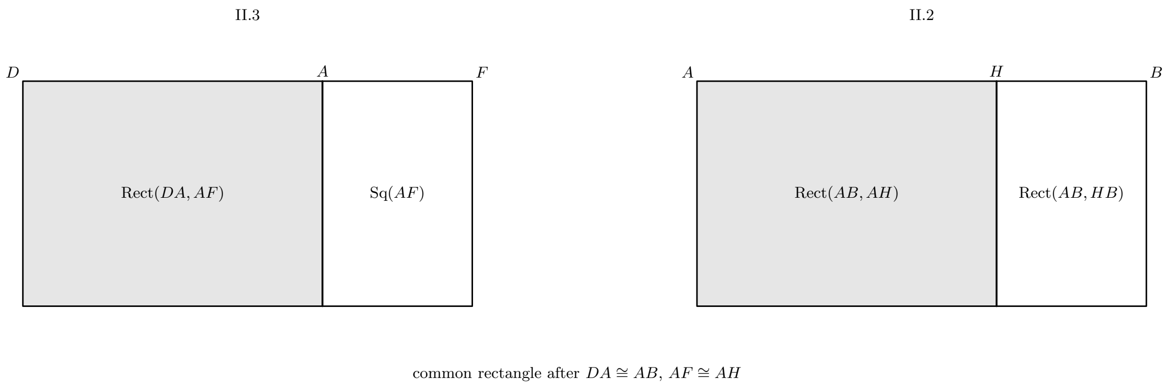


Figure 11.6: The two exact cuts used before the second Common-Notion-3 step. The II.3 rectangle on DF, AF is cut at A, leaving $\mathrm{Rect}(DA, AF)$ and the square on AF. The II.2 square on AB is cut at H, leaving the rectangles contained by AB, AH and AB, HB. Because $DA \cong AB$ and $AF \cong AH$, the shaded regions are scissors-equal representatives of the same common rectangle. Removing that common content leaves $S_{AF} \simeq_{\mathrm{ec}} R_{AB,HB}$.

Together with the central equicomplementability

$$R_{DF,AF} \simeq_{\mathrm{ec}} S_{AB},$$

these yield

$$R_{DA,AF} + S_{AF} \simeq_{\mathrm{ec}} R_{DA,AF} + R_{AB,HB}.$$

The second Common-Notion-3 helper cancels the common rectangle:

$$S_{AF} \simeq_{\mathrm{ec}} R_{AB,HB}.$$

11.7.9 Step 9: final square transport

I.46 constructs the target square on AH. Since $AF \cong AH$, `hilbert_square_transport` gives

$$S_{AF} \simeq_{\mathrm{ec}} S_{AH}.$$

Reversing the relation from Step 8 and composing by transitivity gives

$$R_{AB,HB} \simeq_{\mathrm{ec}} S_{AH}.$$

This is exactly the public II.11 conclusion.

11.8 Lean Formalization

11.8.1 The public theorem signature

The following is an ASCII transcription of the current compiling source:

```
theorem euclid_proposition_2_11
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B : Geo.Point)
    (hAB : Ne A B) :
    exists
      H
      R0 R1 R2 R3
      Q0 Q1 : Geo.Point,
      Geo.Between A H B /\
      IsRectangleContainedBy Geo
        R0 R1 R2 R3 A B H B /\
      IsSquare Geo A H Q0 Q1 /\
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo R0 R1 R2 R3)
        (hilbertParallelogramTerm Geo A H Q0 Q1) := by
```

The theorem is deliberately existential at the representation level. It does not expose the auxiliary square $ABCD$, the midpoint E, the extension point F, the intermediate squares used by I.47, or the cut witnesses required by II.2, II.3, and II.6.

11.8.2 The proposition-local theorem chain

The production module is factored into

```
proposition2_11_rectangle_cut_core
proposition2_11_rectangle_cut_between
proposition2_11_diagonal_cut_intersection
proposition2_11_rectangle_cut_exists
proposition2_11_II6_exists
proposition2_11_common_notion_3
proposition2_11_central_equality
proposition2_11_common_notion_3_second
proposition2_11_second_subtraction
proposition2_11_final_transport
proposition2_11_EA_lt_EB_aux
proposition2_11_construct_F_aux
proposition2_11_AF_lt_AB_aux
proposition2_11_construct_H
euclid_proposition_2_11
```

The source order places the content infrastructure before the final construction helpers, but the mathematical dependency graph is more naturally read in the construction-first order used in the preceding section.

11.8.3 The public wrapper

The public theorem first calls `proposition2_11_construct_H`, obtaining

$$C, D, E, F, H$$

together with the square, midpoint, the orders $D - E - A$ and $E - A - F$, $EF \cong EB$, $AF < AB$, $A - H - B$, $AF \cong AH$, noncollinearity, and the right angle EAB.

It then combines $D - E - A$ and $E - A - F$ to derive $D - A - F$ and passes the complete geometric package to `proposition2_11_final_transport`. Only the point H, the target rectangle representative, the target square representative, and the final equicomplementability witness escape the wrapper.

11.9 Relationship with Earlier Book II Propositions

Proposition II.11 directly reuses three earlier Book II theorems:

$$\text{II.2}, \qquad \text{II.3}, \qquad \text{II.6}.$$

II.6 supplies the first major normal form:

$$S_{AE} + R_{DF,AF} \simeq_{\text{sc}} S_{EF}.$$

II.3 decomposes the rectangle $R_{DF,AF}$ at A, and II.2 decomposes the square S_{AB} at the final cut H. Thus II.11 is not merely a Book I construction followed by Pythagoras. Its final content argument genuinely depends on the rectangle calculus developed earlier in Book II.

The role of II.6 is especially source-faithful: Euclid himself invokes II.6 in the classical proof of II.11. The uses of II.2 and II.3 are more architectural. They provide current formal normal forms for the final subtraction that Euclid performs directly in his configured square-and-rectangle diagram.

There is no direct theorem-level dependency on II.9 or II.10. II.11 therefore forms a new branch: it combines the earlier exact rectangle calculus of II.1–II.8 with the I.47 equicomplementability layer that became prominent in II.9–II.10.

11.10 Relationship with Book I Infrastructure

The construction and proof use a broad but classical Book I layer.

- I.46 constructs the initial square on AB and later the target square on AH.
- I.10 supplies the midpoint E of DA.
- I.16 and I.19 prove the strict metric comparison $EA < EB$ needed to orient the point F.
- I.20 proves the comparison leading to $AF < AB$.
- I.47 supplies the Pythagorean content identity in triangle AEB.
- standard congruence, right-angle, same-ray, betweenness, and parallelogram facts transport these statements between the concrete representatives.

The generic rectangle-cut helper additionally uses Euclidean parallel uniqueness, Pasch, parallelogram opposite-side congruences, and a parallel crossing-order theorem. These are library-level geometric facts, not new axioms introduced by II.11.

11.11 Formalization Notes and Hidden Geometric Data

11.11.1 The source order D-E-A-F is proved

The printed construction invites the reader to see immediately that the point F lies beyond A. Lean must prove this. It first establishes $EA < EB$, constructs F with $EF = EB$, and then uses same-ray information and betweenness trichotomy to isolate the branch $E - A - F$.

Consequently the four-point order

$$D - E - A - F$$

is a theorem of the construction, not an assumption copied from a diagram.

11.11.2 The final point H is not obtained by a diagrammatic intersection

The source construction naturally transfers AF onto AB. Hartshorne notes that the line-circle intersection property is relevant to Euclid's II.11 construction [5]. The present formal proof instead proves $AF < AB$ and then obtains H by unpacking the strict segment comparison.

This avoids importing an additional line-circle construction into the public proof while preserving the intended synthetic geometry.

11.11.3 Noncollinearity carries the right-angle information

The proof of I.47 requires the right triangle AEB to be formally nondegenerate. The point E lies on the side DA of the square, and the square already gives that D, A, B are noncollinear. Collinearity transport therefore yields

$$\neg\,\mathrm{Collinear}(E, A, B).$$

The right angle at A is then transported from the square arm AD to the same ray AE.

11.11.4 Rectangle cuts carry hidden witnesses L and N

The imported Book II APIs do not accept only the statement that a rectangle is cut at a point. They require a complete configured diagram: an upper cut point, a diagonal intersection, two strict betweenness relations, and two parallelograms.

The private rectangle-cut constructor creates these witnesses by parallel geometry and Pasch. They are mathematically meaningful because they certify that the formal scissors split corresponds to an actual geometric cut, but they do not belong in the public II.11 statement.

11.11.5 Common Notion 3 is not numerical subtraction

II.11 uses content cancellation twice. This is the most important semantic difference from II.10, whose final bridge was forward-only composition.

The local Common-Notion-3 theorem does not introduce additive inverses for scissors terms. It enlarges the existing equicomplementability witnesses so that the common figure is absorbed into the complements. The operation is therefore legitimate inside the positive finite-content calculus and should not be described as subtraction of real-valued areas.

11.11.6 Square and rectangle representatives are noncanonical

The construction, II.6, I.47, II.2, and II.3 may create different concrete figures representing the same side data. The formal proof must explicitly move among these representatives with square and rectangle transport. This is why the public theorem existentially quantifies the final figures rather than pretending that "the rectangle" or "the square" is a canonical object.

11.12 Axiomatic and Semantic Status

The production theorem introduces no new proposition-specific geometric axiom and no new content axiom. It contains no `sorry` or `admit`. The full project build was run after integrating II.11 and completed successfully with 1502 jobs.

The transitive axiom audit

```
#print axioms Geometry.euclid_proposition_2_11
```

reported

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are inherited project dependencies. The precise statement is therefore: II.11 adds no new axiom and no `sorry`; it is not literally axiom-free.

The semantic layers are

$$
\begin{array}{c}
\text{incidence/order/congruence and Euclidean parallel geometry} \\
\Downarrow \\
\text{square, midpoint, segment construction, strict segment comparison} \\
\Downarrow \\
\text{concrete rectangle and square representatives} \\
\Downarrow \\
\text{exact II.6 / II.3 / II.2 scissors identities} \\
\Downarrow \\
\text{I.47 equicomplementability + representative transport} \\
\Downarrow \\
\text{two Common-Notion-3 cancellations} \\
\Downarrow \\
\text{II.11 equicomplementability conclusion.}
\end{array}
$$

No numerical area function is used. No segment multiplication is used. The modern golden-ratio equation is an interpretive shadow only.

11.13 Dependency Summary

The actual production chain is

$$
\begin{gathered}
\text{Hilbert incidence, order, congruence, Euclidean parallel layer} \\
\Downarrow \\
\text{I.46 square + I.10 midpoint} \\
\Downarrow \\
\text{I.16 + I.19: } EA < EB \\
\Downarrow \\
\text{segment construction: } EF = EB, \quad E - A - F \\
\Downarrow \\
\text{I.20 + segment-construction uniqueness: } AF < AB \\
\Downarrow \\
\text{construct } H: \ A - H - B, \ AF = AH \\
\Downarrow \\
\text{local rectangle-cut constructor for configured Book II APIs} \\
\Downarrow \\
\text{II.6 + I.47 + square transport} \\
\Downarrow \\
\text{Common Notion 3: } R_{DF,AF} \simeq_{\mathrm{ec}} S_{AB} \\
\Downarrow \\
\text{II.3 + II.2 + rectangle transport} \\
\Downarrow \\
\text{Common Notion 3: } S_{AF} \simeq_{\mathrm{ec}} R_{AB,HB} \\
\Downarrow \\
\text{square transport } AF = AH \\
\Downarrow \\
\texttt{euclid\_proposition\_2\_11}.
\end{gathered}
$$

The implementation checkpoint is:

New geometric axiom in II.11:	no
New content axiom in II.11:	no
New proposition-local helper theorem:	yes
New reusable public helper theorem/module:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.2, II.3, II.6
Conclusion is exact `HilbertScissorsEq`:	no
Conclusion is equal-content / equicomplementability:	yes
Uses exact scissors internally:	yes
Uses content cancellation:	yes, twice
Uses numerical area:	no
Uses segment multiplication:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Full project build:	clean, 1502 jobs
Proposition-local `sorry/admit`:	no
Transitive `#print axioms`:	`propext`, `Classical.choice`, `bookZero_nullSegment1`, `Quot.sound`

11.14 Conclusion

Proposition II.11 combines the two main proof modes developed so far in Book II. Its geometric construction returns to the classical square, midpoint, produced segment, and right-triangle machinery of Book I. Its content proof then reuses the exact rectangle calculus of II.2, II.3, and II.6 together with the I.47 equicomplementability and square-transport machinery that became central in II.9 and II.10.

The most important new formal feature is cancellation. Twice, Euclid removes a common figure. Lean realizes each instance directly inside the equicomplementability definition; no additive inverse, numerical area, or segment field is introduced.

The second important feature is constructional. Euclid's diagram locates the cut point by the classical extension/layoff mechanism. The current Lean proof makes the required order explicit: it proves $EA < EB$, then $AF < AB$, and only then obtains H with $A - H - B$ and $AH = AF$. This is not a line-by-line replay of the source construction, but it is a fully synthetic reconstruction of the same geometric result.

The final theorem therefore states exactly the appropriate Book II content claim:

$$\mathrm{Rect}(AB, HB) \simeq_{\mathrm{ec}} \mathrm{Sq}(AH),$$

with no strengthening to exact scissors equality and no reduction to a numerical golden-ratio equation.

Chapter 12

Proposition II.12

12.1 Introduction

Proposition II.12 is the first proposition of Book II in which Euclid starts from a triangle with an *obtuse* angle and compares the square on the opposite side with the squares on the two sides containing that angle. In the lettering used by the current formal theorem, the triangle is ABC, the obtuse angle is BAC, and the perpendicular from B meets the carrier of AC at a point D lying beyond A:

$$D - A - C, \qquad BD \perp AC.$$

Euclid's content statement is

$$\mathrm{Sq}(BC) = \mathrm{Sq}(BA) + \mathrm{Sq}(AC) + 2\ \mathrm{Rect}(CA, AD).$$

Equivalently, in Euclid's wording, the square on BC is *greater by* twice the rectangle contained by CA, AD than the two squares on BA, AC.

The current Lean theorem reconstructs this statement without numerical area, segment multiplication, coordinates, trigonometry, or a numerical angle measure. It constructs concrete square representatives on AB, AC, and CB, a concrete rectangle representative contained by DA, AC, and proves the exact additive relation in the `HilbertScissorsEquicomplementable` layer.

A modern algebraic shadow is the obtuse case of the law of cosines. That interpretation is useful only as a mnemonic check. It is not the proof architecture of either Euclid or the Lean reconstruction. The actual source DAG is:

$$\text{I.12} \longrightarrow \text{II.4} \longrightarrow \text{Common Notion 2} \longrightarrow \text{I.47 twice.}$$

The production source begins with

```
import CGJteamLab.HilbertRightAngle
import CGJteamLab.Proposition12
import CGJteamLab.Proposition16
import CGJteamLab.Proposition47
import CGJteamLab.HilbertSquareTransport
import CGJteamLab.Proposition2_4
```

Thus Proposition II.12 has one direct earlier Book II theorem dependency, namely II.4. The essential Book I dependencies are I.12, I.16, I.46, and I.47, together with the established Hilbert incidence, order, congruence, parallel, and scissors infrastructure.

12.2 Euclid's Statement and Classical Configuration

12.2.1 The statement

In Heath's wording, Proposition II.12 states:

> In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.

For the triangle ABC with obtuse angle BAC, produce the side CA beyond A and let BD be perpendicular to the produced carrier. The source order is

$$D - A - C.$$

The final content identity is

$$\mathrm{Sq}(CB) = \mathrm{Sq}(AB) + \mathrm{Sq}(AC) + 2\ \mathrm{Rect}(CA, AD).$$

The orientation of the final rectangle is not mathematically essential, but it is formally visible. The production theorem chooses a representative contained by DA, AC; the side reversal and rectangle-representative independence are handled inside the established content transport layer.

12.2.2 The classical construction

The construction is short.

(1) Start with the nondegenerate triangle ABC and assume that $\angle BAC$ is obtuse.

(2) Produce CA through A.

(3) From B draw BD perpendicular to the produced line CA by I.12.

(4) The resulting order is
$$D - A - C.$$

The rest of the proposition is a content argument. The squares and rectangle appearing in the proof are not additional geometric hypotheses; they are the figures "on" or "contained by" the corresponding segments in the standard Book II sense.

Diagrammatic audit. The starting configuration is not sufficient to explain the proof. Three further figures are retained below because the geometric state genuinely changes: the reductio branch used to prove the position of the perpendicular foot, the pair of right triangles sharing the DB square, and the II.4 square dissection on DC cut at A. No separate figure is introduced merely for representation transport or for the final associative-commutative regrouping.

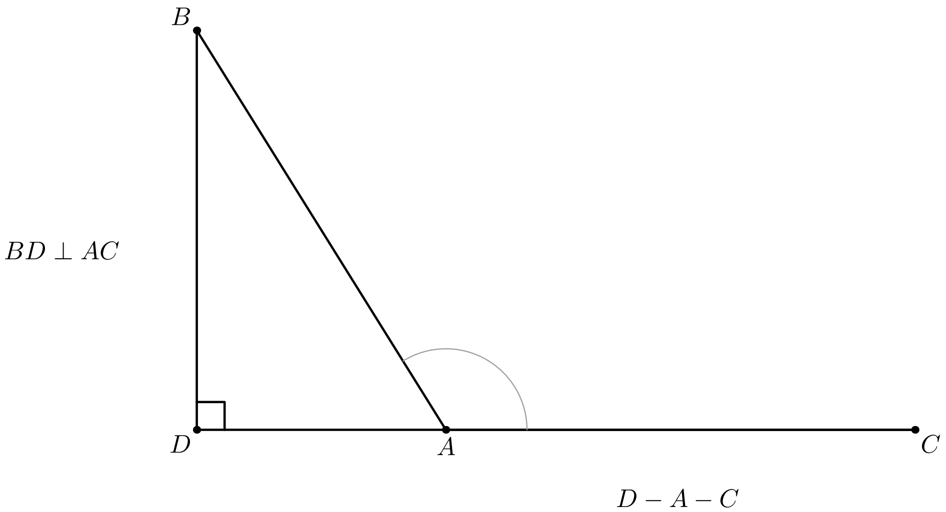


Figure 12.1: The classical starting configuration for Proposition II.12 in the lettering of the Lean source. The angle BAC is obtuse, BD is perpendicular to the carrier AC, and the source order is $D - A - C$. The light arc records only the qualitative obtuse-angle hypothesis. No square or rectangle is shown yet: this figure states the geometric input rather than the later content calculation.

12.3 Classical Proof

Euclid's proof is a direct comparison of one II.4 decomposition with two Pythagorean decompositions.

Since CD is cut at A, Proposition II.4 gives

$$\mathrm{Sq}(DC) = \mathrm{Sq}(DA) + \mathrm{Sq}(AC) + 2\ \mathrm{Rect}(DA, AC).$$

The rectangle may equally be read as contained by CA, AD; the present orientation anticipates the Lean representative.

Add the square on DB to both sides. This is Euclid's use of Common Notion 2:

$$\mathrm{Sq}(DC) + \mathrm{Sq}(DB) = \mathrm{Sq}(DA) + \mathrm{Sq}(DB) + \mathrm{Sq}(AC) + 2\ \mathrm{Rect}(DA, AC).$$

The triangle DCB is right at D, so I.47 gives

$$\mathrm{Sq}(CB) = \mathrm{Sq}(DC) + \mathrm{Sq}(DB).$$

The triangle DAB is also right at D, so I.47 gives

$$\mathrm{Sq}(AB) = \mathrm{Sq}(DA) + \mathrm{Sq}(DB).$$

Substituting these two equalities into the previous identity yields

$$\mathrm{Sq}(CB) = \mathrm{Sq}(AB) + \mathrm{Sq}(AC) + 2\ \mathrm{Rect}(DA, AC),$$

which is the required proposition.

The local classical dependency picture is therefore

$$\begin{gathered}
\text{triangle } ABC,\ \angle BAC \text{ obtuse} \\
\Downarrow \\
\text{I.12: construct } BD \perp CA \text{ produced} \\
\Downarrow \\
D - A - C \\
\Downarrow \\
\text{II.4 on } D - A - C \\
\mathrm{Sq}(DC) = \mathrm{Sq}(DA) + \mathrm{Sq}(AC) + 2R \\
\Downarrow \\
\text{Common Notion 2: add } \mathrm{Sq}(DB) \\
\Downarrow \\
\text{I.47 on } DCB \quad + \quad \text{I.47 on } DAB \\
\Downarrow \\
\mathrm{Sq}(CB) = \mathrm{Sq}(AB) + \mathrm{Sq}(AC) + 2R \\
\Downarrow \\
\text{II.12.}
\end{gathered}$$

Two features are worth emphasizing. First, there is no cancellation step in II.12. The entire content calculation is forward composition and addition. Second, Euclid's short proof presupposes the positional fact $D - A - C$. In the formal reconstruction that fact is a theorem, not a visual convention.

12.4 The Geometric Identity in the Current Formalization

The public theorem takes

```
(A B C : Geo.Point)
(hABC : Not (PrimCollinear Geo A B C))
(hObtuse : HilbertObtuseAngle Geo B A C)
```

and constructs all other geometric and representation data internally.

At the final interface Lean produces a point D satisfying

$$D - A - C,$$

the two right angles

$$\angle ADB, \qquad \angle CDB,$$

concrete square representatives on

$$AB, \qquad AC, \qquad CB,$$

and a concrete rectangle representative R contained by DA, AC.

The conclusion is

$$S_{CB} \simeq_{\mathrm{ec}} (S_{AB} + S_{AC}) + (R + R),$$

where $\simeq_{\mathrm{ec}}$ denotes `HilbertScissorsEquicomplementable`.

This is not an inequality in an ordered area field. Euclid's phrase "greater by twice the rectangle" is represented by an exact additive equal-content statement. The theorem therefore records not merely that one content is larger than another, but an explicit positive decomposition of the larger content into the smaller content plus two copies of the rectangle.

The conclusion is also not strengthened to `HilbertScissorsEq`. The existing I.47 interface returns equicomplementability, and the final theorem remains at that natural content level. Exact scissors equalities occur internally inside the II.4 block.

12.5 Reusable Book II Infrastructure Used

12.5.1 Proposition II.4

The sole direct earlier Book II theorem is `euclid_proposition_2_4`. Its production API is configured: it expects a square, two rectangle representatives, two auxiliary II.3 rectangles, their cut data, squares on the two parts, and the two cross rectangles.

The local wrapper

```
proposition2_12_ii4_DAC
```

specializes that API to the order

$$D - A - C.$$

Under the substitution

$$(A, B, C)_{\mathrm{II.4}} = (D, C, A),$$

the II.4 conclusion becomes

$$S_{DC} \simeq_{\mathrm{ec}} (S_{AD} + S_{AC}) + (R + R).$$

The two cross-rectangle arguments of II.4 represent the same "twice the rectangle" term. The wrapper therefore supplies the same concrete representative twice and uses only associative–commutative regrouping to place the result in the II.12 order.

12.5.2 Existential packaging of the II.4 diagram

The higher-level helper

```
proposition2_12_ii4_exists
```

starts only from

$$D - A - C$$

and constructs the complete geometric data required by the configured II.4 API. It constructs:

- a square on DC;
- the internal cut of that square at A;
- arbitrary representatives of the rectangles contained by DC, DA and DC, AC;
- the two auxiliary rectangles used by the II.3 blocks inside II.4;
- their internal cut diagrams;
- squares on AD and AC;
- one representative of the rectangle contained by DA, AC.

All these witnesses are hidden from the public theorem. The wrapper returns only the square representatives and the final II.4 content relation needed by the subsequent Pythagorean substitution.

12.5.3 Square representative transport

I.47, I.46, and the II.4 construction create independent concrete squares. The formal proof must therefore identify their contents explicitly.

Three representative problems occur:

(1) the two I.47 calls construct two different squares on DB;

(2) I.47 and II.4 construct different squares on DC;

(3) I.47 naturally uses a square on DA, while II.4 returns a square on the oppositely oriented segment AD.

The theorem `hilbert_square_transport` resolves all three situations. No independently constructed witnesses are identified by definitional equality.

12.5.4 Equicomplementability addition and reassembly

The helper imported from I.47,

```
i47_aux_equicomplementable_add
```

lifts equal-content relations through formal addition. Together with `equicomplementable_symm`, `equicomplementable_trans`, and reflexivity, it implements the Common-Notion-2 part of the proof.

The proposition-local helper

```
proposition2_12_scissors_final_bridge
```

contains no new geometry. It is the pure content form of Euclid's final substitution.

12.6 New Proposition-Local Parallelogram-Cut Infrastructure

A major formal feature of II.12 is hidden behind the short public theorem. The low-level II.4 API still requires complete geometric cut diagrams. Proposition II.12 therefore develops a general internal cut constructor for an arbitrary parallelogram.

The proposition-local chain is

```
proposition2_12_parallelogram_L_on_DE
proposition2_12_left_cut_parallelogram
proposition2_12_upper_cut_between
proposition2_12_diagonal_cut_intersection
proposition2_12_parallelogram_cut_exists
```

Start with a parallelogram $BCED$ and an interior point M:

$$B - M - C.$$

The target is to construct points L, X such that

$$D - L - E, \qquad D - X - C, \qquad M - X - L,$$

and the two cut pieces are parallelograms

$$LDBM, \qquad CELM.$$

The first helper proves that the upper cut point lies on the opposite carrier. Completing CEM to the parallelogram $CELM$ gives

$$EL \parallel MC.$$

The original parallelogram gives

$$ED \parallel CB.$$

Because B, M, C are collinear, both EL and ED are parallel to the same base carrier through E. Euclidean parallel uniqueness forces their carriers to coincide, yielding

$$D, L, E \text{ collinear.}$$

The next helper constructs the complementary parallelogram $LDBM$. The delicate point is $L \neq D$. From $B - M - C$ one has the strict segment comparison

$$CM < CB.$$

Opposite-side congruences in the two parallelograms transport this to

$$EL < ED,$$

hence $L \neq D$. The required two pairs of opposite parallel sides then follow by collinearity and parallel transport.

The order helper strengthens collinearity to

$$D - L - E.$$

The three relevant corresponding segment congruences are

$$BM \cong DL, \qquad BC \cong DE, \qquad MC \cong LE.$$

Hilbert's three-point order transport sends the known order $B - M - C$ to the opposite side.

Finally

```
proposition2_12_diagonal_cut_intersection
```

uses Pasch in triangle CBD. The cut line ML enters through CB at M and cannot meet BD because

$$DB \parallel ML.$$

It must therefore meet CD at an interior point X. Parallel crossing order then gives

$$D - X - C, \qquad M - X - L.$$

The wrapper

```
proposition2_12_parallelogram_cut_exists
```

packages exactly the witnesses required by the existing Book II configured cut APIs.

This theorem is structurally more general than II.12 because it applies to an arbitrary parallelogram, not merely a square or rectangle. It remains in `Proposition2_12.lean` under a proposition-specific name. Promoting it to a permanent general rectangle/parallelogram API should be a separate architectural decision rather than an automatic consequence of its use here.

12.7 Proof Architecture of the Reconstruction

12.7.1 Step 1: encode the obtuse angle synthetically

The production module begins with

```
def HilbertObtuseAngle
    [HilbertIncidence Geo]
    (A O B : Geo.Point) : Prop :=
  Exists fun X : Geo.Point =>
    And
      (HilbertRightAngle Geo A O X)
      (HilbertAngleLess Geo A O X A O B)
```

Thus AOB is obtuse when a right angle with the same first arm OA is strictly smaller than AOB. The definition uses only the established synthetic angle-order relation. There is no numerical 90° measure.

The preliminary theorem

```
proposition2_12_obtuse_not_right
```

shows that an obtuse angle cannot itself be right. If both were right, all right angles would be congruent; transport of the strict inequality would produce an angle strictly smaller than itself.

12.7.2 Step 2: construct the perpendicular foot

The helper

```
proposition2_12_perpendicular_foot_exists_full
```

applies Euclid I.12 to the carrier AC and the external point B. Because ABC is noncollinear, B is off that carrier. The theorem returns points D, R with the incidence data

$$A, D, C \text{ collinear}, \qquad R, D, A \text{ collinear}, \qquad R, A, C \text{ collinear},$$

and a right angle

$$\angle RDB.$$

The auxiliary point R belongs to the interface of I.12. It witnesses the carrier arm of the constructed right angle but does not occur in the final II.12 statement.

12.7.3 Step 3: prove the source-critical order $D - A - C$

The central geometric theorem is

```
proposition2_12_obtuse_perpendicular_foot_between
```

First Lean rules out $D = A$. If the foot coincided with A, right-angle transport would show that BAC is right, contradicting `proposition2_12_obtuse_not_right`.

The proof then excludes the possibility that D and C lie on the same ray from A. Suppose

$$\mathrm{SameRay}(A, D, C).$$

Extend AD beyond D to a point E. In triangle BAD, I.16 gives an exterior-angle inequality

$$\angle BAD < \angle BDE.$$

Because D and C are on the same ray from A,

$$\angle BAD = \angle BAC.$$

Because ADB is right and $A - D - E$, the exterior angle BDE is right. The obtuse hypothesis also supplies a right angle BAX with

$$\angle BAX < \angle BAC.$$

All right angles are congruent, so

$$\angle BDE \cong \angle BAX.$$

Transporting the first strict inequality therefore yields

$$\angle BAC < \angle BAX.$$

Combining this with

$$\angle BAX < \angle BAC$$

gives a strict angle-order cycle, hence an angle strictly smaller than itself, contradicting irreflexivity.

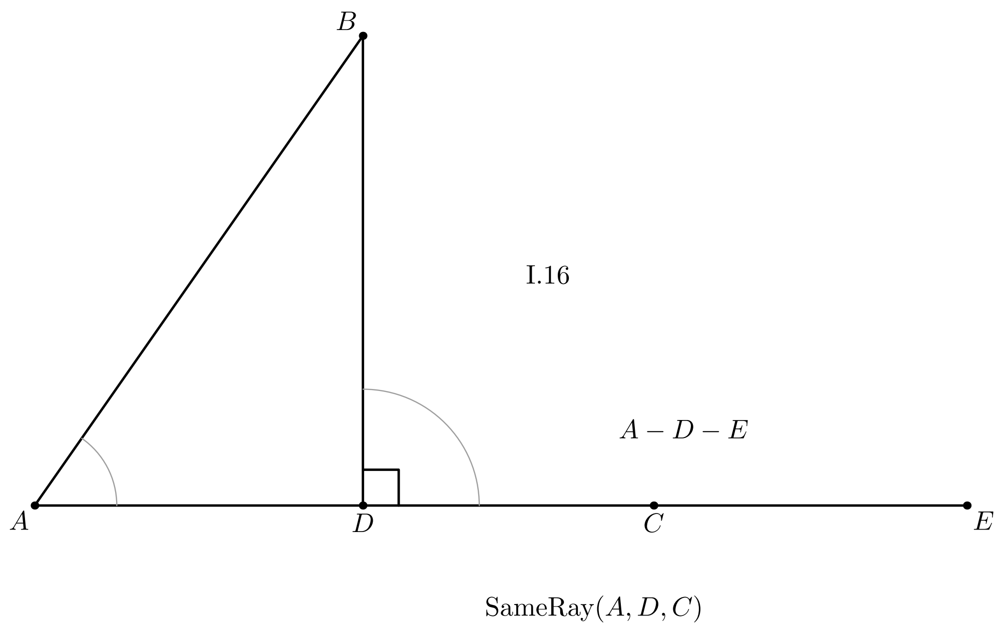


Figure 12.2: The wrong-position branch excluded by the Lean proof. Assuming that D lies on the same ray from A as C, extend AD through D to E. Triangle BAD then has the exterior angle BDE, which is right, and I.16 gives the strict comparison between the interior angle BAD and that exterior angle. Because AD and AC are the same ray, BAD is the angle BAC relevant to the obtuse hypothesis. The separate obtuse-witness ray AX is deliberately omitted: the displayed branch is a reductio configuration, so no Euclidean picture can realize all contradictory assumptions simultaneously.

Betweenness trichotomy on the collinear triple A, D, C now leaves exactly one possible order:

$$\boxed{D - A - C}.$$

This is the principal proof obligation that is implicit in Euclid's printed diagram but explicit in the formal reconstruction.

12.7.4 Step 4: obtain both right triangles

The I.12 construction gives a right angle at D with one carrier arm. The helper

```
proposition2_12_right_angle_collinear_transport
```

transports it to

$$\angle ADB.$$

Since $D - A - C$, the rays DA and DC coincide. Same-ray angle transport therefore gives

$$\angle ADB \cong \angle CDB,$$

and right-angle transport yields

$$\angle CDB$$

right as well.

The formal proof also derives the noncollinearity of the two triangles DAB and DCB from the original noncollinearity of ABC and the strict order $D - A - C$.

12.7.5 Step 5: apply I.47 to triangle DAB

Euclid I.47 gives concrete squares and the equal-content identity

$$S_{AB} \simeq_{\mathrm{ec}} S_{DA} + S_{DB}^{(1)}.$$

The first square on DB is only one representative. Nothing in the type system identifies it with a later square independently erected on the same segment.

12.7.6 Step 6: apply I.47 to triangle DCB

The second I.47 call gives

$$S_{CB} \simeq_{\mathrm{ec}} S_{DC} + S_{DB}^{(2)}.$$

The helper

```
proposition2_12_normalize_second_pythagoras_DB
```

uses square transport to replace $S_{DB}^{(2)}$ by the representative $S_{DB}^{(1)}$ from the first Pythagorean block.

The wrapper

```
proposition2_12_pythagorean_package_common_DB
```

therefore returns the normalized pair

$$S_{AB} \simeq_{\mathrm{ec}} S_{DA} + S_{DB},$$

$$S_{CB} \simeq_{\text{ec}} S_{DC} + S_{DB}.$$

This is exactly the pair of I.47 identities used in Euclid's classical substitution, now synchronized at the representation level.

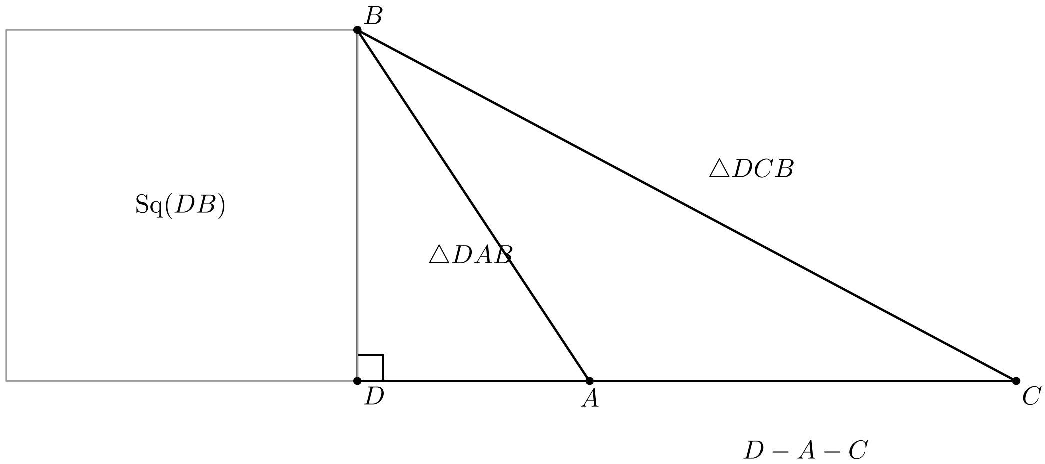


Figure 12.3: The two right triangles to which I.47 is applied. The strict order $D - A - C$ places DAB inside DCB, and both triangles use the same perpendicular leg DB. The single square drawn on DB makes visible the common summand that must be normalized between the two independent I.47 calls. The other Pythagorean squares are omitted because drawing them would obscure, rather than clarify, the shared-DB geometry.

12.7.7 Step 7: construct and apply the II.4 block

From the already proved order

$$D - A - C,$$

the theorem

```
proposition2_12_ii4_exists
```

constructs the complete configured II.4 diagram and returns

$$S_{DC}^{(2)} \simeq_{\text{ec}} (S_{AD}^{(2)} + S_{AC}) + (R + R).$$

Here $S_{DC}^{(2)}$ and $S_{AD}^{(2)}$ are independent representatives constructed inside the II.4 block.

12.7.8 Step 8: normalize the II.4 squares

The theorem

```
proposition2_12_normalize_ii4_squares
```

performs two transports.

First,

$$S_{DC} \simeq_{\text{ec}} S_{DC}^{(2)}$$

$$D - A - C$$

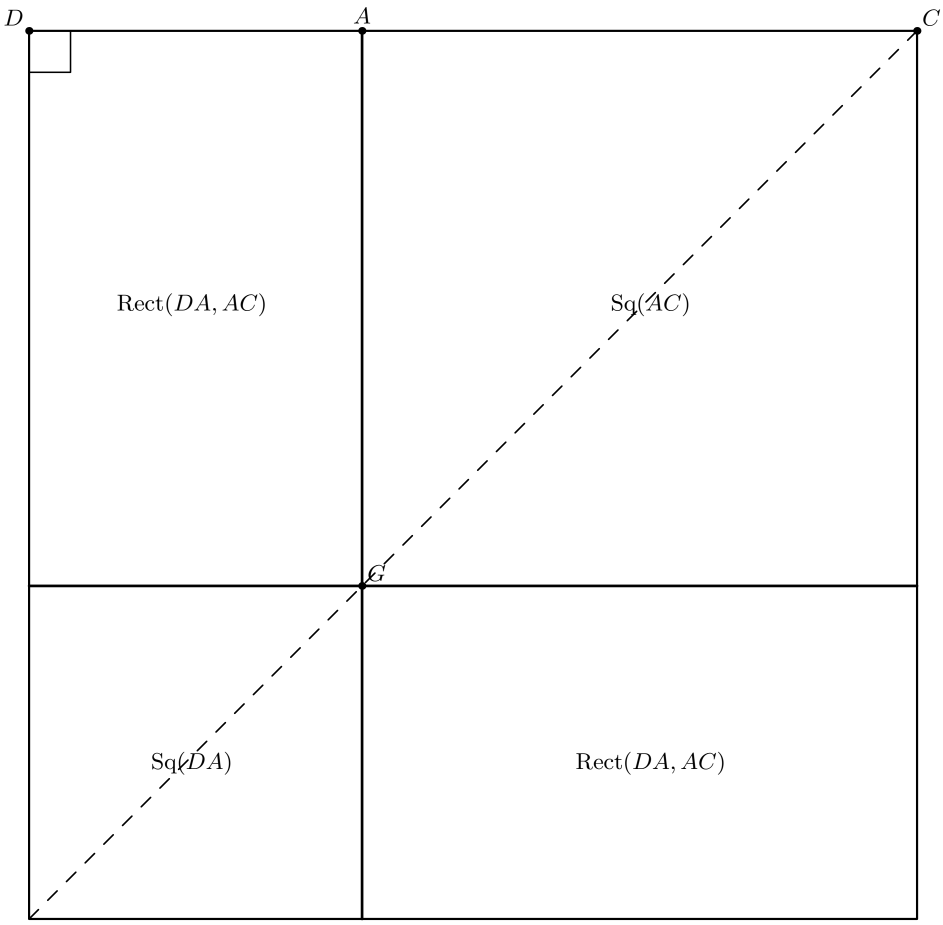


Figure 12.4: Proposition II.4 applied to the segment DC cut at A. The outer square is the square on DC; the vertical through A and the horizontal through its intersection with the diagonal divide it into the square on DA, the square on AC, and two rectangles contained by DA, AC. The dashed diagonal records the classical II.4 construction. Only the cut point A and the diagonal intersection G are named, because the additional configured cut witnesses constructed by Lean certify this same dissection but do not define new content regions.

because both are squares on the same oriented segment DC.

Second,

$$S_{DA} \simeq_{\text{ec}} S^{(2)}_{AD}.$$

This time the endpoint order is reversed, so the proof explicitly constructs the congruence

$$DA \cong AD$$

before applying square transport.

The normalized II.4 statement is therefore

$$S_{DC} \simeq_{\text{ec}} (S_{DA} + S_{AC}) + (R + R).$$

12.7.9 Step 9: perform Euclid's final Common-Notion-2 substitution

The pure content helper

```
proposition2_12_scissors_final_bridge
```

takes exactly the three normalized input relations

$$S_{AB} \simeq_{\text{ec}} S_{DA} + S_{DB},$$

$$S_{CB} \simeq_{\text{ec}} S_{DC} + S_{DB},$$

$$S_{DC} \simeq_{\text{ec}} (S_{DA} + S_{AC}) + (R + R).$$

It first adds the unchanged DB square to the II.4 relation:

$$S_{DC} + S_{DB} \simeq_{\text{ec}} ((S_{DA} + S_{AC}) + (R + R)) + S_{DB}.$$

Composing with the second I.47 block gives the same right-hand side for S_{CB}.

The next step is only a formal reassociation and commutation of the finite multiset sum:

$$((S_{DA} + S_{AC}) + (R + R)) + S_{DB} = (S_{DA} + S_{DB}) + (S_{AC} + (R + R)).$$

Lean proves this with

```
ac_rfl
```

which is purely syntactic bookkeeping. It contributes no geometric or content theorem.

Finally the first I.47 block, used symmetrically, replaces

$$S_{DA} + S_{DB}$$

by S_{AB}. Transitivity gives

$$S_{CB} \simeq_{\text{ec}} (S_{AB} + S_{AC}) + (R + R).$$

No cancellation theorem is used anywhere in this final bridge.

Diagrammatic audit. Steps 8 and 9 introduce no additional geometric state. Square normalization changes representatives, not the underlying square-on-a-segment geometry, and `ac_rfl` only reassociates and commutes a finite scissors sum. For that reason there is no Figure 12.5 for the final substitution: Figures 12.3 and 12.4, together with the dependency DAG, already display all spatial information used by the final composition.

12.8 Lean Formalization

12.8.1 The public theorem signature

The following is an ASCII transcription of the current compiling source:

```
theorem euclid_proposition_2_12
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C : Geo.Point)
    (hABC : Not (PrimCollinear Geo A B C))
    (hObtuse : HilbertObtuseAngle Geo B A C) :
    exists D : Geo.Point,
    exists AB0 AB1 AC0 AC1 CB0 CB1 : Geo.Point,
    exists T0 T1 T2 T3 : Geo.Point,
      Geo.Between D A C /\
      HilbertRightAngle Geo A D B /\
      HilbertRightAngle Geo C D B /\
      IsSquare Geo A B AB1 AB0 /\
      IsSquare Geo A C AC0 AC1 /\
      IsSquare Geo C B CB1 CB0 /\
      IsRectangleContainedBy Geo
        T0 T1 T2 T3 D A A C /\
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo
          C B CB1 CB0)
        ((hilbertParallelogramTerm Geo
            A B AB1 AB0 +
          hilbertParallelogramTerm Geo
            A C AC0 AC1) +
         (hilbertParallelogramTerm Geo
            T0 T1 T2 T3 +
          hilbertParallelogramTerm Geo
            T0 T1 T2 T3)) := by
```

The theorem is deliberately existential at the representation level. The public interface does not expose:

- the I.12 auxiliary carrier point R;
- the two intermediate squares on DA, DB, DC used by I.47;
- the independent II.4 square representatives;
- the square on DC used to instantiate II.4;
- the auxiliary rectangle representatives inside II.4;
- the upper cut points and diagonal intersections required by the configured rectangle-split APIs.

Only the geometrically meaningful foot D, the three target squares, the target rectangle, and the final content relation escape the wrapper.

12.8.2 The proposition-local theorem chain

The production module is factored into

```
HilbertObtuseAngle
proposition2_12_obtuse_not_right
```

```
proposition2_12_right_angle_collinear_transport
proposition2_12_perpendicular_foot_exists
proposition2_12_right_angle_swap
proposition2_12_perpendicular_foot_exists_full
proposition2_12_obtuse_perpendicular_foot_between
proposition2_12_pythagorean_package
proposition2_12_same_base_squares_equicomplementable
proposition2_12_normalize_second_pythagoras_DB
proposition2_12_pythagorean_package_common_DB
proposition2_12_ii4_DAC
proposition2_12_scissors_final_bridge
proposition2_12_final_from_normalized_blocks
proposition2_12_normalize_ii4_squares
proposition2_12_final_with_independent_ii4_squares
proposition2_12_parallelogram_L_on_DE
proposition2_12_left_cut_parallelogram
proposition2_12_upper_cut_between
proposition2_12_diagonal_cut_intersection
proposition2_12_parallelogram_cut_exists
proposition2_12_ii4_exists
euclid_proposition_2_12
```

The source order reflects the incremental proof development. The mathematical dependency graph is more naturally read in the construction-first order of the preceding section:

$$\text{obtuse foot} \longrightarrow \text{two I.47 blocks} \longrightarrow \text{II.4 block} \longrightarrow \text{final scissors bridge.}$$

12.8.3 The public wrapper

The public theorem first calls

```
proposition2_12_pythagorean_package_common_DB
```

and obtains:

$$D - A - C,$$

the two right angles, concrete squares on AB, DA, DB, CB, DC, and the two normalized I.47 relations sharing one DB square.

It then calls

```
proposition2_12_ii4_exists
```

on the same point D. This is an important witness-continuity property: the II.4 block is built on the very perpendicular foot already used by the two I.47 blocks, not on an independently reconstructed collinear configuration.

Finally it calls

```
proposition2_12_final_with_independent_ii4_squares
```

which normalizes the independent II.4 square representatives and invokes the pure final content bridge.

Only the target witnesses are returned.

12.9 Relationship with Earlier Book II Propositions

Proposition II.12 directly reuses exactly one earlier Book II theorem:

$$\boxed{\text{II.4}}.$$

This is source-faithful. Euclid explicitly invokes II.4 after constructing the perpendicular foot. In the formal proof II.4 supplies the content expansion of the square on DC:

$$S_{DC} \simeq_{\text{ec}} (S_{DA} + S_{AC}) + (R + R).$$

There is no direct dependency on II.11. Proposition II.12 therefore begins a new branch immediately after the construction problem II.11. Its content proof returns to the square-expansion theorem II.4 and combines it with Pythagoras.

There is also no theorem-level shortcut through II.13. The acute-angle analogue is later in Euclid and is not used to prove the obtuse case.

The large configured API of II.4 internally embodies earlier rectangle decomposition machinery, but II.12 does not replace the source proof by calls to II.2 or II.3. Its direct Book II proof dependency remains II.4.

12.10 Relationship with Book I Infrastructure

The main Book I roles are classical.

- I.12 constructs the perpendicular from B to the carrier AC.
- I.16 is used in the additional formal proof that the foot lies beyond A, giving the exact order $D - A - C$.
- I.47 is applied twice, once to DAB and once to DCB.
- I.46 constructs the concrete squares needed by the internally instantiated II.4 block.

The formal reconstruction also uses the established Hilbert layer:

- strict betweenness and betweenness trichotomy;
- same-ray construction and angle-arm replacement;
- strict angle order, transitivity, transport, and irreflexivity;
- congruence symmetry and transitivity;
- right-angle transport and congruence of all right angles;

- Euclidean parallel uniqueness;
- parallelogram fourth-vertex construction and opposite-side congruences;
- Pasch and parallel crossing order.

These are infrastructure theorems, not new axioms introduced by II.12.

12.11 Formalization Notes and Hidden Geometric Data

12.11.1 Obtuse means strict angle order, not numerical measure

The predicate `HilbertObtuseAngle` does not assign a real number to an angle. It gives a synthetic witness: a right angle with the same first arm that is strictly smaller.

Consequently the proof of II.12 never invokes 90°, a cosine, or a trigonometric comparison. The only operations on angles are geometric: congruence, strict order, same-ray transport, opposite extension, and transitivity.

12.11.2 The order $D - A - C$ is proved, not assumed

This is the source-critical hidden datum of II.12. The classical diagram makes the order visually immediate once the angle at A is obtuse. Lean cannot import that picture as a theorem.

The proof derives $D - A - C$ from:

obtuseness + I.12 perpendicularity + I.16 exterior-angle order + betweenness trichotomy.

Thus the position of the perpendicular foot is part of the mathematical proof object.

12.11.3 The second right angle needs ray transport

I.12 directly provides one perpendicular direction. For the second Pythagorean triangle the proof needs

$$\angle CDB$$

right. This is not a separate perpendicular construction. Since $D - A - C$, the points A, C lie on the same ray from D, so the right angle is transported from ADB to CDB.

12.11.4 The II.4 cut witnesses are real geometry

The current II.4 API does not merely accept the assertion that DC is divided at A. It expects explicit upper cut points, diagonal intersections, betweenness data, and parallelograms for the component rectangle splits.

The local parallelogram-cut constructor creates those witnesses by parallel geometry and Pasch. They disappear from the public theorem, but they certify that the formal scissors decomposition corresponds to an actual synthetic geometric dissection.

12.11.5 Square representatives are noncanonical

Every phrase "the square on XY" hides an existential representative in the formal development. Two separate calls to I.47 may therefore return different squares on DB, and the II.4 construction may return another square on DC than the one already produced by I.47.

The formal proof never assumes uniqueness of square points. It uses `hilbert_square_transport` to prove content equivalence from segment congruence.

12.11.6 The orientation DA versus AD is not definitional

The first I.47 block naturally produces the square on DA. The II.4 block, under its configured parameter order, produces a square on AD.

These two squares represent the same unoriented classical side, but Lean requires the endpoint reversal to be justified explicitly by segment congruence before square transport can be applied.

12.11.7 "Greater by" is exact additive content

The final theorem does not introduce a strict order on contents. Instead it proves

$$S_{CB} \simeq_{\text{ec}} S_{AB} + S_{AC} + R + R.$$

This directly witnesses what the excess consists of.

Therefore II.12 does not require:

- an area function;
- positivity of area;
- cancellation;
- an order relation on scissors contents;
- subtraction of figures.

12.11.8 `ac_rfl` is only bookkeeping

The final bridge uses `ac_rfl` once to expose the subterm

$$S_{DA} + S_{DB}$$

inside a larger formal sum. This operation proves only equality modulo associativity and commutativity of the multiset addition.

It is not a replacement for Euclid's geometric proof. All geometric and content equalities required for the substitution have already been established before this regrouping step.

12.12 Axiomatic and Semantic Status

The production theorem introduces no new proposition-specific geometric axiom and no new content axiom. It contains no `sorry` or `admit`. The production source compiled cleanly, and the final theorem was subjected to a transitive axiom audit.

The command

```
#print axioms Geometry.euclid_proposition_2_12
```

reported

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are inherited project dependencies. The precise statement is therefore: II.12 adds no new axiom and no `sorry`; it is not literally axiom-free.

The semantic layers are

$$
\begin{gathered}
\text{incidence/order/congruence and Euclidean parallel geometry}\\
\Downarrow\\
\text{synthetic obtuse-angle order} + \text{I.12 perpendicular construction}\\
\Downarrow\\
\text{proof of the source order } D - A - C\\
\Downarrow\\
\text{two right triangles and two I.47 equicomplementabilities}\\
\Downarrow\\
\text{configured II.4 construction and its scissors content identity}\\
\Downarrow\\
\text{square representative transport}\\
\Downarrow\\
\text{equicomplementability addition, regrouping, and transitivity}\\
\Downarrow\\
\text{II.12 additive equal-content conclusion.}
\end{gathered}
$$

No numerical area function is used. No segment multiplication is used. No trigonometric function or numerical angle measure is used. The modern law-of-cosines formula is an interpretive shadow only.

12.13 Dependency Summary

The actual production chain is

$$
\begin{gathered}
\text{Hilbert incidence, order, congruence, Euclidean parallel layer} \\
\Downarrow \\
\texttt{HilbertObtuseAngle}\text{: right angle strictly below } BAC \\
\Downarrow \\
\text{I.12: perpendicular foot } D \text{ on the carrier } AC \\
\Downarrow \\
\text{I.16 + angle-order transport + trichotomy} \\
\Downarrow \\
D - A - C, \quad \angle ADB \text{ right} \\
\Downarrow \\
\text{same-ray transport: } \angle CDB \text{ right} \\
\Downarrow \\
\text{I.47 on } DAB \quad + \quad \text{I.47 on } DCB \\
\Downarrow \\
\text{square transport: one common } DB \text{ representative} \\
\Downarrow \\
\text{local parallelogram-cut constructor} \\
\Downarrow \\
\text{construct the complete configured II.4 block on } D - A - C \\
\Downarrow \\
\text{II.4: } S_{DC} \simeq_{\mathrm{ec}} (S_{DA} + S_{AC}) + (R + R) \\
\Downarrow \\
\text{normalize } DC \text{ and } DA/AD \text{ square representatives} \\
\Downarrow \\
\text{Common Notion 2 in equicomplementability: add } S_{DB} \\
\Downarrow \\
\text{I.47 substitution + formal regrouping} \\
\Downarrow \\
\texttt{euclid\_proposition\_2\_12}.
\end{gathered}
$$

The implementation checkpoint is:

New geometric axiom in II.12:	no
New content axiom in II.12:	no
New proposition-local helper theorem:	yes
New reusable public helper theorem/module:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.4
Conclusion is exact `HilbertScissorsEq`:	no
Conclusion is equal-content / equicomplementability:	yes
Uses exact scissors internally:	yes, through II.4
Uses content cancellation:	no
Uses numerical area:	no
Uses segment multiplication:	no
Uses numerical angle measure / trigonometry:	no
Production source compile:	clean (user verified)
Production theorem axiom audit:	clean baseline
Proposition-local `sorry/admit`:	no
Transitive `#print axioms`:	`propext`, `Classical.choice`, `bookZero_nullSegment1`, `Quot.sound`

12.14 Conclusion

Proposition II.12 preserves Euclid's short top-level proof DAG:

$$\boxed{\text{I.12} \longrightarrow \text{II.4} \longrightarrow \text{add } DB^2 \longrightarrow \text{I.47 twice} \longrightarrow \text{II.12.}}$$

The formal development is longer for two reasons, neither of which changes the mathematical argument.

First, Lean proves the geometric position of the perpendicular foot. The statement

$$D - A - C$$

is derived synthetically from the obtuse-angle hypothesis using I.16, angle order, and betweenness trichotomy. This is the principal hidden geometric fact in the classical diagram.

Second, the existing II.4 interface is deliberately concrete and diagram-explicit. The II.12 module therefore constructs the required parallelogram cuts and normalizes the independently chosen square representatives before performing Euclid's final content substitution.

Once these two formal obligations are discharged, the final scissors calculation is essentially the classical proof. No subtraction, cancellation, numerical area, segment field, coordinate model, cosine, or numerical angle measure is introduced.

The final theorem states the appropriate Book II content relation:

$$\boxed{S_{CB} \simeq_{\text{ec}} (S_{AB} + S_{AC}) + (R + R)}$$

for a rectangle representative R contained by DA, AC. This is the direct synthetic meaning of Euclid's assertion that the square on the side subtending the obtuse angle is greater by twice the indicated rectangle.

Appendix A

Greenberg: Foundational and Semantic Control

A.1 Purpose and scope

This appendix records the Greenberg-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.12. As in Book I, the purpose is not to force a one-to-one correspondence between Euclid's proposition numbering and Greenberg's organization. The relevant question is which parts of the formal architecture Greenberg controls and where the synthetic content theory must be supplied by another source.

A.2 Greenberg's role at the beginning of Book II

Greenberg distinguishes numerical area from equal content. He describes the latter through finite dissection and reassembly of polygonal figures but does not develop the full synthetic equal-content calculus; he refers the reader to Hartshorne for that theory [4, 5].

This source boundary, already important in the area propositions of Book I, continues unchanged at the opening of Book II. Greenberg's Euclidean parallel and parallelogram theory is well suited to the geometric shell of rectangle constructions and cuts. It is not the source for the exact scissors relation used by the Lean theorems.

A.3 Comparison table

Table A.1: Euclid II.1–II.12 in Greenberg's Hilbert-style framework

Euclid	**Greenberg counterpart**	**Status in Greenberg**	**Source-audit remarks and relevance to formalization**
II.1	Rectangle dissection supported geometrically by Euclidean parallel and parallelogram theory.	No dedicated Greenberg proof of II.1 was located in the current audit. Greenberg supplies the geometric setting and the distinction between equal content and numerical area, but not the project's exact content calculus.	The formal theorem should therefore be read as Greenberg-compatible geometry plus Hilbert/Hartshorne content semantics plus Lean scissors infrastructure. The exact `HilbertScissorsEq` conclusion is not a theorem attributed to Greenberg.
II.2	Square cut into two rectangles; same Euclidean rectangle/parallelogram shell as II.1.	No separate Greenberg counterpart was located. The source-level role remains the same as for II.1.	Greenberg controls the geometry of the configured square and parallel cut, but the side-order transport and exact scissors equality belong to the formal rectangle/content layer.
II.3	Rectangle and square configuration assembled from the same parallel, parallelogram, and right-angle geometry.	No separate Greenberg counterpart was located. II.3 introduces no new Greenberg-side semantic principle beyond the boundary already identified at II.1.	The formal reduction of II.3 to the binary II.1 theorem is a library architecture fact. Greenberg remains a geometry control source rather than a proof source for the content identity.
II.4	Square on a divided line assembled from square, rectangle, parallel, and parallelogram geometry.	No dedicated Greenberg proof of II.4 was located in the current Book II audit. The source continues to supply the Euclidean geometric shell and the distinction between equal content and numerical area.	The formal theorem's exact `HilbertScissorsEq` and its composition of earlier rectangle identities are project-level content infrastructure, not a result attributed to Greenberg.
II.5	Bisected-segment configuration with an additional internal cut.	Greenberg supplies the Hilbert-style order, congruence, and Euclidean geometry needed to formulate midpoint and rectangle configurations, but no dedicated synthetic content proof of II.5 was located in the current audit.	The Lean midpoint transport and exact rectangle/scissors composition remain inside the formal project. No numerical area interpretation is imported from Greenberg.
II.6	Bisected segment extended beyond an endpoint, with the corresponding square and rectangle configuration.	The source remains useful for order, midpoint, parallel, and rectangle geometry. No dedicated Greenberg exact-content counterpart to II.6 was located in the current audit.	The formal use of II.1, II.3, II.4 and exact scissors addition is a Lean-library architecture fact. Greenberg is not cited as supplying that proof.
II.7	Arbitrary internal cut of a segment, square on the whole, squares on the parts, and a rectangle contained by the whole and one part.	No separate Greenberg proposition-level proof was located in the current audit. The same geometry/content boundary identified at II.1 remains in force.	The current Lean proof composes exact II.4 and II.3 and performs geometric side-order transport. Its exact scissors conclusion and representative reuse belong to project infrastructure rather than to a Greenberg content calculus.

Continued on next page

Table A.1 – continued from previous page

Euclid	Greenberg counterpart	Status in Greenberg	Source-audit remarks and relevance to formalization
II.8	Arbitrary internal cut followed by an endpoint extension $BD \cong BC$, with a square on the produced whole AD and rectangle/square representatives.	No separate Greenberg proposition-level proof was located in the current audit. The same Euclidean geometry/content boundary remains in force.	Greenberg controls the order, congruence, parallel, and rectangle geometry and the warning against numerical-area substitution. The exact II.4/II.7 composition, square representative transport, and fourfold scissors sum are project infrastructure.
II.9	Bisected segment with a second internal cut, perpendicular and parallel construction, auxiliary parallelogram, right triangles, and squares.	No separate Greenberg proposition-level proof of II.9 was located in the current audit. Greenberg remains a control source for Hilbert-style order, congruence, perpendicular/parallel geometry, and for the distinction between equal content and numerical area.	The Lean proof returns to a Book I geometric construction and ends in `HilbertScissorsEquicomplementable`, not exact `HilbertScissorsEq`. The Pythagorean/scissors composition and square representative transport are project infrastructure and are not attributed to Greenberg.
II.10	Bisected segment extended beyond an endpoint, with perpendicular/parallel construction, auxiliary parallelogram, produced intersection, right triangles, and squares.	No separate Greenberg proposition-level proof of II.10 was located in the current audit. Greenberg remains a control source for Hilbert-style order, congruence, separation, perpendicular/parallel geometry, and for the distinction between equal content and numerical area.	The formal proof is the external-section companion to II.9. Its directed intersection argument and right-triangle geometry remain on the Greenberg-side geometric layer, while the four I.47 blocks, square transport, and final `HilbertScissorsEquicomplementable` composition are project infrastructure.
II.11	Golden-ratio / extreme-and-mean-ratio construction built from a square, midpoint, extension, perpendicular/rectangle geometry, and a Pythagorean comparison.	Greenberg gives a straightedge-and-compass construction of a golden rectangle using a square, the midpoint of a side, an extension determined by an equal segment, and the Pythagorean theorem. This is not presented as a synthetic equal-content proof of Euclid II.11, but it is a direct geometric control on the same metric configuration.	The Lean construction similarly uses a square, midpoint, segment layoff, strict order, and I.47-type geometry. The final equicomplementability cancellation calculus remains project/Hilbert content infrastructure and is not attributed to Greenberg.
II.12	Obtuse triangle with a perpendicular to the produced side, explicit order of the perpendicular foot, two right triangles, and square/rectangle representatives.	No separate Greenberg proposition-level proof of II.12 was located in the current audit. Greenberg remains a control source for Hilbert-style order, congruence, perpendicular geometry, and for the distinction between equal content and numerical area.	The Lean proof makes the source-critical order $D - A - C$ explicit by angle-order and exterior-angle geometry. The II.4/I.47 content composition, `HilbertScissorsEquicomplementable`, and square-representative transport remain project/Hilbert content infrastructure and are not attributed to Greenberg.

A.4 Structural observations from the audit

Three observations should be retained.

- The Euclidean geometry needed to construct rectangles and parallel cuts is naturally expressible in Greenberg's Hilbert-style framework.
- Equality of content is a separate semantic layer. Greenberg's decision not to develop its full synthetic calculus is a source boundary, not a defect to be repaired by importing numerical area formulas.
- The formal project's exact scissors and representative-independence theorems should therefore be documented as project infrastructure, not as Greenberg results.

A.5 II.1–II.8: one stable geometry/content split

Nothing in II.2–II.7 changes Greenberg's role established at II.1. The block uses elementary Euclidean rectangle, square, midpoint, order, and parallel geometry, while the equality-of-content conclusions belong to the separate content calculus. The formal distinctions between binary splitting, side-order rotation, square expansion, midpoint transport, the additive II.7 bridge, and the II.8 endpoint-extension/square-transport composition occur inside the Lean API and do not create new Greenberg source layers.

This is precisely why the appendix should grow by structural blocks rather than by one narrative section per proposition.

A.6 II.9: the same semantic boundary with a different geometric engine

II.9 changes the internal Lean proof architecture without changing Greenberg's source role. The rectangle-normal-form chain of II.1–II.8 is no longer used. Instead the proof develops a perpendicular/parallel construction, isosceles and right-triangle angle geometry, a parallelogram, and four Pythagorean content blocks. These geometric ingredients remain compatible with Greenberg's Hilbert-style Euclidean framework.

The semantic boundary is, however, even more visible than before. The public II.9 theorem lands at `HilbertScissorsEquicomplementable`. Greenberg's distinction between equal content and numerical area continues to be the relevant warning, while the particular formal equicomplementability calculus is not a theorem attributed to Greenberg.

A.7 II.10: the external companion keeps the same semantic boundary

II.10 confirms that the geometric engine introduced at II.9 is not tied to an internal cut. The new point lies beyond the endpoint, so the formal proof must control the produced intersection in the strict orders $E - B - G$ and $F - D - G$. This makes plane separation, line uniqueness, order, and same-ray transport especially visible, all within the Hilbert-style Euclidean geometry for which Greenberg remains a useful control source.

The content conclusion does not move toward numerical area. As in II.9, the public type is `HilbertScissorsEquicomplementable`. The current audit therefore records II.9 and II.10 as a companion pair with different order configurations but the same geometry/content boundary.

A.8 II.12: the perpendicular-foot order is geometry, not content

II.12 adds a new source-critical geometric obligation without changing Greenberg's semantic role. Euclid's diagram displays the perpendicular foot on the produced side in the order

$$D - A - C.$$

The formal reconstruction proves that order from the obtuse-angle hypothesis, the perpendicular construction, angle order, and the exterior-angle theorem. This belongs entirely to the Hilbert-style incidence/order/congruence side of the architecture.

Once the order and the two right triangles are available, the proof passes to the separate content layer: II.4 expands the square on DC, two I.47 blocks supply the Pythagorean comparisons, and the final result is assembled in `HilbertScissorsEquicomplementable`. Greenberg remains a control source for the geometric shell and for the warning that equal content is not numerical area; the particular scissors calculus is project infrastructure.

A.9 Result of the Greenberg audit

At the II.12 checkpoint Greenberg remains a foundational and geometric control source. He supports the distinction

$$\text{synthetic content} \neq \text{numerical area}$$

and the Euclidean geometry of rectangles, parallels, and parallelograms. For II.1–II.8 the exact finite-dissection conclusions, and for II.9–II.12 the more general equicomplementability conclusions, belong to the formal development and to the Hilbert/Hartshorne content side of the source hierarchy.

A.10 II.11: golden-ratio geometry and content

For II.11 Greenberg contributes more than a generic background framework. His exercises give a straightedge-and-compass construction of a golden rectangle beginning with a square, then a midpoint of one side, an extension determined by an equal segment, and a Pythagorean argument [4]. This is a direct geometric relative of the metric shell used by Euclid II.11 and by the Lean construction.

The source role must nevertheless remain limited. Greenberg's construction does not supply the project's two Common-Notion-3 equal-content cancellations. Those are part of the separate synthetic content layer. II.11 therefore strengthens Greenberg's relevance on the geometry/construction side without changing the geometry-versus-content boundary already recorded in this appendix.

A.11 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.12. No new Greenberg-specific boundary appears between these propositions. Later entries should extend the comparison table proposition by proposition, while new narrative sections should be added only when a genuinely new construction, axiomatic, or semantic block appears.

Appendix B

Hartshorne: Equal Content and the Algebra of Areas

B.1 Purpose and scope

This appendix records the Hartshorne-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.12. Its purpose is not to create a second proposition-by-proposition commentary. Instead, it records what Hartshorne contributes to the semantic interpretation of Euclid's rectangle and square equalities, which propositions he singles out, and which source-level distinctions should remain stable as Book II grows.

Hartshorne is especially important here because he separates Euclid's equality of plane figures from numerical area. In Section 22 he defines equal content synthetically and states that the propositions of Book II are valid when Euclid's equality of figures is read in that sense [5].

B.2 Hartshorne's organization of Book II

Hartshorne's treatment places Book II inside the theory of equal content rather than inside a prior arithmetic of numerical lengths. A rectangle described by two segments can therefore play a product-like geometric role without those segments first becoming numbers. This is the background of the familiar interpretation of Book II as an "algebra of areas" [5].

The phrase is structural, not a claim that Euclid has introduced a field of segment lengths. Modern formulas such as

$$a(b+c) = ab + ac$$

are useful mnemonics for the pattern of a geometric decomposition, but the objects compared by Euclid remain plane figures.

Hartshorne's brief Euclid summary explicitly lists II.1 and II.4, but does not list II.2 or II.3 among its selected Book II entries. In particular, II.4 is recorded as the statement that the square on the whole line equals the squares on its two segments plus twice the rectangle on the two segments [5]. The omission of II.2 and II.3 should not be read as a negative judgment about them: his general area discussion states that the results of Book II are valid for equal content.

B.3 Comparison table

Table B.1: Euclid II.1–II.12 in Hartshorne's equal-content framework

Euclid	Hartshorne counterpart	Status in Hartshorne	Source-audit remarks and relevance to formalization
II.1	The rectangle contained by two lines is the sum of the rectangles contained by one of them and the segments of the other.	Hartshorne lists II.1 explicitly in his brief Euclid summary and, more generally, places Book II inside the theory of equal content.	This is the cleanest source match for the formal rectangle-distribution API. The Lean reconstruction strengthens this direct-dissection instance to exact `HilbertScissorsEq`, but that strengthening is a project-level fact and should not be attributed to Hartshorne's general equal-content relation.
II.2	Square on the whole line decomposed into the two rectangles pairing the whole with the two parts.	II.2 is not separately listed in Hartshorne's selected brief Book II summary; it is covered by his general statement that the Book II equalities are valid for equal content.	The source supports the synthetic content reading, while the Lean proof exposes an additional representation issue: the two orders of the containing segments are reconciled geometrically by rectangle rotation rather than by numerical commutativity.
II.3	Rectangle on the whole and one part decomposed into the rectangle on the two parts together with the square on that part.	II.3 is likewise not isolated in the selected brief summary; its semantic status is supplied by Hartshorne's general Book II equal-content result.	The formal theorem is an exact-scissors specialization of the binary II.1 API. The square is used as a representative of the rectangle contained by the same segment twice. Unlike II.2, no side-order transport is required.
II.4	Square on the whole decomposed into the two squares on the parts together with twice the rectangle contained by the parts.	Hartshorne explicitly lists II.4 in his brief Euclid summary, and Section 22 supplies the equal-content semantics for the equality of the figures.	The Lean theorem strengthens the configured case to exact `HilbertScissorsEq`. It composes II.2 with two II.3 instances and treats "twice the rectangle" as two independent representatives of the same rectangle content, without numerical area or segment multiplication.
II.5	The rectangle on the unequal parts of an internally divided bisected segment, together with the square on the displacement from the midpoint, equals the square on the half.	Hartshorne explicitly singles out II.5, together with II.6 and II.11, in his discussion of the area identities used later in the construction of the regular pentagon. The proposition remains within the synthetic equal-content theory rather than numerical area.	For the formal reconstruction the significant point is that the midpoint condition and the rectangle identity still live below segment arithmetic. The current Lean proof reaches exact `HilbertScissorsEq` through earlier Book II rectangle decompositions and geometric transport; the modern difference-of-squares formula is only an interpretive shadow.

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Table B.1 – continued from previous page

Euclid	**Hartshorne counterpart**	**Status in Hartshorne**	**Source-audit remarks and relevance to formalization**
II.6	The rectangle contained by the whole extended line and the added segment, together with the square on the half, equals the square on the half plus the added segment.	Hartshorne explicitly mentions II.6 in the same geometrical-algebra block as II.5 and II.11. He also notes that Proposition III.36 is proved by several applications of II.6 together with I.47.	This gives II.6 a source role stronger than a generic Book II example: it is a reusable equal-content identity in Hartshorne's later Euclidean development. The Lean theorem again strengthens the configured statement to exact `HilbertScissorsEq`, but does so by II.1, II.3, II.4, midpoint transport, and formal scissors addition rather than by numerical algebra.
II.7	Square on the whole together with the square on one chosen part equals twice the rectangle contained by the whole and that part, together with the square on the remaining part.	II.7 is not separately singled out in Hartshorne's brief selected list in the same way as II.4–II.6; its semantic status is nevertheless covered by his general statement that the propositions of Book II are valid for equal content.	The current Lean theorem proves the configured identity by exact `HilbertScissorsEq`, composing II.4 with II.3 on the reversed divided segment and a geometric rectangle swap. The modern formula is only a documentary shadow; no numerical area or segment multiplication is used.
II.8	Four times the rectangle contained by the whole and one part, together with the square on the remaining part, equals the square on the produced segment made from the whole and a congruent copy of the chosen part.	No separate II.8 entry has been identified in the brief selected Book II list used for the present audit; its semantic status is covered by Hartshorne's general statement that Book II equalities are valid for equal content.	The Lean theorem sharpens the configured identity to exact `HilbertScissorsEq`. It composes II.4 with II.7 on the reversed cut and transports the square on BD to the square on CB through rectangle representative uniqueness, without numerical area or segment multiplication.
II.9	Sum of the squares on the unequal parts of a bisected line, expressed through Euclid's right-triangle construction and four uses of the Pythagorean theorem.	Hartshorne's equal-content framework remains the semantic control: Book II square identities are statements about content of plane figures, not numerical area formulas or a pre-existing multiplication of segment lengths.	The current Lean theorem is the first configured Book II proposition in this sequence whose final relation is `HilbertScissorsEquicomplementable` rather than exact `HilbertScissorsEq`. The modern formula $AD^2 + DB^2 = 2(AC^2 + CD^2)$ is therefore only a mnemonic for the geometric-content statement.
II.10	External-section companion to II.9: squares on the produced whole and on the added segment are compared with twice the squares on the half and on the half-plus-added segment.	No separate II.10 entry has been identified in the brief selected Book II list used for the present audit; its semantic status is covered by Hartshorne's general statement that Book II equalities are valid for equal content.	The current Lean theorem, like II.9, concludes `HilbertScissorsEquicomplementable` rather than exact `HilbertScissorsEq`. The modern identity $(2a + x)^2 + x^2 = 2a^2 + 2(a + x)^2$ is only a mnemonic; no numerical area or segment multiplication is used.

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Table B.1 – continued from previous page

Euclid	Hartshorne counterpart	Status in Hartshorne	Source-audit remarks and relevance to formalization
II.11	Division of a segment in extreme and mean ratio: the rectangle on the whole and the smaller part has the same content as the square on the larger part.	Hartshorne discusses II.11 explicitly, identifies it as the area/content form of extreme-and-mean-ratio division, and records its use in the construction leading to the regular pentagon. He also notes the relevant line–circle intersection property in the construction.	The Lean theorem matches Hartshorne's equal-content semantics and concludes at the equicomplementability level. Internally it combines exact II.6, II.3, and II.2 scissors identities with I.47, representative transport, and two Common-Notion-3 cancellations. The final cut point is obtained through synthetic segment comparison rather than by importing the source's line–circle construction.
II.12	Obtuse-angle square identity: the square on the side subtending the obtuse angle equals the two squares on the containing sides together with twice the rectangle cut off by the perpendicular on the produced side.	No separate II.12 entry has been identified in the brief selected Book II list used for the present audit. Its semantic status is nevertheless covered by Hartshorne's general statement that the propositions of Book II are valid for equal content.	The Lean theorem expresses Euclid's phrase "greater by twice the rectangle" as the positive additive relation $S_{CB} \simeq_{\mathrm{ec}} (S_{AB} + S_{AC}) + (R + R)$. This stays in Hartshorne's equal-content semantics: no numerical area order, segment multiplication, or law-of-cosines calculation is introduced.

B.4 Semantic observations retained from Hartshorne

Several points should remain stable as later propositions are added.

- Euclid's equality of plane figures in the area block should not be silently replaced by equality of real-valued areas.
- The phrase "rectangle contained by" two segments names a geometric figure through its adjacent side magnitudes; it is not itself a numerical multiplication operation.
- Hartshorne's equal content is the appropriate general semantic control for Book II. Exact finite equidecomposability is a stronger local conclusion and must be justified proposition by proposition.
- Modern algebraic identities are useful documentary shadows of the synthetic configurations, not substitutes for the geometric proof.
- Hartshorne's discussion of property (Z) belongs to the converse direction in the theory of content. Nothing in II.1–II.4 requires recovering metric data from a hypothesis of equal content.

B.5 II.1–II.8: from distribution through midpoint identities to endpoint-extension composition

The first eight propositions now display three stages of Book II structure.

II.1–II.4 establish and compose the basic rectangle calculus. II.1 supplies the general additive mechanism, II.2 exposes geometric side-order symmetry, II.3 gives a direct rectangle-plus-square specialization, and II.4 composes those results into the square-of-a-sum pattern.

II.5 and II.6 then add midpoint/order information. They are companion difference-of-squares identities: II.5 treats an unequal cut inside a bisected segment, whereas II.6 treats a segment added beyond the endpoint. Hartshorne explicitly singles out both propositions in his later discussion of geometrical algebra. II.7 removes the midpoint condition again and composes the already established square and rectangle identities for an arbitrary cut. II.8 then adds a produced endpoint segment congruent to one part and combines II.4 with II.7, plus representative transport for the square on that produced segment. The formal project keeps all four results in the same synthetic content layer while allowing their Lean DAGs to differ.

For II.6 the modern mnemonic

$$a^2 + (2a + x)x = (a + x)^2$$

for II.7 the mnemonic

$$(a + b)^2 + a^2 = 2(a + b)a + b^2,$$

and for II.8

$$4(a + b)b + a^2 = (a + 2b)^2$$

should all be read exactly as mnemonics. In the current Lean theorems the repeated rectangles are concrete scissors terms, the square terms are concrete square representatives, and the final equalities are obtained inside the synthetic content layer rather than by numerical algebra.

B.6 II.9: from exact rectangle dissections to equicomplementable square identities

II.9 is the first proposition in the current Book II sequence for which the formal conclusion itself makes the distinction between exact scissors equality and the broader equal-content relation unavoidable. The configured theorems II.1–II.8 land at `HilbertScissorsEq`; II.9 lands at `HilbertScissorsEquicomplementable` because its proof is assembled from the current formal I.47 Pythagorean blocks and square-representative transports.

This does not weaken Hartshorne's semantic role. On the contrary, his interpretation of Euclid's Book II equalities as equal content is exactly the level at which II.9 should be read. The familiar algebraic shadow

$$AD^2 + DB^2 = 2(AC^2 + CD^2)$$

remains explanatory notation only; neither Euclid nor the Lean theorem has introduced numerical area or multiplication in a segment field at this stage.

B.7 II.10: the external member of the Pythagorean pair

II.10 preserves the semantic lesson of II.9 while changing the order configuration. The second point of section is moved beyond the endpoint, and the source proof again builds a common square through four applications of I.47. Hartshorne's equal-content interpretation remains exactly the right level for the result: the identity concerns contents of concrete square figures, not values of a prior numerical area function.

The current Lean implementation makes the II.9–II.12 block precise without making one proposition depend on the other. Each theorem reconstructs its own geometric configuration and lands at `HilbertScissorsEquicomplementable`; only lower-level geometric and representation infrastructure is shared.

B.8 II.11: extreme and mean ratio as a content construction

II.11 is more directly represented in Hartshorne than II.9 or II.10. Hartshorne explicitly describes the division point K on a segment AB by the condition that the rectangle on BK, BA has the same content as the square on AK, and explains that this is the area-language form of division in extreme and mean ratio [5]. He then uses precisely this content identity in the route to the regular pentagon.

This source observation aligns closely with the public Lean type: the conclusion is not numerical equality of areas and not a ratio equation, but equicomplementability between a rectangle contained by the whole and smaller part and a square on the remaining part.

The construction layer is not identical. Hartshorne notes the line–circle intersection property used in Euclid's construction of II.11. The current Lean proof proves $AF < AB$ and obtains the cut point by unpacking `HilbertSegmentLess`. The content layer is nevertheless strongly source-aligned: II.6 and I.47 produce the first common-figure comparison, and two positive-content cancellation arguments implement Euclid's removal of common figures.

B.9 II.12: "greater by" as positive equal-content decomposition

II.12 is a useful test of Hartshorne's equal-content reading because Euclid's statement uses the comparative phrase "greater by". The present formalization does not interpret this through an ordered numerical area function. Instead it exhibits the excess explicitly:

$$S_{CB} \simeq_{\mathrm{ec}} (S_{AB} + S_{AC}) + (R + R).$$

Thus the content of the square on CB is represented as the content of the two indicated squares together with two copies of the rectangle.

This is exactly the level at which Hartshorne's Section 22 framework is useful: the figures are compared by synthetic equal content, while the familiar algebraic or law-of-cosines formula remains only an interpretive shadow. The specific proof architecture—II.4, two I.47 blocks, representative transport, and formal addition—belongs to the current Lean development rather than to a separate Hartshorne proof script for II.12.

B.10 Result of the Hartshorne audit

At the II.12 checkpoint Hartshorne supplies the main semantic control for the opening of Book II. His contribution is not a separate proof script for each of II.1–II.12, but a framework in which their equalities are understood as equalities of content between plane figures. II.5, II.6, and II.11 are additionally singled out by Hartshorne for their later use in geometrical constructions.

The Lean development refines this framework in two ways that are specific to the current implementation. First, the configured II.1–II.8 theorems land at exact `HilbertScissorsEq`, while II.9–II.12 land at `HilbertScissorsEquicomplementable`. Second, the formal predicate

`IsRectangleContainedBy` makes representative independence and side order explicit. These refinements should remain distinguished from claims made by Hartshorne himself.

B.11 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.12. The structural picture now branches at the strength of the formal content relation:

$$\begin{array}{c}\text{rectangle/square language}\\ \Downarrow \\ \text{synthetic equal content}\\ \Downarrow \\ \begin{cases}\text{exact } \texttt{HilbertScissorsEq}, & \text{II.1–II.8},\\ \texttt{HilbertScissorsEquicomplementable}, & \text{II.9–II.12.}\end{cases}\end{array}$$

Within the Lean rectangle calculus, II.1 is the reusable distributive core, II.2 tests geometric side-order symmetry, II.3 returns to direct specialization of the binary II.1 theorem, and II.4 demonstrates composition. II.5 and II.6 add midpoint/order geometry, II.7 returns to an arbitrary cut, and II.8 adds an endpoint extension and square representative transport. II.9 is the first architectural break in this sequence: it does not reuse a Book II proposition, but reconstructs Euclid's Book I geometric route and then composes four I.47 content blocks. Hartshorne continues to control the semantic reading as equal content; the stronger exact-scissors status of II.1–II.8 remains a proposition-specific feature of the Lean development.

Appendix C

Hilbert: Content and Segment Arithmetic

C.1 Purpose and scope

This appendix records the Hilbert-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.12. Hilbert is relevant in two distinct ways: Chapter IV supplies the synthetic theory of plane content, whereas Section 15 later develops an arithmetic of segments. The documentary purpose is to keep those layers separate and to identify which one is actually used by the current formal reconstruction.

C.2 Hilbert's organization of the relevant theory

Hilbert's Chapter IV distinguishes finite equidecomposability from the more general relation of equicomplementability. The theory of plane area begins on the basis of Groups I–IV and does not need to be introduced as a numerical area function [2].

For the present project the useful hierarchy is

$$\text{exact finite dissection} \Longrightarrow \text{equicomplementability} \Longrightarrow \text{later numerical area semantics},$$

with no converse inserted merely for convenience.

Section 15 is a different layer. Hilbert there defines segment addition and a geometric product, using a fixed unit segment and parallel construction, and then proves the familiar algebraic laws, including commutativity and distributivity [2]. That later arithmetic is a valuable control source precisely because it shows what would be required if Book II were to be formalized through segment multiplication. The current Lean development does not take that route.

C.3 Comparison table

Table C.1: Euclid II.1–II.12 against Hilbert's content and segment-arithmetic layers

Euclid	**Hilbert counterpart**	**Status in the Grundlagen**	**Source-audit remarks and relevance to formalization**
II.1	Chapter IV finite decomposition and content theory; later Section 15 distributivity of segment multiplication is a structural analogue.	No one-to-one numbered Hilbert theorem reproducing Euclid II.1 was located in the current audit. The relevant material is organized by content theory and, later, by segment arithmetic rather than by Euclid's Book II numbering.	The formal proof belongs to the earlier content/dissection layer. Additivity is obtained from an actual rectangle cut and representative transport, not by invoking the later distributive law for a segment product.
II.2	Chapter IV content plus the geometric symmetry of a rectangle; Section 15 commutativity is only a later arithmetic analogue.	Again the source correspondence is structural rather than proposition-level. Hilbert's later commutative segment multiplication must not be read backward into the scissors proof.	The Lean theorem makes the distinction concrete: the required exchange of the two containing sides is implemented by cyclic rotation of a rectangle representative, preserving exact scissors content.
II.3	Chapter IV content/dissection; later Section 15 distributivity gives the arithmetic pattern only after segment multiplication has been defined.	No separate proposition-level Hilbert counterpart was located. The source supports the distinction between geometric content operations and the later arithmetic of segments.	II.3 is a direct specialization of the binary II.1 cut with the chosen segment used as the fixed side. The square is packaged as a rectangle contained by the same segment twice. No segment product and no side-order transport are needed.
II.4	Chapter IV finite content composition; later Section 15 distributivity and commutativity give an arithmetic analogue of the square expansion.	No one-to-one Hilbert proposition corresponding to Euclid II.4 was located in the current audit. Hilbert's source role remains architectural.	The formal theorem composes exact II.2 with two exact II.3 instances and a geometric rectangle rotation. Despite the modern shadow $(a+b)^2 = a^2 + b^2 + 2ab$, no segment product, unit segment, or numerical area is used.
II.5	Chapter IV content/dissection; Section 15 segment multiplication supplies only a later arithmetic analogue of the identity.	No one-to-one Hilbert proposition corresponding to Euclid II.5 was located in the current audit. The source role remains the separation of synthetic content from the later arithmetic of segments.	The formal theorem combines midpoint and order information with exact II.1 and II.3 rectangle decompositions, representative transport, and scissors addition. The modern shadow $(a-x)(a+x) + x^2 = a^2$ is not used as a proof object; no segment product, unit segment, numerical area, or cancellation axiom is introduced.

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Table C.1 – continued from previous page

Euclid	Hilbert counterpart	Status in the Grundlagen	Source-audit remarks and relevance to formalization
II.6	Chapter IV content/dissection; Section 15 segment multiplication supplies only a later arithmetic analogue of the external-section identity.	No one-to-one Hilbert proposition corresponding to Euclid II.6 was located in the current audit. The relevant source role remains the separation between synthetic content and later segment arithmetic.	The formal theorem combines order from the midpoint/extension configuration with exact II.1, II.3, and II.4 decompositions, rectangle transport, and a purely additive scissors bridge. The modern shadow $a^2 + (2a + x)x = (a + x)^2$ is not a proof object; no segment product, unit segment, numerical area, subtraction, or cancellation axiom is introduced.
II.7	Chapter IV content/dissection; Section 15 segment multiplication supplies only a later arithmetic analogue of the arbitrary-cut identity.	No one-to-one Hilbert proposition corresponding to Euclid II.7 was located in the current audit. The relevant source role remains the separation between synthetic plane content and later segment arithmetic.	The formal theorem composes exact II.4 with II.3 on the reversed divided segment, uses a geometric rectangle rotation, and finishes by additive scissors rearrangement. No unit segment, segment product, numerical area, subtraction, cancellation, or new content principle is introduced.
II.8	Endpoint extension with $BD \cong BC$, square on AD, and four copies of a rectangle contained by the whole and the selected part.	The extension, congruence, rectangle, and square geometry belongs below the later segment-arithmetic layer; Chapter IV supplies the synthetic content setting.	The production theorem remains in exact scissors calculus: II.4 and II.7 are composed, the square on congruent sides is transported through rectangle representative uniqueness, and four copies are a finite formal sum. Hilbert's Section 15 multiplication is not used.
II.9	Right-angle, parallel, parallelogram, square, and Pythagorean configuration for a bisected segment with a second internal cut.	Hilbert's separation between elementary geometry, the Chapter IV theory of content/equicomplementability, and later segment arithmetic is directly relevant.	The current Lean proof stays below segment multiplication and numerical area. Unlike II.1–II.8, its public conclusion is `HilbertScissorsEquicomplementable`; this is a natural Chapter IV-style content statement assembled from four I.47 blocks and square transport, not an exact-dissection theorem asserted by the code.
II.10	External endpoint extension, directed intersection, right-angle and parallelogram geometry, concrete squares, and four Pythagorean content blocks.	Hilbert's separation between Groups I–IV geometry, Chapter IV content/equicomplementability, and later Section 15 segment arithmetic remains the relevant source architecture; no one-to-one numbered counterpart is required.	The Lean proof stays below segment multiplication and numerical area. As in II.9, the public conclusion is `HilbertScissorsEquicomplementable`; the directed intersection and square-representative transports are geometric/content proof obligations, not an invocation of later segment arithmetic.

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Table C.1 – continued from previous page

Euclid	**Hilbert counterpart**	**Status in the Grundlagen**	**Source-audit remarks and relevance to formalization**
II.11	Extreme-and-mean-ratio construction with a square, midpoint, strict segment order, rectangle representatives, I.47, and removal of common content.	Hilbert Chapter IV directly supports the distinction between exact equidecomposability and equicomplementability; Section 18 explicitly records that removing equidecomposable polygons from equidecomposable polygons leaves equicomplementable remainders. Section 15 segment arithmetic is a separate, later layer.	The formal proof uses this Chapter IV-style positive content logic twice: Common Notion 3 is implemented inside the equicomplementability relation, not as subtraction in a segment field. Exact II.6/II.3/II.2 dissections feed the argument, while the public conclusion remains equicomplementability.
II.12	Obtuse-angle order, perpendicular construction, two right triangles, concrete squares, one rectangle representative, II.4, and additive equicomplementability.	Hilbert's Groups I–IV geometry controls incidence, order, congruence, right angles, and the Euclidean construction shell, while Chapter IV supplies the synthetic content/equicomplementability setting. No one-to-one numbered Hilbert counterpart to Euclid II.12 is required; Section 15 segment arithmetic is a separate later layer.	The Lean proof remains wholly below segment multiplication and numerical area. Unlike II.11, it needs no content cancellation: the "greater by" clause is implemented as a positive additive equicomplementability $S_{CB} \simeq_{\mathrm{ec}} (S_{AB} + S_{AC}) + (R + R)$, with square transport handling noncanonical representatives.

C.4 Structural observations retained from Hilbert

The following distinctions should be preserved as Book II grows.

- Plane content and segment arithmetic are separate theories in the Grundlagen and should remain separate in the documentation.
- Exact finite dissection is stronger than the general content relation; when Lean proves `HilbertScissorsEq`, that fact comes from the local proof architecture rather than from a blanket identification of all equal-content figures with equidecomposable ones.
- Segment multiplication is not primitive. It requires a unit segment, proportion geometry, and parallel constructions before algebraic laws such as commutativity and distributivity can even be stated.
- Consequently, a distributive-looking Euclidean proposition can be formalized below the arithmetic layer when its geometry is literally a finite cut-and-paste argument.

C.5 II.1–II.8: exact dissection below segment arithmetic

The opening eight propositions give a useful stress test of the separation. II.1 establishes a reusable exact rectangle split. II.2 shows that the apparent exchange of factors can still be implemented geometrically by rotating a rectangle presentation. II.3 then shows that even this extra symmetry operation is unnecessary when the ordered side data already match the binary II.1 interface. II.4 is the first compositional step: it combines II.2 with two II.3 expansions and uses only exact scissors addition and representation transport. II.5 then adds midpoint and order data and composes the same rectangle/scissors mechanisms into the internal unequal-parts identity. II.6 moves the second cut outside the original segment and, in the current Lean DAG, uses II.4 directly to normalize the square on the produced segment. II.7 then returns to an arbitrary internal cut and combines II.4 with a reversed II.3 normal form. II.8 adds an external congruent segment and composes II.4 with II.7; the only new local bridge is square representative transport. All four remain entirely below segment arithmetic.

The resulting formal chain is therefore

$$\begin{gathered}
\text{rectangle geometry}\\
\Downarrow\\
\text{exact scissors dissection}\\
\Downarrow\\
\text{II.1}\\
\Downarrow\\
\text{II.2 and II.3 specializations}\\
\Downarrow\\
\text{II.4 by composition}\\
\Downarrow\\
\text{II.5 by midpoint/order transport and composition}\\
\Downarrow\\
\text{II.6 by II.1} + \text{II.3} + \text{II.4} + \text{midpoint transport}\\
\Downarrow\\
\text{II.7 by II.4} + \text{reversed II.3} + \text{rectangle rotation}\\
\Downarrow\\
\text{II.8 by II.4} + \text{II.7 on reversed cut} + \text{square representative transport}
\end{gathered}$$

not

$$\text{segment field} \longrightarrow \text{multiplication} \longrightarrow \text{algebraic simplification.}$$

C.6 II.9: equicomplementability becomes visible in the public type

Through II.8 the configured Book II theorems could be documented at the stronger exact-dissection level `HilbertScissorsEq`. II.9 is the first current proposition for which the public type returns to the more general content relation `HilbertScissorsEquicomplementable`. This makes Hilbert's architectural separation especially useful: the theorem belongs to the plane-content layer, while the later arithmetic of segments remains uninvoked.

The formal route is also geometrically Hilbertian in spirit. Betweenness, noncollinearity, perpendiculars, parallels, right angles, congruent segments, and parallelograms are established first; concrete square representatives are then compared in the content layer. No numerical area function is used to bridge those levels.

C.7 II.10: the same Chapter IV level with an external order configuration

II.10 strengthens the evidence that the return to general equicomplementability at II.9 is an architectural feature of the current Pythagorean proof route. The new configuration adds a produced point beyond the endpoint and requires explicit plane-separation, order, and same-ray control before the four I.47 content blocks can be composed.

Nothing in this change crosses Hilbert's later arithmetic boundary. The proof still proceeds from Groups I–IV geometry to concrete square representatives and then to the Chapter IV-style content relation. No unit segment, segment product, numerical area, subtraction, or cancellation principle is introduced.

C.8 II.11: cancellation inside the positive content layer

II.11 adds a feature absent from the final bridge of II.10: common figures are removed twice. Hilbert's Chapter IV gives exactly the semantic control needed for this step. In Section 18, after defining equidecomposability and equicomplementability, Hilbert observes that if equidecomposable polygons are removed from equidecomposable polygons, the remaining polygons are equicomplementable [2].

The Lean helper called Common Notion 3 is a direct formal analogue at the level of the project's finite scissors terms. It does not form negative figures and does not use the later arithmetic of segments. Instead it enlarges the complement witnesses already present in the equicomplementability relation. This keeps II.11 entirely inside the positive Chapter IV-style content layer.

The proposition also shows why exact dissection and general equal content must remain distinct. II.6, II.3, II.2, and rectangle transport are used as exact `HilbertScissorsEq` inputs, but I.47 and the cancellation steps lead to the broader public equicomplementability relation.

C.9 II.12: positive excess after the II.11 cancellation step

II.12 immediately illustrates a different Chapter IV-style content mechanism from II.11. II.11 removes common figures twice; II.12 removes nothing. After the geometric order $D - A - C$ and the two right triangles have been established, the proof is forward-only:

$$S_{DC} \simeq_{\mathrm{ec}} (S_{DA} + S_{AC}) + (R + R),$$

then the same square S_{DB} is added, and the two I.47 relations replace the corresponding Pythagorean sums.

This remains entirely inside Hilbert's synthetic plane-content side of the architecture. The proposition does not require a unit segment, a segment product, an ordered numerical area, or subtraction in the later arithmetic layer. The phrase "greater by twice the rectangle" is certified by the explicit positive summand $R + R$.

The proposition also makes the separation between geometry and content especially clear. The difficult new fact $D - A - C$ is proved before any content composition begins. Only after the order and right-angle data are fixed do concrete square and rectangle representatives enter the Chapter IV- style equicomplementability calculation.

C.10 Result of the Grundlagen audit

At the II.12 checkpoint Hilbert's main contribution remains architectural. Chapter IV validates the use of a synthetic content layer independently of numerical area, while Section 15 marks a later arithmetic boundary. The current Book II formalization through II.12 remains on the earlier side of that boundary.

The rectangle API therefore has a clear foundational interpretation: congruent segments and right angles determine representative rectangles, and the scissors layer compares their content. The algebra-like behavior observed in II.1–II.3 is derived from that geometry.

C.11 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.12. At this stage the durable Hilbert-side picture is

$$\begin{array}{c}\text{Groups I–IV geometry}\\ \Downarrow \\ \text{plane-content/dissection layer}\\ \Downarrow \\ \text{exact rectangle calculus for II.1–II.8}\\ \Downarrow \\ \text{equicomplementable square identities for II.9–II.12}\end{array}$$

while Section 15 segment arithmetic remains a later comparison layer rather than a dependency of the proofs.

Appendix D

Borsuk–Szmielew: Axiomatic Layers and Representation

D.1 Purpose and scope

This appendix records the Borsuk–Szmielew-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.12. The source is used primarily to control the axiomatic layering of incidence, order, congruence, polygons, parallels, and parallelograms. The question at the beginning of Book II is how far that geometry reaches before a separate theory of content is needed.

D.2 The relevant axiomatic boundary

Borsuk–Szmielew develop rigorous polygonal and Euclidean parallelogram infrastructure, but the current audit has not located a synthetic Euclidean theory of equal content or equicomplementability parallel to Hilbert Chapter IV or Hartshorne Section 22 [3].

That absence is structurally useful. It confirms that one can separate the representation and Euclidean geometry of rectangles from the content relation used to compare them. A rectangle is a special parallelogram, and its subdivision is ordinary polygonal geometry; the assertion that the whole and the pieces have the required common content belongs to a different layer.

D.3 Comparison table

Table D.1: Euclid II.1–II.12 against the Borsuk–Szmielew axiomatic layers

Euclid	**Borsuk–Szmielew counterpart**	**Status / layer**	**Structural significance for the reconstruction**
II.1	Rectangle as a special parallelogram together with polygonal subdivision.	Geometry available; no direct content counterpart located.	The geometric configuration is natural in the source architecture, but the conclusion that the whole rectangle equals the sum of the subrectangles belongs to the separate scissors/-content layer of the formal project.

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Table D.1 – continued from previous page

Euclid	**Borsuk–Szmielew counterpart**	**Status / layer**	**Structural significance for the reconstruction**
II.2	Square/rectangle geometry and subdivision by a parallel through the division point.	Geometry available; no direct content counterpart located.	The source can support the representation of the square and its two rectangular parts. Exact scissors equality and the formal side-order transport are not supplied by the audited Borsuk–Szmielew theory.
II.3	Rectangle plus square configuration built from the same Euclidean parallelogram/right-angle infrastructure.	Geometry available; no direct content counterpart located.	No new Borsuk–Szmielew layer appears. The reduction to binary II.1 and the identification of a square with a rectangle contained by the same segment twice belong to the formal rectangle/content API.
II.4	Square and rectangle decomposition supported by Euclidean parallelogram, parallel, right-angle, and polygonal geometry.	Geometry available; no direct content counterpart located.	The source supports the geometric representation layer. The exact scissors composition of II.2 and II.3 and representative rotation remain external formal content infrastructure.
II.5	Midpoint, betweenness, square, and rectangle configuration for an internal unequal cut.	Midpoint, order, congruence, and Euclidean geometry available; no direct content counterpart located.	Borsuk–Szmielew are a useful control source for the separation of midpoint placement from segment congruence. The exact content identity and rectangle transport are supplied by the Lean scissors layer.
II.6	Midpoint plus endpoint extension, with rectangle and square configurations.	Order/congruence and Euclidean representation geometry available; no direct content counterpart located.	The source can control the one-dimensional order and parallelogram geometry, but not the exact II.1/II.3/II.4 scissors combination used by the current Lean proof.
II.7	Arbitrary internal cut with square and rectangle representatives.	Order, congruence, polygon, parallel, and rectangle geometry available; no direct synthetic equal-content counterpart located.	The formal reversal of betweenness and cyclic rectangle representation are compatible with the source's geometric layering. Exact `HilbertScissorsEq`, representative reuse, and the final additive bridge remain separate project infrastructure.
II.8	Arbitrary cut, endpoint extension by a congruent segment, and square/rectangle representatives.	Order, congruence, polygon, parallel, and Euclidean rectangle geometry are available; no direct synthetic equal-content counterpart was located.	The source layering supports the explicit $A - C - B - D$ order and the transport $BD \cong BC$. Exact II.4/II.7 scissors composition, square representative uniqueness, and the fourfold formal sum remain separate project infrastructure.

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Table D.1 – continued from previous page

Euclid	Borsuk–Szmielew counterpart	Status / layer	Structural significance for the reconstruction
II.9	Bisected segment and unequal cut with perpendicular construction, parallel through the cut point, auxiliary parallelogram, strict order on the constructed intersections, and square representatives.	Incidence, order, congruence, perpendicular/parallel, polygonal, and parallelogram layers are relevant; no direct synthetic equal-content counterpart was located in the audited source.	II.9 makes the modular boundary especially clear: the source can control the geometric representation and construction shell, while the four Pythagorean `HilbertScissorsEquicomplementable` blocks and their additive composition remain external formal content infrastructure.
II.10	Bisected segment with an endpoint extension, perpendicular and parallel construction, auxiliary parallelogram, produced intersection with strict order, and square representatives.	Incidence, order, congruence, perpendicular/parallel, polygonal, and parallelogram layers are again the relevant source-side controls; no direct synthetic equal-content counterpart was located in the audited source.	The external order $A - C - B - D$ and the directed positions $E - B - G$, $F - D - G$ belong naturally to the geometry/representation layer. The four I.47 `HilbertScissorsEquicomplementable` blocks, square transport, and their additive composition remain separate formal content infrastructure.
II.11	Square, midpoint, strict segment order, rectangle cuts, parallel uniqueness, Pasch intersection, and congruent-segment transfer.	The source contains a developed Euclidean rectangle layer and a separate ordered arithmetic of free segments, including subtraction only when a strict order $a > b$ has first been established. No direct synthetic equal-content counterpart to the project's scissors cancellation was located.	This is a useful control on the Lean separation: $EA < EB$ and $AF < AB$ belong to the ordered-segment geometry, while the two Common-Notion-3 cancellations act on plane-content representatives and must not be identified with Borsuk–Szmielew free-segment subtraction.

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Table D.1 – continued from previous page

Euclid	Borsuk–Szmielew counterpart	Status / layer	Structural significance for the reconstruction
II.12	Obtuse-angle order, perpendicular to a produced carrier, proof of the strict order $D - A - C$, right-angle transport, parallelogram cuts, Pasch intersection, and square/rectangle representatives.	Incidence, order, congruence, perpendicular/parallel, polygonal, and parallelogram layers are relevant source-side controls. No direct synthetic equal-content counterpart to the project's `HilbertScissorsEquicomplementable` calculation was located in the audited source.	II.12 sharply separates the layers. The position of the foot and the generic parallelogram-cut constructor are geometry/representation facts; II.4/I.47 content composition, square transport, and the final positive equicomplementability sum remain separate project infrastructure.

D.4 Structural observations retained from the audit

The beginning of Book II reinforces the following separation:

$$\text{polygon/rectangle geometry} \quad + \quad \text{scissors/content relation} \quad + \quad \text{representative independence.}$$

Borsuk–Szmielew are strongest on the first term. The second and third are additional layers of the present formal development. This prevents the source audit from silently attributing to the book a theory of equal content that the current audit has not located there.

D.5 II.1–II.8: representation is stable, content is external

Across II.1–II.8 the source role is stable. Rectangle, square, midpoint, order, parallel, and parallelogram geometry remain within the axiomatic geometric layers. What changes from proposition to proposition is the way the formal scissors API packages and reuses that geometry: general split in II.1, side-order transport in II.2, direct specialization in II.3, composition in II.4, midpoint configurations in II.5–II.6, and the arbitrary-cut II.4/II.3 composition in II.7, and the endpoint-extension II.4/II.7 composition with representative transport in II.8.

These are important formal distinctions, but they are not reasons to invent different Borsuk–Szmielew content counterparts. The proper source conclusion is that the same geometry/content boundary persists throughout the opening eight propositions.

D.6 II.9: a richer construction while the content layer remains external

II.9 uses substantially more of the geometric side of the architecture than the preceding rectangle identities. The formal proof tracks midpoint and strict betweenness data, erects a perpendicular, constructs parallels and an auxiliary parallelogram, proves the order of two constructed intersection points, and transports segment congruences through that configuration. These are precisely the kinds of incidence/order/congruence and Euclidean representation questions for which Borsuk–Szmielew remain a useful control source.

What does not change is the content boundary. The present audit has not located in Borsuk–Szmielew a synthetic equal-content calculus matching the project's `HilbertScissorsEquicomplementable` layer. The four Pythagorean content blocks and their final composition must therefore continue to be documented as formal project infrastructure rather than as a Borsuk–Szmielew theorem.

D.7 II.10: external order adds geometry, not a new content theory

II.10 is a particularly clear example of the source role assigned to Borsuk–Szmielew. Compared with II.9, the second point lies beyond the endpoint, so the formal proof must establish a richer one-dimensional order package and prove that the constructed intersection lies on the intended extensions. Incidence, order, congruence, parallelism, and parallelogram representation are therefore central.

The source boundary nevertheless stays fixed. The current audit has not located a Borsuk–Szmielew theory matching the project's `HilbertScissorsEquicomplementable` calculus. II.10's four Pythagorean content blocks and square-representative transports are consequently documented as project infrastructure, exactly as for II.9.

D.8 II.11: strict segment order and content cancellation must not be conflated

II.11 makes the source boundary unusually sharp. The Lean construction proves two strict segment inequalities, $EA < EB$ and $AF < AB$, before it can place the points F and H in the required order. Borsuk–Szmielew's ordered theory of free segments is a relevant control here; in particular, their subtraction of free segments is defined only after a strict comparison has been established [3].

The later content argument is a different operation. When Lean cancels a common square or rectangle from two equicomplementable formal sums, it is not subtracting one free segment from another. The operands are plane-content terms, and the proof works by enlarging scissors complements. The audited Borsuk–Szmielew source does not supply this content calculus.

The proposition-local generic rectangle-cut constructor also fits the source's geometry-and-representation role: rectangle structure, parallelism, strict betweenness, and polygonal intersection belong below the scissors relation.

D.9 II.12: order of the perpendicular foot and cut geometry stay below content

II.12 gives a particularly clean example of the Borsuk–Szmielew role in this documentation. The main new source-critical fact is the one-dimensional order

$$D - A - C.$$

Lean does not read this from the diagram. It derives the position of the perpendicular foot from the obtuse-angle hypothesis, right-angle geometry, angle order, and betweenness trichotomy. This is an incidence/order/congruence problem, not an equality-of-content problem.

The proposition-local `proposition2_12_parallelogram_cut_exists` is likewise entirely on the geometry-and-representation side. It constructs an internal cut of an arbitrary parallelogram, proves the opposite-side order, and obtains the diagonal intersection by Pasch. These witnesses are then consumed by the configured II.4 API.

Only after this geometric shell is complete does the proof enter the separate scissors layer. The audited Borsuk–Szmielew source does not supply the project's equicomplementability calculus, so the II.4/I.47 content composition and representative transports remain documented as formal project infrastructure.

D.10 Consequences for the formal library

The audit supports a modular architecture in which

- Euclidean incidence, parallelism, right-angle, and parallelogram facts live below the Book II content layer;
- rectangle representatives are built from that geometry;
- exact scissors comparison and representative independence are supplied separately;
- later Book II propositions may reuse the rectangle/content API without rebuilding the underlying Euclidean geometry.

D.11 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.12. No direct Borsuk–Szmielew content counterpart has been located for the block. The source continues to function as a geometry-and-representation control source, while the equality-of-content semantics is supplied elsewhere.

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